

Original Mixed Profile Proof Supplement

Period-zero calculation, Gamma rows, conductor transport, and complete returns

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This proof supplement reproduces the complete new period, conductor, Gamma-row, numerator-kernel, and connecting-return calculations, followed by their backward receiving arguments. Original roots, masses, period coefficients, multiplicities, signs, frames, and quotient fibres are retained. The appendices reproduce selected complete immediate prerequisites: RPC, PCL, DNE, PDS, CIT, SGT, CPQ, SFB, NTE, CPR, and BTK, including their inherited receiving calculations. Remaining unchanged foundations and every earlier proof cited by its original equation tag are supplied in the accompanying complete cumulative source and PDF edition. This supplement therefore has an explicitly stated dependency scope; the two artifacts together preserve the complete proof corpus.

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1 An evaluated original-period domain with zero conductor vanishing order

Retain the complete simple-quartet data of PCL1–11 and RPC1–26:

$$x = \delta + i\gamma, \quad y = \delta - i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad U = \text{diag}(a, \bar{a}, \bar{a}, a), \quad a \neq 0. \quad (\text{PZ1})$$

Here $a = v_h(1/2 + x)$ is the original entire-unit value; its phase is not chosen. Set $\beta = -(x^2 + y^2) = 2(\gamma^2 - \delta^2)$ and $\eta = x^2 y^2 = (\delta^2 + \gamma^2)^2$. Write $\tau = u^{-1}$ for the original inverse period, reserving w for the exponential-symbol variable. The original matrix is exactly

$$\mathbf{H}(u) = e^{a/u} G_u \mathcal{R}(\tau) V^{-1} U, \quad L_j(\tau) = U^{-1} V \mathcal{R}(\tau)^{-1} D^{-j} \mathcal{R}(\tau) V^{-1} U, \quad j = 1, 2, 3, 4. \quad (\text{PZ2})$$

The cancellation of $e^{a/u}$ and G_u in this conjugation is the literal identity $L_j = \mathbf{H}^{-1} D^{-j} \mathbf{H}$, since G_u and D commute. Thus no amplitude, period column, contour factor, or original phase has been replaced. The matrix $\mathcal{R}$ is the entire PCL2 series, with determinant one and inverse its adjugate. Put

$$\begin{aligned} d_j(\tau) &= Q_0(L_j(\tau) \mathbf{1}), \quad Q_0(t) = t_1 t_4 - t_2 t_3, \quad \mathbf{1} = (1, 1, 1, 1)^T, \\ \mathcal{E}(\tau) &= \prod_{j=1}^4 d_j(\tau), \quad E_{A,u}(w) = \sum_{b,c=0}^8 a_{bc}(\tau) e^{[4+(2b-8)\delta+i(2c-8)\gamma]w}. \end{aligned} \quad (\text{PZ3})$$

The complete coefficient product PCL8–10 proves $E_{A,u}(0) = A_\tau(1, 1) = \mathcal{E}(\tau)$, including all four nonreduced conductor factors and their actual multiplicities.

1.1 The exact limiting value, including both exceptional amplitude phases

Define

$$\begin{aligned} r &= \frac{a}{\bar{a}}, \quad T = \frac{x^2}{y^2}, \\ A_e &= \frac{2T - 1}{3(T - 1)}, \quad B_e = \frac{2 - T}{3(T - 1)}, \\ C_e &= (r - 1)(A_e/r + B_e), \quad D_e = (r - 1)(A_e/r - B_e). \end{aligned} \quad (\text{PZ4})$$

All denominators are nonzero: $a, y \neq 0$ and $x^2 - y^2 = 4i\delta\gamma \neq 0$. Neither A_e nor B_e vanishes, because $|T| = 1$, whereas their putative roots are $1/2$ and 2 . One has $A_e - B_e = 1$ and $B_e = -\bar{A}_e$, by direct substitution of $\bar{T} = T^{-1}$. Hence

$$\begin{aligned} C_e &= 1 - 2 \operatorname{Re}(A_e/r) \in \mathbb{R}, \\ D_e &= 2i(\operatorname{Im} A_e - \operatorname{Im}(A_e/r)) \in i\mathbb{R}. \end{aligned} \quad (\text{PZ5})$$

For $\rho_j = \zeta^{-j}$ and $h_j = \rho_j^3 - \rho_j$, the full value is

$$\boxed{d_j(0) = h_j D_e (2\rho_j + h_j C_e)}. \quad (\text{PZ6})$$

Here is a direct proof retaining both off-diagonal entries in PCL4. On even centered coefficients (p_0, p_2) , its middle conjugation is

$$\begin{pmatrix} \rho_j & \beta(\rho_j^3 - \rho_j)/3 \\ 0 & \rho_j^3 \end{pmatrix}.$$

Evaluation at (x, y) gives the matrix $\begin{pmatrix} \rho_j + h_j A_e & -h_j A_e \\ h_j B_e & \rho_j - h_j B_e \end{pmatrix}$. Indeed $(x^2 + \beta/3)/(x^2 - y^2) = A_e$ and $(y^2 + \beta/3)/(x^2 - y^2) = B_e$. Conjugation by the original $\text{diag}(a, \bar{a})$ gives

$$\begin{pmatrix} \rho_j + h_j A_e & -h_j A_e/r \\ r h_j B_e & \rho_j - h_j B_e \end{pmatrix}.$$

The input $\mathbf{1}$ has even evaluation vector $(1, 1)^T$ and zero odd evaluation vector. Its two output values are therefore $p_j = \rho_j + h_j A_e(1 - r^{-1})$ and $q_j = \rho_j + h_j B_e(r - 1)$, each repeated at the opposite root. The original quadratic is exactly $p_j^2 - q_j^2$. Its difference is $h_j D_e$ and its sum is $2\rho_j + h_j C_e$, proving PZ6. This calculation is an evaluation through explicit invertible coefficient and primary maps, not an assignment of the unit.

No factor $2\rho_j + h_j C_e$ vanishes. In fact its vanishing would imply $2 + (\rho_j^2 - 1)C_e = 0$. The imaginary part forces $C_e = 0$ because ρ_j^2 is nonreal, and the real part would then be $2 = 0$. Also $h_j \neq 0$. Consequently

$$d_j(0) = 0 \iff r = 1 \text{ or } r = A_e/B_e, \quad j = 1, 2, 3, 4. \quad (\text{PZ7})$$

The second ratio is a unit complex number and is distinct from one. Thus this is an exhaustive two-phase statement about the actual amplitude, with neither phase omitted.

Multiplying all four PZ6 values gives the exact real expression

$$\boxed{\mathcal{E}(0) = 5D_e^4(5(C_e - 1)^4 + 10(C_e - 1)^2 + 1).} \quad (\text{PZ8})$$

To verify the constant, $\prod_j \rho_j = 1$ and $\prod_j (\rho_j^2 - 1) = 5$. The remaining product is

$$\prod_{j=1}^4 (2 - C_e + C_e \rho_j^2) = \frac{(2 - C_e)^5 + C_e^5}{2} = 5(C_e - 1)^4 + 10(C_e - 1)^2 + 1.$$

The first equality follows by including the missing fifth root one in $\prod_{\xi^5=1} (s + C_e \xi) = s^5 + C_e^5$, then dividing by $s + C_e = 2$, so it remains valid at $C_e = 0$. Since D_e is purely imaginary, its fourth power is positive when it is nonzero. PZ8 is therefore strictly positive off exactly the two phases in PZ7. At those phases its value is zero; this inverse-period limit alone makes no assertion of zero value at a finite original period.

1.2 A finite original-period domain and the complete zero-order test

The PCL22 identity gives $L_j(-\tau) = P_{\text{opp}} L_j(\tau) P_{\text{opp}}$. Since $P_{\text{opp}} \mathbf{1} = \mathbf{1}$ and $Q_0(P_{\text{opp}} t) = Q_0(t)$, each d_j is even entire. Thus write

$$d_j(\tau) = \sum_{n \geq 0} d_{j,2n} \tau^{2n}, \quad C_j^{(0)} = \sum_{n \geq 1} |d_{j,2n}| < \infty. \quad (\text{PZ9})$$

Every coefficient is explicitly obtained by multiplying PCL2 and its finite adjugate in PZ2 and then the two products in PZ3. Absolute convergence at radius one proves the displayed finiteness. For an actual amplitude with $r \notin \{1, A_e/B_e\}$, define

$$R_0 = \max \left\{ R_*, 1, \max_{1 \leq j \leq 4} \sqrt{2C_j^{(0)} / |d_j(0)|} \right\}. \quad (\text{PZ10})$$

Here R_* is exactly RPC21 or RPC23 for this original quartet and unit, so this domain already lies in the proved five-orbit locus. For every original logarithm branch and $|u| \geq R_0$,

$$|d_j(u^{-1}) - d_j(0)| \leq C_j^{(0)} |u|^{-2} \leq \frac{1}{2} |d_j(0)|, \quad \frac{\mathcal{E}(0)}{16} \leq |E_{A,u}(0)| \leq \frac{81\mathcal{E}(0)}{16}, \quad \text{ord}_{w=0} E_{A,u}(w) = 0. \quad (\text{PZ11})$$

The error follows by summing the entire series at $|u|^{-1} \leq 1$; if $C_j^{(0)} = 0$ it is identically zero. Multiplication of the four lower and upper bounds gives the factors 2^{-4} and $(3/2)^4$. PZ8 supplies $\prod_j |d_j(0)| = \mathcal{E}(0) > 0$. A holomorphic function with nonzero value has zero vanishing order, proving the last equality without a genericity hypothesis on the period in this domain. Single-valuedness follows from the proved conjugation PZ2.

At arbitrary original periods on the five-orbit locus, the full test is still explicit: expand each actual $(2, 2)$ factor $q_j(\tau; Z, W) = \sum_{b,c=0}^2 c_{j,bc}(\tau) Z^b W^c$ and set

$$E_{j,u}(w) = \sum_{b,c=0}^2 c_{j,bc}(\tau) e^{[1+(2b-2)\delta+i(2c-2)\gamma]w},$$

$$v_j = \min \left\{ 0 \leq n \leq 8 : \sum_{b,c=0}^2 c_{j,bc}(\tau) [1 + (2b-2)\delta + i(2c-2)\gamma]^n \neq 0 \right\}, \quad (\text{PZ12})$$

$$v = \sum_{j=1}^4 v_j, \quad E_{A,u} = \prod_{j=1}^4 E_{j,u}, \quad \frac{E_{A,u}^{(v)}(0)}{v!} = \prod_{j=1}^4 \frac{E_{j,u}^{(v_j)}(0)}{v_j!} \neq 0.$$

The nine exponents in each factor are distinct by their real and imaginary parts. Their nine-by-nine derivative Vandermonde is invertible. On the five-orbit locus the factor is nonzero by the Segre restriction argument in BTK26, so the displayed minimum exists. This proves the exact retained test, including all multiplicities, and $v \leq 32$ on every remaining fibre.

For positive real original u , the actual real-period identity PCL15 also proves v is even. Indeed for real w put $Z = e^{2\delta w}$ and $W = e^{2i\gamma w}$. The full exponential factorization gives

$$E_{A,u}(w) = e^{(4-8\delta-8i\gamma)w} A_\tau(Z, W) = e^{(4-8\delta)w} W^{-4} A_\tau(Z, W) \geq 0. \quad (\text{PZ13})$$

The last expression is the exact squared modulus in PCL15. The function is nonzero analytic on the five-orbit locus. Its first nonzero Taylor term at zero must have even degree and positive real coefficient, because otherwise its real values would change sign on the two sides of zero. Hence $v \in \{0, 2, 4, \dots, 32\}$ on that positive-real-period locus. This retains the actual curve in PCL15, with one real coordinate and one unit-circle coordinate.

On the evaluated domain PZ11 the low-coordinate space $\mathbb{C}^v$ in ROG26 is zero, $Q_{N,s}$ in ROG23 is the zero map, and the full ROG27 shear is exactly the identity for both original source orders $s = 1, k$. Its determinant and every exterior power have norm one; the ROG32 graph-shear error is therefore exactly zero. The polynomial relation quotient and its remaining original metric still retain the full conductor, complete action, and the original residual map. No connecting-rank conclusion has been inferred from this zero-order calculation; the exact target relation is the unchanged RDS8 sequence with $\mathcal{N}_0 = \mathcal{E}_k$ and dimension $q - q'$.

2 The two exceptional unit phases have exact second inverse-period order

Retain the original roots

$$x = \delta + i\gamma, \quad y = \delta - i\gamma, \quad X = x^2, \quad Y = y^2, \quad \beta = -(X + Y), \quad \eta = XY, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2.$$

The original ordered evaluation roots are $(x, y, -y, -x)$. Let V have rows $(1, w, w^2, w^3)$ at those four roots, let $U = \text{diag}(a, \bar{a}, \bar{a}, a)$ with $a \neq 0$, and put $r = a/\bar{a}$. This notation retains the exact scalar map $U = \bar{a} \text{diag}(r, 1, 1, r)$; the scalar $\bar{a}$ cancels between U^{-1} and U in the original conjugation. For $j = 1, 2, 3, 4$, put $\rho = \zeta^{-j}$ and $D_\rho = \text{diag}(\rho, \rho^2, \rho^3, \rho^4)$. The original PCL2 coefficient matrix is denoted by $R(z) = J + R_1 z + R_2 z^2 + \dots$, and the original expression is

$$q_\rho(z) = Q_0 (U^{-1} V R(z)^{-1} D_\rho R(z) V^{-1} U \mathbf{1}), \quad Q_0(t) = t_1 t_4 - t_2 t_3.$$

Every map in this expression has its original domain, codomain and order. In particular $V^{-1} U \mathbf{1} = \bar{a} c_r$, where

$$c_r = \left(\frac{X - rY}{X - Y}, 0, \frac{r - 1}{X - Y}, 0 \right)^\top.$$

Evaluation of the even polynomial represented by c_r at $x, y, -y, -x$ gives $(r, 1, 1, r)$, proving this identity directly.

2.1 Finite coefficients obtained from the complete original series

In the displayed matrices below write $b = \beta$ and $e = \eta$. PCL2 gives, with row indices $a = 1, 2, 3, 4$ and column indices $s = 0, 1, 2, 3$,

$$(R_n)_{a,s} = \sum_{\substack{p,h \geq 0 \\ 2p+4h=5n+s+1-a}} \frac{b^p e^h}{3^p p! h!} (-1)^{p+h-n} 5^{p+h-n} (a/5)_{p+h-n}.$$

For $n = 0, 1, 2$ the finite sum yields respectively

$$J = \begin{pmatrix} 1 & 0 & -b/3 & 0 \\ 0 & 1 & 0 & -2b/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$R_1 = \begin{pmatrix} 0 & b^3/27 - be/3 & 0 & -11b^4/324 + b^2e/3 - e^2/2 \\ -b^2/9 + e & 0 & 7b^3/81 - 2be/3 & 0 \\ 0 & -b^2/6 + e & 0 & 4b^3/27 - be \\ b/3 & 0 & -2b^2/9 + e & 0 \end{pmatrix},$$

and R_2 has its eight nonzero entries

$$\begin{aligned} (R_2)_{1,0} &= -11b^5/4860 + b^3e/27 - be^2/6, & (R_2)_{1,2} &= 22b^6/10935 - 11b^4e/324 + b^2e^2/6 - e^3/6, \\ (R_2)_{2,1} &= -7b^5/1215 + 7b^3e/81 - be^2/3, & (R_2)_{2,3} &= 119b^6/21870 - 7b^4e/81 + 7b^2e^2/18 - e^3/3, \\ (R_2)_{3,0} &= b^4/81 - b^2e/6 + e^2/2, & (R_2)_{3,2} &= -13b^5/1215 + 4b^3e/27 - be^2/2, \\ (R_2)_{4,1} &= b^4/54 - 2b^2e/9 + e^2/2, & (R_2)_{4,3} &= -7b^5/405 + 2b^3e/9 - 2be^2/3. \end{aligned}$$

Thus the second-order calculation uses every term of the original series that contributes at these orders. Multiplication of power series gives the inverse coefficients

$$S_0 = J^{-1}, \quad S_1 = -J^{-1}R_1J^{-1}, \quad S_2 = J^{-1}R_1J^{-1}R_1J^{-1} - J^{-1}R_2J^{-1}.$$

Set

$$f_n = \sum_{i=0}^n S_i D_\rho R_{n-i} c_r \quad (0 \leq n \leq 2).$$

The parity visible in PCL2 implies that f_0, f_2 have only entries indexed 0, 2, while f_1 has only entries indexed 1, 3. For a vector f in polynomial coefficient coordinates define $E_f(T) = f_0 + Tf_2$ and $O_f(T) = f_1 + Tf_3$. Expansion of the original two products defining Q_0 now proves

$$[z^2]q_\rho(z) = \frac{2E_{f_0}(X)E_{f_2}(X) - XO_{f_1}(X)^2}{r^2} - 2E_{f_0}(Y)E_{f_2}(Y) + YO_{f_1}(Y)^2. \quad (\text{PZX1})$$

This is also a short finite certificate for every coefficient below: substitute the displayed matrices and $b = -(X + Y)$, $e = XY$ into PZX1, and reduce powers by $1 + \rho + \rho^2 + \rho^3 + \rho^4 = 0$.

For completeness the absence of odd orders holds to every order. Let $S = \text{diag}(1, -1, 1, -1)$ in polynomial coordinates. PCL2 gives $R(-z) = SR(z)S$, and S commutes with D_ρ and fixes c_r . Consequently replacing z by $-z$ replaces the resulting polynomial $f(w)$ by $f(-w)$. The original products $f(x)f(-x)/r^2$ and $f(y)f(-y)$ are unchanged. Hence $q_\rho(-z) = q_\rho(z)$.

2.2 The phase $r = 1$

Substitution of $r = 1$ in PZX1 gives

$$[z^2]q_\rho(z) = -\frac{(X - Y)\beta^2(\beta^2 - 9\eta)}{81}(\rho^3 + 2\rho^2 + 3\rho - 1). \quad (\text{PZX2})$$

The constant coefficient vanishes by the same matrices. Multiplication by $\rho - 1$ and use of the fifth-root relation gives

$$(\rho - 1)(\rho^3 + 2\rho^2 + 3\rho - 1) = -5\rho.$$

The coefficient PZX2 is nonzero: $X - Y = 4i\delta\gamma \neq 0$, $\beta = 2(\gamma^2 - \delta^2) > 0$, and

$$\beta^2 - 9\eta = -5\gamma^4 - 26\delta^2\gamma^2 - 5\delta^4 < 0.$$

Therefore $q_\rho(z) = ([z^2]q_\rho)z^2 + O(z^4)$ has exact order two. Multiplication over $\rho = \zeta^{-1}, \dots, \zeta^{-4}$ gives

$$\prod_{j=1}^4 q_{\zeta^{-j}}(z) = 125 \left(\frac{(X - Y)\beta^2(\beta^2 - 9\eta)}{81} \right)^4 z^8 + O(z^{10}), \quad (\text{PZX3})$$

because $\prod_j \rho = 1$ and $\prod_j (\rho - 1) = 5$.

2.3 The phase $r = (2X - Y)/(2Y - X)$

Put $t = \beta^2/\eta$, retaining this exact original ratio. Define four real polynomials

$$\begin{aligned} A_3(t) &= 52t^3 - 1053t^2 + 7776t - 17496, \\ A_2(t) &= 4t^3 + 324t^2 - 3888t + 8748, \\ A_1(t) &= 96t^3 - 999t^2 + 3888t - 8748, \\ A_0(t) &= 18t^3 - 297t^2 + 1944t - 4374, \\ P_\rho(t) &= A_3(t)\rho^3 + A_2(t)\rho^2 + A_1(t)\rho + A_0(t). \end{aligned}$$

The constant coefficient again vanishes. PZX1 gives the exact value

$$[z^2]q_\rho(z) = -\frac{(X - Y)(\beta^2 - 9\eta)\eta}{1458(2t - 9)^2} P_\rho(t). \quad (\text{PZX4})$$

An explicit way of checking the homogeneous re-expression in PZX4 is to use

$$(X - 2Y)(2X - Y) = 2(X + Y)^2 - 9XY = \eta(2t - 9)$$

and the symmetric numerator

$$\begin{aligned} \mathcal{P} &= (52\rho^3 + 4\rho^2 + 96\rho + 18)(X^6 + Y^6) \\ &\quad + (-741\rho^3 + 348\rho^2 - 423\rho - 189)(X^5Y + XY^5) \\ &\quad + (4344\rho^3 - 2532\rho^2 + 1332\rho + 1026)(X^4Y^2 + X^2Y^4) \\ &\quad + (-7222\rho^3 + 2996\rho^2 - 5046\rho - 1908)X^3Y^3. \end{aligned}$$

The direct PZX1 numerator is $-(X - Y)(X^2 - 7XY + Y^2)\mathcal{P}$ over $1458(X - 2Y)^2(2X - Y)^2$. The identities for $X^6 + Y^6$, $X^5Y + XY^5$ and $X^4Y^2 + X^2Y^4$ in the elementary symmetric polynomials give $\mathcal{P} = \eta^3 P_\rho(t)$, proving PZX4 without changing the original roots.

Here is an explicit nonvanishing certificate valid for every real t . Set $c = \rho + \rho^{-1}$; then c is real and $c^2 + c - 1 = 0$. Reduction by $\rho^2 - c\rho + 1 = 0$ gives

$$P_\rho(t) = F(t, c) + G(t, c)\rho, \quad F = A_0 - A_2 - cA_3, \quad G = A_1 + c(A_2 - A_3).$$

Define

$$\begin{aligned} U(t, c) &= (1536 - 768c)t^2 + (18432c - 8784)t + 28755 - 104085c, \\ W(t, c) &= (832c - 224)t^2 + (8886 - 12948c)t + 67905c - 52785. \end{aligned}$$

Polynomial multiplication, collecting coefficients of $t^0, \dots, t^5$ and replacing c^2 by $1 - c$, gives the full identity

$$-U(t, c)F(t, c) - W(t, c)G(t, c) = 6495390(11c - 7). \quad (\text{PZX5})$$

The right side is nonzero: $c = 7/11$ would give $c^2 + c - 1 = 5/121 \neq 0$. Since F, G are real and ρ is nonreal, $P_\rho(t) = 0$ would force $G = 0$ by imaginary parts and then $F = 0$, contradicting PZX5. Thus $P_\rho(t) \neq 0$ for every real t . For the actual parameters,

$$t = 4 \left(\frac{\gamma^2 - \delta^2}{\gamma^2 + \delta^2} \right)^2, \quad \frac{900}{289} < t < 4.$$

The lower bound follows from $0 < \delta/\gamma < 1/4$ and the strict decrease of $(1-v)/(1+v)$ for $0 < v < 1$; the upper bound is strict since $\delta > 0$. In particular $2t - 9 < 0$. All factors in PZX4 are therefore nonzero, and each original factor has exact order two. Their product has exact order eight, with its complete nonzero leading coefficient

$$\left(\frac{(X - Y)(\beta^2 - 9\eta)\eta}{1458(2t - 9)^2} \right)^4 \prod_{j=1}^4 P_{\zeta^{-j}}(t). \quad (\text{PZX6})$$

The last product is strictly positive: conjugate roots pair the factors into $|P_{\zeta^{-1}}(t)|^2 |P_{\zeta^{-2}}(t)|^2 > 0$. Also $(X - Y)^4 = (4\delta\gamma)^4 > 0$. Hence PZX6 is positive. No secondary exceptional locus occurs in either of these two original unit phases anywhere on the stated (δ, γ) domain.

2.4 One finite radius for every actual amplitude and the original receiving maps

Use the complete coefficient calculation PZX1–PZX6, which proves that both exceptional phases in PZ7 have nonzero inverse-period coefficient $d_{j,2}$ for every j . Define, from the actual value a ,

$$\ell_A = \begin{cases} 0, & a/\bar{a} \notin \{1, A_e/B_e\}, \\ 2, & a/\bar{a} \in \{1, A_e/B_e\}, \end{cases} \quad b_j = d_{j, \ell_A} \neq 0, \quad C_j = \sum_{n \geq 1} |d_{j, \ell_A + 2n}| < \infty. \quad (\text{PZU1})$$

The symbol ℓ_A describes vanishing at the inverse-period limit; it is not the symbol-variable vanishing order v . The leading b_j is exactly PZ6 or the applicable PZX formula, retaining the original amplitude phase and every constant. Put

$$R_A = \max \left\{ R_*, 1, \max_{1 \leq j \leq 4} \sqrt{2C_j/|b_j|} \right\}. \quad (\text{PZU2})$$

The Taylor series of $\tau^{-\ell_A} d_j(\tau)$ is entire after its removable value at zero is included, proving convergence of C_j . For $|u| \geq R_A$, every original branch satisfies

$$\left| u^{\ell_A} d_j(u^{-1}) - b_j \right| \leq C_j |u|^{-2} \leq \frac{1}{2} |b_j|. \quad (\text{PZU3})$$

Consequently the complete, original zero-symbol value has the bounds

$$\boxed{\frac{\prod_j |b_j|}{16|u|^{4\ell_A}} \leq |E_{A,u}(0)| \leq \frac{81 \prod_j |b_j|}{16|u|^{4\ell_A}}, \quad v = \text{ord}_{w=0} E_{A,u}(w) = 0.} \quad (\text{PZU4})$$

All inequalities follow by multiplying four nonzero factors from PZU3. This proves an original-period domain for every original nonzero amplitude, with neither of the two exceptional phases excluded. The two phases merely change the order of decay at infinite period from zero to eight. No unit value or source order has been assigned.

In the previously proved five-orbit domain $|u| \geq R_*$, the exact exceptional set is

$$\mathcal{Z}_A^{\geq R_*} = \left\{ u \in \mathbb{C}^* : |u| \geq R_*, \quad \prod_{j=1}^4 Q_0(L_j(u^{-1})\mathbf{1}) = 0 \right\}. \quad (\text{PZU5})$$

It is a finite set, with every point satisfying $R_* \leq |u| < R_A$. For a proof, $\mathcal{E}(\tau)$ is a nonzero entire function, with zero order exactly $4\ell_A$ at $\tau = 0$. Its zeros in the compact closed disk $|\tau| \leq R_*^{-1}$ are finite: any infinite sequence would have an accumulation point, contradicting the first nonzero local Taylor coefficient of a nonzero holomorphic function. Removing zero and applying $u = \tau^{-1}$ proves the finite-set assertion. PZU4 proves the strict upper-radius bound. On every point of this finite set the complete PZ12 derivative test retains the actual multiplicity and $v \leq 32$; no transcendental root is discarded. On the complement of this set in $|u| \geq R_*$, $v = 0$ exactly.

For clarity the consequences are now substituted into the original spaces and maps. For every degree $n = 4l + 1 \geq 9$, the original low space is $L_n = \mathbb{C}[S]_{<v} = 0$. Therefore

$$\begin{aligned} t_{0,n} &= \dim(K_n \cap L_n) = 0, & m_n &= q_n - q'_n - v = 16n - 48, \\ t_n^{\text{EFP}} &= 8n - 16 - t_{0,n} = 8n - 16, & j_{0,n} &= 8n - 32 - v + t_{0,n} = 8n - 32. \end{aligned} \quad (\text{PZU6})$$

Here t_n^{EFP} denotes the kernel-image rank called t_n in EFP1. The separate CFR common-factor target rank in RPK remains its proved value zero on its original domain; the two rank symbols have different defining maps. The exact RDS8 quotient becomes

$$0 \longrightarrow \mathcal{E}_{n-8} \xrightarrow{M_{E_A}} \mathcal{E}_n \xrightarrow{\mathfrak{F}_A} (\mathbb{C}[S']/\mathcal{UC}[S])^\vee \longrightarrow 0, \quad f_{\mathfrak{F}_A(N)} = \chi'(D)(N/E_A).$$

Multiplication is the original conductor map and the last quotient has dimension $16n - 48$. The division is holomorphic at zero because PZU4, or membership in the complement in PZU5, proves $E_A(0) \neq 0$. The exact prefactor is $\alpha_A = 1/E_A(0)$, which remains in every original moment map; it has not been set to one.

For the original source degree $k \geq 9$ and target $j = k + 4 \geq 13$, at the two original endpoint indices $e = 0, 1$, BTK25 gives $0 \leq \kappa_e \leq 3 \max(v - e - 2, 0) = 0$. The exact RRC14 sequence then has zero middle space, so its two end spaces vanish. Equivalently

$$\begin{aligned} \mathcal{Z}_e &= \mathcal{S}_e, & \kappa_e &= d_{K,e} = d_{T,e} = 0, & \text{rank } \Psi_e &= c_e, \\ \omega_e(N) &= \omega_{\text{conn},e}(N) \end{aligned} \quad (\text{PZU7})$$

for both unchanged native sources and every retained later target cutoff. To verify the metric assertion, both quotient-value maps now have exactly the same coefficient domain and fibres; minimizing the same full positive native form on each fibre gives the same attained minimum. Thus the first, third and fourth returns in RRC16 are exactly zero, and its second bound becomes $|\Delta \mathcal{R} \log \omega_e| \leq c_e W_j$ with its same four signs and full original width. The connecting rank itself remains

$$c_0 = \dim \mathcal{G}_k, \quad c_1 = \dim \mathcal{G}_k - \dim(\mathcal{G}_k \cap \mathcal{C}_k), \quad (\text{PZU8})$$

through the exact RRC3–11 morphism. None of PZU1–7 assigns a value to these two dimensions. The surviving connecting map, its native minimum sections and its complete signed return remain in the original mixed-state calculation.

The scalar in the same original maps has a quantitatively evaluated finite-period range. Taking reciprocals in PZU4 gives exactly

$$\frac{16|u|^{4\ell_A}}{81 \prod_j |b_j|} \leq |\alpha_A| = |E_{A,u}(0)|^{-1} \leq \frac{16|u|^{4\ell_A}}{\prod_j |b_j|}. \quad (\text{PZU9})$$

In particular its size is of order $|u|^8$ at either exceptional amplitude phase. The vanishing-order result does not remove this scalar. For the original CPQ17 analytic coefficient columns put $\mu_d = E_{A,u}^{(d)}(0)$ and retain $\nu_{a,m} = \sum_{b,c} (Z_{\text{exp},j})_{bc,a} \beta_{j,b,c}^m$. Since $v = 0$ and $\mu_0 \neq 0$, the literal division recursion is

$$f_{a,m} = \mu_0^{-1} \left(\nu_{a,m} - \sum_{d=1}^m \binom{m}{d} \mu_d f_{a,m-d} \right), \quad m \geq 0. \quad (\text{PZU10})$$

Here the empty sum for $m = 0$ is zero. To prove the identity, differentiate $E_{A,u}(w)F_a(w) = N_a(w)$ exactly m times at zero. Leibniz' rule gives $\nu_{a,m} = \sum_{d=0}^m \binom{m}{d} \mu_d f_{a,m-d}$; solving its single nonzero leading term gives PZU10. This is CPQ17 with the proved value of v , not a new choice of rows. The full unchanged Gamma recurrence and row metrics are

$$\begin{aligned}
b_{0,m} &= \mathbf{1}_{m=0}, & b_{-1,m} &= 0, \\
b_{n+1,m} &= i^{-1}(b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2)b_{n-1,m}, \\
h_{n,s_1} &= (2\pi)^{s_1/2} n! (s_1/2)_n, & c'_j &= j/2 - 4, \\
(t_n)_a &= h_{n,s_1}^{-1/2} \sum_m b_{n,m} f_{a,m}, & (v_n)_{b,c} &= h_{n,s_1}^{-1/2} \sum_m b_{n,m} \beta_{j-8,b,c}^m.
\end{aligned} \tag{PZU11}$$

All arrays outside their original polynomial range are zero. The symbols $b_{n,m}$ in this formula are the original polynomial coefficients and retain both source orders $s_1 = 1, j$; the b_j in PZU9 are instead the explicitly evaluated inverse-period leading coefficients from PZU1. PZU10–11 feed the unchanged matrices CPQ15–18, preserving every old minimum section, zero row, full source mass, and signed endpoint return. Thus the period computation sets the low vanishing order to zero while carrying its complete division coefficient into the actual mixed-state rows.

3 The original three-equation transfer of every connecting return

Fix the simple quartet, its actual amplitude, period and logarithm branch on the original five-orbit locus. Keep $k = 4l + 1 \geq 9$, $j = k + 4$, $q = (k + 1)^2$, $Q = (j + 1)^2$, $d = Q - q = 8k + 24$, and the full nonreduced conductor

$$E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{s,a,b} z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v. \quad (\text{PRT1})$$

No conductor factor is removed. Write $B = B_{\text{cof}}$, $\mathcal{U} = \mathbb{C}^{4m_j}$, $\mathbb{J} = \ker \mathbb{B}$, $\mathcal{B} = \text{im } B$ and $W = \mathcal{B} + \mathbb{J}$, as in CPQ2. Define, in this same coefficient space,

$$S_e = \text{im}[B, F_e], \quad K_e = S_e \cap W, \quad T_e = S_e + W \quad (e = 0, 1). \quad (\text{PRT2})$$

These are the CPQ spaces, with S_e abbreviated only to make the two endpoint subscripts legible. The later norm cutoff N is independent of the defining endpoint e . At every $Q - 1 \leq N \leq 2Q$ use exactly

$$D = N - v, \quad \mathcal{D}_N^{(s)} = \text{diag}(D_{j,D}^{(s)}, 4), \quad c'_j = j/2 - 4, \quad h_{n,s} = (2\pi)^{s/2} n! (s/2)_n, \quad s = 1 \text{ or } j. \quad (\text{PRT3})$$

Thus both actual source pairs $(1, 1)$ and (k, j) are covered in their target order. Their degree- k proper source retains its separate form; none of the identities below substitutes that form for A_{inv} .

3.1 The actual three rows in the original period coefficients

Retain the original polynomials $L_i = (\mathbf{H}(ZW, Z, W, 1)^\top)_i$, $b_{i,a,b} = [Z^a W^b] L_i^4$, the exponential $B_i(z)$ and the translation maps $\mathcal{T}_i, \mathcal{T}_A$ of CPR2. In particular

$$R_i(S) = \frac{\sum_{a,b=0}^4 b_{i,a,b} \chi_j(S + \beta_{4,a,b})}{\chi_k(S)}, \quad b_i = B_i(0), \quad \chi_n(S) = \prod_{a,b=0}^n (S - \beta_{n,a,b}). \quad (\text{PRT4})$$

The quotient is the complete polynomial quotient: root addition makes every numerator vanish at every distinct root of χ_k . Choose the first original index i_* with $b_{i_*} \neq 0$, and put

$$C_i = R_i - (b_i/b_{i_*}) R_{i_*} \quad (i \neq i_*), \quad V = ([S^a] C_i)_{0 \leq a < d, i \neq i_*}. \quad (\text{PRT5})$$

Existence of i_* and independence of these three columns are the complete PDS10–11/CPR17 calculation: the original Bezout identity at zero gives a nonzero b_i ; the four R_i are independent; their leading coefficients are b_i . Subtracting their exact leading multiples therefore gives three independent columns of degree at most $d - 1$. No period coefficient is replaced in this operation.

Let h_b be the original primitive germs used to define the d columns of F_0 in CIT7. They are specified modulo the entire raw analytic source. Define their relation-value matrix by

$$(M_0)_{a,b} = \Lambda_{h_b}(\mathcal{T}_A(\chi_k S^a)), \quad 0 \leq a < d, \quad L = V^\top M_0. \quad (\text{PRT6})$$

Every entry uses the ordinary, complex-linear transpose. The complete annihilator identity CPR9–10 proves that the displayed d independent relation polynomials are the full dual of the raw source quotient. The h_b are a basis of that quotient, so M_0 is invertible. Thus L has rank exactly three. The matrix is computable from the original period arrays by finite polynomial division and pairing; it contains neither a new numerical model nor a selected nonzero period minor.

Choose once a full column basis Z of $\ker L$ and a three-column complement Y so that $[Z, Y]$ is invertible. Then the literal coefficient spaces obey

$$\boxed{S_1 = \mathcal{B} + F_0 \ker L \subset S_0, \quad S_0/S_1 \text{ has the three lifts } F_0 Y.} \quad (\text{PRT7})$$

Here equality is of subspaces of the original $\mathcal{U}$, not only of abstract quotient dimensions. To prove it, use PDS11: truncation of the endpoint-one primitive quotient identifies it with exactly $\ker V^\top$ in endpoint-zero relation-value coordinates. The missing top value is uniquely $-u_*^\top x_0/b_{i_*}$ with the complete u_* from PDS10. For a representative h , set $f = E_A h$. The original target numerator in component i is

$$N_i(f) = \mathbb{V}_j^{-1}(r![z^r](B_i f))_{0 \leq r < Q}, \quad (\mathbb{V}_j)_{r,(a,b)} = \beta_{j,a,b}^r. \quad (\text{PRT8})$$

Those values use only the truncation through degree $Q - 1$. At endpoint one, the additional four equations assert exactly that the same interpolants also have the required next jet; they do not alter a coefficient in PRT8. Division by the complete E_A gives the same target analytic functions, modulo precisely the original lower exponentials. An analytic source change has numerator d and changes the four target numerators by $(L_i^4 d)_i$, hence changes their DNE coordinates by Bx with the full original scalar α_A . This is the coefficient square RRC5–9/CIT8. It proves both inclusions in PRT7, including every analytic-source and lower-gauge change. Finally F_0 induces an injection modulo $\mathcal{B}$ by CIT9; the invertible matrix $[Z, Y]$ therefore proves the asserted dimension and the stated three lifts.

The coefficient change can itself be written explicitly. For the original endpoint-one primitive columns $h_{1,b}$ put $(M_1)_{a,b} = \Lambda_{h_{1,b}}(\mathcal{T}_A(\chi_k S^a))$ for $0 \leq a \leq d$, and let $P_{<d}$ select the first d rows. Then

$$\Gamma = M_0^{-1} P_{<d} M_1, \quad \text{im } \Gamma = \ker L, \quad F_1 = F_0 \Gamma + B H \quad (\text{PRT7a})$$

for the unique matrix H with the displayed equality. Indeed PDS11 proves the first two assertions, including that Γ has rank $d - 3$. The preceding numerator interpolation calculation shows that each difference column lies in $\mathcal{B}$. The original injectivity of B gives its unique coordinate in H . Equivalently $\Gamma = ZT$ for an invertible $(d - 3)$ -square matrix T , so every original fixed frame change in PRT7 is specified by a literal coefficient equation. No conjugation occurs in M_0, M_1, L, Γ . Hermitian adjoints enter only when evaluating the actual metric below.

3.2 An exact sequence of these three directions

PRT7 gives $K_1 \subset K_0$ and $T_1 \subset T_0$. The identity on coefficient representatives gives the exact sequence

$$0 \longrightarrow K_0/K_1 \xrightarrow{[x] \mapsto [x]} S_0/S_1 \xrightarrow{[x] \mapsto [x]} T_0/T_1 \longrightarrow 0. \quad (\text{PRT9})$$

The first map is injective because $K_0 \cap S_1 = K_1$. If $x \in S_0$ maps to zero on the right, write $x = s + w$ with $s \in S_1$, $w \in W$; then $w = x - s \in K_0$, so its class is the image of K_0/K_1 . Conversely every such class maps to zero. Every element of $T_0 = S_0 + W$ has the class of its S_0 summand modulo $T_1 = S_1 + W$, proving surjectivity. Let

$$a = \dim(T_0/T_1) = \dim(\mathcal{G}_k \cap \mathcal{C}_k) \quad \text{from CPR17–18.} \quad \dim(K_0/K_1) = 3 - a, \quad 0 \leq a \leq 3. \quad (\text{PRT10})$$

The equality with CPR's actual polynomial intersection follows because $\dim(T_e/W) = c_e$ and CPR18 proves $c_0 - c_1 = a$. It can also be calculated directly in the three columns $F_0 Y$: take their images modulo T_1 , retaining the full $\mathbb{B}$, B and L of PRT2/PRT6. Their exact rank is a and their kernel has dimension $3 - a$. This proves a complete original-data rank computation without assuming a generic minor is nonzero. Rank strata, including $a = 0$, are retained.

Give the three terms of PRT9 their original attained metrics

$$A_N = \text{Gram}_{\mathcal{D}_N}(S_0/S_1), \quad B_N = \text{Gram}_{\mathcal{D}_N}(K_0/K_1), \quad C_N = \text{Gram}_{\mathcal{D}_N}(T_0/T_1). \quad (\text{PRT11})$$

Every value frame is fixed in N and in the source-order comparison. For example if X_1 is any fixed independent frame of S_1 , the first one, in the exact three lifts $X_A = F_0 Y$, is

$$A_N = X_A^* \mathcal{D}_N X_A - X_A^* \mathcal{D}_N X_1 (X_1^* \mathcal{D}_N X_1)^{-1} X_1^* \mathcal{D}_N X_A. \quad (\text{PRT12})$$

Select the first independent images of F_0Y modulo T_1 to obtain the a lifts X_C . A full basis of $\ker(F_0Y \bmod T_1)$, followed by subtracting an S_1 representative as in PRT9, gives $3 - a$ lifts X_B in K_0 . Applying PRT12 with (X_1, X_A) replaced by the frames of (K_1, X_B) and (T_1, X_C) gives B_N and C_N . All three are positive definite on their stated quotient values; the determinant of an empty quotient is one.

For completeness every fixed frame factor can be retained before taking a return. For a fixed inclusion $K \subset Y_0$, choose frames P_K, P_{Y_0} and quotient lifts X . Put

$$[P_K, X] = P_{Y_0} R_{Y_0/K}, \quad d_{Y_0/K} = |\det R_{Y_0/K}|^2, \quad g_{Y_0}(N) = \det(P_{Y_0}^* \mathcal{D}_N P_{Y_0}). \quad (\text{PRT13})$$

The full block Schur determinant identity is

$$\det \text{Gram}_{\mathcal{D}_N}(Y_0/K) = d_{Y_0/K} g_{Y_0}(N) / g_K(N). \quad (\text{PRT14})$$

It follows by expressing the restricted form in $[P_K, X]$, taking its determinant, and eliminating its upper-left block. This defines and evaluates every factor from the stated original frames. It does not identify a varying minimizing representative with a fixed lift. CPQ's frames for each quotient are used in PRT13 when that quotient appears below.

3.3 Evaluation of the difference of the actual CPQ returns

Use the unchanged target return $\mathcal{R}f = -f(Q - 1) - f(Q) + f(2Q - 1) + f(2Q)$ and write

$$U_A = \mathcal{R} \log \det A_N, \quad U_B = \mathcal{R} \log \det B_N, \quad U_C = \mathcal{R} \log \det C_N. \quad (\text{PRT15})$$

These symbols are returns of precisely the three evaluated quotient matrices PRT11–12. They are not freely assignable parameters. The exact CPQ differences are

$$\begin{aligned} V_{\text{conn},0} - V_{\text{conn},1} &= U_A, & V_{h,0} - V_{h,1} &= U_B, \\ V_{s,0} - V_{s,1} &= U_A - U_B, & V_{t,0} - V_{t,1} &= U_C, \\ V_{\text{rem},1} - V_{\text{rem},0} &= U_C, & \mathcal{L}_0 - \mathcal{L}_1 &= U_C - U_A + U_B. \end{aligned} \quad (\text{PRT16})$$

Here V_b, V_I, V_A are independent of the defining primitive endpoint e : their spaces, coefficient maps and frames are the same CPQ3/9 objects. To prove every line, apply PRT14 to $S_e/\mathcal{B}$, $K_e/\mathcal{B}$, S_e/K_e , T_e/W and $\mathcal{U}/T_e$, respectively. For example the logarithmic determinant difference of the latter two endpoint quotients is $\log g_{T_0} - \log g_{T_1}$ plus the explicitly defined fixed frame-factor difference; PRT14 for T_0/T_1 identifies its nonconstant term with $\log \det C_N$. Each frame factor is independent of N , and $\mathcal{R}1 = -1 - 1 + 1 + 1 = 0$, so it cancels only after this full return. The other four differences follow by the same stated determinant identities. Finally substitute them into CPQ10's $\mathcal{L}_e = V_I + V_{t,e} - V_{s,e} - V_A$.

This also gives the exact propagation, without deleting a complement,

$$\begin{aligned} Z_e &= V_b + V_{\text{rem},e} + V_{s,e} + V_A, & Z_1 - Z_0 &= U_C - U_A + U_B, \\ \mathcal{P}_0^{\text{rest}} - \mathcal{P}_1^{\text{rest}} &= U_C - U_A + U_B, & 4S_{K,j} &= \mathcal{L}_0 + Z_0 = \mathcal{L}_1 + Z_1. \end{aligned} \quad (\text{PRT17})$$

In particular a change of primitive endpoint changes the common loss and complementary sum by exactly opposite amounts in their allocation identity. The original total and the complete-action allocation cancellation are preserved.

3.4 Directed finite bounds with the original masses

Use the proved original reconstruction/source constant of FRE1, evaluated at degree j and order s :

$$A_{j,s}^{\text{end}} = \max\{1, Q\mathcal{B}_{2Q-v,s,j,A}^2\} (M_{j,s}^{\text{rec}})^2. \quad (\text{PRT18})$$

The two factors retain DNE13's full mass $(2\pi)^{s/2}$, the full conductor scalar and division coefficients, and LJR16's original fixed-disk reconstruction, phases and Vandermonde. FRE2-4 proves on this exact metric

$$\mathcal{D}_{Q-1}^{(s)} \preceq \mathcal{D}_N^{(s)} \preceq A_{j,s}^{\text{end}} \mathcal{D}_{Q-1}^{(s)}, \quad \log A_{j,s}^{\text{end}} \leq 2Q \log j + O_{\text{actual}}(Q). \quad (\text{PRT19})$$

Substitution in each full affine fibre defining PRT11 and subsequent minimization preserve both inequalities. Original row extension also proves that each of A_N, B_N, C_N increases with N . If G_N is one of these forms of rank r , set $f(N) = \log \det G_N - \log \det G_{Q-1}$. The generalized eigenvalues in PRT19 show $0 \leq f(N) \leq r \log A_{j,s}^{\text{end}}$ and f is increasing. Hence $\mathcal{R} \log \det G = f(2Q-1) + f(2Q) - f(Q)$ is nonnegative and at most $2r \log A_{j,s}^{\text{end}}$. This proves the actual directed intervals

$$\begin{aligned} 0 \leq U_A \leq 6 \log A_{j,s}^{\text{end}}, \quad 0 \leq U_B \leq 2(3-a) \log A_{j,s}^{\text{end}}, \quad 0 \leq U_C \leq 2a \log A_{j,s}^{\text{end}}, \\ \boxed{0 \leq V_{\text{rem},1} - V_{\text{rem},0} \leq 2a \log A_{j,s}^{\text{end}}}, \quad (\text{PRT20}) \\ -2(3-a) \log A_{j,s}^{\text{end}} \leq V_{s,0} - V_{s,1} \leq 6 \log A_{j,s}^{\text{end}}, \\ |\mathcal{L}_0 - \mathcal{L}_1| \leq 6 \log A_{j,s}^{\text{end}}, \quad |Z_1 - Z_0| \leq 6 \log A_{j,s}^{\text{end}}. \end{aligned}$$

The last bound uses $\dim B + \dim C = 3$, keeping the exact sequence rather than separately charging two unrelated rank-three errors. These are finite bounds on actual CPQ expressions. For each fixed original period they are $O_{\text{actual}}(Q \log j) = o(kq)$, for both source pairs. At the actual rank stratum $a = 0$, the remaining-target forms are the same quotient in fixed value frames and their returns are exactly equal. At $a = 3$, the kernel spaces coincide and $U_B = 0$ exactly. All intermediate strata remain governed by their actual original three-column matrix. No claim is made that either endpoint's common leading return has now been evaluated; PRT16 evaluates their difference and reduces its calculation to the stated three original directions.

The proof requires the five-orbit coefficient construction, its original distinct quartet roots and $\mu_v \neq 0$. It does not use the additional pole-coordinate gate minors of RPI/RPK. Consequently a period excluded by a pole-coordinate pivot remains covered when it belongs to the original five-orbit locus: PRT6/8/12 construct all maps without that pivot. Rank changes of the three-row image are handled by PRT10. The value v is retained throughout; neither $v \leq 32$ nor a generic value of v has been substituted for an actual coefficient.

4 The first original row: exact lower cancellation and its finite bound

Fix the actual simple quartet, original conductor coefficients, period, and branch. Write

$$\begin{aligned}
j &\equiv 1 \pmod{4}, \quad j \geq 13, \quad q = (j+1)^2, \quad q' = (j-7)^2, \quad c' = j/2 - 4, \\
E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad \mu_d = E_A^{(d)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \\
\beta_{r,a,b} &= r/2 + (2a-r)\delta + i(2b-r)\gamma, \quad \sigma = (\delta^2 + \gamma^2)^{1/2}, \\
\chi_j(S) &= \prod_{a,b=0}^j (S - \beta_{j,a,b}), \quad \chi'(S') = \chi_{j-8}(S') = \prod_{a,b=0}^{j-8} (S' - \beta_{j-8,a,b}), \\
n &= q - v, \quad m = n - q' = 16j - 48 - v \geq 0, \quad s \in \{1, j\}, \quad \ell = (s-1)/4.
\end{aligned} \tag{FRG1}$$

The prime on χ' names the complete lower root polynomial; it is not differentiation. The original norm is

$$\begin{aligned}
\|P\|_{s,c'}^2 &= \int_{\mathbb{R}} |P(c' + iy)|^2 dm_s(y), \quad dm_s = \frac{M_s 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\
p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{r+1,s} = y p_{r,s} - r(r-1+s/2) p_{r-1,s}, \\
h_{r,s} &= M_s r! (s/2)_r, \quad \phi_r(S') = p_{r,s}((S' - c')/i) / \sqrt{h_{r,s}}.
\end{aligned} \tag{FRG2}$$

These are exactly NSO2,7's native orthonormal polynomials, with the full generating-function and orthogonality proof in SGT2-5; its source mass and recurrence coefficients are unchanged here. Thus the leading coefficient of ϕ_r is $i^{-r}/\sqrt{h_{r,s}}$. All inner products are conjugate-linear in the first variable, whereas every functional below is complex-linear.

4.1 The actual annihilator and the full first-row leverage

Let $F_b = N_b/E_A$ be the original NSO12 numerator germs, with $N_b(z) = \sum_{a,b'} (Z_{\text{exp},j})_{(a,b'),b} e^{\beta_{j,a,b'} z}$. Keep their original fixed NSO12 frame. Let V_0, T_0 be NSO13's lower and numerator matrices on rows $0, \dots, n-1$, and let v_n, t_n be their next rows. There are q' lower columns and m numerator columns; therefore

$$F_0 = [V_0, T_0] \in \mathbb{C}^{n \times n}, \quad w_n = [v_n, t_n]. \tag{FRG3}$$

The NSO12-13 Vandermonde proof applies at cutoff $n-1 = q-v-1$: if such a combined functional has its first n derivatives zero, multiplication of its germ by E_A makes an upper-grid numerator have its first q derivatives zero. The upper Vandermonde kills this numerator, its class in the fixed conductor quotient vanishes, and the lower Vandermonde kills the remaining lower coefficients. Since the triangular change from derivatives to the ϕ_r values has nonzero diagonal, F_0 is invertible.

Define the original translation and polynomial

$$(\mathcal{T}_A P)(S') = \sum_{a,b=0}^8 a_{ab} P(S' + \beta_{8,a,b}), \quad g = \mathcal{T}_A \chi_j, \quad \lambda = \mu_v \binom{q}{v} \neq 0. \tag{FRG4}$$

Expanding every shift shows that all coefficients above degree $q-v$ vanish and the coefficient at degree $q-v$ is λ : for a monomial S^d , its coefficient at S'^{d-r} is $\binom{d}{r} \mu_r$, and $\mu_0 = \dots = \mu_{v-1} = 0$. Hence $\deg g = n$. For any polynomial P the finite Leibniz identity gives the exact pairing

$$\Lambda_{F_b}(\mathcal{T}_A P) = \Lambda_{E_A F_b}(P) = \Lambda_{N_b}(P). \tag{FRG5}$$

The right side for $P = \chi_j$ is a linear combination of evaluations of χ_j at all its upper roots, so is zero. Also $\beta_{j-8,u,w} + \beta_{8,a,b} = \beta_{j,u+a,w+b}$ lies in the original upper grid. Every summand in $g(\beta_{j-8,u,w})$ is zero. Distinctness of all the lower roots proves $\chi' \mid g$. Consequently all columns in $[V, T]$ annihilate g .

Expand $g = \sum_{r=0}^n c_r \phi_r$. The exact top coefficient and row relation are

$$c_n = i^n \lambda \sqrt{h_{n,s}}, \quad c_{<n}^\top F_0 + c_n w_n = 0. \quad (\text{FRG6})$$

Here the transpose is ordinary because of complex linearity. Since F_0 is square and invertible, $F_0(F_0^* F_0)^{-1} F_0^* = I_n$. Thus

$$\begin{aligned} \zeta_{\text{tot}} &= w_n (F_0^* F_0)^{-1} w_n^* = \frac{\sum_{r < n} |c_r|^2}{|c_n|^2}, \\ 1 + \zeta_{\text{tot}} &= \frac{\|\mathcal{T}_A \chi_j\|_{s,c'}^2}{|\lambda|^2 h_{n,s}}. \end{aligned} \quad (\text{FRG7})$$

No factorial, phase, conductor coefficient, or source mass was removed.

4.2 The exact lower ideal norm

Let

$$\nu_{m,s} = \min_{P \text{ monic, deg } P=m} \|\chi' P\|_{s,c'}^2. \quad (\text{FRG8})$$

This is the attained positive norm on the original monic affine fibre; it exists because its finite positive Gram has a unique quadratic minimum. A monic degree- n polynomial $u = \sum_{r=0}^n d_r \phi_r$ has $d_n = i^n \sqrt{h_{n,s}}$. Divisibility by χ' is exactly its vanishing at all lower roots, hence exactly

$$V_0^\top d_{<n} = -d_n v_n^\top. \quad (\text{FRG9})$$

The matrix V_0 has full column rank. In these coefficient coordinates the attained minimum is

$$d_{<n,\min} = -d_n \overline{V_0} (V_0^\top \overline{V_0})^{-1} v_n^\top. \quad (\text{FRG10})$$

Multiplication proves the constraint. Its ordinary Hermitian pairing with every vector in $\ker V_0^\top$ vanishes, proving the minimum and its Pythagorean decomposition. Its squared norm is $|d_n|^2 \overline{v_n} (V_0^\top \overline{V_0})^{-1} v_n^\top = |d_n|^2 \zeta_{\text{low}}$, where $\zeta_{\text{low}} = v_n (V_0^* V_0)^{-1} v_n^*$ is real and nonnegative. Thus

$$\boxed{\nu_{m,s} = h_{n,s} (1 + \zeta_{\text{low}}).} \quad (\text{FRG11})$$

This is a typed monic-fibre map, not a change of the original measure. In y coordinates it retains the phase explicitly: write $\chi'(c' + iy) = i^{q'} Q'(y)$, with monic Q' , and $P(c' + iy) = i^m \tilde{P}(y)$, with monic $\tilde{P}$. The forward and inverse maps on monic fibres are $\tilde{P}(y) = i^{-m} P(c' + iy)$ and $P(S') = i^m \tilde{P}((S' - c')/i)$. Their total phase is i^n and has modulus one; their squared norm is precisely FRG8.

Put $A = V_0^* V_0$, $B = V_0^* T_0$, $D_0 = T_0^* T_0 - B^* A^{-1} B$, and $\rho = t_n - v_n A^{-1} B$. Block factorization of $F_0^* F_0$ gives

$$\zeta_{\text{tot}} = \zeta_{\text{low}} + \rho D_0^{-1} \rho^*, \quad r_n = \frac{\rho D_0^{-1} \rho^*}{1 + \zeta_{\text{low}}}. \quad (\text{FRG12})$$

For instance factor the Gram as $\begin{bmatrix} I & 0 \\ B^* A^{-1} & I \end{bmatrix} \text{diag}(A, D_0) \begin{bmatrix} I & A^{-1} B \\ 0 & I \end{bmatrix}$ and multiply its inverse against $[v_n, t_n]$ and its adjoint. Equations FRG7 and FRG11 now prove

$$\boxed{1 + r_n = \frac{1 + \zeta_{\text{tot}}}{1 + \zeta_{\text{low}}} = \frac{\|\mathcal{T}_A \chi_j\|_{s,c'}^2}{|\mu_v(q)|^2 \nu_{m,s}}.} \quad (\text{FRG13})$$

The scalar λ maps g to the unique monic polynomial g/λ in its original one-dimensional coefficient fibre; it is kept in FRG13. In particular, if $P_{m,s}^{\min}$ is the unique monic minimizer in FRG8, then

$$1 + r_n = 1 + \frac{\|g/\lambda - \chi' P_{m,s}^{\min}\|_{s,c'}^2}{\nu_{m,s}} \geq 1. \quad (\text{FRG14})$$

Indeed the difference belongs to $\chi' \mathbb{C}[S']_{<m}$, to which the minimizer is orthogonal. This also characterizes the exact zero-row case: $r_n = 0$ precisely when $g/\lambda = \chi' P_{m,s}^{\min}$.

4.3 An explicit finite bound from the original root products

Define

$$\begin{aligned} \theta &= \pi/2, \quad h = 1/q, \quad \kappa = \theta - h > 0, \quad \beta_s = \frac{M_s}{\sqrt{2}\Gamma(s/2)}, \\ L_h &= \frac{\pi}{\Gamma(3/4)^2} (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\ Q_- &= q' + \ell, \quad Q_+ = q + \ell, \quad R' = (j - 8)\sigma, \quad M_0 = 2Q_-/\theta, \quad M = \max\{M_0, 2R'\}, \\ B_+ &= \max\{1, (j + 8)\sigma, 2\ell\}, \quad A_{\text{abs}} = \sum_{a,b=0}^8 |a_{ab}|, \\ f_\theta(x) &= 2x \log(2x/\theta) - 2x, \quad F_M = 2Q_- \log M - \theta M, \\ \epsilon_- &= \left(\theta - \frac{2Q_-}{M} \right) + \frac{Q_-}{M^2} + h(M + 1) + \frac{4q'R'}{M}, \quad C_m = (2m + 1) \binom{2m}{m}. \end{aligned} \quad (\text{FRG15})$$

The two inequalities $M \geq 2R'$ and $M \geq M_0$ hold by definition, so the finite bounds below apply at every $j \geq 13$. For each fixed quartet, $M = M_0$ eventually, because M_0 grows quadratically and R' linearly in j . On that eventual range, $F_M = f_\theta(Q_-)$ and $\epsilon_- = \theta^2/(4Q_-) + h(M + 1) + 4q'R'/M$.

For completeness, the Gamma input used in the bound is proved as follows. Gamma recurrence gives

$$dm_s(y) = \beta_s \prod_{r=0}^{\ell-1} (y^2 + (2r + 1/2)^2) d\sigma(y), \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{FRG16})$$

The factor $4^{-\ell}$ from the recurrence is cancelled exactly by the $2^{2\ell}$ in the original density. Rotate Euler's integral for $\Gamma(a + iy/2)$ through $\text{sgn}(y)(\pi/2 - h)$: absolute integration along that ray proves $|\Gamma(a + iy/2)| \leq \Gamma(a)(\sin h)^{-a} e^{-(\pi/2-h)|y|/2}$. The small closing arc tends to zero as its radius to the power a , and the large closing arc tends to zero since its cosine is at least $\sin h$. For the lower bound use the exact reflection product $|\Gamma(1/4 + iy/2)\Gamma(3/4 - iy/2)|^2 = 2\pi^2/\cosh(\pi y)$, the same upper bound on the second Gamma factor, and $\cosh(\pi y) \leq e^{\pi|y|}$. Therefore, also at $y = 0$,

$$L_h e^{-(\theta+h)|y|} \leq \frac{d\sigma}{dy}(y) \leq U_h e^{-\kappa|y|}. \quad (\text{FRG17})$$

For $y \in [M, M + 1]$, every original lower root in the y coordinate has modulus at most R' . Since $R'/y \leq 1/2$, integration of $(1 - t)^{-1} \leq 2$ gives $|\chi'(c' + iy)|^2 \geq y^{2q'} e^{-4q'R'/M}$. Every Gamma recurrence factor is at least y^2 . Writing $y = M + t$, $0 \leq t \leq 1$, the inequality $\log(1 + x) \geq x - x^2/2$ gives

$$2Q_- \log y - \theta y \geq F_M - \left(\theta - \frac{2Q_-}{M} \right) - \frac{Q_-}{M^2}. \quad (\text{FRG18})$$

Thus the complete density $|\chi'(c' + iy)|^2 dm_s/dy$ on this one original-coordinate interval is at least $\beta_s L_h \exp(F_M - \epsilon_-)$.

Here is the exact monic minimum on an interval of length one. Set $L_m(x) = \frac{1}{m!} \frac{d^m}{dx^m} [x^m(x-1)^m]$. Its leading coefficient is $\binom{2m}{m}$. Integrating by parts m times, all boundary terms vanish, proving its orthogonality to degrees less than m . The same integrations give

$$\int_0^1 L_m(x)^2 dx = \binom{2m}{m} \int_0^1 x^m(1-x)^m dx = \frac{1}{2m+1}. \quad (\text{FRG19})$$

The beta integral here is $m!^2/(2m+1)!$, obtained by repeated integration by parts. Subtracting $L_m/\binom{2m}{m}$ from any complex monic polynomial makes an orthogonal lower-degree difference. The least squared norm is exactly C_m^{-1} . Translation of the interval preserves its leading coefficient. The polynomial $P(c' + iy)$ for monic P has leading coefficient i^m ; multiply the displayed minimizing polynomial by that retained phase. Applying this minimum to FRG18 yields

$$\nu_{m,s} \geq \frac{\beta_s L_h}{C_m} \exp(F_M - \epsilon_-). \quad (\text{FRG20})$$

On the entire real line, for every original shift and upper root,

$$|c' + iy + \beta_{8,a,b} - \beta_{j,u,w}| \leq |y| + (j+8)\sigma \leq |y| + B_+. \quad (\text{FRG21})$$

Indeed its constant part is $[(2a-8) - (2u-j)]\delta + i[(2b-8) - (2w-j)]\gamma$. Every original summand in g therefore has modulus at most $(|y| + B_+)^q$, and the complete sum has modulus at most $A_{\text{abs}}(|y| + B_+)^q$. Also $2r + 1/2 \leq 2\ell \leq B_+$ for every recurrence factor when $\ell > 0$, and the empty product for $\ell = 0$ is one. FRG16–17, the substitution $t = y + B_+$ on $y \geq 0$, and enlargement of its integral to $[0, \infty)$ prove

$$\|g\|_{s,c'}^2 \leq 2\beta_s U_h A_{\text{abs}}^2 e^{\kappa B_+} \frac{(2Q_+)!}{\kappa^{2Q_++1}}. \quad (\text{FRG22})$$

The cancellation of coefficients above degree n is still exact in g ; the bound estimated the complete original finite sum before those cancellations and changed no polynomial.

Combining FRG13, FRG20 and FRG22 gives the explicit finite interval

$$\begin{aligned} 0 \leq \log(1 + r_n) &\leq E_{j,s}, \\ E_{j,s} &= 2 \log \frac{A_{\text{abs}}}{|\mu_v(q)|} + \log \frac{2U_h}{L_h} + \kappa B_+ + \log((2Q_+)!) - (2Q_+ + 1) \log \kappa \\ &\quad - F_M + \epsilon_- + \log C_m. \end{aligned} \quad (\text{FRG23})$$

The mass factor β_s cancels between these two displayed bounds on the same original norm. In particular FRG23 has not assumed unit mass. Positivity of the exact ratio proves that this computed upper bound is nonnegative throughout its stated domain.

4.4 Evaluation of the first-row scale and the cancelled leading term

All the following limits fix the actual quartet and conductor, including μ_v , and let $j \equiv 1 \pmod{4}$ tend to infinity. They hold for both $s = 1$ and $s = j$. First, $Q_{\pm} \asymp j^2$, $Q_+ - Q_- = 16j - 48$, $B_+ = O(j)$, $R' = O(j)$, and $m = O(j)$. Eventually $M = M_0$, so $F_M = f_{\theta}(Q_-)$. Directly from FRG15, $\epsilon_- = O(j)$; also $\log C_m \leq \log(2m+1) + 4m \log 2 = O(j + \log j)$. Since $2h/\pi \leq \sin h \leq h$, $\log(U_h/L_h) = O(\log q)$. The fixed- v product for $\binom{q}{v}$ makes $|\log |\lambda|| = O(\log q)$.

For every integer $N \geq 1$, integration bounds on the increasing function $\log x$ give

$$N \log N - N + 1 \leq \log(N!) \leq N \log N - N + 1 + \log N. \quad (\text{FRG24})$$

Consequently

$$\begin{aligned} \log((2Q_+)!) - (2Q_+ + 1) \log \kappa &\leq f_{\theta}(Q_+) + 1 + \log(2Q_+) - \log \theta \\ &\quad + (2Q_+ + 1) \log(\theta/\kappa). \end{aligned} \quad (\text{FRG25})$$

Here $\log(\theta/\kappa) \leq h/(\theta - h)$, so the last term is $O(1)$. Since $f'_\theta(x) = 2 \log(2x/\theta)$, its integral between Q_- and Q_+ is $O(j \log j)$. Every term of the fully finite FRG23 is now controlled, proving

$$\boxed{0 \leq \log(1 + r_n) = O_{\text{actual}}(j \log j).} \quad (\text{FRG26})$$

The leading terms of the two leverages themselves can also be evaluated without taking a difference of uncontrolled remainders. For the upper bound on $\nu_{m,s}$ choose the actual monic trial $P(S') = (S' - c')^m$. Put $B_- = \max\{1, R', 2\ell\}$ and $Q_0 = n + \ell$. The same whole-line integral used in FRG22 now gives

$$\nu_{m,s} \leq 2\beta_s U_h e^{\kappa B_-} \frac{(2Q_0)!}{\kappa^{2Q_0+1}}. \quad (\text{FRG27})$$

The factor $|y|^{2m}$ is bounded by $(|y| + B_-)^{2m}$ and $q' + m = n$, so the exponent is exactly $2Q_0$. Both $Q_- - q$ and $Q_0 - q$ are $O(j)$. Equations FRG20, FRG24–25, FRG27 and the preceding explicit estimates imply

$$\log \nu_{m,s} = \log \beta_s + f_\theta(q) + O_{\text{actual}}(j \log j). \quad (\text{FRG28})$$

To retain the full original factorial normalization, write $b = s/2 = 2\ell + 1/2$ and $u = n + 2\ell$. Then

$$h_{n,s} = \sqrt{2}\beta_s n! \Gamma(u + 1/2), \quad \Gamma(u + 1/2) = \frac{\sqrt{\pi}(2u)!}{4^u u!}. \quad (\text{FRG29})$$

FRG24 in these three factorials shows $\log \Gamma(u + 1/2) = u \log u - u + O(\log u)$. Both $n - q$ and $u - q$ are $O(j)$; integration of $\log x$ between these arguments and q therefore gives

$$\log h_{n,s} = \log \beta_s + 2(q \log q - q) + O_{\text{actual}}(j \log j). \quad (\text{FRG30})$$

Subtracting the two controlled expressions in FRG28 and FRG30, and then using the exact identity FRG13 together with FRG26, proves

$$\boxed{\begin{aligned} \log(1 + \zeta_{\text{low}}) &= 2q \log(4/\pi) + O_{\text{actual}}(j \log j), \\ \log(1 + \zeta_{\text{tot}}) &= 2q \log(4/\pi) + O_{\text{actual}}(j \log j). \end{aligned}} \quad (\text{FRG31})$$

Thus the same positive order- q term belongs to both original leverages and cancels in their exact corrected ratio. This conclusion uses no limiting theorem for a growing-rank determinant and no derivative of an asymptotic error.

Finally the original SFB13–15 four-copy update has budget $\beta_n = 4 \log(1 + r_n)$. Its common signed increment satisfies $|\delta L_n| \leq \beta_n$, either by SFB8 integrated along this positive update, or by the same original projection determinants. For this first original row, FRG26 therefore yields

$$0 \leq \beta_n = O_{\text{actual}}(j \log j), \quad \frac{|\delta L_n|}{jq} \longrightarrow 0. \quad (\text{FRG32})$$

Each original attained quotient in CPQ14 also has its first-row logarithmic determinant increment between 0 and β_n : along the same positive path, its derivative is the trace of the positive relative derivative against the orthogonal projector on its minimum-lift quotient space, which lies between zero and the full trace. Integrating proves the claim with the original quotient kept. Hence each of the five complementary first-row increments is $O_{\text{actual}}(j \log j)$ as well. This evaluates one actual row; FRG32 makes no assertion about the remaining q consecutive rows.

5 The first original mixed row in the full Gamma polynomial ideal

Keep the original hypothetical simple quartet, fixed five-orbit period, logarithm branch, complete nonreduced conductor, and both source pairs of NSO and CPQ. This calculation concerns their actual first appended target row, rather than a freely chosen matrix update. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q &= (j + 1)^2, \\ q' &= (j - 7)^2, & c' &= j/2 - 4, & n &= q - v, & m &= n - q' = 16j - 48 - v. \end{aligned} \quad (\text{FMG1})$$

Here $v = \text{ord}_0 E_A$ is the actual conductor order. The retained BTK estimate is $v \leq 32$ on its stated original domain; the earlier $v \leq 80$ already makes $m > 0$ in this calculation. Let $s = 1$ or $s = j$, according to the prescribed source pair. Throughout,

$$\begin{aligned} \beta_{a,u,w} &= a\left(\frac{1}{2} - \delta - i\gamma\right) + 2u\delta + 2wi\gamma, \\ \chi_a(S) &= \prod_{u,w=0}^a (S - \beta_{a,u,w}), \\ E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \\ \mu_r &= E_A^{(r)}(0), \\ (\mathcal{T}_{Ap})(S') &= \sum_{u,w=0}^8 a_{uw} p(S' + \beta_{8,u,w}), \\ \ell &= \mu_v \binom{q}{v} \neq 0. \end{aligned} \quad (\text{FMG2})$$

The symbol ℓ in FMG2 is a leading coefficient, separate from the integer l in $k = 4l + 1$. No root, period coefficient or conductor factor has been discarded. Define the actual polynomials

$$\chi_- := \chi_{j-8}, \quad g := \mathcal{T}_A \chi_j. \quad (\text{FMG3})$$

For every lower root $\beta_{j-8,a,b}$, each summand defining $g(\beta_{j-8,a,b})$ evaluates χ_j at the upper root $\beta_{j,a+u,b+w}$. Thus χ_- divides g exactly. The first nonzero moment in FMG2 gives $\deg g = n$ and $\text{lc}_{S'} g = \ell$ by the binomial formula. This is a divisibility assertion for the full translated polynomial, including every coefficient.

The original measure and orthonormal polynomials are

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, & M_s &= (2\pi)^{s/2}, \\ p_{0,s} &= 1, & p_{1,s} &= y, & p_{a+1,s} &= y p_{a,s} - a(a-1+s/2) p_{a-1,s}, \\ h_{a,s} &= M_s a! (s/2)_a, & \phi_a(S') &= p_{a,s}((S' - c')/i) / \sqrt{h_{a,s}}, \\ \langle f_1, f_2 \rangle_s &= \int_{\mathbb{R}} \overline{f_1(c' + iy)} f_2(c' + iy) dm_s(y). \end{aligned} \quad (\text{FMG4})$$

This inner product is conjugate-linear in its first argument. In particular the leading coefficient of ϕ_a in S' is $i^{-a} / \sqrt{h_{a,s}}$.

5.1 The exact original row and its two coupled leverages

Use exactly the NSO12 quotient columns $F_b = N_b/E_A$, $1 \leq b \leq m$. Their numerators lie on the original degree- j grid and vanish to order v at zero. The original lower evaluation and numerator matrices at the preceding cutoff are

$$V_0[a, (u, w)] = \phi_a(\beta_{j-8,u,w}), \quad T_0[a, b] = \Lambda_{F_b}(\phi_a), \quad 0 \leq a < n, \quad (\text{FMG5})$$

where $\Lambda_F(S'^a) = F^{(a)}(0)$ is complex-linear. There are $q' + m = n$ columns in $[V_0, T_0]$. To prove their independence, a zero combination of their first n native values has a generating germ F with $F^{(a)}(0) = 0$ for $0 \leq a < n$, since the polynomial change of basis in FMG4 is invertible. Its numerator $E_A F$ is a degree- j grid exponential polynomial and vanishes through order $n + v - 1 = q - 1$. The original upper Vandermonde is invertible, so that numerator is zero. The NSO12 quotient frame then kills the numerator coefficient, and the lower Vandermonde kills the remaining lower coefficient. This proves that $[V_0, T_0]$ is square and invertible, with its complete original frame.

Write the actual next rows as v_n, t_n . Define

$$\begin{aligned} A_L &= V_0^* V_0, & P_V &= V_0 A_L^{-1} V_0^*, & D_0 &= T_0^* (I - P_V) T_0, \\ \zeta &= v_n A_L^{-1} v_n^*, & \rho &= t_n - v_n A_L^{-1} V_0^* T_0, & r &= \frac{\rho D_0^{-1} \rho^*}{1 + \zeta}. \end{aligned} \quad (\text{FMG6})$$

These are precisely the first SFB12 and NTE1 quantities. Inverses exist by the independence just proved. Set $C_0 = [V_0, T_0]$ and $w_n = [v_n, t_n]$. The total leverage is $\zeta_{\text{all}} = w_n (C_0^* C_0)^{-1} w_n^*$.

Expand $g = \sum_{a=0}^n c_a \phi_a$ with column $c_{<} = (c_0, \dots, c_{n-1})^T$. The leading coefficient is $c_n = \ell i^n \sqrt{h_{n,s}}$. Every column functional in FMG5 annihilates g . For the numerator columns, the full Leibniz identity gives

$$\Lambda_{F_b}(\mathcal{T}_A \chi_j) = \Lambda_{E_A F_b}(\chi_j) = \sum_{u,w} (N_b)_{uw} \chi_j(\beta_{j,u,w}) = 0.$$

For the lower columns, FMG3 gives the same zero. These are complex-linear equations, so their coefficient row is an ordinary transpose. Therefore

$$w_n = -c_{<}^{-1} c_{<}^T C_0, \quad 1 + \zeta_{\text{all}} = \frac{\|g\|_s^2}{|\ell|^2 h_{n,s}}, \quad 1 + r = \frac{1 + \zeta_{\text{all}}}{1 + \zeta}. \quad (\text{FMG7})$$

The middle identity uses the invertibility of C_0 , not a replacement of its Gram by the identity. To prove the last identity with every cross term retained, replace the numerator columns in C_0 by $(I - P_V) T_0$ using the invertible triangular column operation which subtracts $V_0 A_L^{-1} V_0^* T_0$. The preceding Gram then has diagonal blocks A_L, D_0 and the new row is $[v_n, \rho]$. Consequently $\zeta_{\text{all}} = \zeta + \rho D_0^{-1} \rho^*$, which gives FMG7.

5.2 A polynomial minimum for the complete lower denominator

For each integer $a \geq 0$ let

$$\nu_a = \min_{P \text{ monic, deg } P=a} \|\chi_- P\|_s^2 > 0. \quad (\text{FMG8})$$

The coefficients here are in the unchanged variable S' . This is an attained finite-dimensional minimum in the original measure; strict positivity follows because a nonzero polynomial cannot vanish on the entire integration line. The transformation $P \mapsto \ell \chi_- P$ is an explicitly invertible map from the monic degree- a affine fibre to the fibre with leading coefficient ℓ and factor χ_- . The norm under that map is multiplied by exactly $|\ell|^2$; no scalar is suppressed.

At degree $n = q' + m$, a monic polynomial with factor χ_- has orthonormal coefficient $a_n = i^n \sqrt{h_{n,s}}$. Its old coefficient column $a_{<}$ must satisfy $V_0^T a_{<} = -a_n v_n^T$. This is the exact evaluation constraint, including the ordinary transpose. Its unique coefficient minimum is

$$a_{<} = -a_n \overline{V_0} (V_0^T \overline{V_0})^{-1} v_n^T. \quad (\text{FMG9})$$

Indeed the displayed column satisfies the constraint, and it is orthogonal to the constraint kernel. Its squared norm is $h_{n,s} v_n A_L^{-1} v_n^* = h_{n,s} \zeta$; the equality follows by conjugating the middle Gram and using that the resulting scalar is real. Adding $|a_n|^2$ proves

$$\boxed{\nu_m = h_{n,s} (1 + \zeta), \quad 1 + r = \frac{\|g\|_s^2}{|\ell|^2 \nu_m}.} \quad (\text{FMG10})$$

Thus the lower denominator in the corrected original row is calculated by an actual lower-root ideal minimum. Replacing it by one would remove the explicitly present factor $\nu_m / h_{n,s}$.

5.3 All eight compressed matrices in exact polynomial coordinates

Put $f_a(S') = \chi_-(S')S'^a$ for $0 \leq a < m$, and write $f_a = \sum_{b < n} C_{ba}\phi_b$. Define

$$G = C^*C = (\langle f_a, f_b \rangle_s)_{a,b < m}, \quad H = C^*c_< = (\langle f_a, g \rangle_s)_{a < m}, \quad A = C^T T_0 = (\Lambda_{F_b}(f_a))_{a < m, b \leq m}. \quad (\text{FMG11})$$

The column span of C is exactly $\ker V_0^T$: every f_a vanishes on the lower roots; their degrees are distinct; and both spaces have dimension m . Thus

$$I - P_V = \overline{C} \overline{G}^{-1} C^T, \quad D_0 = A^* \overline{G}^{-1} A. \quad (\text{FMG12})$$

The first displayed matrix is self-adjoint, idempotent, and has range $\ker V_0^*$, proving it is exactly the required orthogonal projector. It also proves the second identity. The matrix A is invertible: if its product with x is zero, FMG12 gives $x^* D_0 x = 0$, and the positivity already proved forces $x = 0$. This is the original functional pairing on $\chi_- \mathbb{C}[S']_{< m}$, with its row and column orientations specified in FMG11.

The projection of g onto that polynomial ideal is $CG^{-1}H$. Its perpendicular component has leading coefficient ℓ and is exactly the monic-fibre minimizer multiplied by ℓ . Orthogonality and FMG8 therefore give

$$|\ell|^2 \nu_m = \|g\|_s^2 - H^* G^{-1} H, \quad r = \frac{H^* G^{-1} H}{|\ell|^2 \nu_m}, \quad \frac{\rho}{\sqrt{1+\zeta}} = \mathbf{b}A, \quad \mathbf{b} = -\frac{H^T \overline{G}^{-1}}{\ell i^n \sqrt{\nu_m}}. \quad (\text{FMG13})$$

For the row identity, substitute FMG7 in the definition of ρ , then apply FMG12 and $c_{<}^T \overline{C} = H^T$. Its denominator is $c_n \sqrt{1+\zeta} = \ell i^n \sqrt{\nu_m}$ by FMG10. This calculation proves both the phase and the single occurrence of the original leading coefficient.

Retain every fixed original CPQ coefficient subspace of $\mathcal{U} = \mathbb{C}^{4m}$:

$$I = \text{im}(B_{\text{cof}} J_0), \quad B = \text{im } B_{\text{cof}}, \quad S = \text{im}[B_{\text{cof}}, F_e], \quad J = \ker \text{diag}(B_j, 4), \quad (\text{FMG14})$$

$$K_i = I \cap J, \quad W_i = I + J, \quad W = B + J, \quad K_c = S \cap W, \quad T_e = S + J, \quad e = 0, 1.$$

Their matrices are the original period-dependent maps; none is replaced by a plane having merely the same dimension. Let X denote any fixed independent column frame for one of these eight subspaces. Set

$$A = \text{diag}(A, 4), \quad \mathcal{G}^\vee = \text{diag}(\overline{G}^{-1}, 4), \quad \lfloor = \text{diag}(\mathbf{b}, 4), \quad Y_X = AX, \quad (\text{FMG15})$$

$$K_{X,n} = \lfloor Y_X (Y_X^* \mathcal{G}^\vee Y_X)^{-1} Y_X^* \rfloor^*.$$

Here $\text{diag}(\mathbf{b}, 4)$ has four one-row blocks, so FMG15 is exactly a 4×4 matrix. If X is empty the result is the zero matrix. To verify FMG15 in the original metric, FMG12 gives $\mathcal{D}_0 = A^* \mathcal{G}^\vee A$, and FMG13 gives $\mathcal{U}_n = \lfloor A$. Substituting these identities in NTE1 proves FMG15 as literal equality, with no spectral approximation. The map $x \mapsto Ax$ is an isometry from $(\mathcal{U}, \mathcal{D}_0)$ to $(\mathbb{C}^{4m}, \mathcal{G}^\vee)$, with inverse A^{-1} . It carries sums and intersections bijectively; in particular it preserves both full chains

$$J \subset W_i \subset W \subset T_e, \quad K_i \subset I \subset K_c \subset S. \quad (\text{FMG16})$$

All original invariant, conductor and connecting coefficients are retained in Y_X through their exact pairing A .

Writing $b_X = \log \det(I_4 + K_{X,n})$, the first actual signed row is

$$\delta \mathcal{L}_{e,n} = b_{T_e} + b_{K_c} + b_{W_i} + b_{K_i} - b_W - b_S - b_J - b_I. \quad (\text{FMG17})$$

Equations FMG2–15 evaluate every entry using original polynomial inner products and the actual period coefficient maps. They retain all four eigenvalues of each matrix, including zero eigenvalues. The same equality supplies its threshold contribution by replacing b_X by $\sum_{a=1}^4 \log \max(1, \lambda_{X,a})$ with $\lambda_{X,a}$ the four eigenvalues of FMG15. No simultaneous diagonalization is assumed. Every matrix is positive and bounded above by rI_4 , since FMG15 is the unchanged NTE1 compression and FMG13 evaluates its full relative row norm.

5.4 Finite evaluation from the unchanged Gamma moments

The matrices in FMG11 involve no infinite series. For explicit finite evaluation, expand each polynomial on the original line, $f(c' + iy) = \sum_a d_a(f)y^a$. Define $M_r^{(s)} = \int y^r dm_s(y)$. Then

$$\langle f_1, f_2 \rangle_s = \sum_{a,b} \overline{d_a(f_1)} d_b(f_2) M_{a+b}^{(s)}. \quad (\text{FMG18})$$

These moments can be evaluated directly by the retained recurrence: for a requested degree r , form the $(r+1)$ -square real symmetric Jacobi matrix with off-diagonal entries $J_{a,a+1} = J_{a+1,a} = \sqrt{(a+1)(a+s/2)}$, $0 \leq a < r$, and zero diagonal. Then

$$M_r^{(s)} = M_s(J^r)_{00}. \quad (\text{FMG19})$$

Indeed multiplication by y in the orthonormal frame $p_{a,s}/\sqrt{h_{a,s}}$ has exactly these entries by FMG4. Starting in degree zero, r multiplications never visit a degree larger than r , so the finite truncation changes no path contributing to that matrix entry. Its constant component has the displayed integral, including M_s . Odd moments vanish, while all even moments are finite positive sums of these exact recurrence paths.

The entries of $\mathbf{A}$ have the finite original conductor calculation

$$f_{b,a} = \frac{\nu_{b,a+v} - \sum_{d=v+1}^{a+v} \binom{a+v}{d} \mu_d f_{b,a+v-d}}{\binom{a+v}{v} \mu_v}, \quad \nu_{b,r} = \sum_{u,w} (Z_{\text{exp},j})_{uw,b} \beta_{j,u,w}^r, \quad (\text{FMG20})$$

$$A_{ab} = \sum_r [S'^r](\chi_-(S')S'^a) f_{b,r}.$$

The triangular equation is exactly the coefficient of z^{a+v} in $E_A F_b = N_b$, retaining every factorial and μ_v . Thus all entries in FMG15 are finite expressions in the original roots, full conductor, actual chosen quotient frame, actual flag frames and Gamma moments. This gives an executable polynomial evaluation of the original first row, with no new choice of period.

5.5 A conductor-moment witness inside the actual residual

The lower root set is symmetric about c' under negation and conjugation. More explicitly

$$i^{-q'} \chi_-(c' + iy) = \prod_{a,b=0}^{j-8} (y - (2b - j + 8)\gamma + i(2a - j + 8)\delta) \quad (\text{FMG21})$$

is real and even as a polynomial in real y . Therefore $|\chi_-(c' + iy)|^2 dm_s(y)$ is an even positive measure. Its monic degree- a orthogonal polynomial in y has parity $(-1)^a$: reflecting it and multiplying by $(-1)^a$ gives another monic minimizer with the same norm, and uniqueness identifies them. Let P_a be its monic polynomial in S' under the exact map $p(y) \mapsto i^a p((S' - c')/i)$. It has coefficient $[S'^{a-1}]P_a = -ac'$ for $a \geq 1$, while $P_0 = 1$; in every case $\|\chi_- P_a\|_s^2 = \nu_a$.

Define the actual centered conductor moment

$$\eta_A = \frac{\mu_{v+1}}{(v+1)\mu_v} - 4, \quad \beta_m^- = \frac{\nu_m}{\nu_{m-1}} > 0. \quad (\text{FMG22})$$

The coefficient of S^{q-1} in χ_j is $-qj/2$, because its roots have exact mean $j/2$. Using FMG2 gives

$$[S'^{n-1}](g/\ell) = n \left(\frac{\mu_{v+1}}{(v+1)\mu_v} - \frac{j}{2} \right) = -nc' + n\eta_A. \quad (\text{FMG23})$$

The displayed division by ℓ is the explicit affine-fibre map in FMG8; it does not alter the original polynomial g . The coefficient of S'^{n-1} in $\chi_- P_m$ is $-nc'$. Their difference lies in $\chi_- \mathbb{C}[S']_{<m}$ and its

coefficient at S^{m-1} is $n\eta_A$. Its component along the orthogonal column $\chi_{-P_{m-1}}$ is consequently $n\eta_A$, because every other column has smaller degree. Pythagoras therefore proves

$$r \geq \frac{n^2|\eta_A|^2\nu_{m-1}}{\nu_m} = \frac{n^2|\eta_A|^2}{\beta_m^-}. \quad (\text{FMG24})$$

This is an original-data lower witness, which may vanish for the actual period; no nonvanishing is assumed. Even when it vanishes, FMG13 and FMG15 retain all other orthogonal components and the entire signed mixed row. In particular the exact criterion for $r = 0$ is $H = 0$, equivalently $g = \ell\chi_{-P_m}$ in the original Gamma inner product. Equality of a single conductor moment alone does not assert this full polynomial identity.

5.6 Position in the full return

The old target cutoff for this row is $N = q_j - 1$, hence native cutoff $n - 1 = q_j - v - 1$. Its new target cutoff is q_j , so its weight in the original four-endpoint return is one. FMG15–17 apply separately to $e = 0$ and $e = 1$ and both prescribed source orders. They have no effect on the coefficients of the other rows. Bounds for this row must therefore be subtracted from the full signed profile by its exact weight; the remaining interior rows still require their original arithmetic calculation. The complete CPQ identity remains

$$4S_{K,j} = \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A, \quad (\text{FMG25})$$

and all four complementary returns use the same original rows. The calculation above does not replace any of those returns by a dimension or an assigned constant.

5.7 Receiving the proved finite endpoint estimate in every signed term

The complete companion calculation FRG15–31, using the same actual g, ℓ, ν_m , proves the following finite input. Put

$$\begin{aligned} L_s &= (s-1)/4, \quad \theta = \pi/2, \quad h = 1/q, \quad \kappa = \theta - h, \quad Q_- = q' + L_s, \quad Q_+ = q + L_s, \\ \sigma &= (\delta^2 + \gamma^2)^{1/2}, \quad R' = (j-8)\sigma, \quad M_0 = 2Q_-/\theta, \quad M = \max\{M_0, 2R'\}, \\ B_+ &= \max\{1, (j+8)\sigma, 2L_s\}, \quad A_{\text{abs}} = \sum_{u,w} |a_{uw}|, \\ L_h &= \frac{\pi}{\Gamma(3/4)^2} (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\ f_\theta(x) &= 2x \log(2x/\theta) - 2x, \quad F_M = 2Q_- \log M - \theta M, \\ C_m &= (2m+1) \binom{2m}{m}^2, \quad \epsilon_- = \left(\theta - \frac{2Q_-}{M}\right) + \frac{Q_-}{M^2} + h(M+1) + \frac{4q'R'}{M}, \\ E_{j,s} &= 2 \log \frac{A_{\text{abs}}}{|\ell|} + \log \frac{2U_h}{L_h} + \kappa B_+ + \log((2Q_+)!) \\ &\quad - (2Q_+ + 1) \log \kappa - F_M + \epsilon_- + \log C_m. \end{aligned} \quad (\text{FMG26})$$

With these retained finite constants, that proof gives $0 \leq \log(1+r) \leq E_{j,s}$. In particular $E_{j,s}$ is nonnegative at every original $j \geq 13$, as follows also from the two proved bounds on the positive ratio FMG10. For each fixed original quartet and conductor, the actual choice eventually satisfies $M = M_0$ and $E_{j,s} = O_{\text{actual}}(j \log j)$, for both $s = 1, j$. The companion proof includes the complete Gamma density estimate at $h = 1/q$, the monic minimum on an original unit interval, the whole-line upper integral, and all factorials; the estimate is not an assumed analytic input.

Here is its exact effect on the entire signed row. The two chains in FMG16 imply

$$\begin{aligned} 0 &\leq (b_{T_e} - b_W) + (b_{W_i} - b_J) \leq b_{T_e} - b_J \leq 4 \log(1+r), \\ 0 &\leq (b_S - b_{K_e}) + (b_I - b_{K_i}) \leq b_S - b_{K_i} \leq 4 \log(1+r). \end{aligned} \quad (\text{FMG27})$$

Indeed for nested subspaces the orthogonal projectors in their common original metric are ordered; compressing by the same row preserves this order. Monotonicity of $\log \det(I + K)$ follows by differentiating along their positive connecting segment, or by the ordered eigenvalues. The missing middle increment in each chain is nonnegative, proving the middle bound. The last uses $0 \preceq K_X \preceq rI_4$. Subtraction of these two nonnegative quantities gives the exact signed expression FMG17. It follows that

$$|\delta\mathcal{L}_{e,n}| \leq 4E_{j,s}, \quad |\delta\mathcal{P}_{e,n}| \leq 4E_{j,s}, \quad 0 \leq \delta V_{X,n} \leq 4E_{j,s} \quad (X = b, \text{rem}, s, A). \quad (\text{FMG28})$$

For the threshold quantity $\delta\mathcal{P}$, apply the same argument to $\sum_{a=1}^4 \log \max(1, \lambda_a(K_X))$. The min-max characterization orders every eigenvalue when two Hermitian matrices are ordered: restrict the larger Rayleigh quotient to each test subspace, take its extremum there, and then the same extremum over subspaces. The increasing scalar function $\log \max(1, x)$ preserves those four inequalities. Thus the same two chain estimates hold with upper bound $4 \log \max(1, r) \leq 4 \log(1 + r)$. This also retains every zero and every eigenvalue below one.

For the four CPQ returns, their first-row increments are the paired determinant increments on the actual quotients W/W_i , $\mathcal{U}/T_e$, S/K_c , and I/K_i , respectively. For any fixed $Y \supset K$ in this list, the determinant of its attained quotient Gram is $\det G_Y / \det G_K$ times the fixed frame factor, by full block elimination. The factor cancels in a successive increment, leaving $b_Y - b_K$. The same order argument places it between zero and $4 \log(1 + r)$, proving the final part of FMG28. For $Y = \mathcal{U}$ its compression is exactly rI_4 . All four quotient fibres and all their cross terms therefore remain in the row-wise CPQ identity.

Since $j = k + 4$ and $q_k = (k + 1)^2$, each quantity in FMG28 divided by kq_k tends to zero. The original endpoint weight is one. Consequently the first original row can be removed from the leading kq_k calculation with the explicit finite error in FMG28, simultaneously in the signed spectral profile and in all four complementary CPQ returns. This statement concerns this one specified row; it leaves all remaining rows in their original correlated expressions.

6 The original conductor on every native mixed subquotient

This section receives the proved SGT7–28 estimates on the actual DNE6–9 and NSO12–15 spaces, and then on the fixed CIT/CPQ flags. The exterior quotient argument is already present in CEP37–38; we give its complete proof in the form needed here. The new receiver is the full finite numerator-observation metric, with the conductor multiples retained as its entire quotient denominator, and the rank-dependent SGT error on every original native subquotient.

Fix the original actual hypothetical simple quartet, its fixed original five-orbit period and logarithm branch. Keep $0 < \delta < 1/2$, $\gamma > 2$, every coefficient a_{uw} of the nonzero original conductor, and

$$\begin{aligned} \beta_{a,u,w} &= a/2 + (2u - a)\delta + i(2w - a)\gamma, \quad 0 \leq u, w \leq a, \quad a = 4\ell + 1 \geq 9, \\ E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \quad \mu_n = E_A^{(n)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \quad \alpha_A = v!/\mu_v, \\ q_a &= (a+1)^2, \quad q'_a = (a-7)^2, \quad m_a = q_a - q'_a - v, \quad c_0 = a/2, \quad c_1 = a/2 - 4, \\ N &\geq q_a - 1, \quad D = N - v, \quad s \in \{1, a\}. \end{aligned} \tag{NCT1}$$

The original domain of the fixed period, including its original rank conditions, is unchanged. The roots in each displayed grid are distinct: equality of two roots gives separately equality of the indices from their real and imaginary parts. Every polynomial space below has complex coefficients. Its Hermitian inner product is conjugate-linear in the first variable. The complex-linear dual consists of linear, rather than conjugate-linear, functionals.

The full original measures, masses, recurrence and frames are

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad \phi_{n,s,c}(S) = p_{n,s}((S-c)/i)/\sqrt{h_{n,s}}, \\ \|P\|_{s,c}^2 &= \int_{\mathbb{R}} |P(c+iy)|^2 dm_s(y). \end{aligned} \tag{NCT2}$$

SGT3–5 proves that these are orthonormal frames on every indicated finite polynomial space. Their leading coefficients are $i^{-n}/\sqrt{h_{n,s}}$; neither this phase nor the mass is removed.

6.1 The exact dual conductor square

Write $\mathcal{P}_d = \mathbb{C}[S]_{\leq d}$, with $\mathcal{P}_{-1} = 0$, and distinguish the target variable S' . The original, unscaled conductor map is

$$\begin{aligned} (\mathcal{T}_A P)(S') &= \sum_{u,w=0}^8 a_{uw} P(S' + \beta_{8,u,w}), \\ A_{D,s} : (\mathcal{P}_N/\mathcal{P}_{v-1}, \|\cdot\|_{s,c_0,\text{quot}}) &\longrightarrow (\mathbb{C}[S']_{\leq D}, \|\cdot\|_{s,c_1}), \quad [P] \longmapsto \mathcal{T}_A P. \end{aligned} \tag{NCT3}$$

It is the SGT7 isomorphism. For completeness, $\mathcal{T}_A S^n = \sum_{j=0}^n \binom{n}{j} \mu_j S'^{n-j}$. If $n \geq v$, its leading coefficient is $\binom{n}{v} \mu_v \neq 0$, and its degree is $n - v$. Successive elimination in degree gives surjectivity, and the full kernel is $\mathcal{P}_{v-1}$. The quotient is represented isometrically by the original orthogonal complement spanned by $\phi_{v,s,c_0}, \dots, \phi_{N,s,c_0}$.

For an analytic germ F define $\Lambda_F(S'^n) = F^{(n)}(0)$. An upper-variable germ is paired with S^n by the same derivative convention. The contravariant dual map in the direction relevant here is

$$\begin{aligned} A_{D,s}^\vee : (\mathbb{C}[S']_{\leq D})^\vee &\longrightarrow (\mathcal{P}_N/\mathcal{P}_{v-1})^\vee, \quad \Lambda \longmapsto \Lambda \circ A_{D,s}, \\ A_{D,s}^\vee \Lambda_F &= \Lambda_{E_A F}. \end{aligned} \tag{NCT4}$$

To prove the identity, on every $0 \leq n \leq N$ its left side is $\sum_{j=v}^n \binom{n}{j} \mu_j F^{(n-j)}(0)$, precisely $(E_A F)^{(n)}(0)$ by the Leibniz rule. For $n < v$ both sides vanish. The highest derivative of F that occurs is $n - v \leq D$, so the map is well defined on these finite jets; it does not depend on any higher derivative. Conversely the nonzero coefficient $\binom{v+r}{v} \mu_v$ recursively recovers the derivative of order r of F from these products. This also proves the isomorphism on finite jets directly.

Let $w = (\Lambda_F(\phi_{n,s,c_1}))_{n=0}^D$, and let $\tilde{w} = (\Lambda_{E_A F}(\phi_{v+n,s,c_0}))_{n=0}^D$. If $A_{D,s}$ denotes its original orthonormal matrix as in SGT10, then

$$\tilde{w} = A_{D,s}^\top w, \quad \Lambda_{E_A F}(\phi_{n,s,c_0}) = 0 \quad (0 \leq n < v). \quad (\text{NCT5})$$

Indeed the j -th source basis vector has image $\sum_i (A_{D,s})_{ij} \phi_{i,s,c_1}$, and applying the complex-linear functional gives the ordinary transpose in NCT5. There is no complex adjoint in this algebraic square. Squared functional norms are $w^* w$ and $\tilde{w}^* \tilde{w}$. If $A = U \Sigma V^*$ is a singular-value decomposition, then $A^\top = \bar{V} \Sigma U^\top$ is one as well. Thus A and $A^\top$, and their inverses, have exactly the same singular values and every corresponding exterior norm.

No scalar α_A is inserted into NCT3–5. In particular, the literal DNE3 germ is $F_b = \mathcal{N}_b / E_A$, and its image is $\Lambda_{\mathcal{N}_b}$, with coefficient one. The original primary-coordinate maps involving α_A are retained below.

6.2 All numerator observations and all conductor columns

Use the original lexicographic upper-root coefficient order. For $\nu \in \mathbb{C}^{q_a}$ put $\mathcal{N}_\nu(z) = \sum_{u,w} \nu_{uw} e^{\beta_{a,u,w} z}$. Let $\mathcal{N}_{v,a}$ be the coefficient subspace on which $\mathcal{N}_\nu^{(j)}(0) = 0$ for all $0 \leq j < v$. The upper Vandermonde $\mathbb{V}_a = (\beta_{a,u,w}^n)$, $0 \leq n < q_a$, is invertible, so $\dim \mathcal{N}_{v,a} = q_a - v$.

The full original primary conductor C_a , with q'_a rows and q_a columns, is fixed by the literal identity

$$\mathcal{N}_{C_a^\top \eta}(z) = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}. \quad (\text{NCT6})$$

This identity follows by collecting all original shifts, since $\beta_{a-8,u,w} + \beta_{8,r,t} = \beta_{a,u+r,w+t}$. It retains repeated contributions to the same resulting root. The map $C_a^\top$ is injective: its zero image would make the product in NCT6 identically zero; on a neighborhood where $E_A \neq 0$ the lower exponential sum is zero, hence everywhere by analyticity, and its distinct-root Vandermonde makes $\eta = 0$. Its image lies in $\mathcal{N}_{v,a}$, by the zero order of E_A .

Use exactly the DNE2–3/NSO12 frame:

$$\begin{aligned} W_{v,a} &= \mathbb{V}_a^{-1} P_v, & J_{v,a} &= P_v^\top \mathbb{V}_a C_a^\top, & W_{v,a} J_{v,a} &= C_a^\top, \\ Z_{\text{exp},a} &= W_{v,a} E_{\mathcal{J}_a}, & F_{a,b} &= \mathcal{N}_{Z_{\text{exp},a} e_b} / E_A. \end{aligned} \quad (\text{NCT7})$$

Here P_v inserts zero in the first v positions; $\mathcal{I}_a$ is the first ordered set of q'_a rows with nonzero minor of $J_{v,a}$, and $\mathcal{J}_a$ is its complement. Such a minor exists by the proved injectivity in NCT6. Restricting a relation among columns of $[J_{v,a}, E_{\mathcal{J}_a}]$ first to $\mathcal{I}_a$ and then to its complement proves that they are independent. Consequently $[C_a^\top, Z_{\text{exp},a}]$ is the complete frame of $\mathcal{N}_{v,a}$, and $Z_{\text{exp},a}$ represents its quotient by $\text{im } C_a^\top$. Every actual minor is unchanged.

Define the full upper observation matrix and its covariance by

$$\begin{aligned} \mathbb{E}_{a,N,s}[n, (u, w)] &= \phi_{n,s,c_0}(\beta_{a,u,w}), \quad 0 \leq n \leq N, \\ K_{a,N,s} &= \mathbb{E}_{a,N,s}^* \mathbb{E}_{a,N,s}, \\ \hat{D}_{a,D}^{(s)} &= Z_{\text{exp},a}^* [K_{a,N,s} - K_{a,N,s} C_a^\top (\bar{C}_a K_{a,N,s} C_a^\top)^{-1} \bar{C}_a K_{a,N,s}] Z_{\text{exp},a}. \end{aligned} \quad (\text{NCT8})$$

The square covariance $K_{a,N,s}$ is positive definite. Its first q_a observation rows equal an invertible triangular change of the upper Vandermonde, because the polynomials NCT2 have nonzero leading coefficients. Therefore $\mathbb{E}_{a,N,s}$ has full column rank for every stated N , including $N = q_a - 1$. The middle inverse in NCT8 is positive definite by this fact and injectivity of $C_a^\top$.

The same covariance has an exact derivative-coordinate expression. Let $V_{a,N}[n, (u, w)] = \beta_{a,u,w}^n$, and write $S^r = \sum_{n=0}^N B_{s,c_0,N}[n, r] \phi_{n,s,c_0}(S)$. The full monomial Gram is $H_{s,c_0,N} = B^* B$. Applying the complex-linear numerator functional to this identity gives

$$V_{a,N} \nu = B_{s,c_0,N}^\top E_{a,N,s} \nu, \quad K_{a,N,s} = V_{a,N}^* \overline{H_{s,c_0,N}^{-1}} V_{a,N}. \quad (\text{NCT8a})$$

Indeed $(B^\top)^{-*} (B^\top)^{-1} = \overline{(B^* B)^{-1}}$. Thus NCT8 is the full numerator derivative-dual covariance restricted to the vanishing-moment space and minimized over every conductor multiple. This proves the coordinate morphism, including the ordinary transpose and Hermitian conjugation, without changing the original measure or coefficient norm.

For every original coefficient vector $x \in \mathbb{C}^{m_a}$, the exact squared norm represented by NCT8 is

$$x^* \widehat{D}_{a,D}^{(s)} x = \min_{\eta \in \mathbb{C}^{q'_a}} \|E_{a,N,s}(Z_{\text{exp},a} x + C_a^\top \eta)\|_2^2. \quad (\text{NCT9})$$

Differentiate the right side with respect to its real and imaginary coordinates, or complete its positive square. The unique minimizer is

$$\eta_{\min} = -(\overline{C_a} K_{a,N,s} C_a^\top)^{-1} \overline{C_a} K_{a,N,s} Z_{\text{exp},a} x.$$

Substitution gives NCT8. The quotient is positive definite because $[C_a^\top, Z_{\text{exp},a}]$ is independent. The minimum retains every conductor column, including every mixed covariance term.

For $\nu \in \mathcal{N}_{v,a}$ the entries of $E_{a,N,s} \nu$ below row v vanish, since those polynomials use derivatives of order below v . This proves that the ambient polynomial dual norm in NCT9 is exactly the quotient dual norm in NCT4, not an extra truncation of the numerator. All rows $0, \dots, N$ remain in NCT8.

The corresponding original native metric is exactly NSO13:

$$x^* D_{a,D}^{(s)} x = \min_{\eta \in \mathbb{C}^{q'_a}} \left\| \Lambda_{\sum_b x_b F_{a,b}} + \sum_{u,w} \eta_{uw} \text{ev}_{\beta_{a-8,u,w}} \right\|_{s,c_1,D,\vee}^2. \quad (\text{NCT10})$$

NCT4 takes every representative inside this minimum to exactly the representative in NCT9. In particular it takes the complete lower evaluation subspace to the complete conductor-multiple subspace, by NCT6. Let Y be the span of all functionals appearing in NCT10, and K its lower evaluation subspace. Then NCT4 induces the isomorphism

$$Y/K \xrightarrow{A_{D,s}^\vee} \Lambda_{\mathcal{N}_{v,a}} / \Lambda_{\text{im } C_a^\top}, \quad [F] \mapsto [E_A F]. \quad (\text{NCT11})$$

The two quotient Grams in the same fixed value coordinates x are $D_{a,D}^{(s)}$ and $\widehat{D}_{a,D}^{(s)}$. Finite-jet injectivity follows also directly: vanishing numerator derivatives through $N \geq q_a - 1$ kill all upper coefficients by the Vandermonde. Thus this isomorphism holds on every retained endpoint and on every original coefficient fibre, without a genericity or conditioning assumption.

6.3 Rank-dependent exterior bounds on arbitrary subquotients

We prove the finite-dimensional statement in full. Let $B : H_0 \rightarrow H_1$ be an invertible linear map of complex Hilbert spaces, and let $K \subset Y \subset H_0$ be any subspaces. Both quotients have their attained quotient norms. Put $Y_0 = Y \cap K^\perp$ and $Y_1 = BY \cap (BK)^\perp$. Every $y \in Y$ has the unique decomposition $y = x + k$ with $x \in Y_0$, $k \in K$. The squared norm of a lift of $[y]$ is $\|x\|^2 + \|k\|^2$, so x is its minimum lift. Under these isometric identifications the induced quotient map is

$$\overline{B} : Y/K \longrightarrow BY/BK, \quad \overline{B} = P_{Y_1} B|_{Y_0}, \quad \overline{B}^{-1} = P_{Y_0} B^{-1}|_{Y_1}. \quad (\text{NCT12})$$

The second formula follows by applying B^{-1} to any representative and subtracting its K component. It is indeed the inverse because the subtracted components lie respectively in BK and

K . The projections are contractions on every exterior power: in orthonormal bases their singular values are zeros and ones. Restrictions and inclusions of the stated Hilbert subspaces have exterior norm at most one. Consequently for every $0 \leq r \leq \dim(Y/K)$,

$$\|\bigwedge^r \overline{B}\| \leq \|\bigwedge^r B\|, \quad \|\bigwedge^r \overline{B}^{-1}\| \leq \|\bigwedge^r B^{-1}\|. \quad (\text{NCT13})$$

There is no factor depending on $\dim K$. For any ordered independent quotient frame of length r , its squared wedge norm is its Gram determinant. Applying the first inequality to that wedge and the second to its image gives

$$-2 \log \|\bigwedge^r B^{-1}\| \leq \log \frac{\det G_1}{\det G_0} \leq 2 \log \|\bigwedge^r B\|. \quad (\text{NCT14})$$

These statements include $r = 0$, with determinant and exterior norm one. A change of the common value frame by U multiplies both determinants by $|\det U|^2$, which cancels in their ratio.

Successive restrictions and quotients obey the same rank-only bound. Indeed a subspace of Y/K has a full preimage Y' containing K , and its inherited metric is the quotient metric of Y'/K . A subsequent quotient by a subspace with full preimage K' is Y'/K' , since the union of its original affine fibres is precisely the affine fibre modulo K' . Its minimum over those fibres is the same minimum. Induction reduces any finite sequence to NCT12 with the actual full preimages; the distortion does not accumulate a new factor at each intermediate quotient.

Apply this to $B = A_{D,s}^\top$ using NCT5, and to its orthogonal direct sum of b copies, where $b = 1$ or 4. We now specify its finite constants using the original SGT symbol. To avoid confusing the degree a with a circle radius, denote an admissible SGT radius by $\varrho \in (0, 1)$. Set

$$\begin{aligned} w(t) &= -i \arctan t, & g(t) &= t^{-v} e^{-4w(t)} E_A(w(t)), & g(0) &= (-i)^v \mu_v / v!, \\ \rho_n &= \sqrt{n! / (s/2)_n}, & L_{s,D,v} &= \frac{1}{2} \max\{1, s/2 - 1\} \sum_{n=1}^{D+v} \frac{1}{n}. \end{aligned} \quad (\text{NCT15})$$

SGT10 proves the exact original matrix $A_{D,s} = \text{diag}(\rho_0, \dots, \rho_D)^{-1} T_D(g) \text{diag}(\rho_v, \dots, \rho_{D+v})$, where $T_D(g)_{i,j} = g_{j-i}$ for $i \leq j$, and zero otherwise. SGT21 proves that the products of the two forward diagonal operator norms and of the two inverse diagonal operator norms are both at most $e^{2L_{s,D,v}}$.

Here is the full list of original finite constants used in SGT19:

$$\begin{aligned} A_{\text{abs}} &= \sum_{u,w} |a_{uw}|, & B_0 &= \max\{1, \max_{u,w} |\beta_{8,u,w}|\}, & C_v &= A_{\text{abs}} B_0^v / |\mu_v|, \\ R_0 &= B_0^{-1} \min\{1, (v+1)/(2eC_v)\}, & a_0 &= \min\{1/2, \tanh(R_0/2)\}, \\ d_0 &= 1 - \frac{a_0^2}{3(1-a_0^2)}, & M_0 &= \frac{2v!}{|\mu_v|} d_0^{-v} e^{2R_0}, \\ S_D(r) &= \sum_{n=0}^D r^{-n}, & G_{0,D} &= M_0 S_D(a_0), \\ \mathcal{Z}_\varrho &= \{\zeta : |\zeta| < \varrho, g(\zeta) = 0\}, & H_\varrho &= \sum_{\zeta \in \mathcal{Z}_\varrho} m_\zeta, \end{aligned} \quad (\text{NCT16})$$

$$A_{\zeta,h} = \frac{1}{(m_\zeta - h)!} \frac{d^{m_\zeta - h}}{dt^{m_\zeta - h}} \frac{(t - \zeta)^{m_\zeta}}{g(t)} \Big|_{t=\zeta}, \quad 1 \leq h \leq m_\zeta,$$

$$W_\varrho = \text{arctanh } \varrho, \quad m_{E,\varrho} = \min_{|t|=\varrho} |E_A(w(t))|, \quad G_\varrho = A_{\text{abs}} \varrho^{-v} e^{(B_0+4)W_\varrho},$$

$$M_\varrho = \frac{\varrho^v e^{4W_\varrho}}{m_{E,\varrho}} + \sum_{\zeta,h} \frac{|A_{\zeta,h}|}{(\varrho - |\zeta|)^h}, \quad P_\varrho = \sum_{\zeta,h} \frac{|A_{\zeta,h}|}{(1 - |\zeta|)^h},$$

$$F_{\varrho,D} = G_\varrho S_D(\varrho), \quad V_{\varrho,D} = M_\varrho S_D(\varrho) + P_\varrho, \quad V_{\varrho,D}^{\text{cap}} = \min\{G_{0,D}, V_{\varrho,D}\}.$$

Here admissibility means that g has no zero on the indicated circle. The sums retain every pole and its full order. There are finitely many poles in each such closed disk, and there are admissible

radii arbitrarily close to one, as proved in SGT14. SGT11–19 proves from these exact original data that

$$\begin{aligned} \|T_D(g)\| &\leq F_{\varrho,D}, \quad \|T_D(g)^{-1}\| \leq G_{0,D}, \\ T_D(g)^{-1} &= C_{\varrho,D} + R_{\varrho,D}, \quad \text{rank } C_{\varrho,D} \leq H_{\varrho}, \quad \|R_{\varrho,D}\| \leq V_{\varrho,D}. \end{aligned} \quad (\text{NCT17})$$

These are proved internal inputs, including SGT11–13’s explicit zero-free disk, SGT16–18’s exact principal parts and their lower-triangular corrections. In particular NCT17 is a decomposition of the entire inverse; none of its principal parts has been omitted.

For b copies the inverse exceptional rank is at most bH_{ϱ} ; its global norm and its remainder norm are still bounded by $G_{0,D}$ and $V_{\varrho,D}$, respectively. For a rank- r subquotient put $h_b = \min\{r, bH_{\varrho}\}$. On the kernel of the exceptional matrix its inverse has norm at most $V_{\varrho,D}$. The singular-value minimum principle therefore bounds every singular value after the first bH_{ϱ} by that quantity, and every singular value by $G_{0,D}$. Multiplying the largest r singular values, and then the two diagonal factors, gives

$$\begin{aligned} \left\| \bigwedge^r ((A_{D,s}^T)^{\oplus b}) \right\| &\leq e^{2rL_{s,D,v}} F_{\varrho,D}^r, \\ \left\| \bigwedge^r ((A_{D,s}^T)^{\oplus b})^{-1} \right\| &\leq e^{2rL_{s,D,v}} G_{0,D}^{h_b} (V_{\varrho,D}^{\text{cap}})^{r-h_b}. \end{aligned} \quad (\text{NCT18})$$

For clarity, the minimum principle here follows by diagonalizing the positive matrix of the squared norm. Every subspace of codimension $j-1$ meets the span of its first j eigenvectors, and the complement of its first $j-1$ eigenvectors attains the bound. Apply this with the kernel of the exceptional matrix. This proves the estimate even when its rank bound exceeds the whole matrix dimension, when $h_b = r$.

For a fixed admissible ϱ , suppress that parameter in the notation requested for the receiving forms, and define

$$\begin{aligned} \mathcal{U}_{a,D,s,r}^{[b]} &= 4rL_{s,D,v} + 2r \log^+ F_{\varrho,D}, \\ \mathcal{L}_{a,D,s,r}^{[b]} &= 4rL_{s,D,v} + 2h_b \log^+ G_{0,D} + 2(r-h_b) \log^+ V_{\varrho,D}^{\text{cap}}, \\ \mathcal{B}_{a,D,s,r}^{[b]} &= \max\{\mathcal{L}_{a,D,s,r}^{[b]}, \mathcal{U}_{a,D,s,r}^{[b]}\}. \end{aligned} \quad (\text{NCT19})$$

Here $\log^+ x = \max\{0, \log x\}$; all these constants are finite, nonnegative and exactly zero at $r = 0$. The index a records the original grid and centres; the conductor constants themselves depend only on the fixed original conductor and radius.

Let Q be any original rank- r metric obtained by a finite sequence of restrictions and attained quotients from $(D_{a,D}^{(s)})^{\oplus b}$, and let $\widehat{Q}$ be the identical construction, in the identical coefficient and quotient value frames, from $(\widehat{D}_{a,D}^{(s)})^{\oplus b}$. Equations NCT11–14 and NCT18 prove the finite directed statement

$$-\mathcal{L}_{a,D,s,r}^{[b]} \leq \log \det \widehat{Q} - \log \det Q \leq \mathcal{U}_{a,D,s,r}^{[b]}. \quad (\text{NCT20})$$

The full preimage of each restriction and quotient under NCT11 includes the entire conductor-multiple denominator, of dimension bq'_a . NCT13 proves why that dimension contributes no extra volume factor in NCT20. No fixed coefficient map is replaced by an orthogonal surrogate; minimum lifts are used only to prove the bound on its original attained metric.

6.4 The uniform limit and the exact DNE scalar conventions

For fixed $C_D, C_r > 0$, uniformly over $0 \leq D \leq C_D a^2$, $0 \leq r \leq C_r a$, both actual orders, $b = 1, 4$, and every original admissible subquotient,

$$\log \det \widehat{Q} - \log \det Q = o(a^3) = o(aq_a). \quad (\text{NCT21})$$

Here is the limiting proof with the order of choices explicit. Fix an admissible $\varrho < 1$. Since $S_D(\varrho) \leq (D+1)\varrho^{-D}$, NCT16 gives

$$\begin{aligned} \log^+ F_{\varrho,D} &\leq D \log(1/\varrho) + \log(D+1) + \log^+ G_\varrho, \\ \log^+ V_{\varrho,D}^{\text{cap}} &\leq D \log(1/\varrho) + \log(D+1) + \log^+(M_\varrho + P_\varrho), \\ \log^+ G_{0,D} &\leq D \log(1/a_0) + \log(D+1) + \log^+ M_0. \end{aligned}$$

The common term $4rL_{s,D,v}$ is $O(a^2 \log(a+1))$. The exceptional term uses $h_b \leq bH_\varrho$, fixed before the limit, and is $O_\varrho(a^2 + \log(a+1))$. Consequently the limsup of the uniform absolute error divided by a^3 is at most $2C_r C_D \log(1/\varrho)$. Only after this limit choose admissible radii approaching one. The bound tends to zero. This proves NCT21 without any assertion of a zero-free unit disk, bounded pole count near its boundary, or a coefficient condition on the original subquotients.

The original DNE scalar and transpose square is preserved exactly. Let $A_{\ell,a}$ be DNE4's pairing matrix and Z_a the original low-monomial kernel columns. Keep

$$\begin{aligned} D_{a,D}^{(s)} &= A_{\ell,a}^* \overline{(H_{Y,a,N}^{(s)})^{-1}} A_{\ell,a}, & B_{K,a} &= Z_a^\top A_{\ell,a}, \\ Q_{K^\vee} &= (B_{K,a} (D_{a,D}^{(s)})^{-1} B_{K,a}^*)^{-1} = \overline{(Z_a^* H_{Y,a,N}^{(s)} Z_a)^{-1}}, \\ \hat{Q}_{K^\vee} &= (B_{K,a} (\hat{D}_{a,D}^{(s)})^{-1} B_{K,a}^*)^{-1}, & \hat{H}_Y &= \overline{A_{\ell,a} (\hat{D}_{a,D}^{(s)})^{-1} A_{\ell,a}^*}, \\ & & \hat{Q}_{K^\vee} &= \overline{(Z_a^* \hat{H}_Y Z_a)^{-1}}. \end{aligned} \tag{NCT22}$$

For proof, a functional with coordinate column u on a Gram H has norm $u^* \overline{H^{-1}} u$, since its Riesz vector is $H^{-1} \bar{u}$. Substituting this identity and $B_{K,a} = Z_a^\top A_{\ell,a}$ proves every equality in NCT22 by matrix multiplication. The onto map $B_{K,a}$ uses its full fibre, with minimum norm $(B_{K,a} D^{-1} B_{K,a}^*)^{-1}$. This follows by inserting the minimum lift $D^{-1} B_{K,a}^* (B_{K,a} D^{-1} B_{K,a}^*)^{-1} x$ and checking that it is D -orthogonal to $\ker B_{K,a}$. Thus NCT20 applies with $r = \text{rank } B_{K,a}$; inversion changes the sign of the logarithmic determinant difference when it is read on the original restricted kernel Gram.

The primary-coordinate comparison is literally the earlier KRC/RRC map $x \mapsto \alpha_A \mathbf{N}_a Z_{\text{exp},a} x$. For example its four multiplication maps remain

$$M_i = (\alpha_A \mathbf{N}_j Z_{\text{exp},j})^{-1} m_{i,k} (\alpha_A \mathbf{N}_k Z_{\text{exp},k}), \quad j = k + 4. \tag{NCT23}$$

Both scalars and both fixed frame maps are present before their proved cancellation. NCT8–11 acts in the original x coordinates, so the matrices NCT23 are unchanged. If one instead writes the conductor isomorphism as $\alpha_A A_{D,s}$, its ordinary dual is $\alpha_A A_{D,s}^\top$, and its numerator Gram is exactly $|\alpha_A|^2 \hat{D}$. Every rank- r descendant quotient then has determinant factor $|\alpha_A|^{2r}$, because minimization commutes with this positive scalar. NCT3–22 uses the unscaled map and does not introduce this factor into DNE10. This describes the exact scalar map between the two conventions, rather than silently choosing one.

6.5 All original mixed flags and all four-endpoint signs

For $j = k + 4$, take the original coefficient space $\mathcal{U} = \mathbb{C}^{4m_j}$ and maps of EFP18:

$$\begin{aligned} B &= B_{\text{cof}}, \quad \mathbb{B} = \text{diag}(B_j, 4), \quad J = \ker \mathbb{B}, \quad \mathscr{B} = \text{im } B, \quad I = \text{im}(B J_0), \\ S &= \text{im}[B, F_e], \quad W = J + \mathscr{B}, \quad K_e = S \cap W, \quad T_e = S + W, \quad K_i = I \cap J, \quad W_i = I + J. \end{aligned} \tag{NCT24}$$

All spaces and the maps NCT23 are the same coefficient objects as before. Replace only their ambient metric by $\text{diag}(\hat{D}_{j,N-v}^{(s_1)}, 4)$ to define the numerator forms. NCT20 applies to every one of

the following original quotients, with its actual rank:

original form	original quotient	rank	
$G_{h,e}$	$K_e/\mathcal{B}$	h_e	
$G_{s,e}$	S/K_e	c_e	
$G_{t,e}$	T_e/W	c_e	
$Q_{\text{rem},e}$	$\mathcal{U}/T_e$	$r_o - c_e$	
A_{inv}	I/K_i	r_I	(NCT25)
H_{inv}	W_i/J	r_I	
Q_b	W/W_i	r_b	
Q_o	$\mathcal{U}/W$	r_o	
$H_{\text{conn},e}$	$S/\mathcal{B}$	r_e	
$H_{B,j,N-v}$	$\mathbb{C}^{m_j}/\ker B_j$	t_j	

The final row uses one copy; every preceding row uses four. For Q_b , any original EFP21 fixed value-frame factor remains on both forms. The table's maps and ranks are the complete EFP18–21 and CPQ data; NCT12 proves their metric transfer by their full preimages under NCT11. It also applies to every additional restriction of these spaces, including all eight signed-profile spaces $T_e, K_e, W_i, K_i, W, S, J, I$, with their exact original frames. Their dimensions are at most $4m_j = O(k)$, so NCT21 applies uniformly without dropping any zero eigenvalue in a later compressed relative operator.

For a fixed-rank original form at the four cutoffs $N_i = (q_j - 1, q_j, 2q_j - 1, 2q_j)$, write $D_i = N_i - v$ and $\mathcal{R}f = -f(N_0) - f(N_1) + f(N_2) + f(N_3)$. Abbreviate NCT19's constants at each endpoint by $\mathcal{L}_i$ and $\mathcal{U}_i$, retaining its actual degree, order, rank and number of copies. The exact directed return estimate is

$$\begin{aligned}
-\mathcal{U}_0 - \mathcal{U}_1 - \mathcal{L}_2 - \mathcal{L}_3 &\leq \mathcal{R} \log \det \hat{Q} - \mathcal{R} \log \det Q \\
&\leq \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{U}_2 + \mathcal{U}_3.
\end{aligned}
\tag{NCT26}$$

This follows by negating the upper bounds at the two negative endpoints and adding the lower bounds at the two positive endpoints for the lower estimate, and doing the reverse for the upper estimate. Every endpoint has $D_i = O(k^2)$, and all ranks in NCT25 are $O(k)$. Thus each displayed return difference is $o(kq_j)$ in both original source pairs $(1, 1)$ and (k, j) . For a source-degree k form evaluated at the same target cutoffs the identical proof uses its original degree k , centre $k/2$, and $s_0 = 1$ or k ; the uniform domain still holds since $q_j = O(k^2)$.

Every finite CPQ determinant-line identity continues to hold for the numerator metrics, because its proof is the same sequence of Schur minima in the same exact coefficient flags. In particular the original complete complementary returns $V_b, V_{\text{rem}}, V_s, V_A$ all receive NCT26, as do the original equal-rank contraction ratios $\det H_{\text{inv}}/\det A_{\text{inv}}$ and $\det G_t/\det G_s$: subtract the two determinant errors before applying the endpoint signs. For any fixed finite signed sum the absolute error is bounded by the sum of its original summands' $\mathcal{B}$ constants, with multiplicities. No cancellation of an uncomputed summand is asserted. The CPQ affine identities $4S_K = \mathcal{L} + V_b + V_{\text{rem}} + V_s + V_A$ and $P_{\text{rest}} = \mathcal{L} + V_b$ retain their exact forms on these newly specified numerator metrics and differ from their native values by the proved finite errors. The result supplies an exact numerator observation geometry and an $o(kq_j)$ volume transfer; the original mixed numerator matrices and their complete returns remain the quantities to evaluate.

7 The original numerator kernel and the complete mixed spectral return

This calculation receives the full conductor-dual transport proved in NCT. It uses the original DNE numerator frame and the original CPQ flags, with their fixed value coordinates. The source of the construction is the accepted cumulative DNE3,6, SGT7–26, CPQ1–20 and SFB12–16. The period, its logarithm branch, the complete nonreduced conductor and its order at zero are fixed as the original tensor degree varies. No arithmetic coefficient or zero packet is supplied by the following calculation.

7.1 The actual finite numerator observations

Fix the original simple quartet $0 < \delta < 1/2$, $\gamma > 2$, and the original permitted period. For $a \equiv 1 \pmod{4}$, $a \geq 9$, retain

$$\begin{aligned} q_a &= (a+1)^2, & q'_a &= (a-7)^2, & m_a &= q_a - q'_a - v, & c_a &= a/2, \\ \beta_{a,u,w} &= a(1/2 - \delta - i\gamma) + 2u\delta + 2wi\gamma, & 0 &\leq u, w \leq a, \\ E_A(z) &= \sum_{r,t=0}^8 a_{rt} e^{\beta_{8,r,t} z}, & v &= \text{ord}_0 E_A, & \mu_v &= E_A^{(v)}(0) \neq 0. \end{aligned} \tag{NPK1}$$

The proved conductor bound is $v \leq 32$. Let C_a be the original $q'_a \times q_a$ conductor coefficient matrix, so that its ordinary transpose sends the lower exponential coefficient vector to the numerator of its product with E_A . Write $Z_a = Z_{\text{exp},a}$ for DNE3's fixed $q_a \times m_a$ matrix. Thus $N_{Z_a x}/E_A$ is the original analytic germ for coefficient x , and the columns of $[C_a^\top, Z_a]$ are a basis of the numerator subspace whose first v moments vanish. These matrices include their actual chosen minors and every original coefficient; none is replaced by a generic frame.

For $s \in \{1, a\}$ set $b = s/2$, and use the original recurrence and norms

$$\begin{aligned} p_{0,s}(y) &= 1, & p_{1,s}(y) &= y, & p_{n+1,s}(y) &= yp_{n,s}(y) - n(n-1+b)p_{n-1,s}(y), \\ h_{n,s} &= (2\pi)^{s/2} n!(b)_n, & y_{u,w} &= (\beta_{a,u,w} - c_a)/i = (2w-a)\gamma - i(2u-a)\delta, \\ \mathbf{E}_{a,N,s}[n, (u, w)] &= \frac{p_{n,s}(y_{u,w})}{\sqrt{h_{n,s}}}, & 0 \leq n \leq N, & & \mathbf{K}_{a,N,s} &= \mathbf{E}_{a,N,s}^* \mathbf{E}_{a,N,s}. \end{aligned} \tag{NPK2}$$

All square roots in the positive $h_{n,s}$ are positive real. These are functional values of the original Gamma orthonormal polynomials, whose leading coefficient in S is $i^{-n}/\sqrt{h_{n,s}}$. For a numerator coefficient vector d , the column $\mathbf{E}_{a,N,s}d$ is therefore exactly the native functional-value column of N_d at the center $a/2$. It uses the original mass $(2\pi)^{s/2}$.

If $N \geq q_a - 1$, the first q_a rows are an invertible triangular polynomial-frame change times the full original root Vandermonde. The roots are distinct since their real and imaginary index differences vanish separately. Hence $\mathbf{K}_{a,N,s}$ is positive definite. Put

$$\widehat{D}_{a,N}^{(s)} = Z_a^* [\mathbf{K}_{a,N,s} - \mathbf{K}_{a,N,s} C_a^\top (\overline{C}_a \mathbf{K}_{a,N,s} C_a^\top)^{-1} \overline{C}_a \mathbf{K}_{a,N,s}] Z_a. \tag{NPK3}$$

Its exact value interpretation is

$$x^* \widehat{D}_{a,N}^{(s)} x = \min_{\eta \in \mathbb{C}^{q'_a}} \|\mathbf{E}_{a,N,s}(Z_a x + C_a^\top \eta)\|^2. \tag{NPK4}$$

The positive normal equation gives NPK3. Independence of $[C_a^\top, Z_a]$ and injectivity of $\mathbf{E}$ prove strict positivity for $x \neq 0$. This is the complete conductor quotient, including both cross factors and every lower column. By the NCT dual transport, it is the metric on the exact same quotient values as $D_{a,N-v}^{(s)}$ after the map $\Lambda_F \mapsto \Lambda_{E_A F}$. In these literal DNE coordinates the numerator is $Z_a x$. The factor $\alpha_A = v!/\mu_v$ remains in the separate original KRC/RRC primary-coordinate map; it is not inserted into, or removed from, NPK3.

7.2 Evaluation by two original recurrence polynomials

The entire observation kernel in NPK2 has the following exact formula. For two original nodes λ, μ let $z = (\mu - c_a)/i$ and $w = (\lambda - c_a)/i$. Then

$$K_{a,N,s}[\lambda, \mu] = \begin{cases} \frac{p_{N+1,s}(z)p_{N,s}(w) - p_{N,s}(z)p_{N+1,s}(w)}{h_{N,s}(z-w)}, & z \neq w, \\ \frac{p'_{N+1,s}(w)p_{N,s}(w) - p'_{N,s}(w)p_{N+1,s}(w)}{h_{N,s}}, & z = w. \end{cases} \quad (\text{NPK5})$$

The second case occurs among the actual grid entries: it includes $\mu = \beta_{a,a-u,w_0}$ when $\lambda = \beta_{a,u,w_0}$. It must not be omitted as a nongeneric coincidence.

Here is a direct proof with the original normalization. Write $d_n = n(n-1+b) = h_{n,s}/h_{n-1,s}$ for $n \geq 1$. Subtract the recurrence evaluated at z, w to obtain, for $n \geq 1$,

$$(z-w) \frac{p_{n,s}(z)p_{n,s}(w)}{h_{n,s}} = \frac{p_{n+1,s}(z)p_{n,s}(w) - p_{n,s}(z)p_{n+1,s}(w)}{h_{n,s}} - \frac{p_{n,s}(z)p_{n-1,s}(w) - p_{n-1,s}(z)p_{n,s}(w)}{h_{n-1,s}}. \quad (\text{NPK6})$$

At $n = 0$ the separate equality is $(z-w)/h_{0,s} = (p_{1,s}(z)p_{0,s}(w) - p_{0,s}(z)p_{1,s}(w))/h_{0,s}$; there is no $h_{-1,s}$ term. Summing this equality and NPK6 through N telescopes. The recurrence has real coefficients, so $p_{n,s}(\bar{y}) = \overline{p_{n,s}(y)}$; this identifies the sum with the Hermitian kernel NPK2. The numerator in NPK5 vanishes when $z = w$, and its derivative in z gives the displayed second line. This is a polynomial identity, so the limiting line holds without an analytic exclusion. At $N = 0$ both lines give $1/h_{0,s} = 1/(2\pi)^{s/2}$, confirming the mass and sign.

Consequently NPK3 is specified by the two original recurrence polynomials $p_{N,s}, p_{N+1,s}$ and the original finite matrices C_a, Z_a . It does not require a Taylor division by E_A to evaluate this transported form. NCT supplies the exact map back to the divided native form and its complete finite volume error.

7.3 The original eight flags and their numerator row update

Now fix $j = k + 4$, $k = 4\ell + 1 \geq 9$, and one of the original target orders $s = 1, j$. Put $q = q_j$, and retain the original fixed CPQ coefficient maps on $\mathcal{U} = \mathbb{C}^{4mj}$:

$$\begin{aligned} I &= \text{im}(B_{\text{cof}}J_0), \quad B = \text{im } B_{\text{cof}}, \quad S = \text{im}[B_{\text{cof}}, F_e], \quad J = \ker \mathbb{B}, \\ K_i &= I \cap J, \quad W_i = I + J, \quad W = B + J, \quad K_c = S \cap W, \quad T = S + J. \end{aligned} \quad (\text{NPK7})$$

The primitive index remains $e = 0$ or 1 with its original domain. All spaces in NPK7 are the same coefficient subspaces used in SFB and CPQ; the present construction changes none of their maps. For the proof, choose independent fixed columns X for each nonzero subspace, and use determinant one for the empty space. Set $\widehat{\mathcal{D}}_N = \text{diag}(\widehat{D}_{j,N}^{(s)}, 4)$.

For $q \leq N \leq 2q$, let ξ_N be row N of $E_{j,N,s}$, and evaluate all following old matrices at $N-1$. Suppressing their j, s indices only in this display, put

$$\begin{aligned} L &= \overline{C}K_{N-1}C^\top, \quad T_0 = E_{N-1}Z, \quad V_0 = E_{N-1}C^\top, \\ \widehat{t}_N &= \xi_N Z, \quad \widehat{v}_N = \xi_N C^\top, \quad \widehat{\zeta}_N = \widehat{v}_N L^{-1} \widehat{v}_N^*, \\ \widehat{\rho}_N &= \widehat{t}_N - \widehat{v}_N L^{-1} V_0^* T_0, \quad \widehat{U}_N = \text{diag}(\widehat{\rho}_N / \sqrt{1 + \widehat{\zeta}_N}, 4), \\ \widehat{D}_N &= \widehat{D}_{N-1} + \frac{\widehat{\rho}_N^* \widehat{\rho}_N}{1 + \widehat{\zeta}_N}, \quad \widehat{\mathcal{D}}_N = \widehat{\mathcal{D}}_{N-1} + \widehat{U}_N^* \widehat{U}_N. \end{aligned} \quad (\text{NPK8})$$

These formulas retain the complete old conductor Gram L and its scalar denominator. To prove the update, write the old minimizer as $\eta_0 = -L^{-1}V_0^*T_0x$ and replace it by $\eta_0 + h$. The old squared

norm is the old minimum plus h^*Lh ; the new row is $\hat{\rho}_N x + \hat{v}_N h$. Its remaining minimum is $|\hat{\rho}_N x|^2 / (1 + \hat{v}_N L^{-1} \hat{v}_N^*)$, as follows by solving the positive normal equation, or by substitution of $h = -L^{-1} \hat{v}_N^* \hat{\rho}_N x / (1 + \hat{\zeta}_N)$. This proves every line of NPK8.

The eight actual transported compressed matrices are therefore

$$\hat{K}_{X,N} = \hat{U}_N X (X^* \hat{\mathcal{D}}_{N-1} X)^{-1} X^* \hat{U}_N^*, \quad \hat{r}_N = \frac{\hat{\rho}_N \hat{\mathcal{D}}_{N-1}^{-1} \hat{\rho}_N^*}{1 + \hat{\zeta}_N}. \quad (\text{NPK9})$$

For the empty subspace set $\hat{K} = 0_{4 \times 4}$. The isometry by $\hat{\mathcal{D}}_{N-1}^{1/2}$ writes $\hat{K}_X = Y \Pi_X Y^*$, where Π_X is the orthogonal projector onto $\hat{\mathcal{D}}_{N-1}^{1/2} \text{im } X$ and $Y Y^* = \hat{r}_N I_4$. Consequently $0 \preceq \hat{K}_X \preceq \hat{r}_N I_4$ and inclusion of subspaces gives matrix order. All four eigenvalues, including zeros, are retained. The two original chains are exactly $J \subset W_i \subset W \subset T$ and $K_i \subset I \subset K_c \subset S$.

Let $\sigma_X = 1$ for $X = T, K_c, W_i, K_i$ and $\sigma_X = -1$ for $X = W, S, J, I$. The determinant lemma applied to NPK8 gives

$$\begin{aligned} \hat{\ell}_N &= \sum_X \sigma_X \log \det(I_4 + \hat{K}_{X,N}), \\ \hat{\mathcal{L}}_e &= \sum_{N=q}^{2q} \vartheta_N \hat{\ell}_N, \quad \vartheta_q = \vartheta_{2q} = 1, \quad \vartheta_N = 2 \quad (q < N < 2q), \\ \hat{\mathcal{P}}_{k,e} &= \sum_{N=q}^{2q} \vartheta_N \sum_X \sigma_X \sum_{a=1}^4 \log \max(1, \hat{\lambda}_{X,a,N}). \end{aligned} \quad (\text{NPK10})$$

The native row corresponding to this N is exactly $n = N - v$. The weights sum to $2q$, and NPK10 has the unchanged four endpoints $q-1, q, 2q-1, 2q$. The quantity $\hat{\mathcal{L}}_e$ is the complete CPQ common return formed with the transported metric: the same quotient determinant identities prove the signed eight-flag expression. No fixed frame determinant is discarded before taking the return.

7.4 Finite and limiting transfer of the complete signed profile

Denote NCT's nonnegative finite volume-error bounds for b_0 copies, quotient rank r , and an admissible original SGT radius ϱ by $\mathcal{U}_{\varrho;a,D,s,r}^{[b_0]}$ and $\mathcal{L}_{\varrho;a,D,s,r}^{[b_0]}$. They retain the actual SGT poles, residues, zero-free radius, diagonal weights and complete conductor. Define

$$\begin{aligned} B_{a,D,s,r}^{[b_0]} &= \inf_{\substack{0 < \varrho < 1 \\ \varrho \text{ admissible}}} \max(\mathcal{L}_{\varrho;a,D,s,r}^{[b_0]}, \mathcal{U}_{\varrho;a,D,s,r}^{[b_0]}), \\ \mathcal{N} &= (q-1, q, 2q-1, 2q), \quad E_X = \sum_{N \in \mathcal{N}} B_{j,N-v,s,\dim X}^{[4]}, \quad E_{\text{flags}} = \sum_{X \in \{T, K_c, W_i, K_i, W, S, J, I\}} E_X. \end{aligned} \quad (\text{NPK11})$$

At every endpoint NCT gives $|\log \det(X^* \hat{\mathcal{D}}_N X) - \log \det(X^* \mathcal{D}_N X)| \leq B_{j,N-v,s,\dim X}^{[4]}$. Taking the four original signs and then the eight original signs proves

$$|\hat{\mathcal{L}}_e - \mathcal{L}_e| \leq E_{\text{flags}}. \quad (\text{NPK12})$$

The infimum is over the same original function g and its proved admissible radii. It is nonnegative and finite, since SGT supplies at least one such finite bound. The endpoint inequality holds for every admissible radius, hence also for their infimum; attainment is unnecessary. Each $\dim X \leq 4m_j = O(k)$ and $N = O(k^2)$. For each fixed radius the NCT proof bounds the limsup of the displayed infimum divided by kq_k by a constant times $\log(1/\varrho)$, with all remaining terms tending to zero. Only after this limit choose admissible radii tending to one. It follows that $B^{[4]} = o(kq_k)$ and $E_{\text{flags}} = o(kq_k)$. A fixed-radius enclosure itself is not asserted to be little-oh. Any particular finite admissible radius remains a usable, fully specified bound in NPK11–12.

For completeness the threshold error used here has a full direct proof. Put $h(x) = \log(1 + x) - \log \max(1, x) = u(x) - v(x)$, where $u(x) = \log(1 + \min(x, 1))$, and $v(x) = 0$ for $x \leq 1$, $v(x) = \log(2x/(1+x))$ for $x \geq 1$. Both u, v are increasing, vanish at zero and are at most $\log 2$. On any increasing four-term eigenvalue chain, the alternating sum of disjoint increments of u lies between zero and $u(r)$; adding the missing middle increment proves the upper bound. The same argument holds for v . The alternating h sum therefore lies between $-v(r)$ and $u(r)$. Apply this to each of the four ordered eigenvalues on each of the two chains in NPK9; matrix order implies eigenvalue order by the Rayleigh min-max formula. The difference of the two chain sums has absolute value at most $4(u(r) + v(r)) \leq 8 \log 2$ at each row. After the exact weights, both the original and transported threshold errors are at most $16q \log 2$. Combining this with NPK12 proves the finite receiver

$$|\mathcal{P}_{k,e} - \widehat{\mathcal{P}}_{k,e}| \leq E_{\text{flags}} + 32q_j \log 2 = o(kq_k). \quad (\text{NPK13})$$

This transfers the entire signed profile, with its correlations and zero eigenvalues. It does not assert a rowwise approximation of any individual eigenvalue, or a value for the transported profile.

7.5 All complementary returns and the original arithmetic action

For each of the original CPQ forms Q_Y on the specified quotient values (in particular $G_s, Q_{\text{rem}}, Q_b^E, A_{\text{inv}}$), form $\widehat{Q}_Y$ by the same restriction and full affine minimum using $\widehat{\mathcal{D}}_N$. If its rank is r_Y , NCT and the original four signs give the first line below. Define the transported target quotient and its return explicitly by $\widehat{H}_{B,j,N} = (B_j(\widehat{D}_{j,N}^{(s)})^{-1}B_j^*)^{-1}$ and $\widehat{S}_{K,j} = \mathcal{R} \log \det \widehat{H}_{B,j,N}$, in the original onto value frame of B_j . Its full four-copy rank is $4t_j$. Then

$$\begin{aligned} |\widehat{V}_Y - V_Y| &\leq E_Y := \sum_{N \in \mathcal{N}} B_{j,N-v,s,r_Y}^{[4]} = o(kq_k), \\ 4\widehat{S}_{K,j} &= \widehat{\mathcal{L}}_e + \widehat{V}_b + \widehat{V}_{\text{rem}} + \widehat{V}_s + \widehat{V}_A, \\ 4(\widehat{S}_{K,j} - S_{K,j}) &= (\widehat{\mathcal{L}}_e - \mathcal{L}_e) + \sum_{Y=b,\text{rem},s,A} (\widehat{V}_Y - V_Y). \end{aligned} \quad (\text{NPK14})$$

The second line follows by block Schur factorization in precisely CPQ's fixed frames, so the third is its subtraction from CPQ20. Alternatively NCT applies directly to the rank- $4t_j$ target quotient, yielding $|4\widehat{S}_{K,j} - 4S_{K,j}| \leq E_{\text{target}}$, where $E_{\text{target}} = \sum_{N \in \mathcal{N}} B_{j,N-v,s,4t_j}^{[4]}$. Both enclosures hold for the same value, so their intersection is valid. This retains every complementary term and their exact correlation.

The independent proper-source form Q_{inv} uses its original degree- k frame, cutoff and center. Apply NCT there separately. For the exact relative determinant in NFR the identity is

$$\begin{aligned} \log \frac{\det \widehat{A}_{\text{inv}}}{\det \widehat{Q}_{\text{inv}}} - \log \frac{\det A_{\text{inv}}}{\det Q_{\text{inv}}} \\ = (\log \det \widehat{A}_{\text{inv}} - \log \det A_{\text{inv}}) - (\log \det \widehat{Q}_{\text{inv}} - \log \det Q_{\text{inv}}). \end{aligned} \quad (\text{NPK15})$$

The two finite NCT errors add to an enclosure; on the original rank- $O(k)$ and cutoff- $O(k^2)$ domains their four-return is $o(kq_k)$. The two distinct source orders and centers are unchanged in this statement. The EFP same-value maps and HJR inverse-metric identities act on the same quotient coordinates, so NPK14–15 can be substituted into them with their full finite error intervals. In the complete action $J = B + W - P$, transporting only the specified volume term gives exactly $\widehat{J} - J = \widehat{B} - B$ when the original recurrence window W and penalty P are retained. This supplies an explicit error-bearing map into that action, without assigning any missing intrinsic coefficient or cancelling an uncomputed term.

The result is a concrete finite observation problem on the original root grid: NPK3 and NPK5 calculate all transported forms from the original C_a, Z_a and Gamma recurrence; NPK8–10 calculate

their full mixed profile; NPK12–15 prove their quantitative relation to the original divided forms. The surviving large-degree value of these original numerator subquotients is the next arithmetic calculation, and remains present in every displayed identity.

8 Propagation through the original receiving maps

8.1 The evaluated zero-order domain in the original period

The complete PZ, PZX and PZU calculations below evaluate the previously retained value $A_{u^{-1}}(1, 1)$ of this same conductor. In their exact notation put $r = a/\bar{a}$, $T = (\delta + i\gamma)^2/(\delta - i\gamma)^2$, $A_e = (2T - 1)/(3(T - 1))$ and $B_e = (2 - T)/(3(T - 1))$. For the four actual factors $d_j(\tau) = Q_0(L_j(\tau)\mathbf{1})$, the first nonzero inverse-period degree is

$$\ell_A = 0 \quad (r \notin \{1, A_e/B_e\}), \quad \ell_A = 2 \quad (r \in \{1, A_e/B_e\}), \quad b_j = [\tau^{\ell_A}]d_j(\tau) \neq 0. \quad (\text{RMR1})$$

The complete second-order matrices and polynomial nonvanishing certificate at both exceptional phases are PZX1–6; no phase is excluded. Each d_j is even entire by PCL22. Hence, with the actual finite-valued convergent sum $C_j = \sum_{n \geq 1} |[\tau^{\ell_A + 2n}]d_j|$, define $R_A = \max\{R_*, 1, \max_j (2C_j/|b_j|)^{1/2}\}$. For $|u| \geq R_A$ summing that Taylor tail gives

$$\frac{\prod_j |b_j|}{16|u|^{4\ell_A}} \leq |E_{A,u}(0)| \leq \frac{81 \prod_j |b_j|}{16|u|^{4\ell_A}}, \quad v = 0, \quad \alpha_A = 1/E_{A,u}(0). \quad (\text{RMR2})$$

Indeed each $u^{\ell_A}d_j(u^{-1})$ lies within $|b_j|/2$ of b_j , and multiplication of the four complete factors proves both inequalities. The nonzero entire product has only finitely many zeros in $|\tau| \leq 1/R_*$; therefore $v = 0$ also on the complement of the explicitly defined finite set PZU5 in the original domain $|u| \geq R_*$. At its exceptional periods the complete PZ12 derivative test and $v \leq 32$ remain in force.

On this proved $v = 0$ domain, substitution into the original maps gives

$$\begin{aligned} \mathbb{C}[S]_{<v} = 0, \quad t_{0,n} = 0, \quad m_n = 16n - 48, \quad t_n^{\text{EFP}} = 8n - 16, \\ j_{0,n} = 8n - 32, \quad \kappa_e = d_{K,e} = d_{T,e} = 0. \end{aligned} \quad (\text{RMR3})$$

The last three equalities follow from BTK25 and the exact RRC14 sequence. Thus the full low-cycle and connecting-cycle coefficient spaces coincide; their original quotient maps have identical fibres and identical attained metrics. All RRC16 tail corrections vanish. The ROG low-coordinate shear is the identity. The conductor, α_A , every coefficient and the complete CPQ mixed forms remain present: PZU9–11 gives the exact size of this scalar and its unchanged division recursion, including the exceptional phases. In particular these equalities determine the low-order correction; the connecting rank remains the exact CPR18 intersection rank.

8.2 Receiving the conductor estimate on the complete native quotients

The same SGT operator has an exact native-dual application proved in NCT1–26 below. With original degree a , target cutoff $N \geq q_a - 1$ and $D = N - v$, its complex-linear dual is the ordinary transpose $A_{D,s}^\top$. Writing $\mathcal{N}$ for the original numerator germ, Leibniz' rule gives

$$\Lambda_F \circ \mathcal{T}_A = \Lambda_{E_A F}, \quad \Lambda_{\mathcal{N}/E_A} + \sum_{\beta'} \eta_{\beta'} \text{ev}_{\beta'} \longmapsto \Lambda_{\mathcal{N} + E_A \sum_{\beta'} \eta_{\beta'} e^{\beta' z}}. \quad (\text{RMR4})$$

The target functional annihilates every polynomial of degree below v , exactly the dual of the original source quotient in SGT7. The highest required derivative of F is $N - v$, so this is a finite-jet isomorphism with a triangular inverse whose nonzero diagonal is $\binom{n+v}{v} \mu_v$. It maps the entire original lower evaluation fibre onto the entire conductor-multiple fibre.

Let $E_N[n, (u, w)] = \phi_{n,s,a/2}(\beta_{a,u,w})$ through degree N and $K_N = E_N^* E_N$. The original numerator matrix and its attained quotient metric are exactly

$$\hat{D}_N = Z_{\text{exp},a}^* [K_N - K_N C_a^\top (\bar{C}_a K_N C_a^\top)^{-1} \bar{C}_a K_N] Z_{\text{exp},a}. \quad (\text{RMR5})$$

The roots, Gamma mass, centre and conductor coefficients are unchanged. NPK5 evaluates every entry of K_N using the two original recurrence polynomials, including its exact coincident-argument derivative case. The primary scalar α_A stays in the original KRC/RRC coordinate map; RMR4 has coefficient one in the DNE germ frame.

Every rank- r original restriction or iterated affine quotient of $D_{a,D}^{(s)}$ receives the finite NCT20 directed volume bound from the corresponding construction on RMR5. To see why the large conductor denominator introduces no extra rank charge, represent a quotient by its orthogonal minimum lifts: its induced map is a restriction of $A_{D,s}^T$ followed by an orthogonal projection. Apply the same factorization to its inverse. Their r th exterior norms are therefore bounded by the corresponding full r th exterior norms of SGT22. For four copies the exceptional pole rank is $4H_\varrho$; every other SGT constant is retained. For each fixed original period, consequently

$$\log \det \widehat{Q} - \log \det Q = o(aq_a) \quad (D = O(a^2), r = O(a), s = 1 \text{ or } a), \quad (\text{RMR6})$$

uniformly on the original quotient values. The proof first fixes an admissible radius, takes the degree limsup, and then lets admissible radii approach one. A fixed-radius finite enclosure itself is not declared little-oh. NCT supplies every finite constant and the full proof; NPK11 records a finite optimized enclosure by their infimum.

8.3 The actual endpoint inclusion and the numerator return

The complete PRT calculation below identifies the two original connecting coefficient spaces inside this same target. In its exact notation M_0 is the primitive relation-value pairing and V contains the three original polynomials $R_i - (B_i(0)/B_{i*}(0))R_{i*}$. Their transpose matrix $L = V^T M_0$ has rank three, and original Vandermonde interpolation proves

$$S_1 = \text{im } B + F_0 \ker L \subset S_0, \quad 0 \longrightarrow K_0/K_1 \longrightarrow S_0/S_1 \longrightarrow T_0/T_1 \longrightarrow 0. \quad (\text{RMR7})$$

The arrows are the identity on coefficient representatives. The middle rank is three and the outer ranks are $3 - a_k, a_k$, where $a_k = \dim(\mathcal{G}_k \cap \mathcal{C}_k)$ is the actual CPR18 polynomial intersection, including $a_k = 0, 3$. Give these three quotients their original attained metrics, and denote their four-returns by U_B, U_A, U_C , respectively. Block elimination in the same fixed frames proves

$$\begin{aligned} V_{\text{rem},1} - V_{\text{rem},0} &= U_C, & V_{s,0} - V_{s,1} &= U_A - U_B, & \mathcal{L}_0 - \mathcal{L}_1 &= U_C - U_A + U_B, \\ 0 \leq U_C &\leq 2a_k \log A_{j,s}^{\text{end}}, & |\mathcal{L}_0 - \mathcal{L}_1| &\leq 6 \log A_{j,s}^{\text{end}} = O_{\text{actual}}(q_j \log j) = o(kq_k). \end{aligned} \quad (\text{RMR8})$$

All original frame determinants cancel only under the full return. PRT11–20 supplies their complete matrices and the finite constant. In particular $Z_1 - Z_0 = U_C - U_A + U_B$ for $Z_e = V_b + V_{\text{rem},e} + V_{s,e} + V_A$, preserving exactly $4S_{K,j} = \mathcal{L}_0 + Z_0 = \mathcal{L}_1 + Z_1$.

The NCT dual map RMR4 transports every CPQ quotient in these identities, with the same values and fibres, to RMR5 in four copies. The table NCT25 includes $G_s, Q_{\text{rem}}, Q_b^E, A_{\text{inv}}$ and the complete kernel target. Their finite endpoint constants give the exact error identity

$$4(\widehat{S}_{K,j} - S_{K,j}) = \widehat{\mathcal{L}}_e - \mathcal{L}_e + \sum_{Y=b,\text{rem},s,A} (\widehat{V}_Y - V_Y), \quad \widehat{V}_Y - V_Y = o(kq_k). \quad (\text{RMR9})$$

Every term is retained; the directed finite enclosure is NCT26. The independent degree- k proper source is transported on its own centre and cutoffs. Its relative comparison with A_{inv} is the difference of their two determinant errors, exactly NPK15, so the complete EFP/HJR/NFR maps retain their original orientations.

8.4 The evaluated first row and the full transported profile

The first appended native row has $n = q_j - v$. The complete FMG/FRG calculation below evaluates its actual corrected scalar through the original polynomials $g = \mathcal{T}_A \chi_j$, $\chi_- = \chi_{j-8}$ and $\ell = \mu_v \binom{q_j}{v}$.

With $m = q_j - v - q'_j$ and the original Gamma monic minimum $\nu_m = \min_{P \text{ monic, deg } P=m} \|\chi - P\|^2$, it proves

$$1 + \zeta_{\text{all}} = \frac{\|g\|^2}{|\ell|^2 h_{n,s}}, \quad 1 + \zeta_{\text{lower}} = \frac{\nu_m}{h_{n,s}}, \quad 1 + r_n = \frac{\|g\|^2}{|\ell|^2 \nu_m}. \quad (\text{RMR10})$$

The full lower inverse and its denominator remain in this ratio. Both raw logarithms equal $2q_j \log(4/\pi) + O_{\text{actual}}(j \log j)$; their corrected ratio obeys the finite bound $0 \leq \log(1 + r_n) \leq E_{j,s} = O_{\text{actual}}(j \log j)$ with all constants specified in FMG26 and proved in FRG15–31. FMG11–15 also gives a literal equality for each of the eight compressed matrices through the original polynomial-ideal Gram and its exact functional pairing. Both nested chains therefore give

$$|\delta \mathcal{L}_{e,n}| \leq 4E_{j,s}, \quad |\delta \mathcal{P}_{e,n}| \leq 4E_{j,s}, \quad 0 \leq \delta V_{Y,n} \leq 4E_{j,s} \quad (Y = b, \text{rem}, s, A). \quad (\text{RMR11})$$

The endpoint weight is one. This evaluates the entire first-row contribution as $o(kq_k)$ and leaves all the original interior rows.

For every row, NPK3–10 specifies the original numerator-observation metric, its full conductor Schur update, and all eight transported 4×4 compressed matrices. Keep all four eigenvalues and the same signs and weights in its threshold sum $\widehat{\mathcal{P}}_{k,e}$. Applying NCT's volume bounds to the eight fixed flags at the four original endpoints gives

$$|\mathcal{P}_{k,e} - \widehat{\mathcal{P}}_{k,e}| \leq E_{\text{flags}} + 32q_j \log 2 = o(kq_k). \quad (\text{RMR12})$$

Here E_{flags} is the explicitly defined optimized finite enclosure NPK11: take the infimum of the maximum of the two SGT-derived errors over the original admissible radii, separately for each rank and endpoint, and then sum. The bound holds for every admissible radius before taking any infimum. Its little-oh statement follows by the fixed-radius limsup followed by radii tending to one, retaining the full exceptional pole contribution. Thus the surviving leading profile is a precise finite numerator matrix calculation, with all four complementary returns transported by RMR9; its value has not been assigned by a rank count.

A The original period quadric as a residue commutator

Retain $0 < \delta < 1/2$, $\gamma > 2$, the ordered simple quartet $\alpha_1 = 1/2 + \delta + i\gamma$, $\alpha_2 = 1/2 + \delta - i\gamma$, $\alpha_3 = 1/2 - \delta + i\gamma$, $\alpha_4 = 1/2 - \delta - i\gamma$, and $h(s) = \prod_a (s - \alpha_a)$. Put

$$w = s - \frac{1}{2}, \quad r_a = \alpha_a - \frac{1}{2}, \quad \beta = 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad H(w) = w^4 + \beta w^2 + \eta. \quad (\text{RPC1})$$

Thus $r_4 = -r_1$, $r_3 = -r_2$, $r_2 = \overline{r_1}$ and $r_1^2 - r_2^2 = 4i\delta\gamma$. No root or original coordinate is discarded by this translation. Its coefficient isomorphism is $J_c : \mathbb{C}[w]_{<4} \rightarrow E_h$, $J_c p = [p(s - 1/2)]$; its inverse is substitution $s = w + 1/2$ in the unique original representative. The primitive with its original constant is

$$\Phi_h(s) = A + \phi(w), \quad \phi(w) = \frac{w^5}{5} + \frac{\beta w^3}{3} + \eta w, \quad A = \Phi_h(1/2) = \frac{1}{160} + \frac{\beta}{24} + \frac{\eta}{2}. \quad (\text{RPC2})$$

The last equality is forced by $\Phi_h(0) = 0$ and follows by evaluating the displayed odd polynomial at $w = -1/2$.

Let $u \neq 0$ carry the specified original branch of its logarithm. Let Π be the original period map on the rays and orientations of PDE2, and let $F_{ja} = \zeta^{ja} - 1$, $\zeta = e^{2\pi i/5}$, $1 \leq j, a \leq 4$. Its inverse has entries $(F^{-1})_{aj} = \zeta^{-ja}/5$: multiplying the two matrices and using $\sum_{j=1}^4 \zeta^{jb} = 4$ for $5 \mid b$ and -1 otherwise proves this entry by entry. The literal centered coefficient-to-Fourier map is

$$P = F^{-1} \Pi J_c, \quad P_0 = e^{-A/u} P. \quad (\text{RPC3})$$

These are invertible by the complete original period determinant PDE3–13. The contours for P_0 may be taken to be the rays based at $w = 0$ with the original five angles. Indeed translating the old rays gives rays based at $-1/2$. Join corresponding finite endpoints and close each contour at radius R . The integrand is entire. On the bounded-length outer joining curves its exponent is $-R^5/(5|u|) + O(R^4)$, uniformly along the joining curves, so the outer integrals tend to zero. The common finite joining path occurs with opposite coefficients in $-\ell_0 + \ell_j$ and cancels exactly. Cauchy's integral theorem therefore proves this contour identity, including its signs and the factor $e^{A/u}$.

A.1 A flat alternating pairing with all constants

On $\mathbb{C}[w]/(H)$ define the bilinear, rather than sesquilinear, pairing

$$\mathcal{J}(p, q) = \sum_{H(r)=0} \frac{p(r)q(-r)}{H'(r)}. \quad (\text{RPC4})$$

The roots are simple at the original packet. The equivalent polynomial formula is the coefficient of w^3 in the remainder of $p(w)q(-w)$ modulo H . To prove it, use Lagrange interpolation: the coefficient of w^3 in the idempotent at r is $1/H'(r)$. This polynomial formula extends the pairing to every complex value of β, η , including repeated roots, without a division by a repeated-root value. In the increasing frame $1, w, w^2, w^3$, multiplication by w and this pairing are the exact matrices

$$M = \begin{pmatrix} 0 & 0 & 0 & -\eta \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\beta \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & \beta \\ 1 & 0 & -\beta & 0 \end{pmatrix}. \quad (\text{RPC5})$$

For the last matrix, the coefficient functional is zero through degree two, one in degree three, zero in degree four and six, and $-\beta$ in degree five. Inserting the sign $(-1)^j$ from the second argument gives every displayed entry. Thus $J^\top = -J$, $\det J = 1$, and direct multiplication gives $M^\top J + JM = 0$.

Allow the two original odd-phase coefficients to vary complexly, solely to calculate the existing period map. The ray domination $e^{-r^5/(5|u|)+O(r^3+r)}$ on compact parameter sets justifies all coefficient derivatives and integrations by parts below. Ordinary division of w^{j+3} by H , followed by differential reduction, gives

$$\partial_\eta P_0 = \frac{1}{u} P_0 M, \quad \partial_\beta P_0 = \frac{1}{3u} P_0 (M^3 - uN), \quad N = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{RPC6})$$

For clarity, the quotient derivatives for columns $j = 0, 1, 2, 3$ are $0, 0, 1, 2w$, respectively; hence this is the full correction matrix. The two outer boundary terms vanish, and the two values at the common origin cancel. No differential remainder was replaced by an arithmetic remainder in RPC6.

There is also a direct nonvanishing proof on this entire two-parameter family. The displayed matrices satisfy $\text{Tr } M = 0$, $\text{Tr } M^3 = 0$ and $\text{Tr } N = 0$. The first and third follow from their diagonals; the second follows by multiplying RPC5 three times, or from the vanishing third Newton sum of the even monic polynomial H . Multilinearity of the determinant in its columns then gives $\partial_\eta \det P_0 = \partial_\beta \det P_0 = 0$ from RPC6, without first assuming an inverse exists. At the origin the literal Gamma ray integrals give the diagonal matrix G_u defined in RPC8, whose determinant is nonzero. Hence P_0 is invertible at every point of $\mathbb{C}^2$, justifying the inverses throughout this family.

The matrix identity $M^\top J + JM = 0$ also gives $(M^3)^\top J + JM^3 = 0$. Direct multiplication of RPC5–6 gives

$$\partial_\beta J + \frac{1}{3}(N^\top J + JN) = 0. \quad (\text{RPC7})$$

In particular $\partial_\beta J$ has entries $(2, 3) = 1$, $(3, 2) = -1$ when indices begin at zero, while $N^\top J + JN$ has the opposite threefold entries. Differentiate the inverse matrices in $P_0^{-\top} J P_0^{-1}$. RPC6–7 show that both its β and its η derivatives vanish. This matrix is consequently constant on the connected coefficient space $\mathbb{C}^2$.

Define the unchanged positive-Gamma ray constants

$$g_a = 5^{a/5-1} e^{\pi i a/5} \Gamma(a/5), \quad G = \text{diag}(g_1, g_2, g_3, g_4), \quad G_u = \text{diag}(u^{a/5} g_a)_{a=1}^4. \quad (\text{RPC8})$$

At $\beta = \eta = 0$, literal substitution in each original ray integral gives $P_0 = G_u$. Every g_a is nonzero, since its Gamma factor is the integral of a strictly positive function. We have therefore proved the complete flat-pairing identity

$$\begin{aligned} \Omega_0 &= P_0^{-\top} J P_0^{-1} = G_u^{-\top} J|_{\beta=0} G_u^{-1}, \\ (\Omega_0)_{14} &= -\frac{1}{u g_1 g_4}, \quad (\Omega_0)_{23} = \frac{1}{u g_2 g_3}, \quad (\Omega_0)_{ba} = -(\Omega_0)_{ab}, \\ \Omega &= P^{-\top} J P^{-1} = e^{-2A/u} \Omega_0. \end{aligned} \quad (\text{RPC9})$$

All entries not listed are zero; the indices in RPC9 begin at one. This formulation retains the Gamma constants directly and requires no unevaluated normalization of a period or source mass. For $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$, the two paired sums of weights are five. It follows entry by entry that

$$D^\top \Omega D = \Omega, \quad D^\top \Omega_0 D = \Omega_0. \quad (\text{RPC10})$$

A.2 The full arithmetic unit and the exact quadratic form

Retain the original $v_h = 2\xi/h$ and its complete simple-root unit values. Reflection and conjugation of this original function give

$$v_h(\alpha_1) = v_h(\alpha_4) = a, \quad v_h(\alpha_2) = v_h(\alpha_3) = \bar{a}, \quad a \neq 0. \quad (\text{RPC11})$$

Indeed both numerator and denominator are invariant under $s \mapsto 1-s$ and conjugate under $s \mapsto \bar{s}$, first away from the roots and then by the removable values. The complete zero orders give nonzero values at these simple roots. Write

$$c_3 = \frac{a^{-2} - \bar{a}^{-2}}{4i\delta\gamma}, \quad c_1 = \frac{a^{-2} + \bar{a}^{-2}}{2} - (\delta^2 - \gamma^2)c_3, \quad \mathbf{r}(w) = c_1w + c_3w^3. \quad (\text{RPC12})$$

Both coefficients are real, and $(c_1, c_3) \neq (0, 0)$. Directly, $c_1 + c_3r_1^2 = a^{-2}$ and $c_1 + c_3r_2^2 = \bar{a}^{-2}$; the same equalities hold at the respective opposite roots. Thus $\mathbf{r}(M) = M v_h(M + 1/2)^{-2}$ in the original finite algebra, with the entire function evaluated by its complete original arithmetic unit. RPC12 is an exact interpolation of this specific multiplier.

For $Q_0(t) = t_1t_4 - t_2t_3$, let J_{prim} be the original Lagrange coordinate isomorphism and set

$$B = F^{-1}\Pi U J_{\text{prim}}, \quad Q(x) = Q_0(B^{-1}x), \quad \mathcal{H} = P\mathbf{r}(M)P^{-1}. \quad (\text{RPC13})$$

If $p = P^{-1}x$, the primary components of $B^{-1}x$ are exactly $p(r_1)/a, p(r_2)/\bar{a}, p(r_3)/\bar{a}, p(r_4)/a$. Since $H'(r) = 4r(r^2 + \gamma^2 - \delta^2)$, its values give $r/H'(r) = 1/(8i\delta\gamma)$ on roots one and four, and $r/H'(r) = -1/(8i\delta\gamma)$ on roots two and three. Inserting these four values into RPC4 proves the exact identity

$$Q(x) = 4i\delta\gamma p^\top \mathbf{r}(M)^\top J p = 4i\delta\gamma x^\top \mathcal{H}^\top \Omega x. \quad (\text{RPC14})$$

There is no conjugate transpose in this quadratic identity. Because $\mathbf{r}$ is odd, $\mathbf{r}(M)^\top J + J\mathbf{r}(M) = 0$; hence the matrices in RPC14 are symmetric. This also proves $\mathcal{H}^\top \Omega + \Omega \mathcal{H} = 0$. The four eigenvalues of $\mathbf{r}(M)$ are exactly $r_1/a^2, \bar{r}_1/a^2, -\bar{r}_1/a^2, -r_1/a^2$. They are nonzero, although they need not be distinct. Both B and Q_0 are invertible as linear and quadratic matrices, respectively, so Q always has rank four, including the possible eigenvalue collisions.

By RPC10, substituting Dx into RPC14 transports $\mathcal{H}$ to $D^{-1}\mathcal{H}D$. The resulting symmetric quadratic matrices are equal if and only if the corresponding $\mathcal{H}$ matrices are equal, because Ω is invertible. Consequently

$$Q(Dx) = Q(x) \iff [\mathcal{H}, D] = 0 \iff \mathcal{H}_{ab} = 0 \text{ for every } a \neq b. \quad (\text{RPC15})$$

This is an exact test in the actual original period matrix and unit. The final equivalence uses the four distinct diagonal entries of D . It makes no assumption that their off-diagonal tests are independent.

For completeness, a rank-four quadratic semi-invariant of cyclic weight $j \neq 0$ is impossible: its coefficient in row j could pair only with the missing weight zero, so that row is zero. Thus any semi-invariant rank-four quadratic must have weight zero. Since five is prime, the projective orbit of Q under D has size one or five, and size one is exactly RPC15. Every failure of any displayed off-diagonal test proves size five.

A.3 An explicit original-parameter radius forcing orbit five

Here we prove that RPC15 fails for all sufficiently large $|u|$, with a completely specified finite radius for the given original quartet and unit. The argument evaluates the original periods; it does not choose or insert a replacement amplitude.

Set $b = \beta u^{-2/5}$, $c = \eta u^{-4/5}$, and let $T(b, c)$ be the Fourier period matrix for $z^5/5 + bz^3/3 + cz$ at $u = 1$, in columns $1, z, z^2, z^3$ and on the five rays of angles $(\pi + 2\pi j)/5$. Substitution $w = u^{1/5}z$, with exactly the original branch, gives

$$P_0 = T(b, c)S_u, \quad S_u = \text{diag}(u^{a/5})_{a=1}^4, \quad S_u M S_u^{-1} = u^{1/5} M(b, c), \quad T(0, 0) = G. \quad (\text{RPC16})$$

Every column includes its original differential factor. The matrix $M(b, c)$ is RPC5 with β, η replaced by b, c ; in particular $N_0 := M(0, 0)$ has its three subdiagonal entries equal to one.

The following positive constants are explicit finite real integrals or finite expressions in the original Gamma constants:

$$\begin{aligned}
E_j &= \frac{8}{5} \int_0^\infty r^j (r^3/3 + r) e^{-r^5/5 + r^3/3 + r} dr \quad (0 \leq j \leq 3), \\
C_T &= 2 \left(\sum_{j=0}^3 E_j^2 \right)^{1/2}, \quad g_+ = \max_a |g_a|, \quad g_-^{-1} = \max_a |g_a|^{-1}, \\
\kappa &= \max_{1 \leq a < 4} |g_{a+1}/g_a|, \quad K_0 = GN_0 G^{-1}, \\
C_K &= 4C_T g_-^{-1} + 2g_+ g_-^{-1} + 2g_+ g_-^{-2} C_T, \\
C_{K,3} &= C_K(3\kappa^2 + 3\kappa C_K + C_K^2), \quad \epsilon_0 = \min\{1, (2C_T g_-^{-1})^{-1}\}.
\end{aligned} \tag{RPC17}$$

Here g_-^{-2} means $(\max_a |g_a|^{-1})^2$. The negative fifth degree exponent proves convergence of every integral. All norms in the following estimates are the Euclidean operator norm, with the Frobenius norm used explicitly as an upper bound where stated.

Write $\epsilon = |b| + |c|$. When $|b|, |c| \leq 1$, on each ray

$$|e^{bz^3/3 + cz} - 1| \leq \epsilon(r^3/3 + r)e^{r^3/3 + r}.$$

Each contour has two rays, and each entry of F^{-1} has absolute value $1/5$ with four summands. The entry in column j of $T - G$ is therefore at most $E_j \epsilon$. Summing the squares over all four rows proves

$$\|T - G\| \leq \|T - G\|_F \leq C_T \epsilon. \tag{RPC18}$$

For $\epsilon \leq \epsilon_0$, the convergent inverse series for $I + G^{-1}(T - G)$ gives $\|T^{-1}\| \leq 2g_-^{-1}$ and $\|T^{-1} - G^{-1}\| \leq 2g_-^{-2} C_T \epsilon$. Also $\|M(b, c) - N_0\| \leq \epsilon$, $\|M(b, c)\| \leq 2$, and $\|N_0\| = 1$. Subtract the products $K := TM(b, c)T^{-1}$ and K_0 in three terms, changing the left factor, then the middle factor, then the inverse factor. The preceding inequalities give, with the complete RPC17 constants,

$$\|K - K_0\| \leq C_K \epsilon, \quad \|K\| \leq \kappa + C_K \epsilon, \quad \|K^3 - K_0^3\| \leq C_{K,3} \epsilon. \tag{RPC19}$$

For the last assertion use the exact noncommutative identity $K^3 - K_0^3 = (K - K_0)K^2 + K_0(K - K_0)K + K_0^2(K - K_0)$. Its norm is at most $C_K \epsilon$ times $(\kappa + C_K)^2 + \kappa(\kappa + C_K) + \kappa^2$, as claimed.

The original multiplication and full unit in RPC13 now give exactly

$$\mathcal{H} = c_1 u^{1/5} K + c_3 u^{3/5} K^3, \quad (K_0^3)_{41} = g_4/g_1, \quad (K_0)_{21} = g_2/g_1. \tag{RPC20}$$

The scalar $e^{A/u}$ cancels from this conjugation, while it remains in the quadratic form through RPC9. Put $B_* = \beta + \eta > 0$. For $|u| \geq 1$, $\epsilon \leq B_* |u|^{-2/5}$.

If $c_3 \neq 0$, define the completely specified radius

$$T_* = \max \left\{ 1, \frac{B_*}{\epsilon_0}, \frac{2(|c_3| C_{K,3} B_* + |c_1|(\kappa + C_K))}{|c_3 g_4/g_1|} \right\}, \quad R_* = T_*^{5/2}. \tag{RPC21}$$

For every u on its original branch with $|u| \geq R_*$, RPC19–20 imply

$$\left| u^{-3/5} \mathcal{H}_{41} - c_3 g_4/g_1 \right| \leq |u|^{-2/5} (|c_3| C_{K,3} B_* + |c_1|(\kappa + C_K)) \leq \frac{1}{2} |c_3 g_4/g_1|. \tag{RPC22}$$

In particular $\mathcal{H}_{41} \neq 0$. In the quadratic matrix RPC14 this is the nonzero $(1, 1)$ entry $4i\delta\gamma \mathcal{H}_{41} \Omega_{41}$, whose cyclic weight is two.

If $c_3 = 0$, then $c_1 \neq 0$. In this case define

$$T_* = \max \left\{ 1, \frac{B_*}{\epsilon_0}, \frac{2C_K B_*}{|g_2/g_1|} \right\}, \quad R_* = T_*^{5/2}. \tag{RPC23}$$

For every $|u| \geq R_*$, the same estimates give

$$\left| u^{-1/5} \mathcal{H}_{21} - c_1 g_2 / g_1 \right| \leq |c_1| C_K B_* |u|^{-2/5} \leq \frac{1}{2} |c_1 g_2 / g_1|. \quad (\text{RPC24})$$

Thus $\mathcal{H}_{21} \neq 0$. The $(1, 3)$ entry of the quadratic matrix is then $4i\delta\gamma\mathcal{H}_{21}\Omega_{23}$, which has cyclic weight four; symmetry retains its $(3, 1)$ partner. Both cases prove the unconditional conclusion for the actual original parameters and the explicit radius belonging to those parameters:

$$\boxed{|u| \geq R_* \implies \#\{[Q(D^j x)] : 0 \leq j < 5\} = 5.} \quad (\text{RPC25})$$

The bracket denotes the projective class of the quadratic polynomial. RPC21 or RPC23 supplies its radius for every nonzero original unit; neither case assumes a condition on an uncomputed period coefficient.

Finally, the original periods are holomorphic on the connected logarithm cover of $\mathbb{C}^*$, and on each branch domain: locally the contours may be held in common decay sectors and differentiated under the dominated integral, and contour deformation identifies these local definitions. Their determinant never vanishes. Hence all entries of $\mathcal{H}$ are holomorphic there. RPC22 or RPC24 proves that its respective decisive entry is not identically zero on the logarithm cover and, by the identity theorem on that connected cover, on any open branch domain. The zeros of that one entry are isolated, by its convergent local power series. The exact exceptional set

$$\mathcal{E} = \{u : \mathcal{H}_{ab}(u) = 0 \text{ for all } a \neq b\} \quad (\text{RPC26})$$

is a subset of those isolated zeros and satisfies $|u| < R_*$. Thus it has no accumulation inside the branch domain; accumulation at $u = 0$ or a branch boundary has not been excluded. RPC15 is the full calculation specifying this set. No assertion that it is empty at arbitrary finite complex u is made or used.

B The literal original conductor limit and an invertible pivot restriction

Retain all original hypotheses and maps of PQO, RPC, OCF and CEP. In particular $0 < \delta < 1/2$, $\gamma > 2$, the original unit values are $(a, \bar{a}, \bar{a}, a)$ with $a \neq 0$, and the same logarithm branch is used throughout. Write $\beta = 2(\gamma^2 - \delta^2)$, $\eta = (\delta^2 + \gamma^2)^2$, and $\mathfrak{a} = 1/160 + \beta/24 + \eta/2$. No source order is changed.

B.1 An entire coefficient matrix in the inverse original period

Let P_0 be RPC3's centered Fourier period matrix with its factor $e^{a/u}$ removed by the stated exact identity, and retain $G_u = \text{diag}(u^{a/5} g_a)_{a=1}^4$ with RPC8's full Gamma constants. Introduce the coefficient matrix $\mathcal{R}$ by the following identity, rather than replacing the original period:

$$P = e^{a/u} P_0 = e^{a/u} G_u \mathcal{R}(u^{-1}). \quad (\text{PCL1})$$

The row index below is $1 \leq a \leq 4$, and the column index is $0 \leq r \leq 3$. The complete series is

$$\begin{aligned} \mathcal{R}_{ar}(z) &= \sum_{n=0}^{\infty} R_{ar,n} z^n, \\ R_{ar,n} &= \sum_{\substack{p,h \geq 0 \\ 2p+4h=5n+r+1-a}} \frac{\beta^p \eta^h}{3^p p! h!} (-1)^L 5^L (a/5)_L, \quad L = p + h - n. \end{aligned} \quad (\text{PCL2})$$

Every sum at a fixed n is finite. For each contributing pair, $r + 3p + h + 1 = a + 5L$ and $L \geq 0$. To prove the latter assertion, the left side is positive and $1 \leq a \leq 4$, so a negative multiple of five cannot be their difference. The exponent n is nonnegative: its numerator $2p + 4h + a - r - 1$ is an integer multiple of five and is at least -3 . Thus a negative value of n is impossible.

For a direct derivation, divide every term of the absolutely convergent original series PQO9 by $u^{a/5} g_a$. The quotient of its Gamma factors is exactly

$$\frac{5^{(a+5L)/5-1} e^{\pi i(a+5L)/5} \Gamma(a/5 + L)}{g_a} = (-1)^L 5^L (a/5)_L. \quad (\text{PCL3})$$

Its remaining power of u is $u^{L-p-h} = u^{-n}$. These identities prove PCL2 including its signs, factorials and all unit-independent period coefficients. Absolute convergence of PQO9 permits grouping these terms. For each prescribed complex $z \neq 0$, put $u = 1/z$ on any of its original branches. The resulting sum of absolute values is finite. Consequently PCL2 converges absolutely for every complex z , and defines an entire matrix. This also proves that this matrix is independent of the branch; all branch factors remain in G_u in PCL1.

The complete constant matrix in PCL2 is

$$J_{\infty} = \begin{pmatrix} 1 & 0 & -\beta/3 & 0 \\ 0 & 1 & 0 & -2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad J_{\infty}^{-1} = \begin{pmatrix} 1 & 0 & \beta/3 & 0 \\ 0 & 1 & 0 & 2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{PCL4})$$

Indeed $n = 0$ forces $2p + 4h = r + 1 - a \leq 3$, so either $p = h = 0$ or $p = 1, h = 0, r + 1 - a = 2$. PCL3 gives respectively the diagonal entries and the two negative entries displayed. In particular, the original row-scaled period limit includes both cubic-phase corrections. It is not a diagonal coefficient map.

RPC's determinant proof holds on the complete coefficient family and gives $\det P_0 = \det G_u$. Alternatively divide PQO11 by the product of the four unchanged Gamma ray factors. Therefore

$$\det \mathcal{R}(z) = 1, \quad \mathcal{R}(z)^{-1} = \text{adj } \mathcal{R}(z) \quad (z \in \mathbb{C}). \quad (\text{PCL5})$$

The equality first holds for $z \neq 0$ by PCL1 and extends to zero by continuity. Both the matrix and its inverse are entire. For any fixed $R > 0$, the finite, completely specified constant

$$C_R = \sum_{n \geq 1} \|R_n\|_1 R^{n-1} < \infty \quad (\text{PCL6})$$

gives $\|\mathcal{R}(z) - J_\infty\|_1 \leq C_R|z|$ on $|z| \leq R$. Here $\|\cdot\|_1$ is the induced column-sum matrix norm and $R_n = (R_{ar,n})$. Convergence follows from the entire power series in finite dimension, also at any larger radius. Thus this is a bound in the actual coefficients, with no uncomputed nonzero minor used as a premise.

B.2 All eighty-one actual conductor coefficients and their limit

Let V be the original centered evaluation Vandermonde in PQO12, $U = \text{diag}(a, \bar{a}, \bar{a}, a)$, and $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$. The complete period matrix from primary to Fourier coordinates is

$$\mathbf{H}(u) = e^{a/u} G_u \mathcal{R}(u^{-1}) V^{-1} U. \quad (\text{PCL7})$$

For $1 \leq j \leq 4$ define the entire matrix and biform

$$\begin{aligned} L_j(z) &= U^{-1} V \mathcal{R}(z)^{-1} D^{-j} \mathcal{R}(z) V^{-1} U, \\ t(Z, W) &= (ZW, Z, W, 1)^\top, \\ A_z(Z, W) &= \prod_{j=1}^4 Q_0(L_j(z) t(Z, W)) = \sum_{r,s=0}^8 a_{rs}(z) Z^r W^s. \end{aligned} \quad (\text{PCL8})$$

These are precisely all original OCF4 coefficients at $z = u^{-1}$. To prove this, compute

$$\mathbf{H}(u)^{-1} D^{-j} \mathbf{H}(u) = L_j(u^{-1}) : \quad (\text{PCL9})$$

the scalar $e^{a/u}$ cancels with its inverse, and the two diagonal matrices G_u, D^{-j} commute. OCF's definition $Q_j(x) = Q_0(\mathbf{H}^{-1} D^{-j} x)$ followed by $x = \mathbf{H}t$ then proves PCL8. The homogeneous version uses $t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)$ and retains the bidegree $(8, 8)$. Thus no unit or period entry has been omitted by the affine notation.

For an explicit finite formula for the individual coefficients, write $L_{j,be}$ for a matrix entry, and associate the four pairs $(\rho_e, \sigma_e) = (1, 1), (1, 0), (0, 1), (0, 0)$ to its columns. Put

$$\begin{aligned} c_{j,rs}(z) &= \sum_{\substack{1 \leq e, f \leq 4 \\ \rho_e + \rho_f = r, \sigma_e + \sigma_f = s}} (L_{j,1e} L_{j,4f} - L_{j,2e} L_{j,3f}), \\ a_{rs}(z) &= \sum_{\substack{0 \leq r_j, s_j \leq 2 \\ \sum_j r_j = r, \sum_j s_j = s}} \prod_{j=1}^4 c_{j,r_j s_j}(z). \end{aligned} \quad (\text{PCL10})$$

Expanding the two products defining Q_0 proves the first line; expanding its four original orbit factors proves the second. Both lines are finite and retain all ordered-pair multiplicities. Equations PCL2, PCL5 and PCL10 are a convergent formula for every actual coefficient, with no freely assigned amplitude.

Their exact leading values are obtained by the explicit matrices

$$L_j(0) = U^{-1} V J_\infty^{-1} D^{-j} J_\infty V^{-1} U, \quad a_{rs}(u^{-1}) = a_{rs}(0) + O_{\text{actual}}(|u|^{-1}). \quad (\text{PCL11})$$

The equality for $L_j(0)$ follows from PCL4, and the bound follows from the entire coefficient series. More explicitly, if $a_{rs}(z) = \sum_{n \geq 0} a_{rs,n} z^n$, the finite constant $\sum_{r,s} \sum_{n \geq 1} |a_{rs,n}| R^{n-1}$ bounds the sum of all eighty-one errors divided by $|z|$ on $|z| \leq R$. The coefficient functions are single-valued

entire functions of u^{-1} even though the original period itself retains its logarithm branch. This assertion follows from the exact conjugation PCL9, not from a change of the original branch.

At any specified k , the complete central convolution section is therefore the explicitly evaluated entire matrix

$$C_{\text{in}}(z)_{(a,b),(i,j)} = a_{i+4-a,j+4-b}(z), \quad 0 \leq a, b, i, j \leq k-8, \quad (\text{PCL12})$$

with all out-of-range coefficients zero. PCL10–12 give its full constant matrix and all successive coefficients. They do not discard off-diagonal coefficients or replace this matrix by its central coefficient a_{44} . In particular the calculation of the limit has not asserted a sign or diagonal-dominance inequality for this complete original matrix.

B.3 An exact positivity slice for the literal positive period ray

For $u > 0$ on any of its original logarithm lifts, PQO21 gives $P_0 = E_{\text{br}}R$ with R real invertible and E_{br} diagonal unitary commuting with D . Its exact centered quadratic coefficient matrix is S of PQO13; all its entries are purely imaginary. Consequently, for real $y \in \mathbb{R}^4$,

$$\overline{Q_j(E_{\text{br}}y)} = -Q_{5-j}(E_{\text{br}}y) \quad (1 \leq j \leq 4). \quad (\text{PCL13})$$

Indeed the original quadratic matrix in these coordinates is $e^{-2a/u}R^{-\top}SR^{-1}$, which is purely imaginary. Complex conjugation changes its sign and changes D^{-j} into D^j ; this is precisely the displayed formula.

Let P_{conj} be the permutation matrix exchanging primary indices 1, 2 and 3, 4. For real Z and $|W| = 1$, direct substitution gives $P_{\text{conj}}t(Z, W) = W^{-1}t(Z, W)$. The original roots and unit satisfy $\bar{V} = P_{\text{conj}}V$ and $P_{\text{conj}}\bar{U} = UP_{\text{conj}}$. Thus

$$\overline{V^{-1}Ut(Z, W)} = W^{-1}V^{-1}Ut(Z, W). \quad (\text{PCL14})$$

Choose either square root w of W . Then $p = w^{-1}V^{-1}Ut(Z, W)$ is real by PCL14, and $Ht = wE_{\text{br}}y$ for the real vector $y = e^{a/u}Rp$. Homogeneity and PCL13 prove the exact nonnegative slice identity

$$\boxed{W^{-4}A_{u^{-1}}(Z, W) = |Q_1(E_{\text{br}}y)Q_2(E_{\text{br}}y)|^2 \geq 0 \quad (Z \in \mathbb{R}, |W| = 1).} \quad (\text{PCL15})$$

The choice of sign of w has no effect because the quadratic forms have even degree. In particular, comparing the Laurent coefficients on the unit circle, and then the polynomial coefficients in real Z , gives the exact symmetry

$$a_{rs}(u^{-1}) = \overline{a_{r,8-s}(u^{-1})}. \quad (\text{PCL16})$$

PCL11 extends this identity and PCL15 to the coefficient limit at $u^{-1} = 0$ by continuity. The slice in PCL15 has one real coordinate and one unit-circle coordinate; it has not changed either coordinate into a second unit circle.

B.4 A complete invertible restriction supplied by the original pivot

At each actual u in the proved five-orbit locus, use precisely OCF5's lexicographically largest nonzero coefficient (r_*, s_*) , $a_* = a_{r_*s_*} \neq 0$, and its original pivot rectangle $\mathcal{P}_k = \{(r_* + i, s_* + j) : 0 \leq i, j \leq k-8\}$. Let $\mathcal{E}_k$ be its complement and S_k the original insertion of its coordinates. No new coefficient has been selected. The full original maps satisfy

$$CN_k^\top = 0, \quad S_k^\top N_k^\top = I_\Delta, \quad \text{im } N_k^\top = V_A. \quad (\text{PCL17})$$

The first is the transpose of $N_k C^\top = 0$ in OCF7; the second is the transpose of $N_k S_k = I$. Dimension Δ then proves the last equality, since C has full row rank. It follows that restriction to the actual strip is the exact isomorphism

$$S_k^\top : V_A \longrightarrow \mathbb{C}^{\mathcal{E}_k}, \quad (S_k^\top|_{V_A})^{-1} = N_k^\top. \quad (\text{PCL18})$$

Equivalently its square pivot block has nonzero diagonal a_* and is triangular in the elimination order; OCF's complete division supplies its inverse. This is an alternative restriction to the central one in PCL12, with the exact changed set of roots.

Let $P_{\mathcal{P},k}(y) = \prod_{(a,b) \in \mathcal{P}_k} (y - \omega_{ab})$ and $P_{\mathcal{E},k}(y) = \prod_{(a,b) \in \mathcal{E}_k} (y - \omega_{ab})$, where $\omega_{ab} = (2b - k)\gamma - i(2a - k)\delta$. Thus $P_k = P_{\mathcal{P},k}P_{\mathcal{E},k}$ exactly. Let $V_{\mathcal{E}}$ evaluate $1, y, \dots, y^{\Delta-1}$ at those actual outer roots and set $D_{\mathcal{P},\mathcal{E}} = \text{diag}(P_{\mathcal{P},k}(\omega_{ab}))_{\mathcal{E}_k}$. These two matrices are invertible by distinctness of the original roots. The full polynomial-target restriction and its exact inverse are

$$\begin{aligned}\rho_{\mathcal{E}} &= V_{\mathcal{E}}^{-1} D_{\mathcal{P},\mathcal{E}}^{-1} S_k^{\top}|_{V_A}, \\ \rho_{\mathcal{E}}^{-1} &= N_k^{\top} D_{\mathcal{P},\mathcal{E}} V_{\mathcal{E}}.\end{aligned}\tag{PCL19}$$

Multiplying these matrices and using PCL17 proves both inverse identities, with all inner products of root differences retained. In particular the original residual becomes the following specified onto map, with its exact complete kernel:

$$\Xi_{\mathcal{E}} = \Xi_k^{\text{fix}} N_k^{\top} D_{\mathcal{P},\mathcal{E}} V_{\mathcal{E}}, \quad \ker \Xi_{\mathcal{E}} = \rho_{\mathcal{E}}(K).\tag{PCL20}$$

This follows by composition with the proved isomorphism, not by deleting any common-kernel vector.

Finally the original metric and its entire source graph are carried by this isomorphism as follows. Write $J_{\mathcal{E}} = N_k^{\top} D_{\mathcal{P},\mathcal{E}} V_{\mathcal{E}}$. At every original endpoint the actual conductor metric in these outer coordinates is exactly

$$H_{\mathcal{E},N} = J_{\mathcal{E}}^* G_N J_{\mathcal{E}}, \quad Q_{\Xi,N}^{\text{fix}} = (\Xi_{\mathcal{E}} H_{\mathcal{E},N}^{-1} \Xi_{\mathcal{E}}^*)^{-1}.\tag{PCL21}$$

The first formula is the pullback along $\rho_{\mathcal{E}}^{-1}$; the minimum-section proof of GIP6 gives the second. On the full GIP graph its target map is $\rho_{\mathcal{E}} E_N^{\text{graph}}$, with the same complete t ideal columns, and composing with $\Xi_{\mathcal{E}}$ gives exactly B_N^{graph} . Its residual kernel is therefore all $t + r$ columns in GIP4, including the low- v coordinates. The coefficient-space bound for N_k does not remove the displayed root-value and Gamma matrices from PCL21. These formulas retain the full metric transport needed for the next comparison.

B.5 The exact inverse-period reflection of the central section

Let $E_{\text{par}} = \text{diag}(1, -1, 1, -1)$ in the centered coefficient order, and let P_{opp} exchange primary indices 1, 4 and 2, 3. Equation PCL2 implies

$$\mathcal{R}(-z) = E_{\text{par}} \mathcal{R}(z) E_{\text{par}}, \quad V E_{\text{par}} = P_{\text{opp}} V, \quad U P_{\text{opp}} = P_{\text{opp}} U.\tag{PCL22}$$

For the first equality a contributing coefficient satisfies $5n + r + 1 - a = 2p + 4h$, so $n \equiv a - r - 1 \pmod{2}$. The signs from the two parity matrices on entry (a, r) are $(-1)^{a-1+r} = (-1)^n$. Summing the entire series proves the identity. The second equality evaluates each monomial at the opposite original root. The third uses the full original unit values $(a, \bar{a}, \bar{a}, a)$, which agree at opposite roots.

The two diagonal matrices E_{par} and D commute. Substitution into PCL8 and use of PCL22 therefore give the exact maps

$$L_j(-z) = P_{\text{opp}} L_j(z) P_{\text{opp}}, \quad A_{-z}(Z, W) = Z^8 W^8 A_z(Z^{-1}, W^{-1}).\tag{PCL23}$$

For the second identity, $Q_0(P_{\text{opp}} t) = Q_0(t)$ by its two explicit products. Also $P_{\text{opp}} t(Z, W) = ZW t(Z^{-1}, W^{-1})$ when $ZW \neq 0$. Each of the four quadratic factors therefore contributes $(ZW)^2$. This proves the identity on that open set; both sides are polynomials after multiplication by $Z^8 W^8$, so it holds identically including both coordinate axes.

Comparing every coefficient in PCL23 proves

$$a_{rs}(-z) = a_{8-r, 8-s}(z), \quad C_{\text{in}}(-z) = C_{\text{in}}(z)^{\top}, \quad \det C_{\text{in}}(-z) = \det C_{\text{in}}(z).\tag{PCL24}$$

Indeed the $(a, b), (i, j)$ entry on the left of the middle identity is $a_{4+a-i, 4+b-j}(z)$, precisely the transposed entry of PCL12. Out-of-range coefficients are zero on both sides. The determinant equality follows by the permutation expansion of a determinant and its transpose. Thus the exact central determinant is an even entire function of u^{-1} , with every coefficient supplied by PCL2 and PCL10–12. This retains its actual constant matrix and does not assign that determinant a value by deleting any of its entries.

C The original weighted relation metric as a minimum dual extension

Retain the original simple quartet, fixed five-orbit period and logarithm branch, all coefficients and shifts of ROG and RDS, and the same two source orders $s \in \{1, k\}$. Put

$$\begin{aligned} q &= (k+1)^2, & q' &= (k-7)^2, & \Delta &= q - q', & \mathbf{m} &= \Delta - v, & c' &= k/2 - 4, \\ N &\in \{q-1, q, 2q-1, 2q\}, & L &= N - q, & d &= N - v, & M &= N - v - q' = L + \mathbf{m}. \end{aligned} \quad (\text{DNE1})$$

Here $0 \leq v \leq 80$, $\mu_v \neq 0$, and χ' is the complete lower-grid polynomial χ_{k-8} , never a derivative. All spaces of degree at most -1 are zero. The source norm is the original norm at $S' = c' + iy$, with mass $M_s = (2\pi)^{s/2}$.

C.1 A fixed original exponential quotient frame

We specify a finite frame using only the actual original matrices. Let $\mathbb{V}$ be the upper-root Vandermonde with rows $a = 0, \dots, q-1$ and entries $\mathbb{V}_{a,(i,j)} = \beta_{ij}^a$. Let $P_v : \mathbb{C}^{q-v} \rightarrow \mathbb{C}^q$ insert zero in the first v positions, and let C be the exact primary conductor of CEP4, with q' rows and q columns. Define

$$W_v = \mathbb{V}^{-1}P_v, \quad J_v = P_v^\top \mathbb{V} C^\top, \quad W_v J_v = C^\top. \quad (\text{DNE2})$$

The Vandermonde is invertible by the original distinct roots. The first v rows of $\mathbb{V} C^\top$ vanish because the corresponding exponential columns are $E_A e^{\beta'_{ab} z}$, whose zero order is at least v . This proves the last equality. Consequently W_v is an isomorphism onto the coefficient vectors of $\mathcal{N}_v$, and J_v has rank q' , since $C^\top$ is injective.

Choose the first q' row positions $\mathcal{I}$ in a fixed ordering of all subsets for which $\det(J_v)_{\mathcal{I}} \neq 0$, and let $\mathcal{J}$ be their complement. Such a subset exists by the proved rank. Let $E_{\mathcal{J}}$ insert these $\mathbf{m}$ complementary coordinates and put

$$Z_{\text{exp}} = W_v E_{\mathcal{J}}, \quad \mathcal{N}_a(z) = \sum_{i,j} (Z_{\text{exp}})_{(i,j),a} e^{\beta_{ij} z}, \quad F_a(z) = \frac{\mathcal{N}_a(z)}{E_A(z)} \quad (1 \leq a \leq \mathbf{m}). \quad (\text{DNE3})$$

The columns of $[J_v, E_{\mathcal{J}}]$ are independent: restriction of a relation to $\mathcal{I}$ first kills its J_v coefficients, and then its other coefficients vanish. Thus the $\mathcal{N}_a$ give a basis of the exact quotient $\mathcal{N}_v / E_A \mathcal{E}_{k-8}$. Every root value and actual minor defining this frame is retained. The frame is fixed as the endpoint N and source order s change.

For an analytic germ F at zero define the complex-linear functional Λ_F by $\Lambda_F(S'^n) = F^{(n)}(0)$. Let ℓ_a be the functional from RDS8. Then

$$\ell_a(g) = \Lambda_{F_a}(\chi' g), \quad f_{\ell_a} = \chi'(D) F_a, \quad A_{\ell,ba} = \ell_a(S'^b) \quad (0 \leq b < \mathbf{m}). \quad (\text{DNE4})$$

Indeed multiplication by χ' in the functional is precisely the constant-coefficient operator $\chi'(D)$ on its generating series. RDS8 proves that these functionals form the entire dual of the relation quotient. RDS2 identifies that quotient with the increasing monomials below degree $\mathbf{m}$, so A_{ℓ} is invertible. This supplies its existence without assuming a nonzero unevaluated minor.

C.2 The exact original source extension problem

On $\mathbb{C}[S']_{\leq M}$ use the original weighted norm $\|g\|_{\chi',s,c'} = \|\chi' g\|_{s,c'}$. Let G_M be its full monomial Gram, and let $\mathbf{U}_L$ be the complete ROG4 relation matrix. The quotient

$$Y_N = \mathbb{C}[S']_{\leq M} / \text{im } \mathcal{U}_L \simeq \mathbb{C}[S']_{< \mathbf{m}}$$

has Gram, in its fixed low-monomial representative frame,

$$H_{Y,N}^{(s)} = E_{< \mathbf{m}}^* G_M E_{< \mathbf{m}} - E_{< \mathbf{m}}^* G_M \mathbf{U}_L (\mathbf{U}_L^* G_M \mathbf{U}_L)^{-1} \mathbf{U}_L^* G_M E_{< \mathbf{m}}. \quad (\text{DNE5})$$

The second term is absent when $L = -1$. Distinct leading degrees of the columns of $\mathcal{U}_L$ show that the low representatives and all relation columns form a basis. Positivity of G_M and minimization over the full relation columns therefore prove DNE5 and its strict positivity.

For $x \in \mathbb{C}^{\mathbf{m}}$ write $\mathcal{N}_x = \sum_a x_a \mathcal{N}_a$ and $F_x = \mathcal{N}_x / E_A$. Its functional ℓ_x on Y_N has the exact norm

$$\|\ell_x\|_{Y_N^\vee}^2 = \min_{\eta \in \mathbb{C}^{q'}} \left\| \Lambda_{F_x} + \sum_{a,b} \eta_{ab} \text{ev}_{\beta'_{ab}} \right\|_{(\mathbb{C}[S']_{\leq d}, \|\cdot\|_{s,c'})^\vee}^2. \quad (\text{DNE6})$$

Here the dual consists of complex-linear functionals. To prove the equality, multiply by χ' to identify the weighted polynomial space isometrically with $I_N = \chi' \mathbb{C}[S']_{\leq M} \subset \mathbb{C}[S']_{\leq d}$. The restriction of Λ_{F_x} to I_N is exactly $g \mapsto \ell_x(g)$ by DNE4, and it annihilates $\chi' \text{im } \mathcal{U}_L$ by RDS5.

A functional annihilating a subspace has the same norm on the Hilbert quotient: the minimum lift of each quotient class has the same functional value, and every other lift has larger norm. A functional on a Hilbert subspace has a unique extension of minimum norm to the ambient space, namely its composition with the orthogonal projection onto that subspace. Every other extension adds a functional on the perpendicular complement; their squared dual norms add. These two facts reduce its quotient dual norm to the minimum over all ambient extensions of its restriction.

The difference of two extensions annihilates I_N . That annihilator has dimension $(d+1) - (M+1) = q'$. Evaluation at each original lower root annihilates I_N , and these q' evaluation functionals are independent on $\mathbb{C}[S']_{\leq d}$, by the first q' monomials and the distinct-root Vandermonde. They therefore constitute its whole annihilator. This proves DNE6 and retains every permitted extension.

Let

$$\phi_{n,s,c'}(S') = \frac{p_n((S' - c')/i)}{\sqrt{h_n}}, \quad h_n = M_s n! (s/2)_n, \quad 0 \leq n \leq d$$

be the unchanged original orthonormal source frame. Form the complete finite matrices

$$\begin{aligned} T_{N,s;na} &= \Lambda_{F_a}(\phi_{n,s,c'}), & V_{N,s;n,(i,j)} &= \phi_{n,s,c'}(\beta'_{ij}), \\ \mathcal{D}_N^{(s)} &= T_{N,s}^* [I - V_{N,s}(V_{N,s}^* V_{N,s})^{-1} V_{N,s}^*] T_{N,s}. \end{aligned} \quad (\text{DNE7})$$

The matrix $V_{N,s}$ has rank q' by the same Vandermonde argument, since the displayed orthonormal polynomials span all degrees at most d . In an orthonormal basis the squared norm of a complex-linear functional is the sum of the squared moduli of its values. Thus the minimum in DNE6 is $\min_\eta \|T_{N,s}x + V_{N,s}\eta\|_2^2$. The minimizing vector is $-(V_{N,s}^* V_{N,s})^{-1} V_{N,s}^* T_{N,s}x$. Substitution proves DNE7, including both mixed factors, and gives $\|\ell_x\|^2 = x^* \mathcal{D}_N^{(s)} x$. Positivity follows because the ℓ_a are an independent dual frame of Y_N .

C.3 Complex-linear duality and the actual fixed kernel columns

For a positive Gram H and a complex-linear functional $y \mapsto a^\top y$, its squared dual norm is $a^\top H^{-1} \bar{a} = a^* \overline{H^{-1} a}$. Indeed its Riesz vector is $H^{-1} \bar{a}$, and evaluation of its squared norm gives the first expression; the second is its equal real conjugate. Therefore the precise metric identity and determinant consequence of DNE4–7 are

$$\mathcal{D}_N^{(s)} = A_\ell^* (\overline{H_{Y,N}^{(s)}})^{-1} A_\ell, \quad \det \mathcal{D}_N^{(s)} = \frac{|\det A_\ell|^2}{\det H_{Y,N}^{(s)}}. \quad (\text{DNE8})$$

The conjugation records the ordinary complex-linear dual coordinates; it is not discarded by changing the source frame. In the four-endpoint sum, the fixed factor $|\det A_\ell|^2$ cancels with signs $(1, 1, -1, -1)$.

Retain the actual fixed matrix Z_k of FEX27 and FGC9, with $t = r - t_0$ independent columns in the low-monomial frame of dimension $\mathbf{m}$. Multiplying every relation column by the same nonzero

scalar $\alpha_A = v!/\mu_v$ does not change its span or its Schur complement. Consequently its actual native FEX29 matrix and the exact dual quotient map are

$$\widehat{H}_{K,N}^{(s)} = Z_k^* H_{Y,N}^{(s)} Z_k, \quad B_K = Z_k^\top A_\ell, \quad Q_{K^\vee,N}^{(s)} = \left[B_K (\mathcal{D}_N^{(s)})^{-1} B_K^* \right]^{-1} = \overline{(\widehat{H}_{K,N}^{(s)})^{-1}}. \quad (\text{DNE9})$$

The map B_K evaluates each relation-annihilating functional on the actual columns Z_k . It is onto, because A_ℓ is invertible and Z_k has rank t . The first inverse-Gram formula is the minimum dual metric on that same evaluation target. Substitution of DNE8 yields

$$B_K (\mathcal{D}_N^{(s)})^{-1} B_K^* = Z_k^\top \overline{H_{Y,N}^{(s)}} \bar{Z}_k = \widehat{H}_{K,N}^{(s)},$$

which proves the final equality with every transpose in its original direction. Empty rank has determinant one. In particular the full fixed-column return is exactly

$$\widehat{F}_k^{(s)} = - \sum_N \sigma_N \log \det Q_{K^\vee,N}^{(s)}. \quad (\text{DNE10})$$

Its relation to the original $F_K^{(s)}$ retains both low quotient volumes and the full source-map error of FGC12–13 and FEX34–37; DNE10 removes none of those terms.

C.4 A finite source bound before the high-order derivative

Use the actual R, R', B, C_v and R_0 of ROG6 and RDS11, and put

$$\rho = \tanh(R_0/4), \quad C_{F,k,A} = \frac{2R^v}{|\mu_v|} e^{RR_0}. \quad (\text{DNE11})$$

For any $\mathcal{N} = \sum \nu_{ij} e^{\beta_{ij}z} \in \mathcal{N}_v$, RDS12 proves $|\mathcal{N}/E_A| \leq C_{F,k,A} \|\nu\|_1$ on the complete disk $|z| \leq R_0$. Apply the original Gamma generating identity directly to the functional $\Lambda_{\mathcal{N}/E_A}$:

$$\sum_{n \geq 0} \Lambda_{\mathcal{N}/E_A} (p_n((S' - c')/i)) \frac{z^n}{n!} = (1 + z^2)^{-s/4} e^{ic' \arctan z} \frac{\mathcal{N}(-i \arctan z)}{E_A(-i \arctan z)}. \quad (\text{DNE12})$$

The quotient at zero is removable. On $|z| = \rho$, $|\arctan z| \leq R_0/4$. Cauchy's coefficient estimate therefore gives the exact bound

$$\begin{aligned} \|\Lambda_{\mathcal{N}/E_A}\|_{(\mathbb{C}[S']_{\leq d}, \|\cdot\|_{s,c'})^\vee} &\leq \mathcal{B}_{d,s,k,A} \|\nu\|_1, \\ \mathcal{B}_{d,s,k,A}^2 &= \frac{C_{F,k,A}^2}{M_s} (1 - \rho^2)^{-s/2} e^{|c'|R_0/2} \sum_{n=0}^d \frac{n!}{(s/2)_n} \rho^{-2n}. \end{aligned} \quad (\text{DNE13})$$

Indeed the bound for the n th orthonormal coefficient is the square root of its displayed summand times the common factor and $\|\nu\|_1$; summing squared coefficients gives the dual norm. The full Gamma mass remains explicitly in this formula. No application of $\chi'(D)$ and no $q!$ factor is needed in this exact extension norm.

For a fully specified comparison reference, use Euclidean norm in the original upper exponential coefficient order ν_{ij} , with its actual quotient by $\text{im } C^\top$. Its positive Gram in the fixed DNE3 frame is

$$G_{\text{exp}} = Z_{\text{exp}}^* [I - C^\top (\bar{C} C^\top)^{-1} \bar{C}] Z_{\text{exp}}, \quad 0 \prec \mathcal{D}_N^{(s)} \preceq q \mathcal{B}_{d,s,k,A}^2 G_{\text{exp}}. \quad (\text{DNE14})$$

The reference is explicitly specified here for comparison; the actual weighted Gamma quotient remains DNE7. The inverse in G_{exp} exists by the full row rank of C . Its expression is the orthogonal-projection formula, and is positive on this quotient frame by DNE3. Every representative $\nu + C^\top \eta$ remains in $\mathcal{N}_v$ and changes $\mathcal{N}/E_A$ by precisely a lower exponential, by RDS7. Combining DNE6 and DNE13, and then using $\|w\|_1 \leq \sqrt{q} \|w\|_2$, gives

$$x^* \mathcal{D}_N^{(s)} x \leq \mathcal{B}_{d,s,k,A}^2 \inf_\eta \|Z_{\text{exp}} x + C^\top \eta\|_1^2 \leq q \mathcal{B}_{d,s,k,A}^2 x^* G_{\text{exp}} x.$$

This proves DNE14 without a bound on an omitted frame inverse.

At fixed actual period and quartet the finite constant satisfies

$$\log(1 + q\mathcal{B}_{d,s,k,A}^2) = O_{\text{actual}}(q), \quad s = 1, k, \quad d = N - v. \quad (\text{DNE15})$$

For detail, R_0, ρ, v, μ_v are fixed, $R = O(k)$, $|c'| = O(k)$ and $s \leq k$. Thus $\log C_{F,k,A} = O_{\text{actual}}(k + \log(k + 1))$. The original Gamma ratios obey $n!/(s/2)_n \leq 2\sqrt{n}$ for $n \geq 1$ and are one at $n = 0$, as proved by the product estimate in FEX10–11. Hence the sum in DNE13 is at most $(d + 1) \max\{1, 2\sqrt{d}\} \rho^{-2d}$. Since $d \leq 2q$, its logarithm is $O_{\text{actual}}(q)$. The remaining factors have logarithms $O_{\text{actual}}(k)$, with the original positive mass retained before estimation. These bounds prove DNE15.

Restriction of DNE14 to any specified plane, and minimization under any specified onto quotient map, preserve its scalar upper bound. In particular it applies to the actual map B_K with the full reference quotient $(B_K G_{\text{exp}}^{-1} B_K^*)^{-1}$. This is a finite norm comparison of the original metric with a completely retained reference form. The reference contains the full original Vandermonde and conductor data; neither its determinant nor the original return in DNE10 has been assigned a leading value by this upper bound.

D The original primitive source and its adjacent Gamma metric

This calculation continues the complete LCA15–21 polynomial quotient construction. Its new conclusions are the explicit codimension-three truncation, its exact rank-one native metric change, and a finite coefficient-product bound for that change. All spaces and measures below are the original ones. The connecting target metric and its later long return remain the literal target metrics; the source computation here is at the two specified low cutoffs.

D.1 Original objects and the polynomial quotient map

Fix the actual simple quartet, its invertible period matrix H , logarithm branch, and full conductor coefficients. Retain

$$\begin{aligned} \beta_* &= \frac{1}{2} - \delta - i\gamma, & \beta_{n,a,b} &= n\beta_* + 2a\delta + 2bi\gamma, & \delta, \gamma &> 0, \\ E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b}z}, & \mu_r &= E_A^{(r)}(0), & v = \text{ord}_0 E_A &\leq 80, \quad \mu_v \neq 0, \\ L_i(Z, W) &= (H(ZW, Z, W, 1)^\top)_i, & b_{i,a,b} &= [Z^a W^b] L_i^4, & B_i(z) &= \sum_{a,b=0}^4 b_{i,a,b} e^{\beta_{4,a,b}z}, \\ b_i &= B_i(0) = \sum_{a,b=0}^4 b_{i,a,b}, & \sum_{i=1}^4 D_i(z) B_i(z) &= 1. \end{aligned} \tag{PDS1}$$

The D_i are the proved original entire Bezout expressions. In particular at least one b_i is nonzero. Keep the full biform conductor, including repeated components and zero edge coefficients. Set

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q_k &= (k + 1)^2, & Q &= (k + 5)^2, \\ e &\in \{0, 1\}, & D_e &= Q - v - 1 + e, & \ell_e &= 8k + 23 + e, & d_e &= \ell_e + 1, \\ \chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{n,a,b}), & (\mathcal{T}_{AP})(S') &= \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), \\ (\mathcal{T}_{B_i} p)(S) &= \sum_{a,b=0}^4 b_{i,a,b} p(S + \beta_{4,a,b}). \end{aligned} \tag{PDS2}$$

All dual functionals are complex-linear. For a finite germ F through degree D_e , let $\Lambda_F(S'^r) = F^{(r)}(0)$. Let $\mathcal{H}_e = \mathbb{C}[S']_{\leq D_e}^\vee$, and let $\mathcal{S}_e \subset \mathcal{H}_e$ be the jets of the original analytic germs

$$F(z) = \frac{N(z)}{E_A(z)}, \quad N(z) = \sum_{a,b=0}^k n_{ab} e^{\beta_{k,a,b}z}, \quad N^{(r)}(0) = 0 \quad (0 \leq r < v). \tag{PDS3}$$

Define the unchanged relation polynomials and quotient coordinates by

$$r_a(S') = \mathcal{T}_A(\chi_k(S) S^a), \quad 0 \leq a \leq \ell_e, \quad \pi_e(F) = (\Lambda_F(r_a))_{a=0}^{\ell_e}. \tag{PDS4}$$

Then π_e is onto and $\ker \pi_e = \mathcal{S}_e$. Here is the complete kernel argument. The distinct upper k nodes give a Vandermonde matrix, whose first v rows have rank v since $v \leq 80 < q_k$. The numerator space in PDS3 therefore has dimension $q_k - v$. Its jet map is injective: if F vanishes through degree D_e , then $N = E_A F$ vanishes through degree $D_e + v \geq q_k - 1$; the first q_k Vandermonde rows force every $n_{ab} = 0$. The Leibniz identity gives $\Lambda_F(\mathcal{T}_{AP}) = \Lambda_{E_A F}(p)$, so all r_a annihilate PDS3. Finally

$$\mathcal{T}_A S^m = \sum_{r=0}^m \binom{m}{r} \mu_r S'^{m-r}, \quad \deg r_a = q_k + a - v, \quad \text{lc}(r_a) = \mu_v \binom{q_k + a}{v}. \tag{PDS5}$$

These successive degrees prove independence. Their common kernel has dimension $D_e + 1 - d_e = q_k - v$, proving the assertion. This is precisely the LCA16 quotient, with no new row norm. Every original lower degree- $(k - 8)$ exponential is in $\mathcal{S}_e$, since its product with E_A has an allowed degree- k numerator. Thus taking the original lower-gauge quotient first and then quotienting by its analytic-source image gives exactly the same affine fibres as $\mathcal{H}_e/\mathcal{S}_e$.

D.2 The full original Gamma dual metric

The source member of the two original order pairs is $s = 1$ or $s = k$, and its centre is $c = k/2 - 4$. Keep its entire measure and recurrence:

$$M_s = (2\pi)^{s/2}, \quad dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad h_{n,s} = M_s n!(s/2)_n, \\ p_{0,s} = 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \quad \phi_n(S') = p_{n,s}((S' - c)/i)/\sqrt{h_{n,s}}. \quad (\text{PDS6})$$

These are the original orthonormal polynomials for the norm $\|p\|_{s,c}^2 = \int_{\mathbb{R}} |p(c + iy)|^2 dm_s(y)$, as established in the original Gamma construction reproduced in the cumulative providers. The coefficient isometry uses this complete measure, rather than dividing its mass out. Let B_e be the triangular matrix defined by $S'^r = \sum_{n=0}^{D_e} (B_e)_{nr} \phi_n(S')$ and let

$$C_e = (B_e \text{coeff}_{S'} r_a)_{a=0}^{\ell_e}, \quad H_e = C_e^* C_e > 0, \quad G_e = \overline{H_e^{-1}}. \quad (\text{PDS7})$$

The columns of C_e are polynomial coefficients in the original orthonormal frame. For $w = (\Lambda_F(\phi_n))_n$ the literal quotient row map is $x = C_e^T w$. Its minimum-norm lift is $w = \overline{C_e} \overline{H_e}^{-1} x$ and has squared norm $x^* G_e x$. Indeed $C_e^T \overline{C_e} = \overline{H_e}$, and the displayed lift is orthogonal to $\ker C_e^T$. This proves both the full complex-linear dual orientation and the quotient metric, including every conjugation in PDS7. The same metric results from the two successive minima over the original lower gauges and analytic source, by the equality of fibres proved after PDS5.

The degree-one primitive condition at the first endpoint is empty. At the second endpoint its four original rows are

$$R_i(S) = \frac{\mathcal{T}_{B_i} \chi_j(S)}{\chi_k(S)}, \quad W = (\text{coeff}_S R_i)_{i=1}^4 \in \mathbb{C}^{d_1 \times 4}, \quad \mathcal{P}_0 = \mathbb{C}^{d_0}, \quad \mathcal{P}_1 = \ker W^T \subset \mathbb{C}^{d_1}. \quad (\text{PDS8})$$

Here $\mathcal{P}_e$ denotes the primitive quotient source, not a biform section space. All degree- k roots become degree- j roots after every degree-four shift; hence the numerator is divisible by χ_k and $\deg R_i \leq \ell_1$. Translation sums commute, so $\mathcal{T}_A(\chi_k R_i) = \mathcal{T}_{B_i} \mathcal{T}_A \chi_j$. This is exactly the original positive degree-zero exterior differential, and its row is W^T , as in LCA18–19. The full independence proof LCA20–21 applies to the actual invertible period, without a genericity exclusion: the four L_i^4 are independent on the rank-four Segre quadric, and the 25 distinct translations of the degree- $Q - v \geq 116$ polynomial $\mathcal{T}_A \chi_j$ give an injective Vandermonde moment map. Thus $\text{rank } W = 4$. The complete LCA20–21 proof is retained as a provider in the cumulative source, rather than replacing this fact by a new period assumption.

The metric on $\mathcal{P}_e$ is the restriction of G_e . At $e = 1$ its exact polynomial dual is

$$\text{span}(r_0, \dots, r_{\ell_1}) / \text{span}(\mathcal{T}_A(\chi_k R_i))_{i=1}^4.$$

To verify the metric as well as the vector-space statement, choose any fixed full basis matrix Z_1 of $\ker W^T$. A coefficient vector f of a relation polynomial restricts to the functional $Z_1^T f$ on this basis. Minimizing $f^* H_1 f$ over all lifts with $Z_1^T f = a$ gives

$$\min_{Z_1^T f = a} f^* H_1 f = a^* (Z_1^T H_1^{-1} \overline{Z_1})^{-1} a = a^* (\overline{Z_1^* G_1 Z_1})^{-1} a. \quad (\text{PDS9})$$

The minimizer follows by the adjoint of $Z_1^T H_1^{-1/2}$; subtracting it leaves an orthogonal kernel vector. The kernel of Z_1^T is exactly $\text{im } W$, by dimensions and $W^T Z_1 = 0$. This proves the stated original quotient metric and its exact dual morphism, with no deletion of the four row polynomials.

Current full relation and primitive source determinants. In these exact PDS7 coordinates, put $q = (k+1)^2$, $Q = (k+5)^2$, $d_e = 8k + 24 + e$, $N_e = Q - v - 1 + e$, and

$$\mathfrak{F}_{k,e,s} = \prod_{a=0}^{d_e-1} \left| \mu_v \binom{q+a}{v} \right|^2 (2\pi)^{s/2} (q+a-v)! (s/2)_{q+a-v}.$$

The complete input columns are $\chi_k(S)S^a$ at center $k/2$, modulo every polynomial below v . Their quotient coefficient matrix is the actual Y_e of SGT30, and their translated columns are exactly $C_e = A_{N_e,s}Y_e$, with SGT10's full factorial, phase and Gamma diagonal maps. PSG21 and SGT24–31 therefore give, at both defining low cutoffs,

$$\begin{aligned} \log \det H_e &= \log \mathfrak{F}_{k,e,s} + 2d_e q \log(4/\pi) + \varepsilon_{k,e,s}^H, \\ -\mathcal{E}_{k,e,s}^-(a) &\leq \varepsilon_{k,e,s}^H \leq \mathcal{E}_{k,e,s}^+(a), \quad \varepsilon_{k,e,s}^H = o(kq), \\ \log \det G_e &= -\log \mathfrak{F}_{k,e,s} - 2d_e q \log(4/\pi) - \varepsilon_{k,e,s}^H. \end{aligned} \tag{PDS25}$$

The finite $\mathcal{E}^\pm$ are exactly PSR2–4 for any SGT-admissible fixed radius a , with the full finite domain of PSR2; the radius limit is taken after the degree limit. The proof adds the two same-frame logarithmic ratios in PSR5. The dual identity uses the original $G_e = H_e^{-1}$, so its error interval is $[-\mathcal{E}^+, \mathcal{E}^-]$. No polynomial has been replaced by its leading term in either Gram.

At $e = 1$, the complete four-row correction remains the literal $\mathcal{G}_{k,s} = W^*H_1W$: the exact coefficient map is $C_1W = \mathcal{C}$ in PFG5. Its earliest actual moment pivots $t_1 < \dots < t_4 \leq 24$ and the full nonzero minor in PFG10 prove the two-sided original-data estimate

$$0 < \mathcal{L}_{k,s} \leq \det(W^*H_1W) \leq \mathcal{U}_{k,s}, \quad \log \det(W^*H_1W) = 4 \log M_s + 8Q \log Q + O_{\text{actual}}(Q) = o(kq). \tag{PDS26}$$

Here $\mathcal{L}, \mathcal{U}$ are the complete PFG11–13 moment/factorial and coefficient-product expressions. PDS9's quotient by those four columns still carries its fixed determinant factor $|\det[W, J_Z]|^2$, as proved by the full Schur factorization PFG18 and PSR12–20. At $e = 0$ there is no four-row primitive constraint.

D.3 The exact codimension-three truncation

The coefficient of S^{ℓ_1} in R_i is b_i , since both χ_j and χ_k are monic. Choose a fixed original index i_* with $b_{i_*} \neq 0$, write $b_* = b_{i_*}$, and retain

$$R_{i_*}(S) = \sum_{a=0}^{\ell_0} (u_*)_a S^a + b_* S^{\ell_1}, \quad C_i(S) = R_i(S) - \frac{b_i}{b_*} R_{i_*}(S) \quad (i \neq i_*). \tag{PDS10}$$

These displayed combinations are exact coordinates for the intersection of the original four-row span with lower degree; the original rows and coefficients in PDS8 remain fixed. The three C_i have degree at most ℓ_0 and are independent: a relation among them would give a relation among all four R_i , whose coefficients on indices other than i_* are unchanged. Let V have the three coefficient columns C_i , in the original index order with i_* omitted. Then

$$\begin{aligned} \mathcal{Z} &= \ker V^\top \subset \mathbb{C}^{d_0}, \quad \dim \mathcal{Z} = d_0 - 3 = 8k + 21, \quad E = \begin{bmatrix} I_{d_0} \\ -u_*^\top/b_* \end{bmatrix}, \\ \text{pr}_0 : \mathcal{P}_1 &\xrightarrow{\cong} \mathcal{Z}, \quad \text{pr}_0^{-1} = E|_{\mathcal{Z}}. \end{aligned} \tag{PDS11}$$

In fact the i_* row of $W^\top(x_0, x_{\text{top}}) = 0$ gives $x_{\text{top}} = -u_*^\top x_0/b_*$. Substitution in the other three rows gives precisely $V^\top x_0 = 0$. The first d_0 relation polynomials in PDS4 are unchanged at the two cutoffs, so pr_0 is also the actual jet truncation on the primitive quotient source. If an upper analytic source is added its truncated jet is an old analytic source; hence this map is well-defined on the original quotient. Surjectivity and the displayed inverse follow from the equations just proved.

D.4 One rank-one update, with every frame retained

Choose an arbitrary fixed full column basis Z of $\mathcal{Z}$; use EZ at the upper endpoint. For the same source order s and centre c , the old orthonormal polynomials are unchanged when the cutoff increases. In particular the old C_0 is padded by a zero last row in C_1 . Write the exact block matrix and Schur scalar

$$H_1 = \begin{bmatrix} H_0 & b \\ b^* & a \end{bmatrix}, \quad b = C_0^* c_{\text{top}}, \quad a = c_{\text{top}}^* c_{\text{top}}, \quad \sigma = a - b^* H_0^{-1} b > 0, \quad (\text{PDS12})$$

where c_{top} is the full coefficient column of r_{ℓ_1} . Padded columns are understood in the products. Define

$$\eta = -u_*^\top / b_* - b^\top \bar{H}_0^{-1}, \quad K_0 = Z^* \bar{H}_0^{-1} Z, \quad K_1 = (EZ)^* \bar{H}_1^{-1} (EZ). \quad (\text{PDS13})$$

Then the complete metric comparison is

$$\boxed{K_1 = K_0 + \frac{(\eta Z)^* (\eta Z)}{\sigma}, \quad \frac{\det K_1}{\det K_0} = 1 + \frac{(\eta Z) K_0^{-1} (\eta Z)^*}{\sigma} \geq 1.} \quad (\text{PDS14})$$

For proof, block inversion of $\bar{H}_1$ gives for every (x_0, x_{top})

$$\begin{bmatrix} x_0 \\ x_{\text{top}} \end{bmatrix}^* \bar{H}_1^{-1} \begin{bmatrix} x_0 \\ x_{\text{top}} \end{bmatrix} = x_0^* \bar{H}_0^{-1} x_0 + \frac{|x_{\text{top}} - b^\top \bar{H}_0^{-1} x_0|^2}{\sigma}. \quad (\text{PDS15})$$

Insert PDS11 and then $x_0 = Zz$. This proves the matrix equality. The determinant identity follows by taking the determinant of $I + K_0^{-1/2} (\eta Z)^* (\eta Z) K_0^{-1/2} / \sigma$; this rank-at-most-one operator has its only possible nonzero eigenvalue $(\eta Z) K_0^{-1} (\eta Z)^* / \sigma$. The frame Z has not been changed: replacing it by ZT changes both determinants by the same $|\det T|^2$. PDS14 also proves that the actual truncation $\mathcal{P}_1 \rightarrow \mathcal{Z}$ is a contraction for the displayed native metrics, and locates its entire metric defect in this one explicit row.

D.5 An explicit coefficient-product upper estimate

Set $D = D_1 = Q - v$ and keep the literal original polynomial

$$r_*(S') = \mathcal{T}_A(\chi_k R_{i_*})(S') = \mathcal{T}_{B_{i_*}} \mathcal{T}_A \chi_j(S'), \quad \lambda_* = \text{lc } r_* = b_* \mu_v \binom{Q}{v} \neq 0. \quad (\text{PDS16})$$

There is an exact finite upper bound

$$1 \leq \frac{\det K_1}{\det K_0} \leq \frac{\|r_*\|_{s,c}^2}{|b_*|^2 \sigma} \leq \frac{\|r_*\|_{s,c}^2}{|\lambda_*|^2 h_{D,s}}. \quad (\text{PDS17})$$

To prove the first upper inequality, the orthogonal projector onto $\text{im}(\bar{H}_0^{-1/2} Z)$ gives $Z K_0^{-1} Z^* \leq \bar{H}_0$. Thus the numerator in PDS14 is at most $\eta \bar{H}_0 \eta^*$. Let $v_* = u_*/b_* + H_0^{-1} b$. Then $\eta = -v_*^\top$, so this upper bound is $v_*^* H_0 v_*$. The orthogonal projection of r_* onto the old relation span has coefficient vector $b_* v_*$, whereas its orthogonal remainder is b_* times the remainder of r_{ℓ_1} . Consequently

$$\|r_*\|_{s,c}^2 = |b_*|^2 (v_*^* H_0 v_* + \sigma). \quad (\text{PDS18})$$

This proves the first inequality of PDS17. Every old relation has degree at most $D - 1$. The coefficient of ϕ_D in r_{ℓ_1} is $i^D \mu_v \binom{Q}{v} \sqrt{h_{D,s}}$; this uses its actual leading coefficient and the phase $i^{-D} / \sqrt{h_{D,s}}$ in PDS6. This coefficient remains in its orthogonal remainder. Hence $\sigma \geq |\mu_v \binom{Q}{v}|^2 h_{D,s}$, proving the second inequality of PDS17 with its original scalar intact.

For a completely finite coefficient bound define

$$L = \max\{1, |c| + 2(D + 1 + s/2)\}, \quad \Theta_{a,b,r,t;u,w} = \beta_{8,a,b} + \beta_{4,r,t} - \beta_{j,u,w},$$

$$P_* = \sum_{a,b=0}^8 \sum_{r,t=0}^4 |a_{ab} b_{i_*,r,t}| \prod_{u,w=0}^j (L + |\Theta_{a,b,r,t;u,w}|), \quad U_* = \frac{M_s P_*^2}{|b_* \mu_v(Q)|^2 h_{D,s}}. \quad (\text{PDS19})$$

The L in this subsection is a numerical bound and is not the jet length denoted L in LCA2. Then

$$0 \leq \log \frac{\det K_1}{\det K_0} \leq \log U_*. \quad (\text{PDS20})$$

Here is a direct proof that PDS19 bounds the original Gamma norm. In the orthonormal polynomial basis in y , multiplication by y has raising coefficient $\sqrt{(n+1)(n+s/2)}$ and lowering coefficient $\sqrt{n(n-1+s/2)}$, by PDS6 and $h_{n+1,s}/h_{n,s} = (n+1)(n+s/2)$. On polynomials of degree at most $D-1$ both weighted shifts have norm at most $D+1+s/2$; their sum has norm at most $2(D+1+s/2)$. Thus multiplication by $S' = c + iy$, through degree D , has norm at most L . Starting with the constant polynomial, whose squared norm is M_s , proves

$$\|p\|_{s,c} \leq \sqrt{M_s} \sum_{m=0}^D |[S'^m]p| L^m \quad (\deg p \leq D). \quad (\text{PDS21})$$

The exact expansion of PDS16 is

$$r_*(S') = \sum_{a,b=0}^8 \sum_{r,t=0}^4 a_{ab} b_{i_*,r,t} \prod_{u,w=0}^j (S' + \Theta_{a,b,r,t;u,w}). \quad (\text{PDS22})$$

The weighted coefficient ℓ^1 norm is submultiplicative: for two polynomials its coefficient convolution and the triangle inequality give $\sum_m |(fg)_m| L^m \leq (\sum_m |f_m| L^m)(\sum_m |g_m| L^m)$. Applying this identity to every factor in PDS22 bounds the weighted coefficient sum of r_* by P_* . Its coefficients above degree $D = Q - v$ vanish by the exact moment identities in PDS5; bounding the full degree- Q expansion before those cancellations is still an upper bound for the unchanged degree- D polynomial. PDS21 now gives $\|r_*\|_{s,c}^2 \leq M_s P_*^2$, and PDS17 proves PDS20. In particular $U_* \geq 1$ is a consequence, rather than an extra assumption on the original coefficients.

There is also an explicit uniform growth bound in either original source-order family. Define the fixed positive numbers

$$R_* = \sqrt{(1/2 + \delta)^2 + \gamma^2}, \quad A_{\text{abs}} = \sum_{a,b=0}^8 |a_{ab}|, \quad B_{\text{abs}} = \sum_{r,t=0}^4 |b_{i_*,r,t}|,$$

$$C_* = 6 + 17R_*, \quad C_{\text{abs}} = \max \left\{ 1, \frac{2(A_{\text{abs}} B_{\text{abs}})^2}{|b_* \mu_v|^2} \right\}. \quad (\text{PDS23})$$

The original nodes satisfy $|\text{Re } \beta_{n,a,b}| \leq n(1/2 + \delta)$ and $|\text{Im } \beta_{n,a,b}| \leq n\gamma$. Both bounds are attained together at $(a, b) = (n, 0)$ and (n, n) ; thus $\max_{a,b} |\beta_{n,a,b}| = nR_*$, and therefore $|\Theta| \leq (k+16)R_*$. For $s = 1$ or $s = k$, $D \leq Q$, $k \leq Q$, and $|c| \leq k/2 + 4$, so $L \leq 2Q + 3k/2 + 6 \leq 6(Q+1)$. Also $k+16 \leq 17(Q+1)$. Every factor in PDS19 is consequently at most $C_*(Q+1)$, giving $P_* \leq A_{\text{abs}} B_{\text{abs}} [C_*(Q+1)]^Q$. The full factorial denominator gives a sharper bound than dropping its growth. For $s \geq 1$,

$$D!(s/2)_D \geq D!(1/2)_D = \frac{(2D)!}{4^D} \geq \left(\frac{D}{\exp(1)} \right)^{2D}.$$

The product identity follows by separating the even and odd factors of $(2D)!$. The last inequality follows from $\sum_{r=1}^{2D} \log r \geq \int_1^{2D} \log x \, dx = 2D \log(2D) - 2D + 1$. Since $D = Q - v$, $Q \geq 196$ and

$v \leq 80$ imply $Q + 1 \leq 2D$, the exact PDS19 bound therefore satisfies

$$\begin{aligned} U_* &\leq \frac{(A_{\text{abs}} B_{\text{abs}})^2}{|b_* \mu_v|^2} C_*^{2Q} (Q + 1)^{2Q} \left(\frac{\exp(1)}{D} \right)^{2D} \\ &\leq C_{\text{abs}} (Q + 1)^{2v} (2 \exp(1) C_*)^{2Q}. \end{aligned}$$

Here $\binom{Q}{v} \geq 1$ was used only in an upper bound; the full binomial, mass and factorial denominator remain in PDS19. For the second inequality split $(Q + 1)^{2Q}$ into its $2v$ and $2D$ powers and use $Q + 1 \leq 2D$, $D \leq Q$ and $C_* \geq 6$. Consequently the proved finite bound is

$$\begin{aligned} 0 &\leq \log \frac{\det K_1}{\det K_0} \leq \log U_* \\ &\leq \log C_{\text{abs}} + 2v \log(Q + 1) + 2Q \log(2 \exp(1) C_*) \\ &= O_{\text{actual}}(Q) = o(kq_k). \end{aligned} \tag{PDS24}$$

The last limit uses $Q/q_k \rightarrow 1$, fixed v , and $1/k \rightarrow 0$ for $k \equiv 1 \pmod{4}$. The constants are fixed by the actual quartet and period, independent of k and of the choice between the two original source-order families. The sharp finite PDS19 retains the full mass, phase-dependent nodes, factorials, conductor coefficients and leading scalar; PDS24 proves the scale of its upper bound without assuming an inverse estimate.

The maps PDS11 and PDS14 now give the primitive part of the low-cutoff source comparison inside the original LCA connecting sequence. The full cohomology additionally has the retained $\ker U_e$ constituent from LCC. Passing these source metrics through the actual connecting map requires the original target quotient Gram; PDS9 specifies the exact source dual to be used in that calculation, and PDS20–24 control its adjacent source correction. None of these equalities identifies a later target coefficient flag with a common compatible source at all cutoffs.

E The original connecting image and its full native target determinant

Keep the complete CNC, LCC and LCA data, including the nonreduced A^{hom} , every original coefficient and the exact $E_A(z) = \sum a_{ab} e^{\beta_{8,a,b} z}$. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q_a &= (a + 1)^2, & q'_a &= (a - 7)^2, \\ e &\in \{0, 1\}, & D_e &= q_j - v - 1 + e, & L_e &= D_e + 1, & 0 \leq v &\leq 80, & \mu_v &= E_A^{(v)}(0) \neq 0, \\ (s_0, s_1) &= (1, 1) \text{ or } (k, j), & c'_a &= a/2 - 4, & m_a &= q_a - q'_a - v. \end{aligned} \quad (\text{CIT1})$$

The two source orders in this display remain paired throughout; write $s_k = s_0$ and $s_j = s_1$ when a degree subscript is used. All finite functionals are complex-linear. At a fixed D_e let $\mathcal{H}_0, \mathcal{H}_1$ be their original Gamma polynomial duals in degrees k, j , with full orthonormal frames

$$\phi_{r,s,c'}(S') = \frac{p_{r,s}((S' - c')/i)}{\sqrt{(2\pi)^{s/2} r! (s/2)_r}}, \quad 0 \leq r \leq D_e. \quad (\text{CIT2})$$

Use the exact CNC10 matrices $T_0 = T_{k,D_e}$, $T_1 = T_{j,D_e}$ for its original analytic numerators divided by E_A , and V_0, V_1 for every original lower evaluation. Thus $[V_0, T_0]$ has its original full column order.

E.1 The literal primitive domain and target coefficient matrix

The four multiplier matrices are completely specified without an unspecified evaluator. Let $\mathcal{B}_a$ be the original monomial-to-Gamma matrix defined by $S'^r = \sum_n (\mathcal{B}_a)_{nr} \phi_{n,s_a,c'_a}$ and set

$$\begin{aligned} U_a &= \text{diag}(1/r!)_{r=0}^{D_e} \mathcal{B}_a^\top, & (\text{Mult}_{L_e}(B_i))_{rt} &= \begin{cases} B_i^{(r-t)}(0)/(r-t)!, & r \geq t, \\ 0, & r < t, \end{cases} \\ \mathsf{T}_i &= U_j^{-1} \text{Mult}_{L_e}(B_i) U_k, & \mathsf{T} &= \text{stack}_{i=1}^4 \mathsf{T}_i, & \mathcal{V} &= \text{diag}(V_1, 4), & \mathcal{T}_1 &= \text{diag}(T_1, 4). \end{aligned} \quad (\text{CIT3})$$

Here $\text{diag}(P, 4)$ means four copies. The map U_a sends the actual native functional values to Taylor coefficients: ordinary derivatives are $\mathcal{B}_a^\top w$. This proves CIT3 and preserves every factorial and the original leading phase i^r inside $\mathcal{B}_a$. Its full inverse tuple is the analogous multiplication by the original entire D_i ; $\sum D_i B_i = 1$ gives $\mathsf{ST} = I$.

Define the raw source analytic space

$$\mathcal{S}_k^{\text{raw}} = \text{im}[V_0, T_0] \subset \mathcal{H}_0, \quad \mathcal{L}_0 = \text{im } V_0, \quad \mathcal{L}_1 = \text{im } \mathcal{V}. \quad (\text{CIT4})$$

It is exactly the set of native functionals of N/E_A for all original upper numerators N vanishing to order v . Indeed T_0 gives the original DNE quotient frame and V_0 gives exactly its allowed numerator changes by conductor multiples. The columns are independent: if $T_0 x + V_0 \eta = 0$, the CNC13 injectivity at this cutoff gives $x = 0$, and the lower Vandermonde gives $\eta = 0$. Thus $\dim \mathcal{S}_k^{\text{raw}} = q_k - v$.

For the upper endpoint set

$$\begin{aligned} P_A &= \mathcal{T}_A \chi_j, & (\mathcal{T}_A p)(S') &= \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), \\ R_i(S') &= \mathcal{T}_{B_i} P_A(S'), & (\mathcal{T}_{B_i} p)(S') &= \sum_{a,b=0}^4 b_{i,a,b} p(S' + \beta_{4,a,b}), \\ \mathsf{R}_1 &= \text{stack}_{i=1}^4 (\mathcal{B}_k \text{coeff}_{S'} R_i)^\top, & \mathsf{R}_0 &: \mathcal{H}_0 \longrightarrow 0. \end{aligned} \quad (\text{CIT5})$$

All coefficient columns are padded through D_1 . Here $\deg P_A = q_j - v = D_1$ and $\text{lc } P_A = \mu_v \binom{q_j}{v}$. The complete LCC9–12 proof shows that R_1 has rank four: the four original L_i^4 are independent on the

rank-four Segre quadric, and translation of this polynomial is injective on the 25 distinct original shifts. The native row in CIT5 is an ordinary transpose because it evaluates the complex-linear functional on R_i .

The complete primitive domain is

$$\mathcal{Y}_e^{\text{raw}} = \ker R_e, \quad \mathcal{S}_k^{\text{raw}} \subset \mathcal{Y}_e^{\text{raw}}, \quad r_e = \dim(\mathcal{Y}_e^{\text{raw}}/\mathcal{S}_k^{\text{raw}}) = 8k + 24 - 3e. \quad (\text{CIT6})$$

For $e = 0$ it is all of $\mathcal{H}_0$. To prove the target condition, at that endpoint the raw analytic image in each $\mathcal{H}_1$ has dimension $q_j - v = L_0$ and fills it. At $e = 1$ its only missing coordinate is $\Lambda_F(P_A)$ by LCC10. Therefore $\mathsf{T}h$ belongs to the fourfold raw target analytic image precisely when $R_1 h = 0$. The exact numerator cochain square kills $\mathcal{S}_k^{\text{raw}}$ in those rows. Subtracting $\dim \mathcal{S}_k^{\text{raw}} = q_k - v$ from $\dim \ker R_e = L_e - 4e$ proves CIT6.

Choose a full primitive lift matrix P_e so that $[V_0, T_0, P_e]$ is a frame of $\mathcal{Y}_e^{\text{raw}}$. This is an exact computable choice: first express CIT4–5 in ordinary Taylor coordinates, take the first nonzero pivot minors in their original order to obtain the kernel and a complement, then transport those complement columns by U_k^{-1} . It depends on the actual coefficients and imposes no metric conditioning estimate. The chosen Taylor columns are fixed; the original norm remains CIT2.

In the original target coefficient coordinates define

$$\begin{aligned} P_{\mathcal{L}_1} &= \mathcal{V}(\mathcal{V}^* \mathcal{V})^{-1} \mathcal{V}^*, \quad K_1 = (I - P_{\mathcal{L}_1}) \mathcal{T}_1, \quad \mathcal{D}_e = K_1^* K_1 = \text{diag}(D_{j,D_e}^{(s_1)}, 4), \\ F_e &= \mathcal{D}_e^{-1} K_1^* (I - P_{\mathcal{L}_1}) \mathsf{T} P_e, \quad B_{\text{cof}} = \text{stack}_{i=1}^4 M_{i,k}. \end{aligned} \quad (\text{CIT7})$$

The inverses exist by the full lower Vandermonde and CNC13. By CIT6 the projected columns on the right belong to $\text{im } K_1$, so the displayed left inverse gives their unique original coefficient coordinates. Consequently

$$K_1 F_e = (I - P_{\mathcal{L}_1}) \mathsf{T} P_e, \quad K_1 B_{\text{cof}} = (I - P_{\mathcal{L}_1}) \mathsf{T} T_0. \quad (\text{CIT8})$$

These are literal original finite matrices. In particular the $M_{i,k}$ retain the exact KRC scalar-bearing conjugation CNC12.

E.2 Complete connecting kernel and attained target norm

We prove the full kernel of the connecting construction. The exact finite intersection needed here is

$$\text{im } \mathsf{T} \cap \mathcal{L}_1 = \mathsf{T} \mathcal{L}_0. \quad (\text{CIT9})$$

For a direct proof, write $\mathsf{T}h = \mathcal{V}\eta$. Apply the six original compatibility rows. Their coefficient square is $\mathcal{R}\mathcal{V} = \mathcal{V}_2 \mathcal{K}$, so $\mathcal{V}_2 \mathcal{K}\eta = 0$. At the present cutoffs $D_e + 1 - q_k = 8k + 24 - v + e \geq 16$; hence the original degree- k evaluation $\mathcal{V}_2$ is injective. The complete GKC kernel gives $\eta = \mathcal{P}\xi$. The lower root-addition square says $\mathcal{V}\mathcal{P} = \mathsf{T}V_0$, whence $\mathsf{T}h = \mathsf{T}V_0\xi$. Since $\mathsf{S}\mathsf{T} = I$, $h = V_0\xi$. The reverse inclusion is exactly that same square. This proves CIT9 without a transversality or inverse-norm assumption.

For $h \in \mathcal{Y}_e^{\text{raw}}$ let $c(h)$ be the coefficient column defined as in CIT7. The native exterior differential of $\mathsf{T}h$ is zero, since $B_i B_t = B_t B_i$. Replacing its representative by lower evaluations changes its next differential by lower evaluations, by root addition. Therefore $c(h)$ is a cycle in the actual LCC image complex. The complete map is

$$\begin{aligned} \partial_e : \mathcal{Y}_e^{\text{raw}}/\mathcal{S}_k^{\text{raw}} &\longrightarrow H^1(\mathcal{I}_e), \quad [h] \longmapsto [c(h)], \\ \text{im } \partial_e &= \mathcal{C}_e, \quad \mathcal{C}_e = \mathcal{S}_e/\mathcal{B}, \quad \mathcal{S}_e = \text{im}[B_{\text{cof}}, F_e], \quad \mathcal{B} = \text{im } B_{\text{cof}}. \end{aligned} \quad (\text{CIT10})$$

If $h = T_0 x + V_0 \xi$, its target coefficients are the boundary $B_{\text{cof}} x$ by CIT8. Conversely, if $c(h) = B_{\text{cof}} x$, then $\mathsf{T}(h - T_0 x) \in \mathcal{L}_1$ and CIT9 gives $h - T_0 x \in \mathcal{L}_0$. Thus the kernel before quotient is exactly $\mathcal{S}_k^{\text{raw}}$. This proves injectivity and the stated presentation; $[B_{\text{cof}}, F_e]$ is independent. It is exactly the positive connecting image of $H^0(\mathcal{R}_e) = \ker t_e$ in LCC13, since its construction lifts the original

degree-zero quotient cycle and takes its differential with the same positive sign. No splitting of the remaining $\ker U_e$ has been introduced.

Its full cohomology restriction Gram in the P_e image coordinates is

$$H_{\text{conn},e} = F_e^* \mathcal{D}_e F_e - F_e^* \mathcal{D}_e B_{\text{cof}} (B_{\text{cof}}^* \mathcal{D}_e B_{\text{cof}})^{-1} B_{\text{cof}}^* \mathcal{D}_e F_e. \quad (\text{CIT11})$$

The exact least norm over every original boundary gives this Schur complement. All original lower coefficients were already minimized simultaneously in $\mathcal{D}_e$. Equivalently set

$$Z_e^{\text{raw}} = [\mathcal{V}, \mathbb{T}T_0], \quad H_{\text{conn},e} = (\mathbb{T}P_e)^*(I - P_{Z_e^{\text{raw}}})\mathbb{T}P_e, \quad P_Z = Z(Z^*Z)^{-1}Z^*. \quad (\text{CIT12})$$

The columns of Z_e^{raw} are independent by CIT9 and the independence in CIT4. The same argument with the added P_e proves independence of $[Z_e^{\text{raw}}, \mathbb{T}P_e]$. Indeed the complete minimization in CIT11 allows exactly $\mathbb{T}T_0x + \mathcal{V}\eta$. This proves CIT12 and positivity. It also gives the full original raw determinant

$$\det H_{\text{conn},e} = \frac{\det([Z_e^{\text{raw}}, \mathbb{T}P_e]^* [Z_e^{\text{raw}}, \mathbb{T}P_e])}{\det((Z_e^{\text{raw}})^* Z_e^{\text{raw}})}. \quad (\text{CIT13})$$

These are the same fixed combined frames, so their ordinary Schur determinant introduces no missing coordinate factor.

E.3 The original source-to-image determinant at its low cutoff

Give the primitive source quotient its original source metric and its correlated image metric, respectively:

$$H_{\text{src},e} = P_e^*(I - P_{[V_0, T_0]})P_e, \quad H_{\text{cor},e} = (\mathbb{T}P_e)^*(I - P_{\mathbb{T}[V_0, T_0]})\mathbb{T}P_e. \quad (\text{CIT14})$$

These are two explicit attained minima. The original NBG $\|\mathbb{T}\| \leq K_+$ and $\|\mathbb{S}\| \leq K_-$ with $\mathbb{S}\mathbb{T} = I$ imply

$$K_-^{-2} H_{\text{src},e} \preceq H_{\text{cor},e} \preceq K_+^2 H_{\text{src},e}. \quad (\text{CIT15})$$

For each fixed primitive value, apply the two norm inequalities to every representative in $P_e b + \mathcal{S}_k^{\text{raw}}$ and take the minima. This proves both directions on the exact same fibre. The finite constants are the complete NBG12 ones at base k and cutoff D_e , including source order $(1, 1)$ or (k, j) and center shift $(\beta_{4,a,b} - 2)/i$.

There is also an exact independent-gauge quotient identity. In the original source quotient put

$$X_k = \mathcal{S}_k^{\text{raw}}/\mathcal{L}_0, \quad Y_e = \mathcal{Y}_e^{\text{raw}}/\mathcal{L}_0, \quad X_k \subset Y_e.$$

Their correlated target images are the original NKC9 images. The original independent-gauge form restricted to Y_e and then quotiented by X_k allows precisely the total fibre $\mathbb{T}T_0x + \mathcal{L}_1$; its correlated quotient allows $\mathbb{T}T_0x + \mathbb{T}\mathcal{L}_0$. These are CIT12 and CIT14. Hence the literal PCJ7 source quotient gives

$$\frac{\det H_{\text{conn},e}}{\det H_{\text{cor},e}} = \det \Omega_{Y_e/X_k, D_e}. \quad (\text{CIT16})$$

If R transports the original primitive lifts to an adapted correlated quotient frame, both Grams are changed by $R^*(\cdot)R$, and their two factors $|\det R|^2$ cancel in CIT16. This is the exact coordinate map behind that determinant equality.

Use the actual compatibility-plus-complement expressions χ_{X_k, D_e} and χ_{Y_e, D_e} of PCJ1. The complete shared-update proof PCJ13–16 gives

$$\begin{aligned} \log \det H_{\text{conn},e} - \log \det H_{\text{src},e} &= -(\chi_{X_k, D_e} - \chi_{Y_e, D_e}) + \epsilon_e, \\ -2r_e \log \max(1, K_-) - r_e \log c_+ &\leq \epsilon_e \leq 2r_e \log \max(1, K_+) - r_e \log c_-. \end{aligned} \quad (\text{CIT17})$$

Indeed write the left side as the correlated/source logarithm plus $\log \det \Omega_{Y_e/X_k}$. The first is bounded by CIT15. PCJ16 writes the negative of the latter as $\chi_{X_k} - \chi_{Y_e} + \nu$, with $r_e \log c_- \leq \nu \leq r_e \log c_+$. Subtraction gives exactly CIT17. Here $c_- = \min(1, L_-^{-2})$ and $c_+ = \max(1, L_+^2)$ are the original NKC constants at this cutoff. Consequently this specified error is $O_{\text{actual}}(k \log q_k) = o(kq_k)$ in both original source pairs. At $e = 0$, Y_0 is the full ambient source quotient, so its source-complement projector vanishes and $\chi_{Y_0} = 0$ exactly. At $e = 1$ its four actual polynomial constraints remain; χ_{Y_1} is not set to zero.

Current evaluated source and the surviving independent gauges. Here $q = q_k = (k+1)^2$, $e = 0, 1$, $d_e = 8k + 24 + e$, and $N_e = D_e = q_{k+4} - v - 1 + e$. Keep the exact primitive coordinates P_e in CIT6 and $\Theta_e = C_e^\top P_e$ in PDI1–2. With PSR18’s unchanged Γ_e and the full factorial product $\mathfrak{F}_{k,e,s_0}$, set

$$\tilde{Z}_e = \Gamma_e - \log \mathfrak{F}_{k,e,s_0} - 2d_e q \log(4/\pi) + e m_{k,s_0}^{(4)} - \chi_{X_k, D_e} + \chi_{Y_e, D_e}.$$

The full connecting-image determinant now satisfies

$$\begin{aligned} \epsilon_e^{\text{lo}} + e b_{k,s_0}^{\text{lo}} - \mathcal{E}_{k,e,s_0}^+(a) &\leq \log \det H_{\text{conn},e} - \tilde{Z}_e \\ &\leq \epsilon_e^{\text{hi}} + e b_{k,s_0}^{\text{hi}} + \mathcal{E}_{k,e,s_0}^-(a), \end{aligned} \quad (\text{CIT27})$$

$$\log \det H_{\text{conn},e} = \Gamma_e - 2d_e q \{\log(4q/\pi) - 1\} - \chi_{X_k, D_e} + \chi_{Y_e, D_e} + o(kq).$$

Here all finite quantities are exactly PSR2–4 and PSR16–22, at the domain stated in PSR2, with $\epsilon_e^{\text{lo}}, \epsilon_e^{\text{hi}}$ equal to CIT17’s two original endpoints. The proof substitutes PSR19 into the exact CIT17 identity. Its residual is $\epsilon_e + e \varepsilon_{k,s_0}^{(4)} - \varepsilon_{k,e,s_0}^H$; adding their full intervals gives every displayed sign. The two gauge terms have not been combined with the source determinant or omitted. The $e = 0$ complement term vanishes for the exact reason in CIT17; the $e = 1$ term retains all four original constraints.

There is also an exact map for comparing the two original source-order pairs. PSR23 gives $\Theta_e = (R_e^{\text{pol}})^\top \text{diag}(r!) P_e^{\text{Tay}}$ in both orders. The original Taylor cochain square

$$\text{stack}_i \text{Mult}(B_i) P_e^{\text{Tay}} = \text{diag}(\tilde{T}_j, 4) F_e + \text{diag}(\tilde{V}_j, 4) E_e^{\text{low}}$$

has a unique analytic coefficient F_e modulo all lower evaluations, by CNC13. It therefore gives the same F_e for $(s_0, s_1) = (1, 1)$ and $(k, k+4)$. On these exact common ordered values, write $\mathcal{X}_e^{(s_0, s_1)} = \chi_{X_k, D_e}^{(s_0, s_1)} - \chi_{Y_e, D_e}^{(s_0, s_1)}$. Then

$$\log \frac{\det H_{\text{conn},e}^{(k,k+4)}}{\det H_{\text{conn},e}^{(1,1)}} = -\mathcal{X}_e^{(k,k+4)} + \mathcal{X}_e^{(1,1)} + o(kq). \quad (\text{CIT28})$$

This is PSR24–27: the exact residual is the source order ratio plus $\epsilon_e^{(k,k+4)} - \epsilon_e^{(1,1)}$. Its finite interval is obtained by adding the PSR26 source interval, including its full D_e^{ord} , to $[\epsilon_{e,(k,k+4)}^{\text{lo}} - \epsilon_{e,(1,1)}^{\text{hi}}, \epsilon_{e,(k,k+4)}^{\text{hi}} - \epsilon_{e,(1,1)}^{\text{lo}}]$. This statement concerns the two defining low-cutoff metrics; CIT18–26 still uses its full fixed-column target extension at every later original cutoff.

Full finite source-order interval in the original primitive values. Keep CIT27–28, including its original finite-domain bounds and both gauge terms. Put $p = 0 : (s_0, s_1) = (1, 1)$ and $p = 1 : (k, j)$, $j = k+4$, $D_e = q_j - v - 1 + e$, $r_e = 8k + 24 - 3e$, and $\Delta Z = Z^1 - Z^0$. The full NSR2 constants are $\mathbf{a}_s = \rho_s \Pi_s$ and $\mathbf{b}_{s,D} = \rho_s U_{s,D}$, where

$$\rho_s = \frac{(2\pi)^{2\ell_s}}{(1/2)^{2\ell_s}}, \quad \Pi_s = \prod_{h < \ell_s} (2h + \tfrac{1}{2})^2, \quad U_{s,D} = \prod_{h < \ell_s} \{4(D + h + 1)^2 + (2h + \tfrac{1}{2})^2\}, \quad \ell_s = (s-1)/4.$$

All factors retain the original measure; the $s = 1$ products are one. Write $L_{\text{src}} = \Delta \log \det H_{\text{src},e}$, $L_{\text{conn}} = \Delta \log \det H_{\text{conn},e}$ and $b_e^- = \epsilon_e^{1,\text{lo}} - \epsilon_e^{0,\text{hi}}$, $b_e^+ = \epsilon_e^{1,\text{hi}} - \epsilon_e^{0,\text{lo}}$, using the complete CIT17 errors

at their original source pair and cutoff. Then the following finite intervals hold in the same ordered values:

$$\begin{aligned} s_e^- &:= -r_e \log \mathbf{b}_{k,D_e} \leq L_{\text{src}} \leq -r_e \log \mathbf{a}_k =: s_e^+, \\ t_e^- &:= -r_e \log \mathbf{b}_{j,D_e} \leq L_{\text{conn}} \leq -r_e \log \mathbf{a}_j =: t_e^+, \\ L_{\text{conn}} &= -\Delta\mathcal{X}_e + \zeta_e, \quad s_e^- + b_e^- \leq \zeta_e \leq s_e^+ + b_e^+. \end{aligned} \tag{CIT31}$$

Indeed the ordinary-transpose map NSR4 preserves the full Taylor values. It takes the source fibre $P_e x + T_0 \xi + V_0 \eta$ to that same fibre in the other original source order. On the target, NSR5's unique Taylor cochain square preserves F_e and every lower correction. Minimizing over those lower corrections and B_{cof} gives the image fibre, of the same rank r_e . The complete dual Loewner bounds therefore give the first two lines. Subtracting the two exact CIT17 identities gives the last line with $\zeta_e = L_{\text{src}} + \epsilon_e^1 - \epsilon_e^0$. Thus

$$s_e^- + b_e^- - t_e^+ \leq \Delta\mathcal{X}_e \leq s_e^+ + b_e^+ - t_e^-. \tag{CIT32}$$

This proves the finite receiving of CIT28, not only its limit. The old PSR26 source interval can additionally be intersected with the first line on its own stated finite domain and original admissible radius. No source centre or mass is changed. At $e = 0$ the exact complement term vanishes; at $e = 1$ all four original constraints remain inside χ_{Y_e} . The widths and endpoints are $O_{\text{actual}}(k^2 \log q_j) = o(kq_k)$. The later calculation continues to use fixed F_e at every target cutoff; no later primitive germ extension is asserted.

The defining primitive source with its separate exact map. Fix $e = 0$ or 1 , retain $D_e = q_{k+4} - v - 1 + e$ and $r_e = 8k + 24 - 3e$. The complete unscaled columns $r_h = \mathcal{T}_A(\chi_k S^h)$, $0 \leq h < 8k + 24 + e$, satisfy $\psi_k \mid r_h$ by the original root-addition identity. Their leading coefficient is $\binom{q_k+h}{v} \mu_v \neq 0$, so their divided polynomial degree is exactly $a_{\text{src},e} = 24k - 25 - v + e$. The full source relation Gram H_e and the same factorial-bearing matrix Θ_e give

$$H_{\text{src},e}^{(s)} = \Theta_e^* (\overline{H_e^{(s)}})^{-1} \Theta_e. \tag{EFR9}$$

EF32–34 proves the entire affine-fibre equality and its minimizing section: the annihilator kernel is the original analytic/lower plane, and its prescribed value is $\Theta_e = (R_e^{\text{pol}})^{\top} \text{diag}(n!) P_e^{\text{Tay}}$ with R_e^{pol} the unscaled r_h columns. At $e = 1$ the four independent primitive rows remain in the rank- r_e value image. This is the actual source morphism used to apply QOT, even though the degree integer also equals $D_e - q'_k$.

Intersect the source NSR6 interval with QOT27 in this full map, and the target NSR6 interval with QOT30 at degree $a_{\text{tgt},e} = 16k + 15 - v + e$. Call the resulting rank- r_e determinant endpoints $S_e^{\pm}, T_e^{\pm}$, exactly as in EFP35–36, with each original root-radius guard retained separately. For the unchanged NSR7 errors $b_e^{\pm}$ one obtains

$$\begin{aligned} T_e^- - S_e^+ &\leq \mathcal{J}_e^{\text{ord}} \leq T_e^+ - S_e^-, \\ S_e^- + b_e^- &\leq \zeta_e \leq S_e^+ + b_e^+, \quad L_{\text{conn},e} = -\Delta\mathcal{X}_e + \zeta_e, \\ S_e^- - T_e^+ + b_e^- &\leq \Delta\mathcal{X}_e \leq S_e^+ - T_e^- + b_e^+. \end{aligned} \tag{EFR10}$$

Subtracting the exact source and target determinant identities proves these signs; the variables shared by ζ_e and $\Delta\mathcal{X}_e$ stay shared. EFP38 retains the scalar $\log(\kappa_j/\kappa_k)$ with its opposite signs in the Jacobian and compatibility expressions. The original PSR interval may also be intersected on its original guard, with all of its errors.

EOR's necessary parity rectangle guard fails at these defining low degrees: the actual QOT radius with $\gamma > 2$ gives $q_k > 2^{34}k$, whereas that EOR guard would give $q_k \leq 36k - 36$ at the source or $q_j \leq 24k + 24$ at the target. The proof in the EFP primitive section therefore uses QOT's proved low fallback here. It does not insert high-endpoint equilibrium constants into the primitive source. CIT31–32 and their LCC/NCR primitive receivers retain every source form, frame and compatibility error when receiving these narrower finite intervals.

CIT14–17 concern the defining low cutoff D_e . They do not assert that a primitive truncated germ has a common compatible extension to every later cutoff. The target coefficient columns F_e now are fixed, and their complete later metrics are calculated next without that assertion.

E.4 The fixed connecting flag at every original target row

For $q_j - 1 \leq N \leq 2q_j$ put $D = N - v$ and retain the fixed B_{cof}, F_e in the original target coefficients. Define

$$\begin{aligned} \mathcal{D}_N &= \text{diag}(D_{j,N-v}^{(s_1)}, 4), & Z_{B,N} &= [\mathcal{V}_N, \mathcal{T}_{1,N} B_{\text{cof}}], \\ Z_{C,N} &= [Z_{B,N}, \mathcal{T}_{1,N} F_e], & H_{\text{conn},e}(N) &= \text{Gram}_{\mathcal{D}_N}(\mathcal{S}_e/\mathcal{B}), \\ \det H_{\text{conn},e}(N) &= \det(Z_{C,N}^* Z_{C,N}) / \det(Z_{B,N}^* Z_{B,N}). \end{aligned} \quad (\text{CIT18})$$

The original DNE positivity and the independence in CIT10 make both raw frames full column rank at every one of these indices. Eliminating the complete lower frame and then B_{cof} proves the determinant identity. At the defining $N = q_j - 1 + e$, it is exactly CIT11–13 by CIT8.

Every entry is given by the original arithmetic recursion. If $\nu_{a,n} = \sum (Z_{\text{exp},j})_{uv,a} \beta_{j,u,v}^n$, then

$$\begin{aligned} f_{a,n} &= \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \\ b_{0,m} &= \mathbf{1}_{m=0}, \quad b_{-1,m} = 0, \quad b_{n+1,m} = i^{-1} (b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2) b_{n-1,m}, \\ t_n &= \left(\frac{\sum_m b_{n,m} f_{a,m}}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_a, \quad v_n = \left(\frac{\sum_m b_{n,m} \beta_{j-8,u,v}^m}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_{u,v}. \end{aligned} \quad (\text{CIT19})$$

Arrays outside their polynomial range are zero. Differentiating $E_A F_a = N_a$ gives the first line; substituting the full original Gamma recurrence gives the next. Thus these are exactly the unchanged IRR7–9 rows, retaining all original moments and phases. The new four-row blocks in CIT18 are $z_{B,n} = [\text{diag}(v_n, 4), \text{diag}(t_n, 4) B_{\text{cof}}]$ and $z_{C,n} = [z_{B,n}, \text{diag}(t_n, 4) F_e]$. For $X = B, C$ define $G_{X,n} = Z_{X,n+v}^* Z_{X,n+v}$. In degree-major row order each raw Gram increases by $z^* z$. The fixed row permutation from copy-major order is unitary.

For $q_j - v \leq n \leq 2q_j - v$, the exact connecting-volume increment is

$$\log \frac{\det H_{\text{conn},e}(n+v)}{\det H_{\text{conn},e}(n+v-1)} = \log \det(I_4 + z_{C,n} G_{C,n-1}^{-1} z_{C,n}^*) - \log \det(I_4 + z_{B,n} G_{B,n-1}^{-1} z_{B,n}^*) \geq 0. \quad (\text{CIT20})$$

The determinant lemma proves the equality. For its sign, transform the old extra columns to their full least-norm section over $Z_{B,n+v-1}$. The larger row covariance is the smaller one plus $RQ^{-1}R^* \succeq 0$, by block inversion. This proves the inequality and retains the actual old quotient denominator. No independent rowwise gauge minimum is taken.

E.5 Its exact place in the common low invariant loss

Let $\mathbb{J} = \ker \mathbb{B}$ be the full original fourfold DNE invariant kernel. At each later N use the same actual target metric $\mathcal{D}_N$ and fixed flags

$$\begin{aligned} W &= \mathcal{B} + \mathbb{J}, & K_{\text{conn},e} &= \mathcal{S}_e \cap W = \mathcal{B} + (\mathcal{S}_e \cap \mathbb{J}), \\ T_{\text{conn},e} &= \mathcal{S}_e + W = \mathcal{S}_e + \mathbb{J}, \\ \omega_{\text{conn},e}(N) &= \frac{\det \text{Gram}_{\mathcal{D}_N}(T_{\text{conn},e}/W)}{\det \text{Gram}_{\mathcal{D}_N}(\mathcal{S}_e/K_{\text{conn},e})}. \end{aligned} \quad (\text{CIT21})$$

The two Grams use the same quotient-value coordinates under $[z] \mapsto [\mathbb{B}z]$. Its original fibres are exactly the two spaces in the denominator and numerator, as in LCC20–23. The source is the corresponding quotient of the connecting cohomology image; its full kernel is $(\mathcal{S}_e \cap \mathbb{J})/(\mathcal{B} \cap \mathbb{J})$. Thus CIT21 is its actual further target loss, and is in $(0, 1]$.

The full low cycles contain $\mathcal{S}_e$, and LCC13 gives

$$\dim(\mathcal{Z}_e/\mathcal{S}_e) = \kappa_e \leq 3 \max(v - e, 0). \quad (\text{CIT22})$$

Both spaces are fixed original coefficient flags for the later metric comparisons. The difference of their invariant loss returns has the finite bound

$$|\mathcal{R} \log \omega_e - \mathcal{R} \log \omega_{\text{conn},e}| \leq 2\kappa_e \log A_{j,s_1}^{\text{end}} = O_{\text{actual}}(q_j \log j) = o(kq_k). \quad (\text{CIT23})$$

Here $\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j)$. For a complete verification, apply the actual inclusions $\mathcal{S}_e \subset \mathcal{Z}_e$ with their common W . The kernel/intersection quotient and sum quotient fit into the exact sequence $0 \rightarrow K_{\text{full}}/K_{\text{conn}} \rightarrow \mathcal{Z}_e/\mathcal{S}_e \rightarrow T_{\text{full}}/T_{\text{conn}} \rightarrow 0$ by the identity on representatives. Their two ranks sum to κ_e . The full adapted-Gram factorization LFC7–12 expresses the loss difference as the sum of those two quotient log determinants minus the $\mathcal{Z}_e/\mathcal{S}_e$ log determinant and a fixed frame constant. FRE's same native form sandwich puts the three monotone returns between zero and twice their ranks times $\log A_{j,s_1}^{\text{end}}$. Both sides of the resulting signed difference are bounded by $2\kappa_e \log A$. This proves CIT23, including $\kappa_e = 0$, with all frame constants canceled only after the original four signs.

The sharper original tail bound in CIT23. Retain the original simple-quartet five-orbit domain, its invertible period matrix, the full conductor and both native families. In particular this calculation imposes no additional pole-coordinate gate or restriction on that domain. Let $k = 4l + 1 \geq 9$, $j = k + 4$, $e \in \{0, 1\}$, and use exactly the coefficient spaces of CIT21–23. Put $\bar{\kappa}_e = 3 \max(v - e - 2, 0)$. The complete original four-factor calculation BTK26 gives $v \leq 32$, and BTK25 gives $\kappa_e \leq \bar{\kappa}_e$. The actual spaces and the strengthened finite estimate are

$$\begin{aligned} 0 \longrightarrow \frac{\mathcal{Z}_e \cap \mathbb{J}}{\mathcal{S}_e \cap \mathbb{J}} \xrightarrow{[j] \mapsto [j]} \frac{\mathcal{Z}_e}{\mathcal{S}_e} \xrightarrow{[z] \mapsto [z]} \frac{\mathcal{Z}_e + W}{\mathcal{S}_e + W} \longrightarrow 0, \\ d_{K,e} + d_{T,e} = \kappa_e \leq \bar{\kappa}_e, \quad \bar{\kappa}_0 \leq 90, \quad \bar{\kappa}_1 \leq 87, \\ \left| \mathcal{R} \log \omega_e^{(s_1)} - \mathcal{R} \log \omega_{\text{conn},e}^{(s_1)} \right| \leq 2\kappa_e \log A_{j,s_1}^{\text{end}} \leq 6 \max(v - e - 2, 0) \log A_{j,s_1}^{\text{end}}. \end{aligned} \quad (\text{RKR1})$$

Here $W = \mathcal{B} + \mathbb{J}$, and $d_{K,e}, d_{T,e}$ are the dimensions of the first and last nonzero terms. To prove exactness, the first arrow is injective since its zero class means $j \in \mathcal{S}_e \cap \mathbb{J}$. A middle class maps to zero precisely when $z = s + b + j$ with $s \in \mathcal{S}_e$, $b \in \mathcal{B} \subset \mathcal{S}_e$, $j \in \mathbb{J}$. Then $j = z - s - b \in \mathcal{Z}_e \cap \mathbb{J}$ and $[z] = [j]$. Every last class has a representative $z \in \mathcal{Z}_e$, because $W \subset \mathcal{S}_e + W$. This proves the full sequence. The identity on representatives also gives the isomorphism

$$\frac{\mathcal{Z}_e \cap \mathbb{J}}{\mathcal{S}_e \cap \mathbb{J}} \xrightarrow{\sim} \frac{\mathcal{Z}_e \cap W}{\mathcal{S}_e \cap W}, \quad [j] \mapsto [j].$$

Indeed $z \in \mathcal{Z}_e \cap W$ is $b + j$ with $b \in \mathcal{B}$, hence $j \in \mathcal{Z}_e \cap \mathbb{J}$; the kernel is exactly $\mathcal{S}_e \cap \mathbb{J}$. Thus these are exactly the two quotient ranks in the adapted determinant factorization already proved in CIT23.

For each fixed $s_1 = 1$ or j , use its unchanged form $\mathcal{D}_N = \text{diag}(D_{j,N-v}^{(s_1)}, 4)$. The two quotient logarithms of ranks $d_{K,e}, d_{T,e}$ enter with positive signs and the quotient of rank κ_e with a negative sign, together with a fixed frame constant. FRE's complete form sandwich and the same attained affine minima place each rank- r quotient return in $[0, 2r \log A_{j,s_1}^{\text{end}}]$. With the unchanged operator $\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j)$, the constant cancels. Both sides of the resulting signed difference are at most $2\kappa_e \log A_{j,s_1}^{\text{end}}$, since $d_{K,e} + d_{T,e} = \kappa_e$. The logarithm is nonnegative, so BTK25 proves the last bound. If $v - e \leq 2$, this proves $\mathcal{Z}_e = \mathcal{S}_e$, and the quotient-value maps and their attained minima coincide: $\omega_e^{(s_1)}(N) = \omega_{\text{conn},e}^{(s_1)}(N)$ at every retained target cutoff in each native family.

There is a further exact relation to the ORIGINAL ITL source. Write $F_{\text{inv}} = \text{stack}_i(M_{i,k}J_0)$. Since $B_{\text{cof}} = \text{stack}_i M_{i,k}$,

$$\text{im } F_{\text{inv}} \subset \mathcal{B}, \quad \mathbb{J} \subset W_{\text{inv}} := \mathbb{J} + \text{im } F_{\text{inv}} \subset W \subset T_{\text{conn},e}. \quad (\text{CIT24})$$

These inclusions are literal original coefficient maps. The ITL source therefore represents boundaries in low cohomology, but its target Gram remains in the denominator $\text{Gram}(W)$ of the cohomology target quotient.

To exhibit every factor, let $I_R = F_{\text{inv}} Z_R$ be exactly the fixed ITL4 source-kernel complement and choose fixed columns R_b completing $[X_{\mathbb{J}}, I_R]$ to W . In a specified fixed full frame X_W , write

$$[X_{\mathbb{J}}, I_R, R_b] = X_W C_W, \quad Q_b(N) = \text{Gram}_{\mathcal{D}_N}(W/W_{\text{inv}}) \quad \text{in the lifts } R_b.$$

The first quotient of this chain has exactly ITL's target Gram $H_{\text{inv}}(N)$: its fibre is $I_R b + \mathbb{J}$. Its source pullback quotient is the full ITL $A_{\text{inv}}(N)$, and $\det H_{\text{inv}} = \det A_{\text{inv}} \omega_{\text{inv}}$. Sequential Schur complements consequently give

$$|\det C_W|^2 g_W(N) = g_{\mathbb{J}}(N) \det H_{\text{inv}}(N) \det Q_b(N), \quad g_X(N) = \det(X_X^* \mathcal{D}_N X_X). \quad (\text{CIT25})$$

All matrices in this formula are in their stated independent frames; $r_{\text{inv}} = 0$ gives the empty first quotient.

Finally retain the full LFC coordinate constant $c_{\text{conn},e}$ from the adapted frames for

$$(\mathcal{S}_e, K_{\text{conn},e}, T_{\text{conn},e}, W).$$

Its cross ratio and CIT25 give the exact original identity

$$\begin{aligned} \log \omega_{\text{conn},e} + \log \omega_{\text{inv}} &= \log g_{T_{\text{conn},e}} + \log g_{K_{\text{conn},e}} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} \\ &\quad - \log \det A_{\text{inv}} - \log \det Q_b + \log c_{\text{conn},e} + 2 \log |\det C_W|. \end{aligned} \quad (\text{CIT26})$$

This relates the two actual losses through their exact common boundary structure; it does not identify their source spaces. Taking the original four-return cancels the last two fixed coordinate constants. Equivalently ITL16's exact matched relative operator satisfies $\det H_{\text{inv}} = \det Q_{\text{inv}} \det \mathcal{R}_X$, where Q_{inv} is the full original source quotient $J_0^* D_{k,N-v}^{(s_0)} J_0 / E$. Substituting this equality instead in CIT25 replaces $(\omega_{\text{inv}}, A_{\text{inv}})$ in CIT26 by $(\det \mathcal{R}_X, Q_{\text{inv}})$, preserving the source-order pair $(1, 1)$ or (k, j) . It retains the original source mass.

Current complete residual quotient and its directed return. In CIT24–26's original coefficient spaces put $\Phi_b = \mathbb{B} B_{\text{cof}}$, $\mathcal{L}_b = \ker \Phi_b$ and $Y_b = \text{im } J_0 + \mathcal{L}_b$. The map $[x] \mapsto [B_{\text{cof}} x]$ identifies the exact quotient $\mathbb{C}^{m_k} / Y_b$ with W / W_{inv} : its zero fibre consists precisely of $x - J_0 a \in \ker \Phi_b$. RQB5 computes this full kernel by the original conductor and invariant coefficient matrices. Its branch constraint is RQB9–12 on the nonreduced component scheme, with every shared factor retained. Let Q_b^E be RQB14's complete attained target metric. For the original CIT25 lifts, $R_b^{\text{CIT}} = R_b^E C_b + Y_{\text{inv}} A_b$, so minimization over the identical full fibre proves

$$\begin{aligned} Q_b^{\text{CIT}} &= C_b^* Q_b^E C_b, \quad |\det C_W|^2 g_W = g_{\mathbb{J}} \det H_{\text{inv}} |\det C_b|^2 \det Q_b^E, \\ \mathcal{V}_b &= \mathcal{R} \log \det Q_b^{\text{CIT}} = \mathcal{R} \log \det Q_b^E, \quad 0 \leq \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} \leq 4U_{j,s_1}^{\text{native}}. \end{aligned} \quad (\text{CIT29})$$

The exact matrix C_b is fixed across the original four target cutoffs; its determinant cancels only under their signs. Every row of Q_b^E is given by the complete division and Gamma recurrences RQB16–18, and its source denominator is the quotient in RQB19–20, including $B_0 \mathcal{L}_b$ and the original low overlap.

Write the remaining exact part of CIT26's four-return as

$$\mathcal{P}_e^{\text{rest}} = \mathcal{R}(\log g_{T_{\text{conn},e}} + \log g_{K_{\text{conn},e}} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} - \log \det A_{\text{inv}}).$$

Substituting the preceding metric congruence in that identity, with all fixed frame factors kept before the return, gives

$$\begin{aligned} \mathcal{R}(\log \omega_{\text{conn},e} + \log \omega_{\text{inv}}) &= \mathcal{P}_e^{\text{rest}} - \mathcal{V}_b, \\ \mathcal{P}_e^{\text{rest}} - 4\hat{F}_j^{(s_1)} + \mathcal{V}_{\text{inv}}^{\text{target}} &\leq \mathcal{R}(\log \omega_{\text{conn},e} + \log \omega_{\text{inv}}) \leq \mathcal{P}_e^{\text{rest}}. \end{aligned} \quad (\text{CIT30})$$

The lower sign is RQB28/32's inequality for $-\mathcal{V}_b$. Its full missing target quotient Q_o remains in $\mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} + \mathcal{V}_o = 4\widehat{F}_j^{(s_1)}$. The pointwise native cap is the complete RQB35 minimum, including the original positive observation and source corrections. Thus no boundary factor is identified with the source or assigned a common long-return value.

The signed remainder as the complete positive quotient decomposition. Retain $W, \mathcal{S}_e, K_e, T_e$ and the full RQB27 target frame $C_T = [U_R, U_b, U_o]$. The literal map $\pi_o = [0 \ 0 \ I_{r_o}]C_T^{-1}\mathbb{B}$ has kernel W and is onto. Indeed its value vanishes exactly when $\mathbb{B}z \in \text{im}(\mathbb{B}B_{\text{cof}})$. It therefore identifies $\mathcal{U}/W$ isometrically with the original missing quotient Q_o : first minimize over $\mathbb{J}$, then over that complete image. Put $E_e = \pi_o F_e$ and $c_e = \text{rank } E_e$. CPQ7–8 retains the original source quotient $G_{s,e}$ on $\mathcal{S}_e/K_e$, the target restriction $G_{t,e}$ on T_e/W , and the remaining quotient $Q_{\text{rem},e}$ on $\mathcal{U}/T_e$. Their values are fixed by E_e , its kernel complement Z_c , and $C_o = [E_e Z_c, R_e^o]$, so all coordinate determinants remain those of the actual coefficient maps.

Write $V_X = \mathcal{R} \log \det G_X$, with V_{rem} from Q_{rem} and V_A from the full original invariant pullback source A_{inv} . Then

$$\begin{aligned} V_o &= V_t + V_{\text{rem}}, & V_{\text{conn}} &= V_h + V_s, & 4\widehat{F}_j^{(s_1)} &= V_I + V_b + V_t + V_{\text{rem}}, \\ \mathcal{P}_e^{\text{rest}} &= 4\widehat{F}_j^{(s_1)} - V_{\text{rem}} - V_s - V_A, & \mathcal{L}_e &:= \mathcal{R} \log(\omega_{\text{conn},e} \omega_{\text{inv}}) = \mathcal{P}_e^{\text{rest}} - V_b, & & (\text{CIT33}) \\ V_I - V_s - V_A &\leq \mathcal{L}_e \leq 4U_{j,s_1}^{\text{native}} - V_b - V_s - V_A. \end{aligned}$$

Every listed V is nonnegative: it is a fixed restriction or attained quotient of the complete increasing original Gamma row form. The first two equalities follow from full adapted Gram determinants; the third is RQB27. Their substitution into $\mathcal{L}_e = V_t - V_s + V_I - V_A$ proves the next line. Before taking the return the constant is exactly

$$\begin{aligned} \mathbf{c}_e &= \frac{|\det C_T|^2 |\det C_o|^2 |\det C_b|^2}{c_{\text{conn},e} |\det C_W|^2}, \\ P(N) &= 4 \log \det H_{B,j,D} - \log \det Q_{\text{rem},e} - \log \det G_{s,e} \\ &\quad - \log \det A_{\text{inv}} + \log \mathbf{c}_e. \end{aligned}$$

This is CPQ12 obtained by literal substitution in CIT26 and RQB15/27. The constant cancels only under the four signs. The last interval in CIT33 uses $0 \leq V_t \leq V_o$ and the existing cap only on its proved finite domain.

For both original source pairs retain all widths in NSR2 and $r_I = j_{0,k} - \epsilon$, $r_b = t_k - \ell + \epsilon$. The same two quotient values for each contraction give

$$\begin{aligned} C_e^{\text{ord}} &= (c_e + r_I)W_j, & |\Delta \mathcal{L}_e| &\leq C_e^{\text{ord}}, \\ -C_e^{\text{ord}} - r_b \Omega_j^{\text{hi}} &\leq \Delta \mathcal{P}_e^{\text{rest}} \leq C_e^{\text{ord}} + r_b \Omega_j^{\text{lo}}. & & (\text{CIT34}) \end{aligned}$$

The connecting contraction has rank c_e , the invariant contraction rank r_I , both with target source order s_1 . Their common scalar factors cancel at each endpoint under the exact value maps; summing gives the first bound. The equality $\Delta \mathcal{P}_e^{\text{rest}} = \Delta \mathcal{L}_e + \Delta V_b$ and NSR19's original rank- r_b interval prove the second with its two signs. Each pair's CIT33 interval can be intersected with the other pair's interval translated by $[-C_e^{\text{ord}}, C_e^{\text{ord}}]$. Both Q_o and the independent degree- k source comparison remain explicit; no replacement of A_{inv} by Q_{inv} has occurred.

The common projection and its evaluated four-angle enclosure. On the unchanged CIT33 simple-quartet domain put $I = \text{im}(B_{\text{cof}} J_0)$, $B = \text{im } B_{\text{cof}}$, $S = \mathcal{S}_e$ and $J = \ker \mathbb{B}$. Retain $K_i = I \cap J$, $W_i = I + J$, $W = B + J$, $K_c = S \cap W$ and $T = S + J$. Here $S + J = S + W$ because $B \subset S$. The identity on representatives gives the exact quotient maps $I/K_i \rightarrow W_i/J$ and $S/K_c \rightarrow T/W$. Their kernels vanish by the intersection definitions, and their images are the whole target quotients by the sum definitions. The source minimum lifts from the orthogonal

sum $(I \ominus K_i) \oplus (S \ominus K_c)$; the target lifts form $(W_i \ominus J) \oplus (T \ominus W)$, in the original metric $\mathcal{D}_N$. Orthogonality follows from $I \subset K_c$ and $W_i \subset W$. Projection between these sums has block matrix $\begin{pmatrix} M_i & C_N \\ 0 & M_c \end{pmatrix}$: the zero block follows from $I \subset W$, and the diagonal blocks are exactly the two quotient maps in orthonormal frames. Thus its squared determinant is $\omega_{\text{conn},e}\omega_{\text{inv}}$, with the cross block retained.

SFB's full variation calculation on these fixed flags and SFC's explicit scalar optimization now apply to this actual product. Let L_-, L_+ be the finite expressions SFC9–10, using precisely $\omega(N_0) = \det G_t \det H_{\text{inv}} / (\det G_s \det A_{\text{inv}})$ at $N_0 = q_j - 1$ and the three full determinant ratios SFC3 at $N_1 = q_j$, $N_2 = 2q_j - 1$, $N_3 = 2q_j$. Then the earlier signed remainder receives

$$\begin{aligned} -4\mathcal{T}_{j,s_1} &\leq L_- \leq \mathcal{L}_e \leq L_+ \leq 4\mathcal{T}_{j,s_1}, \\ V_b + L_- &\leq \mathcal{P}_e^{\text{rest}} \leq V_b + L_+. \end{aligned} \tag{CIT35}$$

Indeed the forward positive metric path gives $|\Delta \text{arcosh}(\omega^{-1/2})| \leq \frac{1}{2} \Delta \log \det \mathcal{D}$ by the complete projector derivative SFC6–7. Its three successive applications give SFC8; the monotone calculation and unique cubic root in SFC9–15 give the first interval. Adding the exact identity $\mathcal{P}_e^{\text{rest}} = \mathcal{L}_e + V_b$ proves the second. All four positive complementary returns in CPQ11 remain. The CIT33 interval and both order translations CIT34 can be intersected with CIT35 in their original domains. The initial angle is the displayed actual determinant, not an assigned value.

Finite transport of the evaluated common-angle enclosure. In CIT35 retain both original quotient isomorphisms $I/(I \cap J) \rightarrow (I + J)/J$ and $S/(S \cap W) \rightarrow (S + W)/W$, given by the identity on representatives. The defining intersections give their zero kernels and the defining sums their full images. Their ranks are r_I, c_e . Each pair of matched quotient determinants receives EFP14, so subtraction gives the same-endpoint width $r w_{j,D}$, with $w_{j,D} = \lambda_{j,D}^+ - \lambda_{j,D}^-$. Hence

$$|\mathcal{L}_e^1 - \mathcal{L}_e^0| \leq B_e = (r_I + c_e) \sum_{i=0}^3 w_{j,D_i}. \tag{EFR3}$$

The two scalar source factors cancel before taking the width. The common projection cross block and all four quotient metrics remain those of CIT35. With its exact SFC endpoints $L_{\pm}^p$ put

$$\begin{aligned} \tilde{L}_-^1 &= \max(L_-^1, L_-^0 - B_e), & \tilde{L}_+^1 &= \min(L_+^1, L_+^0 + B_e), \\ \tilde{L}_-^0 &= \max(L_-^0, L_-^1 - B_e), & \tilde{L}_+^0 &= \min(L_+^0, L_+^1 + B_e). \end{aligned} \tag{EFR4}$$

CIT35 and EFR3 prove $\tilde{L}_-^p \leq \mathcal{L}_e^p \leq \tilde{L}_+^p$. Adding the exact V_b^p proves $V_b^p + \tilde{L}_-^p \leq \mathcal{P}_e^{\text{rest},p} \leq V_b^p + \tilde{L}_+^p$. The original CIT33 interval remains another valid intersection. The target radius and EOR tests in EFP4–9 are applied endpoint by endpoint; an unguarded improvement is never inserted.

Joint boundary cancellation and the original compressed profile. Work on the unchanged simple-quartet domain of CIT1 and CPQ1–2, with the same $e \in \{0, 1\}$, $k = 4l + 1 \geq 9$, $j = k + 4$, conductor order v , full fixed coefficient maps and source pair $(s_0, s_1) = (1, 1)$ or (k, j) . Set

$$\begin{aligned} \mathcal{H} &= \mathbb{C}^{4m_j}, & I &= \text{im}(B_{\text{cof}} J_0), & B &= \text{im } B_{\text{cof}}, \\ S &= \text{im}[B_{\text{cof}}, F_e], & J &= \ker \mathbb{B}, & K_i &= I \cap J, & W_i &= I + J, \\ & & W &= B + J, & K_c &= S \cap W, & T &= S + J. \end{aligned} \tag{PTHCIT1}$$

Since $B \subset S$, the last space is also $S + W$. All these spaces are fixed as the native cutoff varies. Their metric is $\mathcal{D}_N = \text{diag}(D_{j,N-v}^{(s_1)}, 4)$, with all lower evaluations minimized in the original preceding Gram. Use exactly

$$\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j), \quad \mathcal{V}(Y/X) = \mathcal{R} \log \det \text{Gram}_{\mathcal{D}_N}(Y/X). \tag{PTHCIT2}$$

The quotient form is the attained minimum over the complete fibre X , in fixed quotient-value coordinates. The maps $I/K_i \rightarrow W_i/J$ and $S/K_c \rightarrow T/W$ send $[z]$ to the same representative class $[z]$. Their kernels are zero by the defining intersections, and the defining sums make them onto. Thus they are exactly the original invariant and connecting maps; in their respective common value coordinates their squared volume ratios are ω_{inv} and $\omega_{\text{conn},e}$. In particular, with $\mathcal{L}_e = \mathcal{R} \log(\omega_{\text{conn},e} \omega_{\text{inv}})$,

$$\mathcal{L}_e = \mathcal{V}(T/W) + \mathcal{V}(W_i/J) - \mathcal{V}(S/K_c) - \mathcal{V}(I/K_i). \quad (\text{PTHCIT3})$$

Here is the full coordinate calculation for the remaining determinant identity. For the chain $X_0 = J \subset X_1 = W_i \subset X_2 = W \subset X_3 = T \subset X_4 = \mathcal{H}$, choose fixed full frames A_h of X_h and fixed lifts L_h of the specified quotient coordinates. Write $[A_{h-1}, L_h] = A_h C_h$, where C_h is invertible. With $F_h = A_h^* \mathcal{D}_N A_h$ and quotient Gram G_h , full Schur elimination gives

$$\det G_h = |\det C_h|^2 \frac{\det F_h}{\det F_{h-1}}. \quad (\text{PTHCIT4})$$

For the original direct $\mathcal{H}/J$ quotient frame, likewise write $[A_0, L_*] = A_4 C_*$ and obtain $\det G_* = |\det C_*|^2 \det F_4 / \det F_0$. Multiplying the four displayed identities therefore gives $\prod_{h=1}^4 \det G_h = (\prod_{h=1}^4 |\det C_h|^2) |\det C_*|^{-2} \det G_*$. The entire fixed coefficient factor cancels under $\mathcal{R}1 = 0$. This proves the telescoping identity in the actual independent frames, without replacing a varying minimum lift by a fixed orthogonal representative.

The onto map $\mathbb{B}$ identifies $\mathcal{H}/J$ isometrically with its original target quotient: minimizing first over J gives four copies of $(B_j D_{j,N-v}^{-1} B_j^*)^{-1}$. Consequently the HJR/NSR inverse-metric identity is $\mathcal{V}(\mathcal{H}/J) = 4S_{K,j,s_1}$. Substituting (PTHCIT3) into the proved telescoping identity yields

$$\begin{aligned} 4S_{K,j,s_1} &= \mathcal{L}_e + \mathcal{V}(\mathcal{H}/T) + \mathcal{V}(W/W_i) + \mathcal{V}(S/K_c) + \mathcal{V}(I/K_i) \\ &= \mathcal{P}_e^{\text{rest}} + \mathcal{V}(\mathcal{H}/T) + \mathcal{V}(S/K_c) + \mathcal{V}(I/K_i). \end{aligned} \quad (\text{PTHCIT5})$$

For the second equality use the exact CIT boundary quotient $\mathcal{V}(W/W_i) = V_b$ and $\mathcal{L}_e = \mathcal{P}_e^{\text{rest}} - V_b$. Every fixed factor before taking the return is retained in CIT26 and CPQ12. This is their joint cancellation, at the original signs. The spaces and metrics of the three surviving returns still depend on the same boundary flag.

In the explicit CPQ coordinates these three returns are, respectively, V_{rem}, V_s, V_A . The identification of the first quotient is the map $[z] \mapsto [\pi_o z]$ from $\mathcal{H}/T$ to the quotient of the original remaining target by $\text{im } E_e$. The underlying map on $\mathcal{H}$ has kernel precisely T , since $\ker \pi_o = W$ and $\pi_o S = \text{im } E_e$; the displayed induced map on $\mathcal{H}/T$ therefore has zero kernel and is onto by CPQ4. Minimizing first over W and then over T/W proves its isometry with $Q_{\text{rem},e}$. The second is CPQ7's minimum on S/K_c , in the lifts $F_e Z_c$. For the third put $F = B_{\text{cof}} J_0$ and $E = \ker(\mathbb{B}F)$. Then $FE = I \cap J = K_i$, and the minimum over FE in the lifts FZ_R is precisely A_{inv} . These equalities prove all three typed metric identifications; the separate degree- k source form Q_{inv} retains its own relative comparison.

Finally the eight exact matrices and weighted estimate (PTHS1)–(PTHS6) below give, in this same source order,

$$|4S_{K,j,s_1} - (P_{k,e} + V_b + V_{\text{rem}} + V_s + V_A)| \leq 32q_j \log 2. \quad (\text{PTHCIT6})$$

Indeed CPQ20 or the first line of (PTHCIT5) says that the expression inside the absolute value is exactly $\mathcal{L}_e - P_{k,e}$. Thus the finite reduction preserves all four positive complementary returns. It assigns no absolute allocation coefficient or value to their correlated sum.

Thus the surviving connecting target is given by the explicit matrix CIT7, its complete target minimum CIT11–13, the original low source-to-image determinant CIT17, and all later original row formulas CIT18–20. The fixed-rank remainder is controlled by CIT23. The remaining common target loss has the exact original ITL boundary relation CIT24–26, with its whole residual overlap, all absolute native Grams and both source measures still present.

F The original finite Gamma translation on every small-rank plane

Fix the original simple quartet, its actual five-orbit period and logarithm branch. Keep every coefficient of the original nonreduced conductor:

$$\begin{aligned}
0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \quad \beta_* = \tfrac{1}{2} - \delta - i\gamma, \quad \beta_{n,a,b} = n\beta_* + 2a\delta + 2bi\gamma, \\
E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad \mu_m = E_A^{(m)}(0), \quad v = \text{ord}_0 E_A, \quad 0 \leq v \leq 80, \quad \mu_v \neq 0, \quad (\text{SGT1}) \\
k = 4l + 1 \geq 9, \quad s \in \{1, k\}, \quad c_0 = k/2, \quad c_1 = k/2 - 4.
\end{aligned}$$

The coefficients, symbol, order and first nonzero moment are fixed as k varies. No particular coefficient is assumed nonzero except through the stated exact order. Every result below is for this original operator and these two original source orders. An arbitrary integer $N \geq 0$ is allowed in the finite statements.

F.1 Full Gamma measure, recurrence and coefficient isometry

Write $b = s/2 > 0$ and retain

$$\begin{aligned}
M_s &= (2\pi)^{s/2}, \quad dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \\
\|P\|_{s,c}^2 &= \int_{\mathbb{R}} |P(c + iy)|^2 dm_s(y), \quad (\text{SGT2}) \\
p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+b)p_{n-1,s}.
\end{aligned}$$

These are exactly the measure and recurrence in PDS6 and DNE7. For completeness their generating and norm identities are derived here. The recurrence, multiplied by $t^n/n!$ and summed, gives $(1+t^2)\partial_t F = (y-bt)F$, $F(y,0) = 1$. The unique formal solution, also holomorphic for $|t| < 1$ at fixed y , is

$$F(y,t) = \sum_{n \geq 0} p_{n,s}(y) \frac{t^n}{n!} = (1+t^2)^{-b/2} \exp(y \arctan t). \quad (\text{SGT3})$$

Both branches take their value zero or one at $t = 0$ as appropriate.

Here is a direct check of the full mass and norms. With $a = s/4$, the Euler gamma integrals and the change of variables $r = u + v_0$, $x = \log(u/v_0)$ in their product give

$$|\Gamma(a + i\xi)|^2 = \Gamma(2a) \int_{\mathbb{R}} e^{i\xi x} (2 \cosh(x/2))^{-2a} dx.$$

The integrand as a function of x is analytic for $|\Im x| < \pi$. Shifting the integration line by any fixed amount of modulus less than π is justified by its exponential decay at both real ends. It bounds this Fourier transform by a constant times $e^{-h|\xi|}$ for each $0 < h < \pi$. Fourier inversion is therefore absolutely convergent; it may also be obtained by inserting a Gaussian factor and passing to the limit by dominated convergence. Substituting $y = 2\xi$ and the exact multiplier in SGT2 gives

$$\int_{\mathbb{R}} e^{iuy} dm_s(y) = M_s (\cosh u)^{-b}, \quad \int_{\mathbb{R}} e^{uy} dm_s(y) = M_s (\cos u)^{-b} \quad (|u| < \pi/2). \quad (\text{SGT4})$$

For the second equality, the preceding exponential bounds give analytic continuation to the indicated strip; equality on the real Fourier axis then proves it there. In particular the mass is exactly M_s . For sufficiently small real t, z , SGT3-4 and $\cos(\arctan t + \arctan z) = (1 - tz)/\sqrt{(1+t^2)(1+z^2)}$ imply

$$\int_{\mathbb{R}} F(y,t) F(y,z) dm_s(y) = M_s (1 - tz)^{-b}.$$

The exponential moment bound justifies differentiating under the integral in a neighborhood of $(0, 0)$. Comparing coefficients proves

$$\int_{\mathbb{R}} p_{n,s}(y) p_{m,s}(y) dm_s(y) = \mathbf{1}_{n=m} h_{n,s}, \quad h_{n,s} = M_s n! (b)_n, \quad (SGT5)$$

$$\phi_{n,s,c}(S) = \frac{p_{n,s}((S-c)/i)}{\sqrt{h_{n,s}}}, \quad \rho_n = \sqrt{\frac{n!}{(b)_n}} > 0.$$

The $p_{n,s}$ are real monic polynomials by their recurrence. Thus these ϕ_n are orthonormal for the complex Hilbert norm in SGT2 and span each indicated polynomial degree range. Their leading coefficient in S is $i^{-n}/\sqrt{h_{n,s}}$, not $1/\sqrt{h_{n,s}}$.

F.2 The original quotient map and its exact Toeplitz matrix

Let $\mathcal{P}_m = \mathbb{C}[S]_{\leq m}$, with $\mathcal{P}_{-1} = 0$. The original translation is

$$(\mathcal{T}_A P)(S') = \sum_{a,b=0}^8 a_{ab} P(S' + \beta_{8,a,b}), \quad \mathcal{T}_A S^m = \sum_{j=0}^m \binom{m}{j} \mu_j S'^{m-j}. \quad (SGT6)$$

The second equality is the binomial theorem followed by the exact definition of every moment. If $m \geq v$ and P has leading coefficient $p_m \neq 0$, then $\mathcal{T}_A P$ has degree $m - v$ and leading coefficient $p_m \binom{m}{v} \mu_v \neq 0$. If $m < v$ its image is zero. Consequently the following map is an isomorphism, with an onto representative map and the displayed full kernel:

$$\ker(\mathcal{T}_A : \mathcal{P}_{N+v} \rightarrow \mathcal{P}_N) = \mathcal{P}_{v-1},$$

$$A_{N,s} : (\mathcal{P}_{N+v}/\mathcal{P}_{v-1}, \|\cdot\|_{s,c_0,\text{quot}}) \xrightarrow{\simeq} (\mathbb{C}[S']_{\leq N}, \|\cdot\|_{s,c_1}), \quad (SGT7)$$

$$\| [P] \|_{s,c_0,\text{quot}}^2 = \min_{L \in \mathcal{P}_{v-1}} \| P + L \|_{s,c_0}^2.$$

Surjectivity follows successively by degree from the nonzero diagonal coefficients in SGT6. The orthogonal representative of this quotient is precisely the span of $\phi_{v,s,c_0}, \dots, \phi_{N+v,s,c_0}$. It is not a quotient by an omitted arithmetic relation.

On the entire open unit disk put

$$w(t) = -i \arctan t, \quad g(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad g(0) = (-i)^v \mu_v / v! \neq 0. \quad (SGT8)$$

The value at zero is removable by the exact order v . For $|t| < 1$, $|w(t)| \leq \arctanh |t|$ by the arctangent series. This proves holomorphy of the specified g . Applying the finite sum SGT6 to SGT3, with its polynomial argument, gives exactly

$$\sum_{m \geq 0} (\mathcal{T}_A p_{m,s}((S - c_0)/i))(S') \frac{t^m}{m!} = (1 + t^2)^{-b/2} e^{(S' - c_0)w(t)} E_A(w(t)) \quad (SGT9)$$

$$= t^v g(t) (1 + t^2)^{-b/2} e^{(S' - c_1)w(t)}.$$

The center factor is e^{-4w} because $c_0 - c_1 = 4$. No inverse of $\mathcal{T}_A$, or scalar α_A , has been inserted in this identity.

For a holomorphic germ $u(t) = \sum_{j \geq 0} u_j t^j$ define $T_N(u)_{r,n} = u_{n-r}$ for $0 \leq r \leq n \leq N$, and zero for $r > n$. Coefficients of SGT9 and $n!/\sqrt{h_{n,s}} = \rho_n/\sqrt{M_s}$ give the complete matrix of SGT7 in its two orthonormal frames:

$$A_{N,s} = D_{\text{out}}^{-1} T_N(g) D_{\text{in}},$$

$$D_{\text{out}} = \text{diag}(\rho_0, \dots, \rho_N), \quad D_{\text{in}} = \text{diag}(\rho_v, \dots, \rho_{N+v}), \quad (SGT10)$$

$$(A_{N,s})_{r,n} = \begin{cases} g_{n-r} \rho_{n+v} / \rho_r, & r \leq n, \\ 0, & r > n. \end{cases}$$

Each source and target frame contains $1/\sqrt{M_s}$. Those identical factors cancel only in this coefficient matrix, since s is unchanged. All actual squared norms still use the full SGT2 measure. In particular, for $N = 0$ the matrix is the literal scalar $(-i)^v \mu_v \rho_v / v!$.

F.3 A proved original-data zero-free disk

The following constants use every original coefficient and shift:

$$\begin{aligned} A_{\text{abs}} &= \sum_{a,b} |a_{ab}|, & B &= \max\{1, \max_{a,b} |\beta_{8,a,b}|\}, & C_v &= \frac{A_{\text{abs}} B^v}{|\mu_v|} \geq 1, \\ R_0 &= \frac{1}{B} \min\{1, (v+1)/(2 \exp(1) C_v)\}, & a_0 &= \min\{1/2, \tanh(R_0/2)\}, & & \text{(SGT11)} \\ d_0 &= 1 - \frac{a_0^2}{3(1-a_0^2)} \geq 8/9, & M_0 &= \frac{2v!}{|\mu_v|} d_0^{-v} e^{2R_0}. \end{aligned}$$

All are finite, R_0, a_0, d_0, M_0 are positive, and $a_0 < 1$. Indeed $|\mu_v| \leq A_{\text{abs}} B^v$ proves the stated bound on C_v . Define the entire removable quotient $B_\infty(z) = v! E_A(z)/(\mu_v z^v)$, with value one at zero. Its full Taylor remainder is bounded by

$$|B_\infty(z) - 1| \leq \frac{C_v B |z|}{v+1} e^{B|z|} \leq \frac{1}{2} \quad (|z| \leq R_0). \quad \text{(SGT12)}$$

For the first inequality expand each exponential, use $|\mu_m| \leq A_{\text{abs}} B^m$ and $v!/(v+1+j)! \leq 1/((v+1)j!)$, and sum the exponential series. The two inequalities $BR_0 \leq 1$ and $C_v BR_0/(v+1) \leq 1/(2 \exp(1))$ prove the second. This is the complete RDS11 disk proof for the unchanged symbol.

On $|t| \leq a_0$ we have $|w(t)| \leq R_0/2$ and, also at zero,

$$\left| \frac{\arctan t}{t} - 1 \right| \leq \sum_{j \geq 1} \frac{a_0^{2j}}{2j+1} \leq \frac{a_0^2}{3(1-a_0^2)}.$$

Thus $|w(t)/t| \geq d_0$. Factoring SGT8 as $g(t) = (\mu_v/v!)(w(t)/t)^v e^{-4w(t)} B_\infty(w(t))$ shows that it has no zero on that entire closed disk and proves

$$|g(t)| \geq \frac{|\mu_v|}{2v!} d_0^v e^{-2R_0}, \quad |1/g(t)| \leq M_0 \quad (|t| \leq a_0). \quad \text{(SGT13)}$$

This supplies the zero-free disk and the inverse bound from the actual finite symbol rather than assuming them.

F.4 Every actual pole and its finite-rank Toeplitz contribution

Call $0 < a < 1$ admissible when g has no zero on $|t| = a$. Zeros of a nonzero holomorphic function are isolated; therefore there are only finitely many in each compact subdisk. There are admissible radii arbitrarily close to one. For a fixed such a , let $\mathcal{Z}_a$ be the exact zeros ζ of g with $|\zeta| < a$, and let m_ζ be their orders. Zero is not among them. Retain every multiplicity and define the actual Laurent coefficients

$$\begin{aligned} H_a &= \sum_{\zeta \in \mathcal{Z}_a} m_\zeta, & A_{\zeta,h} &= \frac{1}{(m_\zeta - h)!} \frac{d^{m_\zeta - h}}{dt^{m_\zeta - h}} \left(\frac{(t - \zeta)^{m_\zeta}}{g(t)} \right) \Big|_{t=\zeta}, & 1 \leq h \leq m_\zeta, \\ f &= 1/g, & f_a(t) &= f(t) - \sum_{\zeta \in \mathcal{Z}_a} \sum_{h=1}^{m_\zeta} \frac{A_{\zeta,h}}{(t - \zeta)^h}. \end{aligned} \quad \text{(SGT14)}$$

The factors t^{-v} and e^{-4w} in g create no nonzero zero. Moreover $w'(t) = -i/(1+t^2) \neq 0$ and $e^{-2w(t)} = (1+it)/(1-it)$ proves injectivity of w on the disk. Thus these poles are precisely the symbol zeros already identified in PEC3–5, under this exact change of variable, with their same orders. The extra center factor is fully retained in every $A_{\zeta,h}$. The function f_a is holomorphic on a neighborhood of the closed a disk, since all its interior principal parts have been subtracted and no boundary zero exists.

Here are completely specified finite constants on that circle:

$$\begin{aligned}
W_a &= \operatorname{arctanh} a, \quad m_{E,a} = \min_{|t|=a} |E_A(w(t))| > 0, \quad G_a = A_{\text{abs}} a^{-v} e^{(B+4)W_a}, \\
M_a &= \frac{a^v e^{4W_a}}{m_{E,a}} + \sum_{\zeta, h} \frac{|A_{\zeta, h}|}{(a - |\zeta|)^h}, \quad P_a = \sum_{\zeta, h} \frac{|A_{\zeta, h}|}{(1 - |\zeta|)^h}, \\
S_N(a) &= \sum_{j=0}^N a^{-j} = a^{-N} \frac{1 - a^{N+1}}{1 - a}.
\end{aligned} \tag{SGT15}$$

Empty pole sums are zero. The minimum is the same actual finite-circle quantity used in PEC4, and is positive by continuity and admissibility. It is not a new uniform zero-free assertion. The triangle inequality on that circle gives $|g| \leq G_a$ and $|f_a| \leq M_a$. Every constant in SGT15 is defined by the original fixed function and its exact finitely many interior poles; none depends on k, N or s .

For one term $A/(t - \zeta)^h$, its Taylor coefficients at zero are

$$c_n = A(-1)^h \zeta^{-(n+h)} \binom{n+h-1}{h-1} \quad (n \geq 0). \tag{SGT16}$$

Extend the binomial as the polynomial $\binom{x+h-1}{h-1} = \prod_{\ell=1}^{h-1} (x + \ell)/(h-1)!$ to every integer x , also for $h = 1$ by the empty product. The full matrix $C_{i,j} = c_{j-i}$, $0 \leq i, j \leq N$, has rank at most h : after multiplying rows by ζ^{-i} and columns by ζ^j , its entries are a polynomial of total degree at most $h-1$ in $j-i$. Expanding that polynomial in powers of i gives at most h column vectors, independently of N . Equivalently its column space is contained in $\operatorname{span}\{(\zeta^i i^d)_i : 0 \leq d < h\}$. For all terms belonging to one pole, these nested spaces give combined rank at most m_ζ , rather than the sum of the integers h .

For a lower-diagonal gap $\ell = i - j \geq 1$, direct substitution in the polynomial binomial gives

$$c_{-\ell} = \begin{cases} 0, & 1 \leq \ell < h, \\ -A\zeta^{\ell-h} \binom{\ell-1}{h-1}, & \ell \geq h. \end{cases} \tag{SGT17}$$

For $\ell \geq h$ each factor in the binomial is negative, and their $(-1)^{h-1}$ sign combines with $(-1)^h$ to give the displayed minus sign. For $1 \leq \ell < h$ one factor is zero. Consequently the upper triangular Taylor matrix of this pole is $C + R$, where R is strictly lower triangular and its gap- ℓ coefficient is $A\zeta^{\ell-h} \binom{\ell-1}{h-1}$ for $\ell \geq h$. If J is the upper shift with $J_{i,i+1} = 1$, then $(J^*)^\ell$ has norm at most one. The absolutely convergent binomial series therefore proves

$$\|R\| \leq |A| \sum_{\ell=h}^{\infty} |\zeta|^{\ell-h} \binom{\ell-1}{h-1} = \frac{|A|}{(1 - |\zeta|)^h}. \tag{SGT18}$$

The series identity follows by differentiating the geometric series $h-1$ times, or multiplying h copies of it and collecting coefficients. All powers and signs are those of the original pole; no principal part has been deleted from the actual Toeplitz inverse.

Since $T_N(u) = \sum_{j=0}^N u_j J^j$, multiplication of these finite matrices is truncated power-series multiplication. In particular $T_N(g)^{-1} = T_N(f)$, because $g(0) \neq 0$ and $gf = 1$ as germs. Cauchy's coefficient estimate, SGT13 and SGT15–18 prove

$$\begin{aligned}
\|T_N(g)\| &\leq F_{a,N} := G_a S_N(a), \\
\|T_N(f)\| &\leq G_{0,N} := M_0 S_N(a_0), \\
T_N(f) &= C_{a,N} + R_{a,N}, \quad \operatorname{rank} C_{a,N} \leq H_a, \quad \|R_{a,N}\| \leq V_{a,N} := M_a S_N(a) + P_a.
\end{aligned} \tag{SGT19}$$

Indeed sum the coefficient estimates times $\|J^j\| \leq 1$ for the analytic terms, and sum the exact pole ranks and lower-triangular remainders just proved. The total inverse matrix is unchanged.

F.5 Finite exterior bounds, including the expensive directions

Write $d = N + 1$ and arrange singular values in decreasing order. If a matrix C has rank at most H , then $\ker C$ has codimension at most H . For $x \in \ker C$, $\|(C + R)x\| \leq \|R\|\|x\|$. The variational description of singular values therefore gives $\sigma_j(C + R) \leq \|R\|$ for $j > H$. To verify this description directly, diagonalize $(C + R)^*(C + R)$: the complement of its first $j - 1$ eigenvectors attains the upper bound, and every subspace of codimension $j - 1$ intersects the first j eigenvectors for the reverse inequality. Applying this argument to SGT19 and retaining the separate global bound gives

$$h = \min\{r, H_a\}, \quad 0 \leq r \leq d, \quad V_{a,N}^{\text{cap}} = \min\{G_{0,N}, V_{a,N}\}, \quad (\text{SGT20})$$

$$\left\| \bigwedge^r T_N(g) \right\| \leq F_{a,N}^r, \quad \left\| \bigwedge^r T_N(f) \right\| \leq G_{0,N}^h (V_{a,N}^{\text{cap}})^{r-h}.$$

The norm of the r th exterior power is the product of the r largest singular values: this follows by applying a singular-value decomposition to the orthonormal wedge bases. Only at most H_a directions use the zero-free-radius bound. If $H_a > d$, then $h = r$ and there is no tail factor. Every constant is positive; the empty exterior power has norm one.

Retain the exact diagonal constants from the original norms:

$$\alpha_+ = \left(\max_{0 \leq n \leq N} \rho_{n+v} \right) \left(\max_{0 \leq n \leq N} \rho_n^{-1} \right), \quad \alpha_- = \left(\max_{0 \leq n \leq N} \rho_{n+v}^{-1} \right) \left(\max_{0 \leq n \leq N} \rho_n \right), \quad (\text{SGT21})$$

$$L_{s,N,v} = \frac{1}{2} \max\{1, s/2 - 1\} \mathfrak{H}_{N+v}, \quad \mathfrak{H}_m = \sum_{j=1}^m 1/j, \quad \mathfrak{H}_0 = 0.$$

For $s = k \geq 9$, the product defining ρ_n and $\log(1+x) \leq x$ imply $|\log \rho_n| \leq (b-1)\mathfrak{H}_n/2$. For $s = 1$, its logarithm is positive and $\log((j+1)/(j+1/2)) \leq 1/(2j+1) \leq 1/(j+1)$ gives $|\log \rho_n| \leq \mathfrak{H}_n/2$. Thus $|\log \rho_n| \leq L_{s,N,v}$ for all $0 \leq n \leq N + v$, including $n = 0$. In particular $|\log \alpha_{\pm}| \leq 2L_{s,N,v}$. Applying the two diagonal maps in SGT10 on exterior powers gives

$$\left\| \bigwedge^r A_{N,s} \right\| \leq \alpha_+^r F_{a,N}^r, \quad (\text{SGT22})$$

$$\left\| \bigwedge^r A_{N,s}^{-1} \right\| \leq \alpha_-^r G_{0,N}^h (V_{a,N}^{\text{cap}})^{r-h}.$$

This retains the rectangular degree shift $n \mapsto n + v$ exactly.

Let $[P_1], \dots, [P_r]$ be any independent original classes in SGT7 and use that same ordered frame in both Grams:

$$H_{\text{in}} = (\langle [P_i], [P_j] \rangle_{s, c_0, \text{quot}})_{i,j}, \quad H_{\text{out}} = (\langle \mathcal{T}_A P_i, \mathcal{T}_A P_j \rangle_{s, c_1})_{i,j}, \quad \Delta_{N,s}(\mathbf{P}) = \log \frac{\det H_{\text{out}}}{\det H_{\text{in}}}. \quad (\text{SGT23})$$

The inner products are conjugate-linear in their first variable. Both Grams are positive, by independence and SGT7. Exterior volumes and their inverse give the finite, potentially asymmetric bounds

$$-2r \log \alpha_- - 2h \log G_{0,N} - 2(r-h) \log V_{a,N}^{\text{cap}} \leq \Delta_{N,s}(\mathbf{P}) \leq 2r \log \alpha_+ + 2r \log F_{a,N}. \quad (\text{SGT24})$$

For proof, express the classes in their SGT7 orthonormal quotient frame as columns X . Their squared wedge norm is $\det(X^*X)$, and the squared norm of their image wedge is $\det(X^* A_{N,s}^* A_{N,s} X)$. Apply SGT22 to that wedge and then to its image with $A_{N,s}^{-1}$. A change of the original ordered frame by a matrix U multiplies both determinants by $|\det U|^2$; the ratio therefore retains the original frame and is frame invariant. No specially conditioned plane has been selected.

For a nonnegative enclosure write $\log^+ x = \max\{0, \log x\}$ and set

$$\begin{aligned}\mathcal{U}_{a,N,s,r} &= 4rL_{s,N,v} + 2r \log^+ F_{a,N}, \\ \mathcal{L}_{a,N,s,r} &= 4rL_{s,N,v} + 2h \log^+ G_{0,N} + 2(r-h) \log^+ V_{a,N}^{\text{cap}}.\end{aligned}\tag{SGT25}$$

Then $-\mathcal{L} \leq \Delta_{N,s}(\mathbf{P}) \leq \mathcal{U}$. These are finite explicit constants even for $r = 0$, with value zero for that empty Gram ratio. All maxima, minima, poles and residues in the constants refer to the fixed original symbol SGT1.

F.6 Uniform small-rank volume distortion and the actual PDS frame

Fix finite constants $C_N, C_r > 0$. Uniformly over $0 \leq N \leq C_N k^2$, $0 \leq r \leq \min\{C_r k, N+1\}$, $s = 1$ or k , and every original rank- r plane in SGT7,

$$\sup \frac{|\Delta_{N,s}(\mathbf{P})|}{k^3} \longrightarrow 0 \quad (k = 4l+1 \longrightarrow \infty).\tag{SGT26}$$

Here the supremum is over precisely those finite-degree and rank domains, including all coefficient choices for the planes. We give both limiting estimates. For each fixed admissible a , $S_N(a) \leq (N+1)a^{-N}$ and $a^{-N} \geq 1$, so

$$\begin{aligned}\log^+ F_{a,N} &\leq N \log(1/a) + \log(N+1) + \log^+ G_a, \\ \log^+ V_{a,N}^{\text{cap}} &\leq \log^+ V_{a,N} \leq N \log(1/a) + \log(N+1) + \log^+(M_a + P_a), \\ \log^+ G_{0,N} &\leq N \log(1/a_0) + \log(N+1) + \log^+ M_0.\end{aligned}\tag{SGT27}$$

Also $\mathfrak{H}_m \leq 1 + \log(m+1)$, by integral comparison. The common diagonal error $4rL$ is therefore $O_{C_N, C_r, v}(k^2 \log(k+1))$. The global inverse term has $h \leq H_a$, which is fixed before the k limit, so it is $O_{\text{actual}, a, C_N}(k^2 + \log(k+1))$. It follows from SGT25 and SGT27 that

$$\limsup_{k \rightarrow \infty} \sup \frac{|\Delta_{N,s}(\mathbf{P})|}{k^3} \leq 2C_r C_N \log(1/a).\tag{SGT28}$$

The constants in this step may depend on this fixed a . Only after this limit do we take admissible radii arbitrarily close to one, as proved before SGT14. The right side then tends to zero. This proves SGT26 without assuming a zero-free unit disk or a uniform pole count as a tends to one.

For the original PDS quotient set

$$\begin{aligned}q_k &= (k+1)^2, \quad Q = (k+5)^2, \quad e \in \{0, 1\}, \\ N_e &= Q - v - 1 + e, \quad \ell_e = 8k + 23 + e, \quad d_e = \ell_e + 1 = 8k + 24 + e, \\ P_a(S) &= \chi_k(S) S^a, \quad r_a(S') = \mathcal{T}_A P_a(S'), \quad 0 \leq a \leq \ell_e, \\ \chi_k(S) &= \prod_{u,w=0}^k (S - \beta_{k,u,w}).\end{aligned}\tag{SGT29}$$

Then $q_k + \ell_e = N_e + v = Q - 1 + e$. A nonzero linear combination of the P_a has degree at least $q_k > v - 1$ because it is χ_k times a nonzero polynomial. Thus their classes modulo $\mathcal{P}_{v-1}$ are independent, of exactly the original rank d_e , without a minor or genericity assumption.

Let X_e be their full column matrix in the original c_0 orthonormal frame of degrees $0, \dots, N_e + v$, and write $P_{<v}$ and $P_{\geq v}$ for the row selections in that frame. The exact source and image Grams are

$$\begin{aligned}Y_e &= P_{\geq v} X_e, \quad H_{\text{src},e} = Y_e^* Y_e = X_e^* X_e - (P_{<v} X_e)^* (P_{<v} X_e) > 0, \\ C_e &= A_{N_e, s} Y_e, \quad H_e = C_e^* C_e, \quad -\mathcal{L}_{a, N_e, s, d_e} \leq \log \frac{\det H_e}{\det H_{\text{src},e}} \leq \mathcal{U}_{a, N_e, s, d_e}.\end{aligned}\tag{SGT30}$$

The first equality is the attained full quotient minimum over the entire original $\mathcal{P}_{v-1}$. The next equality is the coefficient map SGT10, since those removed low columns are killed by $\mathcal{T}_A$. The resulting C_e, H_e are exactly PDS7, including the target center c_1 and both original orders. For $k \geq 9$, $Q \leq (14k/9)^2 < 3k^2$, so $N_e \leq 3k^2$ and $d_e \leq 11k$. Hence SGT26 applies directly:

$$\log \frac{\det H_e}{\det H_{\text{src},e}} = o(k^3) = o(kq_k) \quad (e = 0, 1, \quad s = 1, k), \quad (\text{SGT31})$$

uniformly over those four original source/endpoint choices. The full original mass has not been removed from either Gram.

The precise dual statement also retains all PDS conjugations: for the complex-linear functional $x^\top u$ on a Gram H , its squared dual norm is $x^* \overline{H^{-1}} x$. Thus the same coefficient coordinates carry

$$G_e = \overline{H_e^{-1}}, \quad G_{\text{src},e} = \overline{H_{\text{src},e}^{-1}}, \quad \log \frac{\det G_e}{\det G_{\text{src},e}} = -\log \frac{\det H_e}{\det H_{\text{src},e}}. \quad (\text{SGT32})$$

This is an equality of the original full quotient-source dual metrics, not a statement about a later connecting-target Gram.

For the endpoint $e = 1$ and the original four PDS8 rows, retain the same full rank-four matrix W and choose any basis Z of $\ker W^\top$. The polynomial quotient by those four columns, in the dual coordinates $a = Z^\top f$, has metric $(Z^\top H^{-1} \bar{Z})^{-1}$ by minimizing $f^* H f$. Consequently it is the conjugate inverse of the restriction $Z^* \overline{H^{-1}} Z$, exactly as in PDS9. For the paired forms $H_e, H_{\text{src},e}$, fixed complementary frames or the same attained Schur decomposition therefore give

$$\begin{aligned} \log \frac{\det(Z^* G_e Z)}{\det(Z^* G_{\text{src},e} Z)} &= -\Delta_{N_e,s}(\mathbf{P}) + \log \frac{\det(W^* H_e W)}{\det(W^* H_{\text{src},e} W)}, \\ -\mathcal{U}_{a,N_e,s,d_e} - \mathcal{L}_{a,N_e,s,4} &\leq \log \frac{\det(Z^* G_e Z)}{\det(Z^* G_{\text{src},e} Z)} \leq \mathcal{L}_{a,N_e,s,d_e} + \mathcal{U}_{a,N_e,s,4}. \end{aligned} \quad (\text{SGT33})$$

Indeed $\ker Z^\top = \text{im } W$. If J is any fixed right inverse of $Z^\top$, then $[W, J]$ is invertible, and block Schur factorization gives $\det((Z^\top H^{-1} \bar{Z})^{-1}) = |\det[W, J]|^2 \det H / \det(W^* H W)$. Taking the ratio cancels this exact frame determinant and proves the first equality. The W columns correspond to the four independent original polynomials $\chi_k R_i$ modulo $\mathcal{P}_{v-1}$; SGT24–25 applied to that actual rank-four plane prove the interval. At $e = 1$ this is the original primitive source, and its determinant distortion is again $o(kq_k)$. At $e = 0$ the primitive condition is empty and SGT32 already gives its full statement.

The earlier PEC1–18 theorem uses actual numerator-jet cancellation at symbol zeros to control a positive dual spectrum. Here SGT7–10 first proves the polynomial source/target morphism, and SGT16–22 retains all principal parts in an exact finite-rank Toeplitz summand, giving both directions on every small-rank plane. Neither statement replaces the original lower quotient, the four primitive rows, or the connecting-target metric. The original-data zero-free disk is the RDS11 disk with its full proof repeated in SGT11–13.

G The connecting image inside the full remaining target quotient

This calculation uses the complete original CIT, RQB and NSR objects. The fixed hypothetical simple quartet, five-orbit period, logarithm branch, conductor E_A , order $v = \text{ord}_0 E_A$, nonzero moment μ_v and scalar $\alpha_A = v!/\mu_v$ remain those of the original construction. Put $k = 4l + 1 \geq 9$, $j = k + 4$, $q_a = (a + 1)^2$, $q'_a = (a - 7)^2$, $m_a = 16a - 48 - v$, $t_a = 8a - 16 - t_{0,a}$ and $j_{0,a} = 8a - 32 - v + t_{0,a}$. Use the original rank-valid domain, including $|u| \geq R_*$. The original source pairs are $p = 0 : (s_0, s_1) = (1, 1)$ and $p = 1 : (s_0, s_1) = (k, j)$. Every form below has its actual pair p ; we suppress this index only in identities valid separately for both.

At each original target cutoff $q_j - 1 \leq N \leq 2q_j$, write $D = N - v$. The native coefficient form is

$$\mathcal{D}_N = \text{diag}(D_{j,D}^{(s_1)}, 4), \quad \mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j). \quad (\text{CPQ1})$$

Here $D_{j,D}$ is the full analytic-numerator form after minimizing over all original lower evaluations, as in DNE and CIT19. Its rows use $M_s = (2\pi)^{s/2}$ and $h_{n,s} = M_s n! (s/2)_n$, the original centres, division moments and phases. Nothing in this calculation changes those rows or replaces their lower Gram. Fixed low-cycle coefficient flags are used at all these later cutoffs.

G.1 All coefficient maps and their kernels

Retain

$$\begin{aligned} \mathcal{U} &= \mathbb{C}^{4m_j}, \quad \mathcal{V} = \mathbb{C}^{m_k}, \quad B = B_{\text{cof}} : \mathcal{V} \longrightarrow \mathcal{U}, \\ \mathbb{B} &= \text{diag}(B_j, 4) : \mathcal{U} \longrightarrow \mathbb{C}^{4t_j}, \quad \mathbb{J} = \ker \mathbb{B}, \quad \Phi = \mathbb{B}B, \\ \mathcal{B} &= \text{im } B, \quad \mathcal{S}_e = \text{im}[B, F_e], \quad W = \mathbb{J} + \mathcal{B}, \quad K_e = \mathcal{S}_e \cap W, \quad T_e = \mathcal{S}_e + W. \end{aligned} \quad (\text{CPQ2})$$

B is the original injective coefficient multiplier and B_j is onto with its full original invariant kernel. The columns F_e , for $e = 0, 1$, are the actual CIT7 connecting columns, determined at $N = q_j - 1 + e$ by the complete Taylor equation with independent target lower corrections. They give an isomorphism $\mathbb{C}^{r_e} \rightarrow \mathcal{S}_e/\mathcal{B}$, $x \mapsto [F_e x]$, with $r_e = 8k + 24 - 3e$. Their independence and equality in the two source orders are proved in CIT9 and NSR5.

Use the original full target frame

$$C_T = [U_R, U_b, U_o], \quad U_R = \Phi J_0 Z_R, \quad U_b = \Phi E_b, \quad \text{im}[U_R, U_b] = \text{im } \Phi. \quad (\text{CPQ3})$$

It is the same invertible RQB27 frame. Its block dimensions are $r_I = j_{0,k} - \epsilon$, $r_b = t_k - \ell + \epsilon$ and $r_o = 4t_j - m_k + \ell$, where $\ell = \dim \ker \Phi$ and $\epsilon = \dim \ker(\Phi J_0)$. Define the actual quotient-coordinate map and its connecting matrix by

$$\pi_o = [0 \ 0 \ I_{r_o}] C_T^{-1} \mathbb{B}, \quad E_e = \pi_o F_e : \mathbb{C}^{r_e} \longrightarrow \mathbb{C}^{r_o}. \quad (\text{CPQ4})$$

The map π_o is onto and $\ker \pi_o = W$. Indeed its value is zero exactly when $\mathbb{B}z \in \text{im } \Phi$, equivalently $\mathbb{B}(z - Bx) = 0$ for some $x \in \mathcal{V}$. Its onto property uses the onto map $\mathbb{B}$ and the full last block of C_T . Consequently the maps

$$\mathcal{U}/W \longrightarrow \mathbb{C}^{r_o}, \quad [z] \mapsto \pi_o z, \quad \mathcal{S}_e/K_e \longrightarrow \text{im } E_e, \quad [z] \mapsto \pi_o z \quad (\text{CPQ5})$$

are isomorphisms; the latter identifies T_e/W with the same image. The exact sequence in the original connecting coordinates is

$$0 \longrightarrow \ker E_e \longrightarrow \mathbb{C}^{r_e} \xrightarrow{E_e} \text{im } E_e \longrightarrow 0, \quad \ker E_e \simeq K_e/\mathcal{B}, \quad c_e = \text{rank } E_e, \quad h_e = r_e - c_e. \quad (\text{CPQ6})$$

For its asserted kernel isomorphism, $F_e x \in W$ means that $F_e x - Bb \in \mathbb{J}$ for some b . Its class lies in $K_e/\mathcal{B}$, and every such class has a unique original connecting coordinate x . This proves both injectivity and onto without removing any conductor or invariant kernel. In particular $c_e = \text{rank}[\Phi, \mathbb{B}F_e] - \text{rank } \Phi$, and $0 \leq c_e \leq \min(r_e, r_o)$.

Choose fixed independent columns Z_h spanning $\ker E_e$ and fixed complement columns Z_c such that $C_P = [Z_h, Z_c]$ is invertible. Set $A_e = E_e Z_c$ and choose R_e^o so that $C_o = [A_e, R_e^o]$ is an invertible r_o -square matrix. These choices concern only the actual fixed coefficient maps. They are made once and used at every N and both orders. Their determinants are retained below. Empty blocks have their ordinary zero-dimensional meanings and determinant one.

G.2 The two original source and target minima

Let $H_e(N) = H_{\text{conn},e}(N)$ be CIT18's full connecting Gram on $\mathcal{S}_e/\mathcal{B}$ in F_e coordinates. It is the minimum of $(F_e x + Bb)^* \mathcal{D}_N(F_e x + Bb)$ over all b . The source kernel restriction and source quotient are exactly

$$\begin{aligned} G_{h,e} &= Z_h^* H_e Z_h, \\ G_{s,e} &= Z_c^* H_e Z_c - Z_c^* H_e Z_h (Z_h^* H_e Z_h)^{-1} Z_h^* H_e Z_c, \\ x^* G_{s,e} x &= \min_{z,b} (F_e Z_c x + F_e Z_h z + Bb)^* \mathcal{D}_N(F_e Z_c x + F_e Z_h z + Bb), \\ |\det C_P|^2 \det H_e &= \det G_{h,e} \det G_{s,e}. \end{aligned} \tag{CPQ7}$$

The kernel fibre in the displayed minimum is exactly K_e by CPQ6. Thus $G_{s,e}$ is the original metric on $\mathcal{S}_e/K_e$ in the values $[F_e Z_c x]$. Block elimination in $C_P^* H_e C_P$ proves its formula and the full determinant identity. Positive definiteness follows from independence in these quotient values.

Write $Q_o(N)$ for the original RQB remaining quotient on $\mathbb{C}^{4t_j}/\text{im } \Phi$, in the lifts U_o . The map CPQ5 is an isometry from $\mathcal{U}/W$ with its attained $\mathcal{D}_N$ metric to this form. To prove it, first minimize over $\mathbb{J}$, giving exactly $\text{diag}(H_{B,j,D}, 4)$ on $\mathbb{B}z$, where $H_{B,j,D} = (B_j D_{j,D}^{-1} B_j^*)^{-1}$. Then minimize over $\text{im } \Phi$. The combined fibre is W ; this is precisely the definition of Q_o in the frame C_T .

It follows that the original connecting target metric and its remaining target quotient are

$$\begin{aligned} G_{t,e} &= A_e^* Q_o A_e, \\ Q_{\text{rem},e} &= (R_e^o)^* Q_o R_e^o - (R_e^o)^* Q_o A_e (A_e^* Q_o A_e)^{-1} A_e^* Q_o R_e^o, \\ x^* G_{t,e} x &= \min_{w \in W} (F_e Z_c x + w)^* \mathcal{D}_N(F_e Z_c x + w), \\ |\det C_o|^2 \det Q_o &= \det G_{t,e} \det Q_{\text{rem},e}, \quad \omega_{\text{conn},e} = \det G_{t,e} / \det G_{s,e}. \end{aligned} \tag{CPQ8}$$

For the target restriction, every representative of a value in T_e/W lies in T_e because $W \subset T_e$. Thus the ambient minimum over W is its intrinsic quotient minimum. Substitution of $\pi_o F_e Z_c = A_e$ proves the first and third lines. The second line is the complete minimum over the connecting image inside Q_o . Block elimination in $C_o^* Q_o C_o$ proves the determinant identity. The last equality uses the identical value coordinates on both sides of the original CIT21 contraction. Its inverse and frame factors have therefore been established, rather than deduced from equality of dimensions. The new quotient has rank $r_o - c_e$ and represents exactly $\mathcal{U}/T_e$.

G.3 Positive returns and the formerly signed remainder

For the forms in CPQ7–8 set

$$V_h = \mathcal{R} \log \det G_{h,e}, \quad V_s = \mathcal{R} \log \det G_{s,e}, \quad V_t = \mathcal{R} \log \det G_{t,e}, \quad V_{\text{rem}} = \mathcal{R} \log \det Q_{\text{rem},e}. \tag{CPQ9}$$

Keep the original $V_I = \mathcal{R} \log \det H_{\text{inv}}$, $V_b = \mathcal{R} \log \det Q_b^E$, $V_o = \mathcal{R} \log \det Q_o$ and $V_A = \mathcal{R} \log \det A_{\text{inv}}$, with the full invariant source fibre $FZ_R x + FE$, $F = BJ_0$ and $E = \ker(\Phi J_0)$. In particular A_{inv} uses the target order s_1 and is not replaced by the independent source form Q_{inv} of NSR25.

All eight returns $V_h, V_s, V_t, V_{\text{rem}}, V_I, V_b, V_o, V_A$ are nonnegative. Indeed the original row extension increases $\mathcal{D}_N$. Restriction preserves this order. Minimizing two ordered forms over the same full affine fibre preserves it too. Every displayed form is such a restriction or quotient on a fixed

flag. Its successive log-determinant increments are nonnegative, and $\mathcal{R}$ is their sum with weights one at both ends and two at each interior row. This proves the assertion for these particular forms, including their complete gauges.

Write $V_{\text{conn}} = \mathcal{R} \log \det H_e$. CPQ7–8 and the original RQB target identity give

$$\begin{aligned} V_{\text{conn}} &= V_h + V_s, & V_o &= V_t + V_{\text{rem}}, \\ 4\widehat{F}_j^{(s_1)} &= V_I + V_b + V_t + V_{\text{rem}}, \\ \mathcal{R} \log \omega_{\text{conn},e} &= V_t - V_s, & \mathcal{L}_e &:= \mathcal{R} \log(\omega_{\text{conn},e} \omega_{\text{inv}}) = V_I + V_t - V_s - V_A. \end{aligned} \quad (\text{CPQ10})$$

All fixed frame determinants in CPQ7–8 cancel only here, since $\mathcal{R}1 = 0$. No varying minimum representative is treated as a fixed frame. Substitution into CIT30's original identity $\mathcal{L}_e = \mathcal{P}_e^{\text{rest}} - V_b$ now evaluates its entire signed Gram combination as

$$\boxed{\mathcal{P}_e^{\text{rest}} = 4\widehat{F}_j^{(s_1)} - V_{\text{rem}} - V_s - V_A, \quad \mathcal{L}_e = 4\widehat{F}_j^{(s_1)} - V_b - V_{\text{rem}} - V_s - V_A.} \quad (\text{CPQ11})$$

This equality retains both new proper quotients. In particular neither Q_o nor the source quotient has been discarded. It expresses the original signed combination by the complete target return and three specified nonnegative returns.

For completeness the fixed factors before taking the return can also be made explicit. Use the original CIT constants $c_{\text{conn},e}$ and C_W , and the fixed change $Q_b^{\text{CIT}} = C_b^* Q_b^E C_b$. The original expression $P(N) = \log g_{T_e} + \log g_{K_e} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} - \log \det A_{\text{inv}}$ satisfies

$$\begin{aligned} P(N) &= 4 \log \det H_{B,j,D} - \log \det Q_{\text{rem},e} - \log \det G_{s,e} - \log \det A_{\text{inv}} + \log \mathfrak{c}_e, \\ \mathfrak{c}_e &= \frac{|\det C_T|^2 |\det C_o|^2 |\det C_b|^2}{c_{\text{conn},e} |\det C_W|^2}. \end{aligned} \quad (\text{CPQ12})$$

To verify every factor, CIT26 gives $P = \log \omega_{\text{conn}} + \log \omega_{\text{inv}} + \log \det Q_b^{\text{CIT}} - \log c_{\text{conn},e} - 2 \log |\det C_W|$. Substitute CPQ8 and $|\det C_T|^2 (\det H_{B,j,D})^4 = \det H_{\text{inv}} \det Q_b^E \det Q_o$. Then substitute $\det G_{t,e} = |\det C_o|^2 \det Q_o / \det Q_{\text{rem},e}$. These substitutions give CPQ12 with the displayed positive constant. Its return is zero, which proves CPQ11 in the original absolute frames as well.

The following directed intervals follow from these exact nonnegative quantities:

$$\begin{aligned} V_I - V_s - V_A &\leq \mathcal{L}_e \leq 4\widehat{F}_j^{(s_1)} - V_b - V_s - V_A, \\ V_I + V_b - V_s - V_A &\leq \mathcal{P}_e^{\text{rest}} \leq 4\widehat{F}_j^{(s_1)} - V_s - V_A, \\ V_I - V_{\text{conn}} - V_A &\leq \mathcal{L}_e \leq 4U_{j,s_1}^{\text{native}} - V_b - V_s - V_A. \end{aligned} \quad (\text{CPQ13})$$

The width of each of the first two intervals is exactly V_o ; the location of its actual value is fixed by the retained V_{rem} . The first follows from $0 \leq V_t \leq V_o = 4\widehat{F}_j - V_I - V_b$ in CPQ10. The second adds V_b . For the third use $V_s \leq V_{\text{conn}}$ on the lower side and the original RQB35 cap, on its proved domain, on the upper. These are finite bounds on the original scalar. They do not assign an uncomputed asymptotic value to a positive return.

G.4 Actual row formulas for each new factor

The new factors can be evaluated by the original arithmetic rows. Here is a full common formula specifying their minima. Let Y be any of the fixed coefficient subspaces and $K \subset Y$ the kernel in one of the following pairs:

$$(Y, K) = (K_e, \mathcal{B}), (\mathcal{S}_e, K_e), (T_e, W), (\mathcal{U}, T_e), (\text{im } F, FE). \quad (\text{CPQ14})$$

They represent respectively $G_h, G_s, G_t, Q_{\text{rem}}, A_{\text{inv}}$. Choose independent coefficient columns X_K for K and lifts X_Q for the specified quotient values. The lifts are respectively $F_e Z_h, F_e Z_c, F_e Z_c$,

any fixed lifts of $U_o R_e^o$ under $\mathbb{B}$, and FZ_R . In the fourth pair the coordinate map to $\mathcal{U}/T_e$ is the last quotient in CPQ8.

Let $\mathcal{V}_N$ be the full original four-copy lower-evaluation matrix through D , and $\mathcal{T}_N$ the four-copy original analytic-numerator matrix through D . Form the raw matrices

$$Z_{K,N} = [\mathcal{V}_N, \mathcal{T}_N X_K], \quad Z_{Y,N} = [Z_{K,N}, \mathcal{T}_N X_Q], \quad G_K^{\text{raw}} = Z_{K,N}^* Z_{K,N}. \quad (\text{CPQ15})$$

Their columns are independent: the full lower frame is independent, and the DNE coefficient Gram is positive after its minimum; the remaining fixed coefficient columns are independent. Thus all the following inverses exist, including at the first cutoff, with the empty-block convention. Sequential elimination gives exactly the quotient form $Q = \text{Gram}_{\mathcal{D}_N}(Y/K)$ in these lifts and $\det Q = \det(Z_{Y,N}^* Z_{Y,N}) / \det G_K^{\text{raw}}$.

At the next original degree $n = D + 1$, append the four rows

$$a = [\text{diag}(v_n, 4), \text{diag}(t_n, 4)X_K], \quad b = \text{diag}(t_n, 4)X_Q, \quad C = (G_K^{\text{raw}})^{-1} Z_{K,N}^* \mathcal{T}_N X_Q, \quad R = b - aC. \quad (\text{CPQ16})$$

These are literally CIT19's v_n, t_n , with the complete arithmetic recursion

$$\begin{aligned} f_{a,m} &= \frac{\nu_{a,m+v} - \sum_{d=v+1}^{m+v} \binom{m+v}{d} \mu_d f_{a,m+v-d}}{\binom{m+v}{v} \mu_v}, \quad \nu_{a,n} = \sum_{u,w} (Z_{\text{exp},j})_{uw,a} \beta_{j,u,w}^n, \\ b_{0,m} &= \mathbf{1}_{m=0}, \quad b_{-1,m} = 0, \quad b_{n+1,m} = i^{-1}(b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2) b_{n-1,m}, \\ (t_n)_a &= h_{n,s_1}^{-1/2} \sum_m b_{n,m} f_{a,m}, \quad (v_n)_{u,w} = h_{n,s_1}^{-1/2} \sum_m b_{n,m} \beta_{j-8,u,w}^m. \end{aligned} \quad (\text{CPQ17})$$

The symbol $b_{n,m}$ in CPQ17 is the original polynomial coefficient, distinct from the new-row block b in CPQ16. Arrays outside their original polynomial range are zero. The divisor μ_v and the full old lower and coefficient Gram occur explicitly. The row formula remains valid for every allowed zero row and for both prescribed source orders.

Put $L = a(G_K^{\text{raw}})^{-1} a^*$. The exact next quotient is

$$Q^+ = Q + R^*(I_4 + L)^{-1} R, \quad \gamma_{Y/K,n} = \log \det(I_4 + L + RQ^{-1} R^*) - \log \det(I_4 + L) \geq 0. \quad (\text{CPQ18})$$

To prove the first equality, subtract the old attained minimum and change the old kernel coefficient by η . The remaining minimum is $\eta^* G_K^{\text{raw}} \eta + \|Rx - a\eta\|^2$. Solving its positive normal equations gives the residual $x^* R^* (I_4 + L)^{-1} Rx$. The determinant lemma gives the second equality. For its sign, conjugate by $(I_4 + L)^{-1/2}$; the added form is positive. Thus each of the five returns in CPQ14 is the sum of these particular γ 's, with endpoint weights one and all interior weights two. This supplies the original row calculation for all terms of CPQ11, including the previously unspecified remainder, without a new row model.

G.5 Back substitution into both source orders and the allocation

The ranks of $(G_h, G_s, G_t, Q_{\text{rem}}, A_{\text{inv}})$ are respectively $(h_e, c_e, c_e, r_o - c_e, r_I)$. NSO's full native same-fibre comparison therefore gives, for each of their returns V_X , the finite interval

$$-\text{rank}(X) \Omega_j^{\text{hi}} \leq V_X^{p=1} - V_X^{p=0} \leq \text{rank}(X) \Omega_j^{\text{lo}}. \quad (\text{CPQ19})$$

The cutoffs inside Ω_j are exactly those in CPQ1 and NSR2. The same coefficient frames occur in both orders by CPQ4 and NSR5. Their determinants therefore cancel in these ratios. Adding the two correlated decompositions gives precisely $\Delta V_o = \Delta V_t + \Delta V_{\text{rem}}$ and $\Delta V_{\text{conn}} = \Delta V_h + \Delta V_s$, preserving the full earlier NSR intervals. All ranks are $O(k)$ on the fixed actual programme, so the new individual order differences are $O_{\text{actual}}(k^2 \log q_j) = o(kq_k)$.

Finally the original degree- j allocation obeys $S_{K,j}^{(s_1)} = \widehat{F}_j^{(s_1)}$ and $S_{R,j}^{(s_1)} = \mathcal{T}_{j,s_1} - \widehat{F}_j^{(s_1)}$ by the exact HJR/NSR31 inverse-metric calculation. Hence CPQ10 gives the complete positive refinement of that kernel allocation:

$$\begin{aligned} 4S_{K,j}^{(s_1)} &= V_I + V_b + V_t + V_{\text{rem}}, & 4S_{R,j}^{(s_1)} &= 4\mathcal{T}_{j,s_1} - V_I - V_b - V_t - V_{\text{rem}}, \\ 4S_{K,j}^{(s_1)} &= \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A. \end{aligned} \tag{CPQ20}$$

The target degree is $j = k + 4$ in this identity. The independent degree- k source quotient Q_{inv} and the proper-source correction in NFR retain their own centre, displaced cutoffs and source order. In particular replacing V_A by $\mathcal{R} \log \det Q_{\text{inv}}$ is not part of CPQ20. The exact relation is $V_A = \mathcal{R} \log \det Q_{\text{inv}} + \mathcal{R} \log(\det A_{\text{inv}} / \det Q_{\text{inv}})$; both forms use the same lifts Z_R , so this ratio is the determinant of their actual relative positive operator. ITL/NFR retains its full proper-source comparison. CPQ20 thus carries the new quotient refinement into the original allocation while preserving that remaining source comparison and every positive term which must enter a quantitative estimate.

Finite equilibrium comparison for all five attained quotients. Retain CPQ2–8’s fixed maps and frames in both actual source pairs, and evaluate EFP11–15 at the original target degree $j = k + 4$ and $D_i = q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v$. For the five forms in CPQ14, whose ranks are exactly $(h_e, c_e, c_e, r_o - c_e, r_I)$, the improved finite receiver is

$$\text{rank}(X)\Phi_j^- \leq V_X^{(k,j)} - V_X^{(1,1)} \leq \text{rank}(X)\Phi_j^+, \quad X = G_h, G_s, G_t, Q_{\text{rem}}, A_{\text{inv}}. \tag{EFR2}$$

To verify its types, the value maps in order are $K_e/\mathcal{B}$, S/K_e , T_e/W , $\mathcal{U}/T_e$ and $\text{im}(BJ_0)/(BJ_0E)$, in the unchanged lifts $F_e Z_h, F_e Z_c, F_e Z_o, U_o R_e^o$ (lifted under $\mathbb{B}$), and $BJ_0 Z_R$. Each is the full original affine minimum EFP13. Thus EFP14 and the four signed additions EFP16 prove EFR2. The identities $V_{\text{conn}} = V_h + V_s$ and $V_o = V_t + V_{\text{rem}}$ remain equalities of these same quantities. The determinants $|\det C_P|^2, |\det C_o|^2, |\det C_T|^2$, the full CPQ12 $\mathfrak{c}_e$, and the original lower-Gram inverse in CPQ18 remain in EFP21–22 before their exact return cancellation. The independent degree- k Q_{inv} is related to A_{inv} by the unchanged relative determinant in CPQ20 and NFR7.

H A single spectral budget for the original common long loss

Source and precision. The complete supplied shared-long-flag proof is reproduced below, including all definitions, calculations and its stated limits. The original source is preserved byte for byte in the source closure. Its independent review supplies three explicit precision completions: the positive metric path is forward, the four update eigenvalues may vanish, and the inherited target centre and source-order pairs remain specified. These change no bound or quotient identity.

Scope of the delivered calculation. The connecting and invariant-target losses share an actual boundary flag. Their product is the squared determinant of one orthogonal projection between two orthogonal sums of minimum-lift spaces. Consequently their joint logarithmic variation costs one native spectral budget, rather than the sum of two separate budgets. A nonlinear finite enclosure retains the actual initial common angle. The original four-row recurrence gives all entries of the enclosure. An additional exact flag inequality couples the common loss to the target kernel allocation while preserving its two complementary quotient returns. The resulting joint long-loss bound is quantitative at scale kq_k . It does not evaluate an individual kernel or residual leading coefficient, nor does it establish a narrower unconditional interval for either individual allocation.

H.1 The original objects and the precise scope

Fix the stipulated actual simple quartet, complete unit, one original five-orbit period and its branch. Retain

$$k = 4l + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \quad \Delta_a = 16a - 48.$$

The source pairs are exactly $(s_0, s_1) = (1, 1)$ or (k, j) . The conductor coefficients a_{ab} are the actual period coefficients. Write

$$E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \beta_{a,r,t} = a/2 + (2r-a)\delta + i(2t-a)\gamma. \quad (\text{SFB1})$$

No period coefficient, source mass or primary multiplicity is replaced. The inherited target centre is $c'_j = j/2 - 4$; every target form uses the original $s_1 = 1$ or j , while its matched degree- k source retains $s_0 = 1$ or k . This is the simple-quartet metric lane; no higher-multiplicity metric theorem is inferred.

Use the complete native target coefficient space $\mathcal{H}$ of CIT and ITL, with the four-copy positive form

$$\mathcal{D}_N = \text{diag}(D_{j,N-v}^{(s_1)}, 4). \quad (\text{SFB2})$$

The D 's are the original full lower-evaluation minimum-extension Grams. All subspaces below are fixed coefficient subspaces as N varies. Put

$$I = \text{im } F_{\text{inv}}, \quad B = \text{im } B_{\text{cof}}, \quad S = \mathcal{S}_e, \quad J = \ker \mathbb{B}, \quad e \in \{0, 1\}, \quad I \subset B \subset S, \quad (\text{SFB3})$$

where the literal matrices are

$$B_{\text{cof}} = \text{stack}_{i=1}^4 M_{i,k}, \quad F_{\text{inv}} = B_{\text{cof}} J_0, \quad S = \text{im}[B_{\text{cof}}, F_e].$$

Here $M_{i,k}$ is the original coefficient map induced by multiplication by x_i^4 , in the complete DNE frames; J_0 is the original invariant-source inclusion, and F_e is the full CIT connecting matrix. In particular these are not independently chosen numerical subspaces. Define

$$K_i = I \cap J, \quad W_i = I + J, \quad W = B + J, \quad K_c = S \cap W, \quad T = S + J. \quad (\text{SFB4})$$

Thus $K_i \subset I \subset B \subset K_c \subset S$ and $J \subset W_i \subset W \subset T$. Subscript i here abbreviates “invariant”, not a summation index.

For a positive form D let $P_X^D = X(X^*DX)^{-1}X^*D$ be the orthogonal projector onto a fixed subspace X , expressed in the original coordinates using any full-rank frame of X . A zero-dimensional space uses its empty inverse and determinant one. The source and target quotient metrics are always their attained minimum norms. The two original losses are

$$\omega_i(D) = \frac{\det \text{Gram}_D(W_i/J)}{\det \text{Gram}_D(I/K_i)}, \quad \omega_c(D) = \frac{\det \text{Gram}_D(T/W)}{\det \text{Gram}_D(S/K_c)}, \quad \omega(D) = \omega_i(D)\omega_c(D). \quad (\text{SFB5})$$

Each ratio uses the same quotient-value coordinates in numerator and denominator. These are exactly ω_{inv} and $\omega_{\text{conn},e}$, not the determinant of the whole connecting space S/B . In particular the connecting kernel K_c/B remains separate.

H.2 The common product is one projection determinant

Theorem H.1. *For every positive form D , set*

$$\mathcal{P}_D = (I \ominus_D K_i) \oplus (S \ominus_D K_c), \quad \mathcal{Q}_D = (W_i \ominus_D J) \oplus (T \ominus_D W).$$

Both sums are orthogonal. They have equal dimension, the orthogonal projection $\mathcal{P}_D \rightarrow \mathcal{Q}_D$ is invertible, and its squared determinant in orthonormal frames is exactly $\omega(D)$. In particular $0 < \omega(D) \leq 1$.

Proof. The first orthogonality follows from $I \subset K_c$; the second from $W_i \subset W$. The natural maps $I/K_i \rightarrow W_i/J$ and $S/K_c \rightarrow T/W$ are isomorphisms. Take their original minimum lifts and then orthonormalize separately in each of the four quotient forms.

In the corresponding source and target orthonormal frames, the projection has block matrix

$$M_D = \begin{pmatrix} M_i & C_D \\ 0 & M_c \end{pmatrix}. \quad (\text{SFB6})$$

The lower-left block is zero because $I \subset W$. The diagonal blocks are precisely the two original quotient contractions. They are invertible by the displayed quotient isomorphisms, and $|\det M_i|^2 = \omega_i$, $|\det M_c|^2 = \omega_c$. Taking the determinant proves the result. Its norm is at most one because it is an orthogonal projection. The off-diagonal block C_D has not been deleted. In particular, deleting the corresponding cross blocks in $M_D^*M_D$, while retaining its diagonal blocks, need not preserve its determinant. $\square$

In Euclidean relative coordinates define $\Pi_X = D^{1/2}P_X^D D^{-1/2}$. The two projectors in Theorem H.1 are

$$\Pi_+ = \Pi_T - \Pi_W + \Pi_{W_i} - \Pi_J, \quad \Pi_- = \Pi_S - \Pi_{K_c} + \Pi_I - \Pi_{K_i}. \quad (\text{SFB7})$$

Each is an orthogonal projector: within each sum the two ranges are orthogonal by the same inclusions. This is the reason a single budget applies. Bounding the two losses independently misses it.

H.3 Variation of the actual common loss

Theorem H.2. *Let $D(t)$ be a continuously differentiable positive form on a forward interval $t_0 \leq t_1$, with $\dot{D}(t) \succeq 0$, and set*

$$L(t) = \log \omega(D(t)), \quad C(t) = D(t)^{-1/2} \dot{D}(t) D(t)^{-1/2}, \quad b'(t) = \text{Tr } C(t).$$

Then

$$L'(t) = \text{Tr}(C(t)(\Pi_+(t) - \Pi_-(t))), \quad |L'(t)| \leq \sqrt{1 - e^{L(t)}} b'(t) \leq b'(t). \quad (\text{SFB8})$$

Consequently, for $a(t) = \text{arcosh}(e^{-L(t)/2})$,

$$|a(t_1) - a(t_0)| \leq \frac{1}{2} \log \frac{\det D(t_1)}{\det D(t_0)}. \quad (\text{SFB9})$$

Proof. For a fixed frame X , Jacobi's determinant formula gives

$$\frac{d}{dt} \log \det(X^*DX) = \text{Tr}(C\Pi_X).$$

Full flag Gram elimination expresses a quotient determinant as a ratio of these determinants, times a fixed coefficient factor. That factor has zero derivative. Applying this identity to (SFB5) gives (SFB8)'s equality and exactly the two projectors in (SFB7). This argument does not differentiate an asymptotic remainder, or assume that the orthogonal minimum sections are constant.

Let $\sigma_a \in (0, 1]$ be the singular values of M_D . An orthogonal principal-plane calculation shows that the nonzero eigenvalues of $\Pi_+ - \Pi_-$ are $\pm\sqrt{1 - \sigma_a^2}$. Explicitly, if source and target principal unit vectors p_a, q_a satisfy $\langle p_a, q_b \rangle = \sigma_a \delta_{ab}$, then $(q_a - \sigma_a p_a)/\sqrt{1 - \sigma_a^2}$, for $\sigma_a < 1$, completes their mutually orthogonal two-planes. The difference of projectors on each plane has the stated eigenvalues. Since $\min_a \sigma_a^2 \geq \prod_a \sigma_a^2 = \omega$, its operator norm is at most $\sqrt{1 - \omega}$. For $C \succeq 0$, diagonalizing this self-adjoint difference gives $|\text{Tr}(CA)| \leq \|A\| \text{Tr } C$. This proves both inequalities.

Where $L < 0$, differentiation of a gives $|a'| = |L|/(2\sqrt{1 - e^L}) \leq b'/2$. To include points with $L = 0$, use $a_\varepsilon = \text{arcosh}(e^{-L/2} + \varepsilon)$. Its derivative obeys the same bound. Integrate and let $\varepsilon \downarrow 0$. This proves (SFB9) without a differentiability assumption at a zero angle. $\square$

For a finite positive update let $\beta = \log(\det D_1/\det D_0)$ and $a_0 = \text{arcosh}(\omega(D_0)^{-1/2})$. Writing $f(a) = -2 \log \cosh a$, the finite enclosure is

$$f(a_0 + \beta/2) - f(a_0) \leq \log \frac{\omega(D_1)}{\omega(D_0)} \leq f((a_0 - \beta/2)_+) - f(a_0). \quad (\text{SFB10})$$

Both endpoints lie in $[-\beta, \beta]$; the upper endpoint is also at most $-\log \omega(D_0)$. Equivalently the new absolute common loss is between $\text{sech}^2(a_0 + \beta/2)$ and $\text{sech}^2((a_0 - \beta/2)_+)$. The initial common angle is an actual finite quotient quantity. It has not been assigned a period-independent value.

H.4 The exact original four-row update

Here are the original entries to which the preceding results apply. For a fixed target numerator column N_a , let

$$\nu_{a,d} = \sum_{u,w} (Z_{\text{exp},j})_{(u,w),a} \beta_{j,u,w}^d.$$

The analytic germ $F_a = N_a/E_A$ has derivatives given exactly by

$$f_{a,n} = \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}. \quad (\text{SFB11})$$

The numerator's first v derivatives vanish by its original coefficient domain. Retain the original Gamma polynomials and norms

$$p_{n+1}(y) = yp_n(y) - n(n-1 + s_1/2)p_{n-1}(y), \quad h_{n,s_1} = (2\pi)^{s_1/2} n! (s_1/2)_n.$$

Expanding $p_n((S' - c'_j)/i) = \sum_a b_{n,a} S'^a$ gives

$$b_{n+1,a} = i^{-1}(b_{n,a-1} - c'_j b_{n,a}) - n(n-1 + s_1/2)b_{n-1,a}.$$

Thus the new numerator and lower-evaluation rows are

$$t_n = \left(\frac{\sum_a b_{n,a} f_{b,a}}{\sqrt{h_{n,s_1}}} \right)_b, \quad v_n = \left(\frac{\sum_a b_{n,a} (\beta')^a}{\sqrt{h_{n,s_1}}} \right)_{\beta'}.$$

There is no removal of a source mass or of the affine phase.

Let T_0, V_0 contain every earlier numerator and lower-evaluation row in the original native source. Put

$$A_L = V_0^* V_0, \quad D_0 = T_0^* [I - V_0 A_L^{-1} V_0^*] T_0, \quad \rho_n = t_n - v_n A_L^{-1} V_0^* T_0, \quad \zeta_n = v_n A_L^{-1} v_n^*. \quad (\text{SFB12})$$

The exact minimum-extension update is

$$D_1 = D_0 + \frac{\rho_n^* \rho_n}{1 + \zeta_n}, \quad \mathcal{D}_1 = \mathcal{D}_0 + \mathcal{U}_n^* \mathcal{U}_n, \quad \mathcal{U}_n = \text{diag} \left(\frac{\rho_n}{\sqrt{1 + \zeta_n}}, 4 \right). \quad (\text{SFB13})$$

These are CIT19/IRR and FAB's same update. Its four potentially nonzero relative-update eigenvalues are four copies of $\rho_n D_0^{-1} \rho_n^* / (1 + \zeta_n)$, so its full budget is

$$\beta_n = 4 \log \left(1 + \frac{\rho_n D_0^{-1} \rho_n^*}{1 + \zeta_n} \right). \quad (\text{SFB14})$$

If the original residual row is zero, all four update eigenvalues and this budget are zero. Both the old lower-evaluation inverse and its $1 + \zeta_n$ denominator remain.

For any one of the eight fixed flag subspaces X , set

$$K_{X,n} = \mathcal{U}_n X (X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{U}_n^*, \quad b_{X,n} = \log \det(I_4 + K_{X,n}).$$

These positive four-by-four matrices give the exact common increment

$$\delta L_n = b_{T,n} - b_{W,n} + b_{K_c,n} - b_{S,n} + b_{W_i,n} - b_{J,n} + b_{K_i,n} - b_{I,n}. \quad (\text{SFB15})$$

Thus the result bounds the original signed difference of paired positive increments. It does not bound just a numerator and drop the paired denominator.

H.5 The common long interval and its evaluated budget

Use the original native target cutoffs

$$d = q_j - v - 1, \quad d, d + 1, d + q_j, d + q_j + 1$$

with native-dual signs $(-, -, +, +)$. Let $\mathcal{R}$ denote this signed return. Its consecutive weights ϑ_n equal one at $d + 1, d + q_j + 1$, and two at the internal degrees. Define

$$\mathcal{L}_e = \mathcal{R}(\log \omega_{\text{conn},e} + \log \omega_{\text{inv}}), \quad \mathcal{B}_{k,s_1} = \sum_n \vartheta_n \beta_n. \quad (\text{SFB16})$$

Then

$$|\mathcal{L}_e| \leq \mathcal{B}_{k,s_1}. \quad (\text{SFB17})$$

A sharper finite interval is obtained by summing the two endpoints of (SFB10), with the same ϑ_n , actual preceding common angles, and β_n .

The budget has an already evaluated scalar expression:

$$\mathcal{B}_{k,s_1} = 4\mathcal{T}_{j,s_1} = 4\Delta_j q_j C_\partial + o(j q_j). \quad (\text{SFB18})$$

This uses the inherited evaluated total, with $j = k + 4$ and source $s_1 = 1$ or j ; no source order is substituted. Consequently

$$\limsup \frac{|\mathcal{L}_e|}{k q_k} \leq 64 C_\partial, \quad 86.67404722080 < 64 C_\partial < 86.67404722084. \quad (\text{SFB19})$$

The numerical interval is just multiplication by four of the supplied interval for $16C_\partial$, not a new elliptic calculation. Separate application of the single-flag budget to the two losses allows twice this joint absolute allowance. The common flag improves their joint interval, not an individual leading allocation. No assertion that the inequality is attained on an actual period is made.

H.6 The complementary terms and the allocation-side constraint

For any fixed flag quotient Y/X , write

$$V_n(Y/X) = b_{Y,n} - b_{X,n} \geq 0, \quad \mathcal{V}(Y/X) = \sum_n \vartheta_n V_n(Y/X).$$

The nonnegativity is monotonicity of the same attained quotient minimum under the positive source update. Let

$$\Gamma_{+,n} = V_n(T/W) + V_n(W_i/J), \quad \Gamma_{-,n} = V_n(S/K_c) + V_n(I/K_i).$$

Then $\delta L_n = \Gamma_{+,n} - \Gamma_{-,n}$ and both $\Gamma_{\pm}$ are nonnegative. Full nested-flag elimination gives

$$\Gamma_{+,n} = V_n(\mathcal{H}/J) - V_n(\mathcal{H}/T) - V_n(W/W_i), \quad \Gamma_{-,n} = V_n(S/K_i) - V_n(K_c/I). \quad (\text{SFB20})$$

Every omitted interval in the flag is retained in this identity. In particular,

$$\mathcal{L}_e \leq \mathcal{V}(\mathcal{H}/J) - \mathcal{V}(\mathcal{H}/T) - \mathcal{V}(W/W_i) = 4S_{K,j,s_1} - \mathcal{V}(\mathcal{H}/T) - \mathcal{V}(W/W_i). \quad (\text{SFB21})$$

The final equality uses the original four-copy kernel quotient: $\mathcal{H}/J$ has metric $(\mathbb{B}\mathcal{D}^{-1}\mathbb{B}^*)^{-1}$, four copies of the original native quotient, with exactly the dual return orientation. This is not an equality with the arithmetic canonical allowance.

The quotient W/W_i is precisely CIT's Q_b quotient:

$$\mathcal{V}(W/W_i) = \mathcal{R} \log \det Q_b. \quad (\text{SFB22})$$

It must not be dropped to claim equality in (SFB21). Likewise the full CIT common-boundary factorization is retained:

$$\begin{aligned} \log \omega_c + \log \omega_i &= \log g_T + \log g_{K_c} - \log g_S - \log g_J \\ &\quad - \log \det A_{\text{inv}} - \log \det Q_b + \log c_{\text{conn},e} + 2 \log |\det C_W|. \end{aligned} \quad (\text{SFB23})$$

Only its fixed coefficient constants cancel under $\mathcal{R}$.

If the new common finite enclosure is $\underline{L}_e \leq \mathcal{L}_e \leq \overline{L}_e$, then (SFB21) gives the valid receiver

$$S_{K,j,s_1} \geq \frac{\underline{L}_e + \mathcal{V}(\mathcal{H}/T) + \mathcal{V}(W/W_i)}{4}. \quad (\text{SFB24})$$

This is to be intersected with the existing actual intervals, and paired with $S_R = \mathcal{T} - S_K$. None of the right-hand finite terms has been numerically evaluated here. Therefore no unconditional numerical narrowing of an individual allocation is claimed.

The arithmetic return remains

$$F_K^{(s)} = S_{K,s} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} - F_{W_0}^{(s)}, \quad \Phi_{k,s} = F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}.$$

Both unequal flag endpoints, the original transfer allowance, masses, low metrics and period domains stay in the SAR receiver. The signed return and its whole action budget retain their exact shared-residual cancellation. A sign of either new log-loss interval does not prove an exclusion of an arithmetic packet.

H.7 What is new and what is still uncomputed

The new result is the single-projection realization of the product of the two original losses, its nonlinear common finite enclosure, the one-budget long estimate, and its explicit allocation-side constraint retaining both complementary flag returns. The original rows, target kernel and connecting flags are used throughout. The low primitive and short connecting corrections, native four-row recurrence, evaluated total and single-flag FAB theorem remain inherited inputs.

The individual coefficients of $S_K/(kq_k)$ and $S_R/(kq_k)$ are not evaluated. Nor is an improved unconditional numerical interval for either coefficient established. The remaining quantity in this particular comparison is the actual shared projection angle along the native update, together with $\mathcal{V}(\mathcal{H}/T)$ and $\mathcal{V}(W/W_i)$ in (SFB24). Replacing either by zero, or comparing the kernel alone with the whole canonical allowance, would not complete that calculation.

No actual period integral, hypothetical zero location, independent referee acceptance, or Lean execution is certified by this note. The manuscript supplies a written mathematical derivation. Any separate build or diagnostic status must be taken from its actual execution record, not inferred from this text.

References

- [1] Split-Zero research programme. (2026, September 15). *Current kernel and multiplicity proofs* (498-page supplied edition). Original ITL, IRR, FAB, NFR, HJR/OVR and SAR providers.
- [2] Split-Zero research programme. (2026, September 15). *Full primitive connecting proofs*. Original PDS, CIT and PDI providers; in particular CIT21 and CIT24–26.

Eight actual rank-four compressions and the retained spectral threshold. Retain exactly SFB’s original flags $I \subset B \subset S$, $J = \ker \mathbb{B}$, $K_i = I \cap J$, $W_i = I + J$, $W = B + J$, $K_c = S \cap W$ and $T = S + J$. The same fixed frames are used for every cutoff and both allowed source-order pairs. At native row n , the preceding single-copy Gram and its complete lower correction are

$$\begin{aligned} A_L &= V_0^* V_0, & D_0 &= T_0^* [I - V_0 A_L^{-1} V_0^*] T_0, \\ \rho_n &= t_n - v_n A_L^{-1} V_0^* T_0, & \zeta_n &= v_n A_L^{-1} v_n^*, & r_n &= \frac{\rho_n D_0^{-1} \rho_n^*}{1 + \zeta_n}, \\ \mathcal{D}_0 &= \text{diag}(D_0, 4), & \mathcal{U}_n &= \text{diag}\left(\frac{\rho_n}{\sqrt{1 + \zeta_n}}, 4\right), & \mathcal{D}_1 &= \mathcal{D}_0 + \mathcal{U}_n^* \mathcal{U}_n. \end{aligned} \tag{PTHS1}$$

Here T_0, V_0 contain every earlier numerator and lower-evaluation row, and t_n, v_n are the exact CIT19/SFB11–13 arithmetic rows, with their original $h_{n,s_1} = (2\pi)^{s_1/2} n! (s_1/2)_n$, center $c'_j = j/2 - 4$, division moments and phases. The notation $\text{diag}(\cdot, 4)$ denotes four copies. In particular $r_n \geq 0$, and the complete $1 + \zeta_n$ denominator is present.

For any of the eight subspaces use its fixed full-rank frame X and define the actual four-by-four matrix

$$K_{X,n} = \mathcal{U}_n X (X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{U}_n^*, \quad \mathcal{X}_+ = (T, K_c, W_i, K_i), \quad \mathcal{X}_- = (W, S, J, I). \tag{PTHS2}$$

For a zero-dimensional space this is the zero matrix. The formula is unchanged by an invertible fixed change of columns in X : the inverse congruence cancels both column changes. Let $Y = \mathcal{U}_n \mathcal{D}_0^{-1/2}$ and $\Pi_X = \mathcal{D}_0^{1/2} X (X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{D}_0^{1/2}$. The displayed matrix Π_X is self-adjoint, idempotent, and has range $\mathcal{D}_0^{1/2} \text{im } X$; these assertions follow by direct multiplication and full column rank. Thus $0 \preceq \Pi_X \preceq I$, and direct substitution gives

$$K_{X,n} = Y \Pi_X Y^*, \quad Y Y^* = r_n I_4, \quad 0 \preceq K_{X,n} \preceq r_n I_4. \tag{PTHS3}$$

The coordinate map $z \mapsto \mathcal{D}_0^{1/2} z$ is an isometry from $(\mathcal{H}, \mathcal{D}_0)$ to the Euclidean coordinate space, since its squared Euclidean norm is $z^* \mathcal{D}_0 z$; its inverse is $\mathcal{D}_0^{-1/2}$. Hence this calculation expresses the original metric and all eight original subspaces.

Let $\lambda_{X,a,n}$, $1 \leq a \leq 4$, be all four eigenvalues of $K_{X,n}$, retaining their zeros. The determinant lemma applied to $X^* \mathcal{D}_1 X$ shows that its log-determinant increment is $b_{X,n} = \log \det(I_4 + K_{X,n})$. Indeed the nonzero product factors of the rectangular maps $\mathcal{U}_n X$ and $(X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{U}_n^*$ give the

same determinant, including the complementary identity eigenvalues. Spectral factorization then gives $b_{X,n} = \sum_{a=1}^4 \log(1 + \lambda_{X,a,n})$. SFB15's exact signed increment is consequently

$$\delta \log \omega_n = \sum_{X \in \mathcal{X}_+} \sum_{a=1}^4 \log(1 + \lambda_{X,a,n}) - \sum_{X \in \mathcal{X}_-} \sum_{a=1}^4 \log(1 + \lambda_{X,a,n}). \quad (\text{PTHS4})$$

The two lists in (PTHS2) therefore retain every sign of the original two contractions.

Put $d = q_j - v - 1$. The original four-endpoint return has consecutive row indices $n = d + 1, \dots, d + q_j + 1$ and weights $\theta_n = 1$ at its two endpoints, $\theta_n = 2$ at the $q_j - 1$ interior indices. This follows by expanding $-f(d) - f(d+1) + f(d+q_j) + f(d+q_j+1)$ in consecutive increments. Thus $\sum_n \theta_n = 2 + 2(q_j - 1) = 2q_j$ exactly. Define the full signed threshold profile by

$$P_{k,e} = \sum_{n=d+1}^{d+q_j+1} \theta_n \left(\sum_{X \in \mathcal{X}_+} \sum_{a=1}^4 \log \max(1, \lambda_{X,a,n}) - \sum_{X \in \mathcal{X}_-} \sum_{a=1}^4 \log \max(1, \lambda_{X,a,n}) \right). \quad (\text{PTHS5})$$

For every real $x \geq 0$, $1 \leq (1+x)/\max(1,x) \leq 2$. Taking logarithms proves $0 \leq \log(1+x) - \log \max(1,x) \leq \log 2$. Each sign list contains four subspaces with four eigenvalues each, so its total error at each row lies in $[0, 16 \log 2]$. The difference of the two errors lies in $[-16 \log 2, 16 \log 2]$. Multiplication by the actual nonnegative weights and their exact sum proves

$$\boxed{|\mathcal{L}_e - P_{k,e}| \leq 32q_j \log 2.} \quad \frac{|\mathcal{L}_e - P_{k,e}|}{kq_k} \leq \frac{32(q_j/q_k) \log 2}{k} \longrightarrow 0. \quad (\text{PTHS6})$$

Here $q_j/q_k = (k+5)^2/(k+1)^2 \longrightarrow 1$. No individual row limit or positive lower bound on the nonzero eigenvalues was used. If $r_n = 0$, (PTHS3) makes every matrix zero, and this row contributes zero on both sides.

For $r_n > 0$ there is also an exact scalar coordinate identity, with its inverse retained:

$$\begin{aligned} \mu_{X,a,n} &= \lambda_{X,a,n}/r_n \in [0, 1], & \lambda_{X,a,n} &= r_n \mu_{X,a,n}, \\ \alpha_n &= \log r_n/k, & b_{X,a,n} &= -\log \mu_{X,a,n}/k, & \log \max(1, \lambda_{X,a,n}) &= k(\alpha_n - b_{X,a,n})_+ \quad (\mu_{X,a,n} > 0). \end{aligned} \quad (\text{PTHS7})$$

The equality follows by substituting the displayed inverse into $\max(0, \log \lambda)$. Zero eigenvalues contribute zero and are retained separately, so no logarithm of zero or deleted kernel occurs. This changes no source form or arithmetic period.

The proved EOR order comparison may now be composed with this finite estimate by the triangle inequality. With superscripts $p = 0$ for $(1, 1)$ and $p = 1$ for (k, j) , it gives, on the same eventual domain and its stated q -scale,

$$\limsup_{k \rightarrow \infty} \frac{|\mathcal{L}_e^{p=1} - P_{k,e}^{p=0}|}{q_k} \leq C_{\text{com}} + 32 \log 2, \quad C_{\text{com}} < 29.311139553141. \quad (\text{PTHS8})$$

Indeed add $|\mathcal{L}_e^{p=1} - \mathcal{L}_e^{p=0}|$ to the error in (PTHS6) for $p = 0$ and use $q_j/q_k \rightarrow 1$. This is the transported finite profile estimate. It leaves the actual signed compressed spectra and the complementary returns in (PTHCIT5)–(PTHCIT6) present.

I A smaller threshold-profile error from the original nested flags

Retain the complete SFB and CPQ coefficient spaces at target degree $j = k + 4$, the two original source pairs and their inherited simple-quartet and fixed-period domains. At each original row, use its full preceding positive form D_0 , its lower-evaluation-corrected numerator row ρ_n , and the original scalar $\zeta_n \geq 0$. Set

$$r_n = \frac{\rho_n D_0^{-1} \rho_n^*}{1 + \zeta_n}, \quad \mathcal{D}_0 = \text{diag}(D_0, 4), \quad \mathcal{U}_n = \text{diag}\left(\frac{\rho_n}{\sqrt{1 + \zeta_n}}, 4\right), \quad (\text{NTE1})$$

$$K_{X,n} = \mathcal{U}_n X (X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{U}_n^*.$$

An empty subspace gives the zero 4×4 matrix. All frames X are the fixed coefficient frames of the actual eight flags. With $Y = \mathcal{U}_n \mathcal{D}_0^{-1/2}$ and Π_X the orthogonal projector onto $\mathcal{D}_0^{1/2} \text{im } X$, this is exactly $K_{X,n} = Y \Pi_X Y^*$, and $Y Y^* = r_n I_4$. In particular $0 \preceq K_{X,n} \preceq r_n I_4$. When $X \subset Z$, the difference $\Pi_Z - \Pi_X$ is an orthogonal projector, so $K_{X,n} \preceq K_{Z,n}$. These are identities in the original metric and preserve both the full old lower inverse and its denominator.

The two nested chains already used in the common-loss proof are

$$J \subset W_i \subset W \subset T, \quad K_i \subset I \subset K_c \subset S. \quad (\text{NTE2})$$

Indeed $I \subset B \subset S$, $K_i = I \cap J$, $W_i = I + J$, $W = B + J$, $K_c = S \cap W$ and $T = S + J$. In particular $I \subset K_c$. There is no required simultaneous diagonalization of the eight compressed matrices.

For $x \geq 0$ put

$$h(x) = \log(1 + x) - \log \max(1, x) = u(x) - v(x),$$

$$u(x) = \log(1 + \min(x, 1)), \quad v(x) = \begin{cases} 0, & 0 \leq x \leq 1, \\ \log \frac{2x}{1+x}, & x \geq 1. \end{cases} \quad (\text{NTE3})$$

The definitions agree at $x = 1$ and give the displayed identity on each interval. Both u and v are increasing and vanish at zero. For $0 \leq x_0 \leq x_1 \leq x_2 \leq x_3 \leq r$, their disjoint increment sums satisfy

$$0 \leq u(x_3) - u(x_2) + u(x_1) - u(x_0) \leq u(r),$$

$$0 \leq v(x_3) - v(x_2) + v(x_1) - v(x_0) \leq v(r).$$

For example the first sum is at most $u(x_3) - u(x_0)$ after adding the nonnegative missing increment $u(x_2) - u(x_1)$; its upper bound is $u(r)$. The other line is identical. Subtracting proves that the alternating h sum lies in $[-v(r), u(r)]$, including $r = 0$ and threshold equalities.

Write the four eigenvalues of each $K_{X,n}$ in increasing order, including every zero, as $\lambda_{X,a,n}$, $1 \leq a \leq 4$. Positive matrix order makes each indexed eigenvalue increase along each chain in NTE2. To see this without a commutativity assumption, use the variational expression $\lambda_a(H) = \min_{\dim L=a} \max_{0 \neq x \in L} (x^* H x) / (x^* x)$ for increasing eigenvalues: an eigenbasis proves the formula, and pointwise ordered Rayleigh quotients preserve the minimum and maximum. Thus NTE3 applies to each of the four indexed scalar chains separately. Consequently

$$A_n = \text{Tr}\{h(K_{T,n}) - h(K_{W,n}) + h(K_{W_i,n}) - h(K_{J,n})\},$$

$$B_n = \text{Tr}\{h(K_{S,n}) - h(K_{K_c,n}) + h(K_{I,n}) - h(K_{K_i,n})\}, \quad (\text{NTE4})$$

$$-4v(r_n) \leq A_n, B_n \leq 4u(r_n), \quad |A_n - B_n| \leq 4\{u(r_n) + v(r_n)\}.$$

Here $h(K)$ is defined by the finite Hermitian spectral calculus; only its trace, the sum of the four scalar values, is used.

Let $\mathcal{P}_{k,e}$ be the original eight-flag threshold profile: at each row replace every $\log \det(I_4 + K_{X,n})$ in the exact SFB increment by $\sum_{a=1}^4 \log \max(1, \lambda_{X,a,n})$, with positive flags (T, K_c, W_i, K_i) and

negative flags (W, S, J, I) . Take precisely the original nonnegative weights ϑ_n , one at the two end rows and two inside, whose sum is $2q_j$. Spectral factorization and NTE3 show that the row error is exactly $A_n - B_n$. Therefore

$$|\mathcal{L}_e - \mathcal{P}_{k,e}| \leq E_{k,e}^{\text{nested}} := \sum_n \vartheta_n \varepsilon(r_n) \leq 16q_j \log 2, \quad (\text{NTE5})$$

$$\varepsilon(r) = 4\{u(r) + v(r)\} = \begin{cases} 4\log(1+r), & 0 \leq r \leq 1, \\ 4\log \frac{4r}{1+r}, & r \geq 1. \end{cases}$$

The last inequality uses $u(r) \leq \log 2$, $v(r) \leq \log 2$. At a zero residual row its contribution is exactly zero. This proves a smaller complete finite error than the independent eight-matrix bound $32q_j \log 2$ while preserving precisely the same threshold profile, ranks, metrics and original row parameters. No equality case on an actual period is asserted.

The exact CPQ return identity is $4S_{K,j} = \mathcal{P}_e^{\text{rest}} + V_{\text{rem}} + V_s + V_A$ and $\mathcal{P}_e^{\text{rest}} = \mathcal{L}_e + V_b$. Thus the new profile interval gives

$$\frac{\mathcal{P}_{k,e} - E_{k,e}^{\text{nested}} + V_b + V_{\text{rem}} + V_s + V_A}{4} \leq S_{K,j} \leq \frac{\mathcal{P}_{k,e} + E_{k,e}^{\text{nested}} + V_b + V_{\text{rem}} + V_s + V_A}{4}. \quad (\text{NTE6})$$

Addition of these retained returns and division by four is the exact affine map; no boundary dependence is removed by the cancellation in $\mathcal{P}_e^{\text{rest}}$. Intersect NTE6 with the original positivity, total and native-cap intervals on their respective domains, then use $S_R = \mathcal{T} - S_K$ to reverse the endpoints for the residual allocation. The same common-loss interval can be intersected with SFC11 before this affine map is applied.

On EOR's proved paired-source domain, put $C_{k,e}^{\text{ord}} = R_e \sum_N (\bar{b}_{j,N-v-q'_j} - \underline{b}_{j,N-v-q'_j})$, with the original four endpoints and all original finite guards. The triangle inequality applied to EOR's exact same-value comparison and NTE5 at order one gives

$$|\mathcal{L}_e^{(k,j)} - \mathcal{P}_{k,e}^{(1,1)}| \leq C_{k,e}^{\text{ord}} + E_{k,e}^{\text{nested},(1,1)}, \quad \limsup \frac{|\mathcal{L}_e^{(k,j)} - \mathcal{P}_{k,e}^{(1,1)}|}{q_k} \leq C_{\text{com}} + 16 \log 2. \quad (\text{NTE7})$$

The last step uses $q_j/q_k \rightarrow 1$ and EOR29's proved bound. The prescribed source orders and the actual affine target centre remain fixed. This is an error bound for the original profile; the profile itself and the complementary quotient returns remain their exact original matrix quantities. It assigns no individual allocation coefficient or actual-period compressed spectrum.

J The original connecting image as a polynomial translation relation space

Retain the original simple quartet, its amplitude, fixed period on the proved five-orbit locus, logarithm branch, and every unit and period coefficient. Set

$$\begin{aligned}
k &= 4l + 1 \geq 9, & j &= k + 4, & q &= (k + 1)^2, & Q &= (j + 1)^2, \\
e &= 0, 1, & D_e &= Q - v - 1 + e, & d &= Q - q = 8k + 24, \\
\beta_* &= \frac{1}{2} - \delta - i\gamma, & \beta_{n,a,b} &= n\beta_* + 2a\delta + 2bi\gamma, \\
E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b}z}, & v &= \text{ord}_0 E_A, & \mu_v &= E_A^{(v)}(0) \neq 0, & \alpha_A &= v!/\mu_v.
\end{aligned} \tag{CPR1}$$

The coefficients are those of the complete original nonreduced conductor, not its radical. All duals are complex-linear. Write

$$\begin{aligned}
\chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{n,a,b}), & L_i(Z, W) &= (\mathbf{H}(ZW, Z, W, 1)^\top)_i, \\
b_{i,a,b} &= [Z^a W^b] L_i^4, & B_i(z) &= \sum_{a,b=0}^4 b_{i,a,b} e^{\beta_{4,a,b}z}, \\
(\mathcal{T}_A p)(S') &= \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), & (\mathcal{T}_i p)(S) &= \sum_{a,b=0}^4 b_{i,a,b} p(S + \beta_{4,a,b}), \\
L &= \mathbb{C}[S]_{<v}.
\end{aligned} \tag{CPR2}$$

The variable of a translated polynomial is determined by the displayed map; translation sums commute on polynomials. In particular no source centre is inserted into these algebraic maps. Their eventual original Gamma norms still use the centres and the two source-order pairs $(1, 1)$ and (k, j) of CIT and NSR.

J.1 A full polynomial model of the original target kernel

Let $K_j \subset \mathbb{C}[S]_{<Q}$ be the literal original kernel, represented by interpolation at all the degree- j roots. Here is an explicit definition using only its original period coordinates. For every $\nu = (\nu_1, \dots, \nu_4)$ with $|\nu| = j$ and $\sum_{i=1}^4 i\nu_i \equiv 0 \pmod{5}$, put

$$m_{\nu,a,b} = [Z^a W^b] \prod_{i=1}^4 L_i(Z, W)^{\nu_i}.$$

Then

$$K_j = \left\{ p \in \mathbb{C}[S]_{<Q} : \sum_{a,b=0}^j m_{\nu,a,b} p(\beta_{j,a,b}) = 0 \text{ for every such } \nu \right\}. \tag{CPR3}$$

Indeed the homogeneous invariant monomials are a basis of the weight-zero part of $\mathbb{C}[x_1, \dots, x_4]_j$. Their substitutions are precisely the displayed coefficient vectors. The original observation dual has this image: restriction of its coefficient functionals to a primary vector $p(\beta_{j,a,b})$ is exactly the sum in CPR3. Thus its annihilator is the original K_j . Equivalently this is KRC3's original observation map, with the full invertible period substitution retained. The distinct-root Vandermonde identifies these values with a unique polynomial of degree $< Q$, so CPR3 gives its full original polynomial representative, rather than an independently chosen plane.

The trace identity for the conductor gives $K_j \subset V_{A,j}$. Explicitly, if C_j is the original full conductor convolution, then

$$(C_j p)_{a,b} = \sum_{r,t=0}^8 a_{rt} p(\beta_{j-8,a,b} + \beta_{8,r,t}) = (\mathcal{T}_A p)(\beta_{j-8,a,b}). \quad (\text{CPR4})$$

The original invariant trace contains every conductor column $C_j^\top$, hence CPR3 annihilates it and CPR4 is zero. It follows that $\mathcal{T}_A p$ is divisible by χ_{j-8} . All coefficients of the original convolution occur in this equation.

Define the exact polynomial space

$$\mathcal{U}_j = \alpha_A \mathcal{T}_A K_j \subset \mathbb{C}[S']_{\leq D_0}. \quad (\text{CPR5})$$

The degree bound and its entire kernel follow from

$$\mathcal{T}_A S^n = \sum_{r=0}^n \binom{n}{r} \mu_r S'^{n-r}, \quad \deg \mathcal{T}_A p = \deg p - v \quad (\deg p \geq v). \quad (\text{CPR6})$$

The latter equality uses the nonzero leading coefficient $\mu_v \binom{\deg p}{v} \text{lc}(p)$. Thus $\ker \mathcal{T}_A = L$ on every finite polynomial space and

$$K_j / (K_j \cap L) \xrightarrow{[p] \mapsto \alpha_A \mathcal{T}_A p} \mathcal{U}_j \quad \text{is an isomorphism,} \quad \dim \mathcal{U}_j = 8j - 16 - t_{0,j}. \quad (\text{CPR7})$$

This is the exact polynomial realization of the dual of the original onto map $\mathbf{B}_j$. To verify its scalar, if $F = N/E_A$, the binomial identity gives

$$\Lambda_F(\alpha_A \mathcal{T}_A p) = \alpha_A \Lambda_{E_A F}(p) = \alpha_A \sum_{a,b} N_{ab} p(\beta_{j,a,b}). \quad (\text{CPR8})$$

KRC5–6 identifies the DNE coordinates with conductor coordinates by $\alpha_A N_j Z_{\text{exp},j}$, so CPR8 is precisely $y^\top \mathbf{B}_j x$ when p represents the original kernel-value coordinate y . It is an ordinary transpose. Changing p by $K_j \cap L$ changes neither side. Changing N by a conductor multiple adds a lower exponential to F ; CPR4 proves that the polynomial on the left annihilates that entire lower span. This checks all representative choices.

J.2 The exact polynomial image of the connecting transpose

Put

$$\begin{aligned} \mathcal{V}_k &= \sum_{i=1}^4 \mathcal{T}_i \mathcal{U}_j \subset \mathbb{C}[S']_{\leq D_0}, \\ \mathcal{R}_0 &= \mathcal{T}_A(\chi_k \mathbb{C}[S]_{\leq d-1}), \quad \mathcal{R}_1 = \mathcal{T}_A(\chi_k \mathbb{C}[S]_{\leq d}), \\ \mathcal{I}_k &= \mathcal{V}_k \cap \mathcal{R}_0. \end{aligned} \quad (\text{CPR9})$$

These are finite polynomial spans and intersections; multiplication by arbitrary polynomials is not implicitly allowed in $\mathcal{V}_k$. The degrees of the successive generators of $\mathcal{R}_e$ are $q + a - v$, with the full leading coefficients in CPR6. Consequently $\dim \mathcal{R}_e = d + e$, and

$$\mathcal{R}_1 \cap \mathbb{C}[S']_{\leq D_0} = \mathcal{R}_0. \quad (\text{CPR10})$$

For $p \in K_j$, CPR4 and root addition imply that $\mathcal{T}_i \mathcal{T}_A p$ vanishes at every original degree- $(k-8)$ lower root: after each degree-four shift it is evaluated at a degree- $(j-8)$ lower root. Hence every element of $\mathcal{V}_k$ annihilates the full source lower-evaluation span.

Let $\mathcal{S}_k^{\text{raw}}$ be the source analytic space of CIT4 and PDS3, including every lower gauge. Its annihilator on polynomials through D_e is exactly $\mathcal{R}_e$. For a direct proof, multiplication by E_A

changes pairing on $\mathcal{T}_{Ap}$ to pairing of the upper numerator on p . Every $\chi_k S^a$ therefore annihilates it. The numerator Vandermonde makes that analytic space have dimension $q - v$; the $d + e$ independent polynomials just displayed have complementary dimension $D_e + 1 - (q - v)$. This proves the full annihilator equality.

Write the unchanged coefficient maps

$$B = B_{\text{cof}}, \quad \mathbb{B} = \text{diag}(\mathbf{B}_j, 4), \quad \Phi = \mathbb{B}B.$$

A tuple $u = (u_1, \dots, u_4) \in \mathcal{U}_j^4$ is the dual of a unique original target coordinate tuple. Its pullback to a finite source germ F is, with the positive degree-zero differential,

$$\sum_i \Lambda_{B_i F}(u_i) = \Lambda_F \left(\sum_i \mathcal{T}_i u_i \right). \quad (\text{CPR11})$$

This follows by the same complete binomial convolution as CPR8. Every source lower evaluation is killed as proved after CPR10. The transpose of Φ therefore kills the tuple exactly when the polynomial $\sum_i \mathcal{T}_i u_i$ belongs to $\mathcal{R}_e$. Since it has degree at most D_0 , this condition is independent of e by CPR10, and its polynomial image is exactly $\mathcal{I}_k$. This identifies the entire boundary annihilator and its image, rather than replacing it by a dimension hypothesis.

For the second endpoint retain the four original polynomials

$$R_i(S) = \frac{\mathcal{T}_i \chi_j(S)}{\chi_k(S)}, \quad \mathcal{C}^{(4)} = \text{span}_{i=1}^4 \mathcal{T}_A(\chi_k R_i). \quad (\text{CPR12})$$

The quotient is polynomial by root addition at every degree- k node. These four polynomials are independent by the full original rank-four quadric and 25-shift Vandermonde proof LCC9–12. They are the four exact defining equations of the primitive source at $e = 1$; at $e = 0$ no such equation is imposed. The transpose of the induced original connecting map consequently has the following complete image:

$$\boxed{(T_0/W)^\vee \simeq \mathcal{I}_k, \quad (T_1/W)^\vee \simeq \mathcal{I}_k/(\mathcal{I}_k \cap \mathcal{C}^{(4)}).} \quad (\text{CPR13})$$

Here $T_e = \mathcal{S}_e + W$, $W = \text{im } B + \ker \mathbb{B}$, with exactly the original NSR10 spaces. More explicitly the target dual $(\mathbb{C}^{4t_{K,j}} / \text{im } \Phi)^\vee$ is $\ker \Phi^\top$; CPR11 sends it into the dual of the primitive source. At $e = 0$ the latter dual is $\mathcal{R}_0$. At $e = 1$ it is $\mathcal{R}_1/\mathcal{C}^{(4)}$. The image of that transpose is the dual of the original onto image map: in finite dimension, a functional on an image extends to the full target, and a target functional restricts to zero precisely when it annihilates that image. This proves both isomorphisms and their kernels. It also proves that all later Gamma cutoffs retain the same spaces: only their metrics change, while the defining coefficient map and polynomial transpose in CPR11 are fixed.

J.3 Eliminating the conductor division and the primitive complement

The preceding image has a direct original polynomial relation model before division by the conductor:

$$\begin{aligned} \mathcal{W}_k &= L + \sum_{i=1}^4 \mathcal{T}_i K_j \subset \mathbb{C}[S]_{<Q}, \\ \mathcal{G}_k &= \{g \in \mathbb{C}[S]_{\leq d-1} : \chi_k g \in \mathcal{W}_k\}, \\ \mathcal{G}_k &\xrightarrow{g \mapsto \alpha_A \mathcal{T}_A(\chi_k g)} \mathcal{I}_k \quad \text{is an isomorphism.} \end{aligned} \quad (\text{CPR14})$$

To prove surjectivity, write an element of $\mathcal{I}_k$ as $\alpha_A \mathcal{T}_A \sum_i \mathcal{T}_i p_i = \mathcal{T}_A(\chi_k g_0)$, where $p_i \in K_j$. Commutation of the literal finite translation sums has been used. CPR6 says that $\sum_i \mathcal{T}_i p_i - \alpha_A^{-1} \chi_k g_0 \in L$; thus $g = \alpha_A^{-1} g_0$ is in $\mathcal{G}_k$. Conversely its defining relation, followed by $\alpha_A \mathcal{T}_A$, gives an element of $\mathcal{V}_k \cap \mathcal{R}_0$. Finally $\chi_k g \in L$ implies $g = 0$, since $q > v$. This proves injectivity with the original scalar

intact. Thus the space is calculated from the original observation polynomials, four translations and literal divisibility by the full χ_k .

There is a finite coefficient construction of every vector of $\mathcal{G}_k$, with no F_e , Gram inverse or assumed nonzero minor. Order monomials and indices as in CPR3. Construct its kernel basis by exact elimination; call its full coefficient matrix P_K . For $0 \leq r \leq n < Q$ form the literal triangular matrices

$$(\mathbf{T}_i)_{rn} = \binom{n}{r} \sum_{a,b=0}^4 b_{i,a,b} \beta_{4,a,b}^{n-r}, \quad (\mathbf{T}_i)_{rn} = 0 \ (r > n), \quad \mathbf{W} = [E_{\{0,\dots,v-1\}}, \mathbf{T}_1 P_K, \dots, \mathbf{T}_4 P_K]. \quad (\text{CPR15})$$

Keep dependent columns. For each column perform ordinary monic Euclidean division by χ_k , recording both complete arrays:

$$\mathbf{W}(S) = \mathbf{A}(S) + \chi_k(S) \mathbf{D}(S), \quad \deg \mathbf{A} < q, \quad \deg \mathbf{D} \leq d-1. \quad (\text{CPR16})$$

If $\mathbf{Z}$ is an exact basis of the kernel of the coefficient matrix of $\mathbf{A}$, the nonzero independent columns of $\mathbf{DZ}$ form a complete basis of $\mathcal{G}_k$. In fact a column combination of $\mathbf{W}$ is divisible by χ_k if and only if its recorded remainder vanishes; its quotient is then precisely $\mathbf{DZ}$ on those coordinates. A zero quotient and zero remainder means that the original column combination was zero, so taking its image, rather than counting all remainder syzygies, removes exactly those dependencies. Every coefficient in this algorithm is specified in CPR2–3 and CPR15; the monic divisions retain all root factors. On an exact coefficient stratum Gaussian elimination selects the first nonzero pivot and terminates after at most Q rows. It neither assumes that a chosen minor is nonzero nor substitutes a generic matrix for the original one.

The next-jet comparison also has a complete scalar-polynomial form. Put $b_i = B_i(0)$, and choose the first original index i_* with $b_{i_*} \neq 0$, which exists by the entire Bezout identity at zero. Retain

$$C_i = R_i - \frac{b_i}{b_{i_*}} R_{i_*} \quad (i \neq i_*), \quad \mathcal{C}_k = \text{span}_{i \neq i_*} C_i \subset \mathbb{C}[S]_{\leq d-1}. \quad (\text{CPR17})$$

The leading coefficient of R_i is b_i ; the leading scalar of $\mathcal{T}_A(\chi_k R_i)$ is $b_i \mu_v \binom{Q}{v}$. Hence the full intersection of the four-row span with degrees through D_0 is exactly $\mathcal{T}_A(\chi_k \mathcal{C}_k)$. Its dimension is three: dependence of the three displayed combinations would contradict independence of all four original rows. It follows from CPR13–14 that the exact ranks and complete kernel dimensions are

$$\begin{aligned} g_k &= \dim \mathcal{G}_k, & a_k &= \dim(\mathcal{G}_k \cap \mathcal{C}_k), & 0 &\leq a_k \leq 3, \\ c_0 &= g_k, & c_1 &= g_k - a_k, & h_0 &= d - g_k, & h_1 &= d - 3 - g_k + a_k, \\ & & 0 &\leq c_0 - c_1 = a_k \leq 3, & h_0 - h_1 &= 3 - a_k. \end{aligned} \quad (\text{CPR18})$$

These quantities use the original common image, not separate new rank choices at the two endpoints. The complete kernels themselves are

$$\begin{aligned} \ker(\mathcal{C}_0 \rightarrow T_0/W) &\simeq \text{Ann}_{(\mathbb{C}[S]_{\leq d-1})^\vee} \mathcal{G}_k, \\ \ker(\mathcal{C}_1 \rightarrow T_1/W) &\simeq \text{Ann}_{(\mathbb{C}[S]_{\leq d-1})^\vee} (\mathcal{G}_k + \mathcal{C}_k). \end{aligned} \quad (\text{CPR19})$$

In this display $\mathcal{C}_e$ denotes the original CIT connecting cohomology image, distinguished by its endpoint subscript from the polynomial space $\mathcal{C}_k$. Use the PDS4 quotient-value coordinates and PDS11's exact truncation to identify its primitive sources. A primitive vector is killed exactly when all the transpose polynomials in CPR13 pair to zero; the common nonzero scalar α_A does not change this zero set. At the upper endpoint it must additionally annihilate precisely the three polynomials of CPR17. This proves both maps, their surjectivity onto the specified kernels, and the dimensions CPR18.

J.4 The full low-cohomology excess and what remains in the metric

The complete LCC positive connecting sequence is retained:

$$0 \longrightarrow \mathcal{C}_e \longrightarrow H^1(\mathcal{I}_e) \longrightarrow \ker U_e \longrightarrow 0. \quad (\text{CPR20})$$

It is a sequence of the defining low-cutoff coefficient spaces. Let $\mathcal{Z}_e$ be its full cycle space, $\mathcal{S}_e$ its connecting cycle subspace, and $\mathbb{J} = \ker \mathbb{B}$. The identity on representatives gives the more precise exact sequence

$$0 \longrightarrow \frac{\mathcal{Z}_e \cap \mathbb{J}}{\mathcal{S}_e \cap \mathbb{J}} \longrightarrow \frac{\mathcal{Z}_e}{\mathcal{S}_e} \longrightarrow \frac{\mathcal{Z}_e + W}{\mathcal{S}_e + W} \longrightarrow 0. \quad (\text{CPR21})$$

For the kernel, if $z = s + w$ with $s \in \mathcal{S}_e$ and $w \in W = \mathcal{B} + \mathbb{J}$, subtract its boundary part, which lies in $\mathcal{S}_e$, and represent the class by $z - s - b \in \mathcal{Z}_e \cap \mathbb{J}$. Two such representatives agree exactly modulo $\mathcal{S}_e \cap \mathbb{J}$. Surjectivity is the sum-space definition. LCC13 identifies the middle space with $\ker U_e$ using its positive double lift, retaining all six compatibility components and every lower evaluation. Thus the additional target rank is the dimension of the last quotient in CPR21; it is zero whenever that exact tail kernel is zero. Its source and target metrics remain the two distinct attained minima of LCC21–23.

The independent low-tail companion in this directory calculates the same $\ker U_e$ through the actual barycentric biform. Its full proof retains each of the four $(2, 2)$ conductor factors and gives

$$v \leq 32, \quad \kappa_e = \dim \ker U_e \leq 3 \max(v - e - 2, 0). \quad (\text{CPR22})$$

The companion supplies the full irreducibility, derivative and next-jet proofs; it is part of this source closure. In particular

$$\begin{aligned} d_{T,e} &= 0 \quad (v - e \leq 2), & \text{rank } \Psi_e &= c_e \quad (v - e \leq 2), \\ 0 &\leq d_{T,e} \leq 3 \max(v - e - 2, 0) & \text{in general.} \end{aligned} \quad (\text{CPR23})$$

All claims here are on the original simple-quartet domain. No original multiplicity has been changed to reach this domain.

At every later target cutoff N and either original source-order pair, the image in CPR13 carries the restriction of the original kernel-target quotient by $\text{im } \Phi$; its source carries the quotient of the full CIT connecting Gram by the exact kernel CPR19. CPR8 and CPR11 give the fixed algebraic transport between these coordinate presentations. If Z_h is its actual kernel basis and Z_c a complementary source lift, those source forms are respectively the restriction to Z_h and the Schur quotient in Z_c . A change of either fixed basis acts by the full congruence $R^*(\cdot)R$; its squared determinant is retained at each endpoint. Neither CPR18 nor the zero excess in CPR23 assigns the determinant of this source-to-target contraction a value. The exact polynomial space $\mathcal{G}_k$ and its three-row intersection are now a complete original algebraic presentation of the connecting rank, without primitive lift or Gamma-frame dependence. No universal numerical value for g_k or a_k is asserted in this calculation.

K The complete original barycentric low-tail calculation

This standalone calculation concerns the actual last differential of LCC5–6 and LCA37–39. It retains the entire original root rectangle, every coefficient of the invertible original period matrix, and the ordered exterior signs. No source or provider is revised by this note.

K.1 Original objects and the coefficient kernel

Fix $k = 4l + 1 \geq 9$, $j = k + 4$, $Q = (j + 1)^2$, $e \in \{0, 1\}$, $0 \leq v \leq 80$, and put

$$L_e = Q - v + e, \quad \ell_e = \max(v - e, 0), \quad \beta_* = \frac{1}{2} - \delta - i\gamma, \quad \alpha = 2\delta, \quad \eta = 2i\gamma, \quad \delta, \gamma > 0. \quad (\text{BTK1})$$

The original roots and their monic polynomial are

$$\lambda_{ab} = j\beta_* + a\alpha + b\eta \quad (0 \leq a, b \leq j), \quad \chi_j(S) = \prod_{a,b=0}^j (S - \lambda_{ab}). \quad (\text{BTK2})$$

They are pairwise distinct: equality of two roots first equates their real index differences and then their imaginary index differences. Let $\mathbf{H}$ be the unchanged invertible original 4 by 4 matrix, and retain

$$L_i(Z, W) = (\mathbf{H}(ZW, Z, W, 1)^T)_i, \quad h_i = L_i^4 = \sum_{r,t=0}^4 b_{i,r,t} Z^r W^t, \quad B_i(z) = e^{4\beta_* z} h_i(e^{\alpha z}, e^{\eta z}). \quad (\text{BTK3})$$

An input of the last coefficient differential is ordered as $(E_{123}, E_{124}, E_{134}, E_{234})$. Its scalar output is exactly

$$-B_4 E_{123} + B_3 E_{124} - B_2 E_{134} + B_1 E_{234} = \sum_{i=1}^4 \varepsilon_i B_i E_{\hat{i}}, \quad \varepsilon_i = (-1)^{i-1}, \quad (\text{BTK4})$$

where $\hat{i}$ denotes the increasing complementary triple. Each $E_{\hat{i}}$ belongs to the full degree- j exponential space and satisfies $E_{\hat{i}}^{(n)}(0) = 0$ for $0 \leq n < L_e$. The dimension of the kernel of BTK4 on these four tail spaces is the original κ_e . Indeed root addition transports the original coefficient convolution to BTK4. The degree- $(j + 4)$ exponential map is injective by its distinct roots and full Vandermonde. Its output has zero first L_e moments by Leibniz's rule; therefore the tail extraction $S_{j+4,e}$ in LCC5 is injective on that output. Thus this is exactly $\ker U_e$, not a larger germ or truncated kernel.

K.2 Exact barycentric tail coordinates

Define the complete original generator and its biform presentation by

$$\phi_j(z) = \sum_{a,b=0}^j \frac{e^{\lambda_{ab} z}}{\chi'_j(\lambda_{ab})}, \quad F_j(Z, W) = \sum_{a,b=0}^j \frac{Z^a W^b}{\chi'_j(\lambda_{ab})}, \quad \phi_j(z) = e^{j\beta_* z} F_j(e^{\alpha z}, e^{\eta z}). \quad (\text{BTK5})$$

Every coefficient in F_j is nonzero. In particular its bidegree is exactly (j, j) and its constant coefficient is nonzero. Lagrange interpolation of S^n for $0 \leq n < Q$ gives

$$S^n = \sum_{a,b} \lambda_{ab}^n \frac{\chi_j(S)}{(S - \lambda_{ab}) \chi'_j(\lambda_{ab})}.$$

Comparison of the coefficient of S^{Q-1} proves

$$\sum_{a,b} \frac{\lambda_{ab}^n}{\chi'_j(\lambda_{ab})} = \begin{cases} 0 & 0 \leq n < Q - 1, \\ 1 & n = Q - 1. \end{cases} \quad \text{Hence } \text{ord}_0 \phi_j = Q - 1. \quad (\text{BTK6})$$

If $\ell = \ell_e > 0$, then $L_e = Q - \ell$. The exact tail map is

$$p(S) \in \mathbb{C}[S]_{<\ell} \mapsto E_p(z) = p(\partial_z)\phi_j(z) = \sum_{a,b} \frac{p(\lambda_{ab})}{\chi'_j(\lambda_{ab})} e^{\lambda_{ab}z}. \quad (\text{BTK7})$$

If p has degree r and leading coefficient $p_r \neq 0$, BTK6 gives

$$\text{ord}_0 E_p = Q - 1 - r, \quad E_p^{(Q-1-r)}(0) = p_r. \quad (\text{BTK8})$$

Thus BTK7 is injective and its image lies in the required tail. That tail has dimension ℓ because its first $Q - \ell$ Vandermonde rows are independent. Consequently BTK7 is an isomorphism onto the entire original tail. For $\ell = 0$, its first Q moments already vanish and the tail is zero, also when $L_e = Q + 1$.

For completeness, the exact conversion to LCC5's remaining-derivative coordinates is as follows. Write $p(S) = \sum_{r=0}^{\ell-1} p_r S^r$ and put

$$m_n = \sum_{a,b} \frac{\lambda_{ab}^n}{\chi'_j(\lambda_{ab})}, \quad y_t = E_p^{(L_e+t)}(0) = \sum_{r=0}^{\ell-1} m_{L_e+t+r} p_r \quad (0 \leq t < \ell). \quad (\text{BTK9})$$

The displayed matrix has zero entries for $t+r < \ell-1$ and ones for $t+r = \ell-1$, so its determinant is $(-1)^{\ell(\ell-1)/2}$. LCA38's Taylor coordinates are precisely $y_t/(L_e+t)!$. This includes every factorial and supplies the inverse coordinate map through the invertible displayed matrix.

Both original native families are retained: at cochain degree p , $s_p = 1$ for every p , or $s_p = k+4p$ for every p , with $c'_p = (k+4p)/2 - 4$. The native evaluation matrix is obtained from the ordinary moment matrix by the full original triangular polynomial frame matrix. Its diagonal at row n is

$$\frac{i^{-n}}{\sqrt{(2\pi)^{s_p/2} n! (s_p/2)_n}} \neq 0. \quad (\text{BTK10})$$

Thus it has exactly the same kernel. This proves the map from both original native presentations to BTK7; no norm, phase or mass is used to change the coefficient kernel.

K.3 Irreducibility from maximal contact

For any nonzero polynomial G of bidegree at most (a, b) , $G(e^{\alpha z}, e^{\eta z})$ is a linear combination of at most $(a+1)(b+1)$ exponentials with distinct exponents. A nonzero such combination has order at most the number of exponents minus one: otherwise its first that many derivatives form an invertible Vandermonde annihilating its entire coefficient vector. Hence

$$\text{ord}_0 G(e^{\alpha z}, e^{\eta z}) \leq ab + a + b. \quad (\text{BTK11})$$

Suppose $F_j = GH$ with both factors nonconstant. Polynomial degrees in each variable add in this integral domain. If G has bidegree (a, b) , H therefore has bidegree $(j-a, j-b)$. Their orders add, and BTK11 gives

$$\text{ord}_0 F_j(e^{\alpha z}, e^{\eta z}) \leq Q - 1 - \{a(j-b) + b(j-a)\} < Q - 1. \quad (\text{BTK12})$$

For $0 \leq a, b \leq j$, the braces are nonnegative and vanish only at $(a, b) = (0, 0)$ or (j, j) ; either case would make a factor constant. This contradicts BTK5–6, proving F_j irreducible in $\mathbb{C}[Z, W]$. The polynomial ring is a unique factorization domain (successive application of Gauss's lemma to $\mathbb{C}[Z]$), so F_j is prime. Its bihomogenization is also irreducible. A bihomogeneous factorization would dehomogenize to one just excluded unless one factor were supported entirely on the two divisors at infinity. Such a factor is a monomial in the homogenizing variables, which cannot divide this biform because its original corner coefficients are all nonzero.

There is also a useful stronger structural statement. Every nonzero tail polynomial $F_p = \sum p(\lambda_{ab})Z^aW^b/\chi'_j(\lambda_{ab})$ is irreducible if $1 \leq \ell \leq j$. Its order is at least $Q - \ell$. If its degree in either variable were at most $j - 1$, BTK11 would bound that order by $j(j + 1) - 1 = Q - j - 2$, a contradiction. Thus its bidegree is (j, j) . For a nontrivial splitting, the positive integer in braces in BTK12 is at least j . To verify this, if $1 \leq a \leq j - 1$, its minimum over $0 \leq b \leq j$ occurs at an endpoint and equals $j \min(a, j - a) \geq j$; the cases $a = 0, j$ give the same bound directly. Such a splitting would bound the order by $Q - j - 1 < Q - \ell$, another contradiction. This structural statement alone asserts no kernel vanishing for $\ell \geq 3$.

K.4 Independence of the four unchanged fourth powers

We include the complete coefficient argument used in LCA20. Put $Q_0(t) = t_1t_4 - t_2t_3$. It is primitive and linear in t_1 over the fraction field of $\mathbb{C}[t_2, t_3, t_4]$, hence irreducible by Gauss's lemma and prime. A homogeneous polynomial vanishing on $(ZW, Z, W, 1)$ vanishes after substitution $t_1 = t_2t_3/t_4$ on $t_4 \neq 0$. Its Laurent remainder is zero, so after clearing powers of t_4 it is divisible by Q_0 ; primality and $Q_0 \nmid t_4$ remove that power. This proves the complete homogeneous Segre ideal assertion.

If $\sum c_i h_i = 0$ with some $c_i \neq 0$, the substitution $x = Ht$ therefore gives a nonzero polynomial identity

$$\sum_{i=1}^4 c_i x_i^4 = Q_H(x) R_2(x), \quad Q_H(x) = Q_0(H^{-1}x), \quad (\text{BTK13})$$

with Q_H a rank-four quadratic and R_2 homogeneous of degree two. If some $c_i = 0$, the left side has degree zero in x_i . Additivity of this polynomial degree forces Q_H independent of x_i , contradicting rank four. Thus all c_i are nonzero. The two quadrics have a nonzero common zero: restrict R_2 to the Segre quadric parametrized after H ; its restriction is a biform of degree $(2, 2)$. If it is identically zero any point works. Otherwise choose a projective w for which it is a nonzero binary quadratic in z ; that quadratic has a projective root over $\mathbb{C}$. At the resulting nonzero common zero, differentiating BTK13 gives $\nabla \sum c_i x_i^4 = 0$. But its components are $4c_i x_i^3$, which cannot all vanish at a nonzero point. This contradiction proves

$$h_1, h_2, h_3, h_4 \text{ are linearly independent over } \mathbb{C}. \quad (\text{BTK14})$$

K.5 Exact vanishing when each tail has dimension at most two

Define the original scalar differential operator on biforms

$$\mathcal{D}_j = j\beta_* + \alpha Z \partial_Z + \eta W \partial_W. \quad \partial_z \phi_j = e^{j\beta_* z} (\mathcal{D}_j F_j)(e^{\alpha z}, e^{\eta z}). \quad (\text{BTK15})$$

The irreducible polynomial F_j does not divide $\mathcal{D}_j F_j$. Indeed this derivative polynomial is nonzero and has bidegree at most (j, j) . Divisibility would force $\mathcal{D}_j F_j = c F_j$ with a constant c . Comparison of every nonzero coefficient in BTK5 would then give $j\beta_* + a\alpha + b\eta = c$ for every a, b , which contradicts the distinct roots. Thus

$$\gcd(F_j, \mathcal{D}_j F_j) = 1. \quad (\text{BTK16})$$

If $\ell_e = 1$, BTK7 says $E_{\hat{e}} = b_i \phi_j$. BTK4 becomes $\phi_j \sum_i \varepsilon_i b_i B_i = 0$. The nonzero analytic factor can be removed, and the exact exponential-to-biform injection gives $\sum_i \varepsilon_i b_i h_i = 0$. BTK14 forces every $b_i = 0$.

If $\ell_e = 2$, write uniquely $E_{\hat{e}} = (a_i \partial_z + b_i) \phi_j$. Set

$$A(Z, W) = \sum_i \varepsilon_i a_i h_i(Z, W), \quad B(Z, W) = \sum_i \varepsilon_i b_i h_i(Z, W).$$

BTK4, BTK15 and the injective degree- $(j + 4)$ exponential map give the exact polynomial identity

$$A \mathcal{D}_j F_j + B F_j = 0. \quad (\text{BTK17})$$

Primality of F_j and BTK16 imply $F_j \mid A$. But A has bidegree at most $(4, 4)$ whereas F_j has bidegree (j, j) and $j \geq 13$. Additivity of degrees makes this impossible for nonzero A , so $A = 0$. BTK17 then gives $B = 0$. BTK14, with every sign ε_i nonzero, proves $a_i = b_i = 0$ for all four components. Including the empty tail gives

$$\boxed{\ell_e = \max(v - e, 0) \leq 2 \implies \kappa_e = 0, \quad \rho_e = 4\ell_e.} \quad (\text{BTK18})$$

The stated domain is exact: both endpoints vanish for $0 \leq v \leq 2$; the $e = 1$ endpoint also vanishes for $v = 3$. For $v = 3$ only, this gives $\kappa_1 = 0$ without assigning a value to κ_0 . The conclusion holds at every permitted k , for every original invertible period matrix, and in both original native families through BTK10.

K.6 An exact lower-degree polynomial presentation for every tail

This final identity retains the complete remaining problem at arbitrary ℓ_e . Put $\nu_{rt} = \beta_{4,r,t} = 4\beta_* + r\alpha + t\eta$ and

$$R_{rt}(S) = \frac{\chi_{j+4}(S)}{\chi_j(S - \nu_{rt})} \in \mathbb{C}[S], \quad \deg R_{rt} = 8j + 24. \quad (\text{BTK19})$$

The denominator roots are exactly the shifted original j -rectangle, a subset of the complete $(j+4)$ -rectangle, so division is exact. For the unique tail polynomials p_i in BTK7, define

$$\mathcal{U}(p_1, p_2, p_3, p_4)(S) = \sum_{i=1}^4 \varepsilon_i \sum_{r,t=0}^4 b_{i,r,t} p_i(S - \nu_{rt}) R_{rt}(S). \quad (\text{BTK20})$$

At any upper root Λ , if $\Lambda - \nu_{rt}$ is a lower root, differentiating $\chi_{j+4} = \chi_j(S - \nu_{rt}) R_{rt}$ at Λ gives $R_{rt}(\Lambda) = \chi'_{j+4}(\Lambda) / \chi'_j(\Lambda - \nu_{rt})$. If it is not a lower root, then $R_{rt}(\Lambda) = 0$ by BTK19. Consequently $\mathcal{U}(p)(\Lambda) / \chi'_{j+4}(\Lambda)$ is exactly the full output coefficient at $e^{\Lambda z}$ in BTK4, including the absent shifted nodes and all repeated contributions to present nodes. Moreover

$$\deg \mathcal{U}(p) \leq 8j + \ell_e + 23 < (j+5)^2, \quad \ker U_e \simeq \ker \mathcal{U}. \quad (\text{BTK21})$$

For the strict inequality, the difference is $Q - \ell_e + 1 > 0$ since $Q \geq 196$ and $\ell_e \leq 80$. Vanishing of the output is therefore equivalent to vanishing of this entire polynomial: its values at all $(j+5)^2$ distinct upper roots determine it. BTK7–9 give the exact input coordinate isomorphism. BTK20–21 is an equality for arbitrary tail size; it does not assign an unproved rank for $\ell_e \geq 3$.

K.7 Successive original moments and the improved uniform bound

Fix the same j , original roots and quartic symbols, and for each integer $0 \leq \ell \leq 80$ let K_ℓ be the kernel of BTK4 restricted to the four original degree- j exponentials whose derivatives of orders $0, \dots, Q - \ell - 1$ vanish. For $\ell = 0$ this is the zero space. These are coefficient subspaces of the same four full original exponential spaces, and

$$K_{\ell-1} \subset K_\ell, \quad J_\ell : K_\ell \longrightarrow \mathbb{C}^4, \quad J_\ell(E) = (E_i^{(Q-\ell)}(0))_{i=1}^4, \quad \ker J_\ell = K_{\ell-1}. \quad (\text{BTK22})$$

The last equality follows from adding exactly the single missing derivative in each input component; the entire convolution identity and all earlier zero derivatives remain unchanged.

Differentiate the exact identity BTK4 precisely $N = Q - \ell$ times at zero. Leibniz's formula is

$$0 = \sum_{i=1}^4 \varepsilon_i \sum_{r=0}^N \binom{N}{r} B_i^{(r)}(0) E_i^{(N-r)}(0) = \sum_{i=1}^4 \varepsilon_i B_i(0) E_i^{(N)}(0). \quad (\text{BTK23})$$

For $r > 0$ the derivative of E is below N and vanishes; the surviving binomial coefficient is $\binom{N}{0} = 1$. The row $(\varepsilon_i B_i(0))_i$ is nonzero. Directly from the full original matrix, $B_i(0) = L_i(1, 1)^4$ and the

column $(L_i(1, 1))_i = \mathbf{H}(1, 1, 1, 1)^\top$ is nonzero because $\mathbf{H}$ is invertible. Thus the image of J_ℓ lies in the specified three-dimensional hyperplane. Rank-nullity gives

$$0 \leq \dim K_\ell - \dim K_{\ell-1} \leq 3. \quad (\text{BTK24})$$

No factorial is suppressed: if the target coordinates were Taylor coefficients, each displayed component would be divided by the common nonzero $N!$, leaving the same hyperplane and rank. The target here is the derivative coordinate specified in BTK22.

BTK18 proves $K_0 = K_1 = K_2 = 0$ on the entire present $j \geq 13$ domain. Summing BTK24 from 3 to ℓ therefore proves

$$\dim K_\ell \leq 3 \max(\ell - 2, 0), \quad \kappa_e \leq 3 \max(v - e - 2, 0). \quad (\text{BTK25})$$

This retains the actual kernel, rather than replacing it by an arbitrary tuple in the hyperplane. In particular for $v = 3$ the two endpoint dimensions satisfy $\kappa_1 = 0$ and $0 \leq \kappa_0 \leq 3$.

Here is the complete original factor argument for the numerical conductor range used with this bound. The original full conductor is the product of the four Segre restrictions $C_a = Q_a(\mathbf{H}t)$, where $t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^\top$ and $1 \leq a \leq 4$, of the other four quadrics in the original five-member orbit. Each Q_a is nonassociate to $Q_{\mathbf{H}}$. The homogeneous Segre ideal argument in BTK13 shows that its degree-two restriction cannot vanish identically: otherwise Q_a would be a constant multiple of $Q_{\mathbf{H}}$. Thus each nonzero C_a is a biform of degree $(2, 2)$, with the entire original coefficient array $C_a(Z, W) = \sum_{r,t=0}^2 c_{a,r,t} Z^r W^t$ retained. The exact original symbol is

$$E_A(z) = \prod_{a=1}^4 E_a(z), \quad E_a(z) = e^{2\beta_* z} C_a(e^{\alpha z}, e^{\eta z}).$$

Each factor has nine distinct possible exponents. Its first nine derivatives cannot all vanish by the invertible nine-node Vandermonde and the nonzero coefficient array. Hence $v_a = \text{ord}_0 E_a \leq 8$. Multiplication of these exact Taylor series gives

$$v = \sum_{a=1}^4 v_a \leq 32, \quad \frac{\mu_v}{v!} = \prod_{a=1}^4 \frac{E_a^{(v_a)}(0)}{v_a!} \neq 0, \quad \boxed{\kappa_0 \leq 90, \quad \kappa_1 \leq 87.} \quad (\text{BTK26})$$

This is the same full conductor product and coefficient identity spelled out in the companion `../ORIGINAL_CONDUCTOR_VANISHING_ORDER.tex`; no factor or original multiplicity is removed. The last two inequalities are BTK25 evaluated at $\ell_0 \leq 32$ and $\ell_1 \leq 31$.